



PERINATAL CARE MANUAL

4TH EDITION

MINISTRY OF HEALTH MALAYSIA

PERINATAL CARE MANUAL

4TH EDITION

*Division of Family Health Development
Ministry of Health Malaysia
2020 (4th Edition)*

FOREWORD

Malaysia has seen incredible progress in perinatal care services since our independence, which was translated into improved survival of mother and child. In the current era of Sustainable Development Goals (SDGs), provision of quality services for mother and child throughout antenatal, intrapartum and postnatal is the main strategy to further reduce the mortality rates.

This Perinatal Care Manual was first published in 2002 and has been reviewed throughout the years with the aim to provide updated guidance for healthcare providers who are involved in caring for mothers and newborn.

This fourth edition has a slightly different approach where it emphasises on the complete pregnancy-cycle for each condition. This is to prioritise person-centred health and wellbeing. Processes and procedures involved during antenatal, intrapartum and postnatal were also specifically outlined in detail.

It is my hope that this manual will complement with other existing related guidelines and improve the quality of care given to mothers and newborn. The energy, effort and commitment of the working group and editorial board to bring forth this manual are greatly appreciated.

We would like to acknowledge and express our gratitude to Jabatan Kesihatan Negeri Sabah for the permission to adopt their Sabah Shared Obstetric Care Guidelines.

Dr. Faridah binti Abu Bakar

Director

Family Health Development Division

Ministry of Health

MEMBERS OF WORKING GROUPS

(In Alphabetical Order)

Pn. Aleijjah bt. Ali

Ketua Penyelia Jururawat Kesihatan
Kanan,
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Cik Amalina bt. Muhamad

Penolong Pengarah,
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Bahagian Pemakanan, KKM

Dr. Amelia Hazreena bt. Hamidon

Pakar Perubatan Keluarga
KK Greentown, Perak

Pn. Amizam bt. Tamby Ahmad

Pengajar Jururawat
Institut Latihan KKM
Sg. Buloh, Selangor

Dr. Amy Nur Diyana bt. Mohamed Nasir

Ketua Penolong Pengarah
Sektor Kesihatan Kanak-Kanak
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dr. Angeline Wan Seng Lian

Pakar Perunding Neonatologi
Ketua Jabatan Pediatrik
Hospital Pakar Sultanah Fatimah
Muar, Johor

Dr. Azianey bt. Yusof @ Abdullah

Pakar Perubatan Keluarga
KK Kepala Batas, Pulau Pinang

Dato' Dr. Bavanandan Naidu Gopal

Pakar Perunding O&G Dan Jabatan O&G
Hospital Sultanah Bahiyah
Alor Setar, Kedah

Dr. Bong Yee Khiun

Pakar Psikiatri
Hospital Kuala Lumpur

Dr. Carol Lim Kar Koong

Pakar Perunding Kanan O&G
Ketua Jabatan O&G,
Hospital Ampang, Selangor

Pn. Chen Phaik Guan

Pengajar Jururawat
Institut Latihan KKM
Sg. Buloh, Selangor

Dr. Faridah bt Abu Bakar

Pengarah
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dato' Dr. Ghazali Bin Ismail

Pakar Perunding Kanan O&G
Ketua Jabatan O&G, Hospital Sultan
Ismail, Johor Bahru, Johor

Dr. Habsoh bt. Hat

Pakar Perunding Perubatan Keluarga,
KK Bandar Sungai Petani
Kedah

Assoc. Prof. Dato' Dr. Hamizah bt. Ismail

Head Department Of O&G
International Islamic University
Malaysia, Kuantan, Pahang

Dr. Haris Njo Suharjono

Pakar Perunding Kanan O&G
Ketua Jabatan O&G
Hospital Umum Sarawak

Dr. Irm Zarina bt. Ismail

Pensyarah Jabatan Perubatan
Keluarga Fakulti Perubatan Sains
Kesihatan, Universiti Putra Malaysia,

Dr. Izwana bt. Hamzah

Pakar Perubatan Kesihatan Awam
Pegawai Kesihatan Keluarga Negeri
JKN Perlis

Dr. J. Ravichandran

Pakar Perunding Kanan O&G,
Ketua Perkhidmatan O&G Kebangsaan,
Ketua Jabatan O&G
Hospital Sultanah Aminah,
Johor Bahru, Johor

Pn. Jamilah bt. Ahmad

Pegawai Sains (Pemakanan)
JKN Melaka

Pn. Jaminah bt. Yahaya

Ketua Penyelia Jururawat Kesihatan
JKN Sarawak

Pn. Kamisah Barukang

Ketua Penyelia Jururawat Kesihatan
Jabatan Kesihatan WP Labuan

Dr. Krishnan Kumar

Pakar Perunding Kanan O&G,
Ketua Jabatan O&G
Hospital Tuanku Jaafar,
Seremban, Negeri Sembilan

Dr. Leow Poy Lee

Pakar Perunding Kanan Neonatologi
Ketua Jabatan Pediatrik
Hospital Melaka

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Ketua Penyelia Jururawat Kesihatan
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Dr. Mairin Dulasi

Pakar O&G
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Pulau Pinang

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Pakar Perubatan Kesihatan Awam
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Bahagian Pembangunan Kesihatan
Keluarga, KKM

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Ketua Penyelia Jururawat Kesihatan
JKN Pahang

Dr. Muniswaran A/L Ganeshan

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Dr. Noraini bt. Jali

Pakar Perunding Perubatan Keluarga
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Pegawai Dietetik,
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Keluarga, KKM

Pn. Norhashimah bt. Hassan

Penyelia Jururawat Kesihatan
KK Tanah Merah, Kelantan

Pn. Norlela bt. Hassan

Ketua Penyelia Jururawat Kesihatan
PKD Tampin, Negeri Sembilan

Dr. Nurly Zahureen bt. Mustapha

Pakar Perubatan Kesihatan Awam,
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Pn. Nurul Ashikin bt. Ag Ismail

Ketua Penyelia Jururawat Kesihatan (KA)
JKN Sabah

Pn. Nurul Huda bt. Ibrahim

Pegawai Dietetik,
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Pn. Nurul Zaiza bt. Zainudin

Penolong Pengarah Kanan
Sektor Pemakanan Ibu,
Bahagian Pemakanan, KKM

Dr. Nurzeiti Yuslinda bt. Yusof

Pakar Perubatan Keluarga
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Dr. Nurzeti bt. Shaikh Ahmad

Pakar Perubatan Keluarga
Hospital Angkatan Tentera Tuanku Mizan
Kuala Lumpur

Pn. Paridah bt. Samad

Penyelia Jururawat Kesihatan
KK Selandar, Melaka

Dr. Radziah bt. Mohamad

Pakar Perubatan Kesihatan Awam,
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Pn. Rohaya bt. Ramli

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PKD Larut Matang & Selama, Perak

Pn. Rohayu bt. Mohamad

Ketua Jururawat Kesihatan,
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Dr. Rokiah bt. Mohd

Pakar Perubatan Kesihatan Awam
Timbalan Pengarah (Keluarga)
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Pn. Rokiah bt. Samin

Penyelia Jururawat
Hospital Sultanah Aminah,
Johor Bahru, Johor

Dr. Rozita bt. Zakaria

Pakar Perubatan Keluarga
Klinik Kesihatan Presint 18
Wilayah Persekutuan, Putrajaya

Pn. S. Kalyani R Sangaran

Ketua Penyelia Jururawat Kesihatan
PKD Kuala Pilah, Negeri Sembilan

Dr. Sarah bt. Awang Dahlan

Ketua Penolong Pengarah
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
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Dr. Sham Su Lia

Pakar Perubatan Keluarga
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Dr. Siti Halimatul Saadiah bt. Hasan

Pakar Psikiatri
Hospital Kuala Lumpur

Dr. Siti Khatijah bt. Abdul Rahim

Pakar Perubatan Kesihatan Awam
JKN Johor

Dr. Siti Zaleha bt. Suleiman

Pakar Perunding Perubatan Keluarga
KK Merlimau, Melaka

Dato' Dr. Sudesan Raman

Pakar Perunding Kanan O&G
Ketua Jabatan O&G, Hospital Tawau

Dr. Suresh Kumar

Pakar O&G
Hospital Sultanah Aminah, Johor Bahru

Pn. Suzana bt. Kipli

Penyelia Jururawat Kesihatan
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dr. Tan Hui Sui

Pakar Pediatrik
Hospital Ampang, Selangor

Dr. Tuty Aridzan Irdawati bt. Mohsinon

Ketua Penolong Pengarah Kanan
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dr. Umi Adzlin bt. Silim

Pakar Psikiatri
Hospital Kuala Lumpur

Pn. Wan Nora bt. Wan Salleh

Penyelia Jururawat Kesihatan
Jabatan Kesihatan Kelantan

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Dr. Zaiton bt. Yahya

Pakar Perubatan Keluarga
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Pn Zaleha bt. Samin

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PKD Melaka Tengah, Melaka

Dr. Zamzaireen bt Zainal Abidin

Ketua Penolong Pengarah
Sektor Kesihatan Remaja
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dato' Dr. Zuraidah bt. Abd Latif

Pakar Perunding Kanan Pediatrik
Ketua Jabatan Pediatrik
Hospital Ampang, Selangor

EDITORIAL COMMITTEE

CHAIRMAN

Dr. J. Ravichandran

Pakar Perunding Kanan O&G,
Ketua Perkhidmatan O&G Kebangsaan,
Ketua Jabatan O&G
Hospital Sultanah Aminah,
Johor Bahru, Johor

MEMBERS

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Jabatan O&G,
Hospital Seri Manjung, Perak

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Pakar Perunding Perubatan Keluarga
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Pakar Perubatan Kesihatan Awam
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dr. Tuty Aridzan Irdawati bt. Mohsinon

Ketua Penolong Pengarah Kanan
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
Keluarga, KKM

Dr. Sarah bt. Awang Dahlan

Ketua Penolong Pengarah
Sektor Kesihatan Ibu
Bahagian Pembangunan Kesihatan
Keluarga, KKM

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Bahagian Pembangunan Kesihatan
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Keluarga, KKM

Pn. Nur Fatin Syahirah bt. Hasbullah

Pegawai Penyelidik
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Bahagian Pembangunan Kesihatan
Keluarga, KKM

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ABBREVIATIONS AND ACRONYMS

AADK	Agensi Anti-Dadah Kebangsaan (National Anti-drugs Agency)	CEMD	Confidential Enquiries into Maternal Deaths
ABC	Alternative Birthing Centre	CHF	Congestive Heart Failure
ACE-I	Angiotensin-converting enzyme inhibitor	CKD	Chronic Kidney Disease
ACR	Albumin-to-Creatinine Ratio	CNS	Central Nervous System
ADHD	Attention Deficit Hyperactivity Disorder	CO	Carbon Monoxide
AED	Anti-Epileptic Drugs	CPD	Cephalopelvic Disproportion
AFI	Amniotic Fluid Index	CPG	Clinical Practice Guidelines
AIDS	Acquired Immunodeficiency Syndrome	CTG	Cardiotocography
AMO	Assistant Medical Officer	CTPA	Computed Tomography Pulmonary Angiogram
ANA	Antinuclear Antibody	CVD	Cardiovascular Diseases
APH	Antepartum Haemorrhage	CVS	Cerebrovascular Diseases
APS	Antiphospholipid Syndrome	CVT	Cerebral Venous Thrombosis
ARB	Angiotensin Receptor Blocker	CXR	Chest X-ray
ART	Antiretroviral Therapy	DBP	Diastolic Blood Pressure
ATD	Anti-Thyroid Drugs	DCDA	Dichorionic Diamniotic
ATT	Anti Tetanus Toxoid	DIVC	Disseminated Intravascular Coagulopathy
BCG	Bacillus Calmette–Guerin	DM	Diabetes Mellitus
BD	<i>Bis in die</i> (Latin) - Twice Daily	DOT	Directly Observed Treatment
BMI	Body Mass Index	DVT	Deep Vein Thrombosis
BP	Blood Pressure	ECG	Electrocardiogram
bpm	Beats per Minute	ECV	External Cephalic Version
BTL	Bilateral Tubal Ligation	EDD	Expected Date of Delivery
C&S	Culture and sensitivity	EEG	Electroencephalogram
CBT	Cognitive Behavioural Therapy	eGFR	Estimated Glomerular Filtration Rate

EOD	Every Other Day	HPT	Hypertension
EPDS	Edinburgh Postnatal Depression Score	HPV	Human Papillomavirus
ESA	Erythropoietin Stimulating Agent	HSG	Hysterosalpingogram
ESR	Erythrocyte Sedimentation Rate	HVS	High Vaginal Swab
ESRF	End Stage Renal Failure	IAP	Intrapartum Antibiotic Prophylaxis
FAS	Foetal Alcohol Syndrome	ICD-10	International Classification of Diseases-10
FASD	Foetal Alcohol Spectrum Disorder	ICSI	Intracytoplasmic Sperm Injection
FBC	Full Blood Count	ID	Infectious Disease
FBS	Fasting Blood Sugar	IDA	Iron-deficiency Anaemia
FHR	Foetal Heart Rate	IgG	Immunoglobulin G
FKC	Foetal Kick Chart	ILI	Influenza-like illness
FMH	Foetal Maternal Haemorrhage	IM	Intramuscular
FMS	Family Medicine Specialist	IOL	Induction of Labour
FPA	Family Planning Association	IPT	Interpersonal Psychotherapy
FRHAM	Federation of Reproductive Health Associations Malaysia	IUCD	Intrauterine Contraceptive Device
GBS	Group B Streptococcus	IUD	Intrauterine Death
GDM	Gestational Diabetes Mellitus	IUGR	Intrauterine Growth Retardation
GINA	Global Initiative for Asthma	IVF	In vitro Fertilisation
GTT	Gestational Transient Thyrotoxicosis	JKN	Jabatan Kesihatan Negeri
GSH	Group Screen and Hold	KD	Klinik Desa
GXM	Group & Cross Matching	KK	Klinik Kesihatan
HCG	Human Chorionic Gonadotropin	KUB	Kidney Ureter & Bladder
HCQ	Hydroxychloroquine	LBW	Low Birth Weight
HDN	Hemolytic Disease of Newborn	LCB	Last Childbirth
HELLP	Haemolytic Elevated Liver Enzymes and Low Platelet	LFT	Liver Function Test
HIV	Human Immunodeficiency Virus	LMWH	Low Molecular Weight Heparin

LNMP	Last Menstrual Period	OT	Operation Theatre
LPC	Labour Progress Chart	PAC	Pregnancy Assessment Centre
LPPKN	Lembaga Pendudukan dan Pembangunan Keluarga Negara	PBM	Patient Blood Management
LRBC	Low-risk Birthing Centre	PEFR	Peak Expiratory Flow Rate
LSCS	Lower Segment Caesarean Section	PHQ-4	Patient Health Questionnaire-4
LVH	Left Ventricular Hypertrophy	PIH	Pregnancy-induced Hypertension
LVS	Lower Vaginal Swab	PKD	Pejabat Kesihatan Daerah
M&HO	Medical and Health Officer	POA	Period of Amenorrhoea
MCDA	Monochorionic Diamniotic	POC	Product of Conception
MCH	Maternal Child Health	POCT	Point of Care Test
MCV	Mean Corpuscular Volume	POG	Period of Gestation
MFM	Maternal Foetal Medicine	PPC	Pre pregnancy Care
MMR	Measles, Mumps and Rubella	PPH	Postpartum Haemorrhage
MOH	Ministry of Health	PPROM	Premature Prelabour Rupture of Membrane
MTAC	Medication Therapy Adherence Clinic	PPT	Postpartum Thyroiditis
NAAT	Nucleic Acid Amplification Test	PR	Pulse Rate
NGO	Non-governmental Organisation	PrEP	Pre-exposure Prophylaxis
NNJ	Neonatal Jaundice	PROM	Prelabour Rupture of Membrane
NRT	Nicotine Replacement Therapy	PTU	Propylthiouracil
NTD	Neural Tube defect	RAADP	Routine Antenatal Anti-D Prophylaxis
NYHA	New York Heart Association	RAI	Radioactive Iodine
O&G	Obstetric & Gynaecology	RBC	Red Blood Cell
OERT	Obstetric Emergency Retrieval Team	RBS	Random Blood Sugar
OGDS	Oesophagogastroduodenoscopy	REDD	Revised Expected Date of Delivery
OGTT	Oral Glucose Tolerance Test	RME	Routine Medical Examination
OPD	Outpatient Department	RP	Renal Profile

RPR	Rapid Plasma Reagin	VDRL	Venereal Disease Research Laboratory
RSAT	Rumah Sakit Angkatan Tentera	VTE	Venous Thromboembolism
RVS	Rectovaginal Swab	WBC	White Blood Cells
SBP	Systolic Blood Pressure	WHO	World Health Organization
SFH	Symphysio-fundal Height		
SGA	Small for Gestational Age		
SI	Shock Index		
SIDS	Sudden Infant Death Syndrome		
SLE	Systemic Lupus Erythematosus		
SSRI	Selective Serotonin Reuptake Inhibitors		
STI	Sexually Transmitted Infections		
Tdap	Tetanus Diphtheria Pertussis (Vaccine)		
TDS	<i>ter die sumendum (Latin)</i> Three Times A Day		
TENS	Transcutaneous Electrical Nerve Stimulation		
TFT	Thyroid Function Test		
TIBC	Total Iron Binding Capacity		
TORCHES	Toxoplasmosis, Others (syphilis, varicella zoster, parvovirus) Rubella, Cytomegalovirus and Herpes Simplex		
TRAb	TSH receptor autoantibodies		
TSH	Thyroid Stimulating Hormone		
TTTS	Twin-to-twin Transfusion Syndrome		
TVS	Transvaginal Scan		
UPT	Urinary Pregnancy Test		
UTI	Urinary Tract Infection		

HOW TO USE THIS MANUAL

This manual is not intended to replace standard textbooks used for teaching. It is to be kept at hand at your workplace, which can be referred for guidance. The manual consists of three main components; maternal, neonatal and perinatal nutrition care. For maternal component, the manual consist of four main sections; section of processes and procedures during pre pregnancy, antenatal, intrapartum and postnatal; section of conditions from pre pregnancy till post puerperium; and section of conditions specific during antenatal, intrapartum and postnatal; and section of obstetric emergencies.

SECTION A: MATERNAL

Chapter 1 – Processes and Procedures of Pre pregnancy Care

- Focuses on specific groups of women in the reproductive age group with counselling on appropriate medical care to optimise pregnancy outcomes. It includes a risk assessment checklist and management of various conditions.

Chapter 2 – Processes and Procedures of Antenatal Care

- Describes activities and screening services for each trimester including contents of routine antenatal contacts.

Chapter 3 – Processes and Procedures of Intrapartum Care

- Provides understanding of the process of normal labour and delivery which allows optimal care for the mother and timely recognition and intervention of abnormal events.

Chapter 4 – Processes and Procedures of Postnatal Care

- Provides information for appropriate care, reassurance and early recognition of postpartum problems.

Chapter 5 – Medical complications in Pregnancy

- Provides information for appropriate and complete care to manage common conditions in each stage throughout pregnancy. It also includes pre-existing conditions prior to pregnancy.

Chapter 6 – Antenatal Complications

- It explains how to diagnose and manage common pregnancy-related conditions, which can be identified during routine examination of the mother. It provides standard operating procedures for quick reference in the management of common complications and high risk cases.

Chapter 7 – Intrapartum Complications

- Provides information on how to diagnose and manage conditions specifically during intrapartum.

Chapter 8 – Postnatal Complications

- Provides information and standard operating procedures to diagnose and manage common conditions specifically during postnatal period.

Chapter 9 – Obstetric Emergencies

- Focuses on timely recognition and management of obstetric emergencies.

SECTION B: NEONATAL CARE

- Outlines the comprehensive approach to neonatal care. Flow charts and checklists are available to aid health care workers to provide quality care and to initiate and facilitate referrals when necessary.

SECTION C: PERINATAL NUTRITION CARE

- Outlines nutrition advises as a guide for health professionals to educate mothers from pre pregnancy stage until postpartum period using holistic approach.

OBJECTIVES

General objective:

- To develop a comprehensive manual and reference for general use by health care providers who are entrusted with the care of mothers and their newborns.

Specific objectives:

- To serve as a guide containing the basic knowledge and skills required in the care for women beginning at pre pregnancy and extending to the neonatal period.
- To provide management of certain common conditions which occur during the different stages of pregnancy and neonatal period.
- To serve as a guide for health care providers to meet the expected standard of care in the delivery of respective services in an endeavour to improve maternal and neonatal outcomes and reduce morbidity and mortality.

SECTION A: MATERNAL CARE

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CHAPTER 1

PROCESSES AND PROCEDURES OF PRE PREGNANCY CARE

1.1 INTRODUCTION

‘Every mother has the right to expect her baby to be born alive and healthy just as every baby has the right to a living and healthy mother.’

Making pregnancy safer is an important component of maternal and child health (MCH) services. The profile of a woman embarking upon pregnancy changes as our nation develops. A greater number of them are being categorised as high risk pregnancies. Pre pregnancy care with early intervention and treatment can reduce the incidence of maternal and neonatal complications in these women.

The couple or women with good physical and psychological health, living in a good socioeconomic environment, will benefit both the mother and child. As such, pre pregnancy assessment and consultation can assist the couple and women in reproductive age to choose the appropriate time to conceive and thus reduce the risk of complications to the mother and baby.

Definition:

Preconception care is the provision of biomedical, behavioural and social health interventions to women and couples before conception occurs. It aims at improving their health status, and reducing behaviours and individual and environmental factors that contribute to poor maternal and child health outcomes. Its ultimate aim is to improve maternal and child health, in both the short and long term.

(Meeting to Develop A Global Consensus on Preconception Care to Reduce Maternal and Childhood Mortality and Morbidity. Geneva, World Health Organization, 2013).

1.2 RATIONALE

In making pregnancy safer, policies are primarily focused on optimising antenatal and intrapartum care. The increase in the number of high risk pregnancies requires readily available formalised pre pregnancy care services. As such, pre pregnancy care should be formalised into our health care services. Evidence

suggests that appropriate pre pregnancy care has improved pregnancy outcomes.

Analysis of maternal death from published reports on Confidential Enquiries into Maternal Deaths (CEMD) 2012-2014 showed that four main causes of maternal deaths are complications from associated medical conditions (20.7%), postpartum haemorrhage (16.7%), hypertensive disorders in pregnancy (11.8%) and pulmonary embolism (11.5%). These causes of deaths are associated with the current prevalence of non-communicable disease (NCDs) in Malaysia. National Health and Morbidity Survey (NHMS) revealed an increasing trend of diabetes prevalence among Malaysians above 18 years old; 11.2% (NHMS 2011), 13.4% (NHMS2015) and 18.3% (NHMS2019). Overweight and obesity prevalence has also increased; overweight prevalence was 29.1% (NHMS 2006), 29.4% (NHMS 2011), 30% (NHMS 2015) and 30.4% (NHMS2019), while obesity prevalence was 14% (NHMS 2006), 15.1% (NHMS 2011), 17.7% (NHMS 2015) and 19.7% (NHMS2019). Overweight and obesity is one of the risk factors linked to adverse obstetric outcomes, as well as contributed to perinatal mortality and morbidity.

1.3 OBJECTIVES

General objective:

To provide couples, men and women in the reproductive age group with an avenue to achieve a planned, safe and successful pregnancy.

Specific objectives:

- i. To identify women with medical conditions and/or at risk if pregnant and offer them with pre pregnancy care
- ii. To enable prospective parents and women in reproductive age group to plan for pregnancy through:
 - Provision of appropriate and adequate information.
 - Health promotion and education
 - Counselling
- iii. To screen and counsel future mothers appropriately for early intervention and treatment, aimed to reduce maternal and perinatal morbidity and mortality.
- iv. To emphasise the practice of healthy lifestyle and initiative in making pregnancy safer to prospective parents and family members.

1.4 TARGET GROUPS

Women in reproductive age group (15-49 years old) with:

i. Chronic medical conditions:

- Hypertension
- Heart disease
- Diabetes mellitus
- Thyroid disease
- Epilepsy
- Bronchial asthma
- Connective tissue diseases such as SLE
- Renal disorders
- Blood disorders
- Malignancy
- Transplant surgery (example liver & renal)
- Mental illnesses (example: mood disorder, schizophrenia)

ii. Infectious diseases:

- HIV
- Active tuberculosis under treatment
- Hepatitis B or C

iii. Lifestyle risks:

- Smoking, alcoholism and substance abuse:
These may have teratogenic effects resulting in foetal abnormalities and growth restriction.
- Obesity: Metabolic disorders have a detrimental effect during pregnancy both on the fetus and mother. It may also affect the mode of delivery.

Note: Clients with inherited structural or genetic abnormalities and family history of genetic disorder require shared care with related specialties.

1.5 ENTRY POINTS

- i. Outpatient Department (OPD)
- Wellness Clinic

- Community Clinic
 - Premarital HIV Screening Programme
 - Adolescent Health Services
 - Community Outreach Programme
 - Non-communicable Diseases Programme
- ii. Maternal and Child Health (MCH) Services
- Family Planning Services
 - Child Health Services
 - Postnatal Services
- iii. Specialised Clinic
- Cardiology Clinic
 - Nephrology Clinic
 - General Medicine Clinic
 - Paediatric Clinic
 - Obstetrics & Gynaecological Clinic
 - Other specialist clinic
- iv. Hospital Inpatient (All disciplines)
- v. Ambulatory Care Centre
- vi. Others:
- LPPKN (Lembaga Penduduk & Pembangunan Keluarga Negara)
 - RSAT (Rumah Sakit Angkatan Tentera)
 - FPA (Family Planning Association)
 - FRHAM (Federation of Reproductive Health Association of Malaysia)
 - University Hospitals
 - General Practitioners
 - Private Medical Centres

1.6 PLACE OF PRE PREGNANCY CARE SERVICES

- O&G Specialist Clinic
 - Coordinator/ provider of pre pregnancy care services at hospital level, preferably under the supervision of Maternal Foetal Medicine Specialist.
 - Other specialist clinics (i.e., medical/surgical/psychiatric etc.) should also involve actively in referring to pre pregnancy care services.
- Health Clinic

- At primary care level, pre pregnancy care will be integrated into current (MCH/OPD) services, headed by Family Medicine Specialists (FMS)/ Medical & Health Officer (M&HO).
- Outpatient services at district hospitals
 - Hospital without specialists (visiting O&G Specialists and other specialists of other disciplines).

1.7 SETTING UP A PRE PREGNANCY CARE SERVICE

- Scope
 - Identification of clients
 - Receive referral
 - Assessment
 - Management :
 - Therapeutics
 - Referrals
 - Counselling
 - Supplementation
 - Health education
- Clinic Schedule
 - Integrated/dedicated
- Specialties (as appropriate)

1.8 FLOW PROCESS (refer **Appendix 1-1** and **Appendix 1-2**)

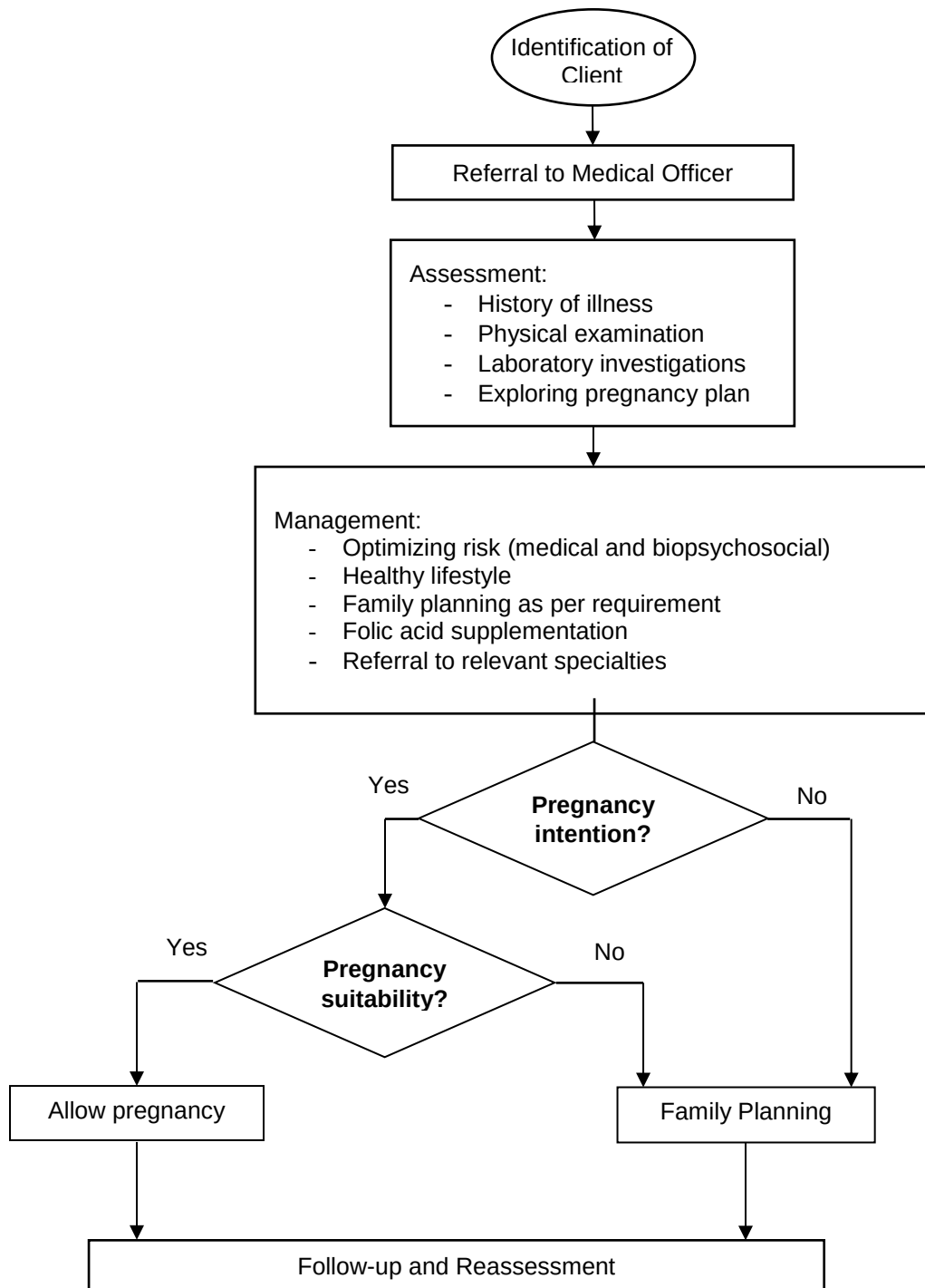
1.9 MAIN ACTIVITIES DURING A PRE PREGNANCY VISIT

- i. Identification of pre pregnancy risk factors (**Appendix 1-3**)
- ii. Assessment of the client:
 - History taking
 - Physical examination
 - Clinical laboratory tests
- iii. Appropriate management according to identified risk factors
 - Appropriate treatment and management
 - Investigations
 - Health education
 - Counselling
 - Appropriate referral

iv. Criteria for referral:

- For determination of fitness for pregnancy in certain medical conditions (e.g., heart disease, connective tissue disease).
- Shared care for preparation of pregnancy especially among clients with poorly optimised conditions.

FLOWCHART FOR PRE PREGNANCY CARE



APPENDIX 1-2

WORK PROCESSES

No.	WORK PROCESS	RESPONSIBILITY
1.	Identification of client; Identify/receive referral from entry points as mentioned in paragraph 1.5 • Outpatient clinic • Maternal and child health clinic • Specialist Clinics in Hospital • Ambulatory care centre • Private Clinic / hospital • NGO and others	Paramedics / Medical Officers / Specialist
2.	Assessment by Medical Officer/ Specialist : • Full history taking • Physical examinations • Laboratory Investigations • Exploring pregnancy intention and plan • Suitability of pregnancy assessment • Fill-up first assessment form (Appendix 1-4)	Medical Officer / Specialist
3.	Management: • Optimising risk (medical and biopsychosocial) • Healthy lifestyle • Family planning as per requirement • Folic acid supplementation • Referral to relevant specialties	Paramedics Medical Officer FMS Dietician/ Nutritionist Specialists
4.	Follow-up schedule Reassessment of risk and pregnancy intention	Paramedics Medical Officer FMS Dietician Specialists

PRE PREGNANCY RISK FACTORS

General Risk factors

- Age
 - Women less than 18 years old - Teenage pregnancies are associated with poor maternal and foetal outcomes.
 - Women above 40 years old - Advanced maternal age is associated with higher prevalence of medical illnesses and foetal chromosomal abnormalities.
- Lifestyle
 - Smoking, alcoholism and substance abuse: These may have teratogenic effect resulting in foetal abnormalities and growth restriction
 - High risk sexual behavior: Increases the risk of maternal and foetal infection.
 - Obesity/underweight: Metabolic disorders have detrimental effects during pregnancy both on the fetus and mother. It may also affect the mode of delivery.
 - Pets: Some household pets such as cats and birds may be associated with infections (e.g., toxoplasmosis, psittacosis and bird flu). Infections or exposure of these allergens to mothers with bronchial asthma can affect a pregnant mother and may result in poor foetal outcome.
- Chronic medical conditions:
 - Hypertension
 - Heart disease
 - Diabetes mellitus
 - Thyroid disease
 - Epilepsy
 - Bronchial asthma
 - Connective tissue disease (eg.; SLE)
 - Renal disorders
 - Blood disorders
 - Malignancy
 - Transplant surgery
 - Mental illness (e.g.; mood disorder, schizophrenia)
- Infectious diseases
 - Communicable diseases (e.g., HIV, active tuberculosis, hepatitis B or C)
- Family history

- Congenital structural abnormalities
- Social history
 - Domestic violence
 - Socioeconomic status
 - Marginalised group
 - Single mothers

APPENDIX 1-4

FIRST ASSESSMENT OF PRE PREGNANCY CARE

O&G History

Menarche
 Cycle:.....
 Para :
 LCB:
 Vaccination:.....

Medical History

DM ☐
 HPT ☐
 Asthma ☐
 CVD ☐
 Mental ☐
 Others ☐

Medications

1.
 2.
 3.
 4.
 5.

Psychosocial Risk:

Smoking ☐
 Alcohol ☐
 Obesity ☐
 Substance abuse ☐
 High-risk sexual behaviour ☐
 Social risk (e.g., ☐
marginalized group,
single mother, domestic
violence)

Contraception:

Pill ☐
 Injection ☐
 IUCD ☐
 Implanon ☐
 Condom ☐
 Others (e.g., rhythm ☐
 method, withdrawal
 etc)
 Nil ☐

Physical Examination:

Height (m) :
 Weight (kg):.....
 BMI (kg/m²):.....
 BP (mmHg):.....
 PR :.....

Pallor:
 CVS:
 Resp:
 Thyroid:

Lab investigations:

Hb (g/dL):
 Pap smear :
 Others :

Pregnancy Intention: Yes/ No

Suitable for pregnancy: Yes / No

Care plan :

1.
2.
3.

Follow up:

REASSESSMENT OF PRE PREGNANCY CARE

Date	PPC Component	
	Family Planning	Method: Not relevant ☐
	Lifestyle Modification	Healthy dietary advise ☐ Exercise and weight loss advise ☐ Stress management advise ☐ Exercise and physical activity ☐ Avoidance of high risk behaviours and stop smoking ☐
	Optimisation of disease	Yes ☐ No ☐
	Referral	Yes ☐ Unit/Department..... No ☐ Not relevant ☐
	Pregnancy intention explored?	Yes ☐ No ☐
	Suitable for pregnancy?	Yes ☐ No ☐
	Folic acid supplementation	Yes ☐ No ☐ Not relevant ☐

**REFERRAL LETTER
PRE PREGNANCY CARE SERVICE**

FROM :
TO :

CLIENT'S IDENTIFICATION:

NAME : **I.C NO.:**
AGE :

BRIEF MEDICAL HISTORY :
.....

RELEVANT INVESTIGATIONS :
.....

MEDICATIONS:
.....

REASONS FOR REFERRAL:

☐ Pregnancy preparation ☐ Contraception
.....
☐ Others :
Date :

Doctor's signature, name, & designation :

FEEDBACK

To :
Client's Name :
Age : IC :
We have assessed the client above and our plan is as below:

The client is **fit /unfit** for pregnancy:

☐ We send her back for your care
☐ We will follow-up the client

Date : Doctor's signature, name & designation:
Chop :

APPENDIX 1-7

STANDARD OPERATING PROCEDURES (SOP)

Standard operating procedure is designed to assist health care providers in managing the patient. The conditions are selected based on risk factors present.

SOP 1: DIABETES MELLITUS

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
• Disease severity • Complications • Comorbidities • Glycemic control and optimisation	• FBS • HbA1c • Lipid profile • Renal profile • LFT • Urine albumin/protein • Funduscopy • ECG • Blood pressure	• Uncomplicated diabetes with complications (i.e., nephropathy, retinopathy, neuropathy, coronary heart disease etc.)	• Management according to CPG Management of Type 2 Diabetes Mellitus (6th ed), 2020 and CPG Management of Diabetes in Pregnancy, 2017 • Refer to appropriate disciplines – multidisciplinary management • Screening for complications and comorbidities • Optimisation of pregnancy risks – stabilisation of blood sugar – add insulin therapy if glycemic control not achieved • Discussion on timeline for pregnancy planning	MO/ FMS/ Physician/ Endocrinologist	Health Clinics / Hospital ± specialist

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
			(to consider patient's pregnancy needs) • Full medication review, discontinue potential teratogenic medications • Folic acid supplementation 5mg/daily at least 3 months before conception • Family planning • Pregnancy risks to mother and fetus.		
Maternal risks: 1. Polyhydramnios 2. Pre-eclampsia 3. Micro and macrovascular complications: nephropathy, retinopathy, vasculopathy, neuropathy etc. 4. Increased risk of operative delivery 5. Recurrent urinary tract infection or vulvo-vaginal infections Foetal risks: 1. Risk of neural tube defect for mothers with pre-existing diabetes (e.g., anencephaly, Arnold-Chiari malformation, spina bifida, ventriculomegaly) 2. Cardiovascular malformation: atrioventricular septal defect, transposition of great arteries 3. Macrosomia leading to shoulder dystocia and possible brachial plexus injury 4. Hypoglycemia, hypocalcemia 5. Jaundice, polycythemia Risk of intrauterine death is up to 50% in uncontrolled diabetes					

SOP 2: HYPERTENSION

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
• Disease severity • Complications • Comorbidities • Blood pressure control and optimisation	• FBS • Lipid profile • Renal profile • Urine microalbumin / protein • ECG • CXR (if indicated) • Ultrasound kidney, ureter & bladder to look for renal artery stenosis and other conditions in young hypertensions	Types: • Essential HPT • Secondary HPT Severity: • Uncomplicated HPT • HPT with target organ damage (TOD) eg: LVH, retinopathy, proteinuria • HPT with target organ complications (TOC) eg: heart failure, renal failure	• Refer to appropriate disciplines – multidisciplinary approach • Management according to CPG Management of Hypertension (5th ed), 2018 • Family planning	MO/ FMS/Physician	Health Clinic/Hospital ± specialist
1. In young HPT, the cause needs to be ascertained because: a. Some causes may be familial and can affect the fetus b. Other causes may be detrimental to the health of the women (e.g. Conn's syndrome and pheochromocytoma) 2. Risks of hypertension in pregnancy include: a. Superimposed pre-eclampsia: Essential hypertension is a predisposing factor for superimposed pre-eclampsia (risk: 15-20%). This risk may be higher in secondary hypertension (especially secondary to a renal cause) b. Intrauterine growth restriction (IUGR) c. Pulmonary oedema * Risks mentioned above are especially associated with severe uncontrolled hypertension at conception and renal hypertension 3. Risk-benefit analysis of therapeutics					

- a. Minimise number of drugs
- b. Switch to safer drug options
- 4. Therapeutics:
 - a. Methyldopa is the safest choice of drug during pregnancy. Most women on antihypertensives before pregnancy will be switched to methyldopa
 - b. Prolonged use of beta-blockers has been found to be associated with small for gestational age (SGA)
 - c. Diuretics are associated with:
 - i. reduction in utero-placental blood flow causing IUGR
 - ii. increase in viscosity of maternal blood leading to VTE
 - d. ACE inhibitors are contraindicated in pregnancy as they are associated with severe neonatal outcomes such as renal agenesis and pulmonary hypoplasia
 - e. Angiotensin receptor blockers (ARBs) are contraindicated in pregnancy and breastfeeding. They are associated with an increased risk of fetopathy (e.g. pulmonary hypoplasia, limb contractures, and calvarial hypoplasia)

SOP 3: HEART DISEASE

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
• NYHA Functional Classification • WHO Risk Classification • Heart disease with comorbidity	• FBS • Lipid profile • ECG • CXR (if indicated) • Echo-cardiography • Renal profile • Exercise stress test • Blood pressure	• NYHA Class I – IV • WHO Class I – IV	• Refer to appropriate disciplines – multidisciplinary approach • Management according to Clinical Practice Guideline Heart Disease in Pregnancy 2016 • Family counselling sessions with emphasis on contraception must be given. • Discuss and educate risk of pregnancy with cardiac disease	MO/ FMS/ Physician/ Cardiologist	Health Clinic / Hospital ± specialist
1. Cardiac disease is one of the leading causes of maternal death. Pregnancy increases cardiac workload by approximately 30% and this further increases by 20% during the intrapartum period. Diseased hearts may not be able to withstand this load. 2. Congenital heart lesions can be inherited and the risk of foetal congenital heart disease is approximately 4%. This is significantly higher than the general malformation rate of 2% 3. All women of reproductive age who attend regular follow-up at cardiology clinics should be given pre pregnancy counselling. 4. Cardiac contraindications for pregnancy: a. Primary pulmonary hypertension b. Severe secondary pulmonary hypertension c. Severe fixed output lesions (e.g., severe mitral stenosis, severe aortic stenosis) d. Cardiac failure of any cause					

e. Non-resolved peripartum cardiomyopathy of pregnancy

Since termination of pregnancy should be considered in unplanned pregnancy in the above conditions, early referral to O&G specialists should be made.

5. Therapeutics:

a. Antiarrhythmic and anti-coagulation drugs should be thoroughly discussed

i. Most antiarrhythmic drugs are well-tolerated and generally safe

ii. Warfarin use is associated with an increased risk of foetal anomalies (estimated at 5-10 %) and a high risk of late foetal loss (approximately 10%). Women requiring lifelong warfarin therapy should be counselled on risk of miscarriage and foetal demise

b. Anti-failure medication (e.g., digoxin and diuretics) can be continued in pregnancy

c. ACE inhibitors are contraindicated in pregnancy but are safe during breastfeeding

d. Statins are generally contraindicated however if it is necessary for patients with coronary artery disease and hypercholesterolemia, hydrophilic agents should be preferred over lipophilic ones

i. Lipophilic: Atorvastatin, lovastatin, and simvastatin

ii. hydrophilic: pravastatin, rosuvastatin, and fluvastatin

*(*Lipophilic statins cross the blood-brain barrier more readily, which may lead to central nervous system complaints. However, this is rare.)*

e. Low dose aspirin (100 – 150 mg OD) is safe in pregnancy and breastfeeding. However, the dose for cardiac disease is usually higher. Dose may need to be lowered during pregnancy towards term (36 weeks) due to increased risk of bleeding and transfusion

f. Beta blockers are the gold standard treatment for fixed output lesions such as mitral and aortic stenosis. In these instances, they should be continued. However, prophylaxis beta blockers should be discontinued.

SOP 4: THYROID DISEASE

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
• Hypothyroid and hyperthyroid symptoms • Stability of thyroid disease on treatment	• FBS • Lipid profile • TSH/free T4/free T3 • ECG • Blood pressure • FBC • Neck ultrasound (if indicated)	• Euthyroid • Hyperthyroid • Hypothyroid	• Refer to appropriate disciplines – MDT approach • Management according to Clinical Practice Guidelines Management of Thyroid Disorders 2019 and other relevant guidelines • Family planning	MO/FMS/ Physician/ Endocrinologist/ surgeon	Health Clinic/ Hospital ± specialist
1. When disease is well-controlled, pregnancy is usually uncomplicated. Uncontrolled disease are associated with miscarriage, IUGR, preterm labor and intrauterine foetal death (IUD) 2. Antibody positive hyperthyroidism can cause foetal thyrotoxicity as IgG can pass through placenta, but it is rare 3. Complications of pregnancy such as UTI, labour or Caesarean section may precipitate thyroid storm especially in uncontrolled disease 4. If conception occurs in uncontrolled disease, therapy such as beta-blockers can increase risk of SGA/IUGR. Achieving control of disease may take months and pose risk to the fetus 5. Women who have been treated for years with antithyroid drugs should be referred to the endocrinologist/ physician for consideration for radioactive iodine (RAI) or surgical management. Women who wish to avoid foetal exposure to drugs can be advised to await resolution of disease and weaning off of treatment. 6. If unable to wean off medications, RAI can be considered to make women euthyroid/hypothyroid. Thyroxine carries negligible risk to the fetus 7. Weigh the risk and benefits of therapeutic agents of choice. Continuation of drugs in pregnancy is vital as uncontrolled disease is detrimental to foetal and maternal health. 8. Explain increased vigilance is required in pregnancy with increased frequency of antenatal visits with shared care between combined clinics and fetomaternal specialists					

SOP 5: EPILEPSY

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
Seizure-free interval	- Neurological examination - EEG test - Regular blood test : FBC, LFT [depends on types of anti-epileptic drugs (AED)] - Therapeutic drug monitoring level if not compliant or poorly controlled seizure, or suspect overdose	Controlled / uncontrolled	- All women in the reproductive age who are diagnosed to have epilepsy and plan to conceive should be referred to the pre pregnancy service. - Refer to appropriate disciplines for multidisciplinary approach - Management according to Consensus Guidelines on the Management of Epilepsy 2017 and other relevant guidelines. - Family planning counselling – Refer WHO Medical Eligibility Criteria for Contraceptive Use 2015 for drug interaction between AEDs and hormonal contraceptives. - Folic acid 5mg daily for at least 3 months prior to conception continue at least until the end of first trimester	MO/ FMS/ Physician/ Neurologists	Hospital with specialist /Health Clinics

			to reduce the incidence of major congenital malformations and to reduce the risk of AED-related cognitive deficits		
1. Ideally, women should be advised against getting pregnant until they become seizure-free and are off AEDs. However, for various personal, cultural or religious reasons, this is seldom possible or practical. Hence, in all women with childbearing potential, the risk of teratogenicity while on AEDs and the risk of recurrent seizures if AEDs were to be withdrawn must be discussed long before they wish to conceive. The latter risk is low if the patient has been seizure-free for more than 2 years and tapering is done gradually. 2. If AED withdrawal is impossible, effort to achieve monotherapy and lowest effective dose should be attempted before conception. Switching to a less teratogenic AED should be done before conception; switching during pregnancy is likely to be not beneficial because most teratogenic effects take place in the first trimester. For the above reason, contraception should be practised till AED adjustment is achieved. 3. Shared decision should be made with the patient regarding the choice and dose of Anti-epileptic Drugs (AEDs), based on the risk to the fetus and control of seizures. 4. Enzyme-inducing AEDs can increase the chance of combined oral contraceptive failure. 5. Teratogenicity risks depend on types of AED and dose: ▪ Monotherapy : 4-8% ▪ Polytherapy : 15% ▪ Combination of valproate, carbamazepine and phenytoin : 50% ▪ Phenytoin monotherapy : not related to increased risk of major foetal malformation ▪ Valproate >1000mg daily: high risk for foetal malformation					

SOP 6: BRONCHIAL ASTHMA

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
Assessment of asthma control according to guidelines (i.e., GINA, Asthma Control Test Scoring)	• PEFR • Spirometry • CXR (if indicated)	• Well-controlled • Partly controlled • Uncontrolled	• Refer to appropriate disciplines – multidisciplinary approach • Management according to Clinical Practice Guidelines Management of Asthma in Adults 2017 or other related guideline • Family planning	MO/FMS/Physician / Respiratory physicians	Health Clinics/ Hospital ± specialist
1. The natural history of asthma during pregnancy is extremely variable. Asthma may worsen, improve or remain unchanged during pregnancy. 2. Asthma is more likely to become severe or worsen during pregnancy in women with pre-existing severe asthma. 3. During PPC counselling, the following should be emphasised: ▪ patient education on good asthma control ▪ frequent monitoring (4 – 6 weeks) ▪ maintenance, reliever and anti-leukotriene should be continued ▪ stepping down medication should be done after delivery if asthma is well controlled 4. Asthma medications are safe in pregnancy. There is a greater risk to both mother and baby if asthma is poorly controlled. These include all inhalers – reliever and preventer, and steroid tablets.					

SOP 7: SYSTEMIC LUPUS ERYTHEMATOSUS

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
• Disease activity • Autoantibody profile • Comorbidities	• Renal profile • Auto-antibodies test (anti-Ro, anti-La, anti-cardiolipin, lupus anticoagulant) • Urinalysis • ESR	N/A	• Refer to appropriate disciplines – multidisciplinary approach • Management according to guidelines (i.e. EULAR Recommendations 2016) • Family planning • Review medications – the goal is to maintain disease control on medications with best safety profile during pregnancy	MO/ FMS/ Physician/ Rheumatologist	Health Clinics/ Hospital ± specialist
1. Assessment of the risk of pregnancy including: ▪ Disease activity and major organ involvement ▪ Hypercoagulability state ▪ Other concurrent medical disorders that may impact pregnancy ▪ Previous obstetric outcome 2. Pregnancy is allowed if: ▪ Disease is quiescent for ≥ 6 months ▪ BP well controlled ▪ eGFR> 60ml/min ▪ proteinuria <1g/day (proteinuria 2+) 3. Use of hydroxychloroquine (HCQ) is recommended in women with SLE preconceptionally and throughout pregnancy. Discontinuation of HCQ is related to an increased risk for SLE exacerbations during pregnancy					

SOP 8: RENAL DISEASE

Assessment	Lab Investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
- CKD Staging (CKD 1 – 5 with or without proteinuria) - Assessment for other concurrent medical conditions	- FBS - Lipid profile - Renal profile - Microalbuminuria 24hrs urine protein/albumin-to-creatinine ratio (ACR) - eGFR - Ultrasound KUB - ECG - CXR (if indicated) - lupus anticoagulant, ANA - HIV, VDRL, Hep B/C (for women on dialysis)	- CKD Stage 1 – 5 - Renal disease with comorbidity	- Refer to appropriate disciplines – multidisciplinary approach - Management according to Clinical Practice Guidelines Management of Chronic Kidney Disease 2018 - Family planning	MO/ FMS/ Physician/ Nephrologist	Health Clinic/ Hospital ± specialist
Maternal Risk - Renal workload is tremendously increased in pregnancy with an increased GFR and a physically enlarged kidney - Rate of renal function deterioration and worsening of proteinuria during pregnancy correlates significantly with CKD stages. - Renal function deteriorates more in CKD stage 3/4 compared with stage 2 (60% vs 14.3%) - Doubling of proteinuria as CKD stage progresses are 20.5%, 86.5% and 70% in stage 1, 3 and 4 – 5 respectively - Women with severe impairment need daily dialysis during pregnancy, which is associated with increased morbidity and mortality. The only risk modifying intervention is renal transplantation. Realistic options should be offered and counselled accordingly. - All women with unexplained proteinuria/hematuria should have a referral to medical for renal work-up and be referred to the pre pregnancy clinic - Adverse maternal outcomes (pre-eclampsia, hypertension and caesarean delivery) are significantly higher as CKD stage advances.					

6. Risks of preterm delivery and IUGR correlate with maternal renal function and level of proteinuria.
7. Pregnancy may be considered in women with mild renal impairment (serum creatinine $<125 \mu\text{mol/L}$), well controlled blood pressure and without significant proteinuria ($<1 \text{ g/day}$).
8. Pregnancy should be avoided in women with either:
 - a. moderate to severe renal impairment
 - b. poorly controlled hypertension
 - c. heavy proteinuria
 - d. active systemic disease
9. Aim to conceive when:
 - a. Cause of renal impairment has been optimised
 - b. Doses of drugs for treatment are minimised
 - c. Hypertension (if present) is well-controlled
10. Women with moderate to severe impairment should be advised for more permanent methods of contraception.
11. Safety of treatment:
 - a. Calcium: safe in pregnancy
 - b. Activated Vitamin D: data is scarce but reassuring
 - c. Erythropoietin is safe

SOP 9: THALASSAEMIA MAJOR

Assessment	Lab investigations and PE findings	Classification	Care plan		
			Management	Level of personnel	Level of care
- Review of transfusion requirements - Assess compliance with chelation therapy and body iron burden - Screen for end-organ damage	- FBP - Renal profile - LFT - Iron studies - Hb electrophoresis for both women and partner - DNA analysis (if indicated) - TFT - Family screening - Look for organomegaly	- Mild - Moderate - Severe - Symptomatic/asymptomatic	- Counselling on risk of pregnancy – she needs to be fully informed about how thalassaemia affects pregnancy and vice versa - Family planning - Refer for genetic counselling if the need arises - Transfusion where indicated	MO/FMS/Physician/Paediatrician	Health Clinics Hospital ± specialist
- Aggressive chelation in the preconception stage - reduce and optimise body iron burden - reduce end-organ damage - As diabetes is common in patients with thalassaemia, women with diabetes should be referred to endocrinologists (Refer SOP for PPC in Diabetes for such women) - Serum fructosamine is preferred for monitoring (target serum fructosamine concentrations < 300 nmol/l for at least 3 months pre-conception) - Assessment by a cardiologist - Echocardiogram, electrocardiogram (ECG) and T2* cardiac MRI - Assessment of liver iron concentration using a FerriScan® or liver T2* (if it is available)					

- Ideal liver iron should be < 7 mg/g (dry weight)
- 5. Bone density scan to document any pre-existing osteoporosis.
- 6. Serum vitamin D concentrations to be optimised with supplements.
- 7. To measure ABO and full blood group genotype and antibody titres
- 8. Iron chelators should be reviewed. Deferasirox or deferiprone should be ideally discontinued 3 months before conception
- 9. Offer genetic counselling if the partner is a carrier of a haemoglobinopathy that may adversely interact with the woman's genotype
- 10. Consider in vitro fertilisation/intracytoplasmic sperm injection (IVF/ICSI) with a pre-implantation genetic diagnosis (PGD) in the presence of haemoglobinopathies in both partners to avoid a homozygous or compound heterozygous pregnancy
- 11. Hepatitis B vaccination is recommended in HbsAg negative women who are transfused or may be transfused.
- 12. Hepatitis C status should also be determined.
- 13. Offer penicillin prophylaxis for all women who have undergone a splenectomy.
- 14. All women who have undergone a splenectomy should be vaccinated for pneumococcus and *Haemophilus influenzae* type B
- 15. Folic acid (5 mg) is recommended to all women who are planning to conceive to prevent neural tube defects.

SOP 10: MALIGNANCY

Assessment	Lab investigations and PE findings	Diagnostic criteria and Differential diagnosis	Care plan		
			Management	Level of personnel	Level of care
- Stage of disease - Disease activity – remission/active	Refer specific guidelines	Refer specific guidelines	Refer to multidisciplinary team (oncology, primary team, O&G team) if planning for pregnancy	MO/ FMS/ other related specialist	Secondary care
- Even though the incidence of malignancy occurring during pregnancy is low ranging from 0.02-0.1%, diagnosing it during a pregnancy is a challenge to the clinicians. The list of common cancers occurring in pregnancy are cervical cancer (1:2,000–1:10,000), breast cancer (1:3,000–1:10,000), ovarian cancer (1:10,000–1:100,000), leukaemia (1:75,000–1:100,000), lymphoma (1:1,000–1:6,000), colon cancer (1:13,000) and malignant melanoma (1:1,000–1:10,000). - With appropriate treatment of the cancer, pregnancy itself does not appear to be associated with worse cancer outcomes. Pregnant women diagnosed with breast cancer can receive treatment comparable with non-pregnant women leading to a similar survival when matched for stage at diagnosis. Foetal Risk - The risk of miscarriage and congenital anomalies does not increase with surgery. Preterm deliveries usually occurred in cases after abdominal surgery and peritonitis - The risk of major malformations, spontaneous abortions, and foetal death may be increased when chemotherapy is instituted during the first trimester and it is advisable that the chemotherapy should be delayed until 14 weeks of gestation. - Even though no evidence of teratogenic effect was demonstrated from chemotherapy exposure in the second and third trimester, the risk for low birth weight and foetal growth restriction may be increased. The reported foetal malformation from exposure to chemotherapy in the second and third trimester rates range from 1.3% to 3.8%, similar to the rate reported for the general population. Chemotherapy exposure of 5-fluorouracil, doxorubicin, and cyclophosphamide (FAC) in the second and third trimester does not cause teratogenic effects. Few studies that evaluated the use of taxanes (T), such as docetaxel (D) and paclitaxel (P), demonstrated no increase in the occurrence of foetal defects and other maternal complications when used in the second and third trimesters of pregnancy.					

4. However, trastuzumab is associated with oligohydramnios and anhydramnios, even when given in the second and third trimesters and not recommended in pregnancy.
5. Methotrexate is teratogenic. The most common abnormalities following methotrexate exposure in utero demonstrated by a few case studies are skull, hand, and limb deformities such as hydrocephalus, hypoplasia of frontal and orbital bones, micrognathia, microcephaly, hypertelorism, low set ears and upsweep of the frontal hairline, dysmorphic features, short proximal limbs, low set ears, small chin and mouth, mongoloid slant and a systolic heart murmur.
6. Women who stopped taking methotrexate even up to 6 months prior to conception are still at risks of developing spontaneous abortion.
7. The foetal central nervous system during 8 to 25 weeks after conception is sensitive to radiation, and a radiation exposure dose of >0.1 Gy could decrease the intelligence quotient. Radiation in the second and third trimesters is correlated with carcinogenic effects within the first decade of life, such as the development of solid tumours and leukaemia.

Pre pregnancy management

1. A woman with malignancy who considers pregnancy has to be managed by multidisciplinary teams taking into consideration the balance of effect on maternal and foetal health.
2. In view of the effect of treatments on fetus, women with malignancy have to consider the problems of diagnostic activities being hindered by pregnancy, the need to delay the institution of chemotherapy to the second or third trimester and also radiotherapy to postpartum, the reduced options of chemotherapy.
3. Even though, therapeutic abortion does not provide a better outcome when appropriate treatment for cancer is given, termination of pregnancy should be considered when there is a need for immediate treatment of the disease especially in abdominal and pelvis malignancy.
4. In women who wish to be pregnant, a concurrent folic acid may be helpful to reduce the neural tube defects in the fetus of women taking methotrexate.

SOP 11: RETROVIRAL DISEASE

Assessment	Lab investigations and PE findings	Diagnostic criteria and Differential diagnosis	Care plan		
			Management	Level of personnel	Level of care
- Asymptomatic - Symptomatic	- FBC - LFT - HBV/HCV - CD4/CD8 ratio - Viral load - VDRL & other STI screening - Renal profile - Physical examination for opportunistic infections/ AIDS defining complex - STI workup	WHO clinical classification criteria of severity	- Antiretroviral therapy (ART) as indicated in accordance to the Malaysian Consensus Guidelines on Antiretroviral Therapy 2017 - Family Planning: Emphasise on dual-protection	MO/FMS/ Infectious Disease Physician/ General Physician	Health Clinic/ Hospital ± specialist
- In general, the existing ART is to be continued throughout pregnancy and after delivery. - Special effort must be made to determine the current CD4 and viral load during the early stage of pregnancy. - Consultation with an infectious-disease physician is strongly recommended if the patient is experiencing virological failure. - Dual protection contraception should be advised - Look for evidence of opportunistic infections, HIV-related illnesses and evaluation for possible STIs - Counsel regarding risk of transmission to the fetus					

SOP 12: POST TRANSPLANT SURGERY

Assessment	Lab Investigations and PE findings	Diagnostic criteria and Differential diagnosis	Care plan		
			Management	Level of personnel	Level of care
- Transplanted organ function - Graft rejection - Dosage and teratogenicity of immunosuppressive drugs - Foetal and maternal risks	- Renal profile - LFT - Lung function tests - EEG, ECG - Urine protein - Blood sugar profile	N/A	- Multidisciplinary team approach - Family planning	MO /FMS /O&G specialist/ Transplantation unit	Tertiary hospitals
- Counselling on family planning and pregnancy including individualised maternal and foetal risks, alternatives, and timing. - Timing of pregnancy: avoid pregnancy during the first year post-transplantation (preferably 2 years) when the risk of rejection is greatest and immunosuppressive therapy is most aggressive. - Contraception should be emphasised during first year post-transplant (fertility is not affected by immunosuppressive medications): Long-acting reversible contraception is effective and options include the etonogestrel implant, copper intrauterine device (IUCD), and levonorgestrel-releasing IUCD. - For renal transplant patients, vaginal delivery should not be impaired, as the pelvic allograft does not obstruct the birth canal in most patients. The obstetrician should review operative notes from the transplant procedure to confirm location of the allograft and ureter. A renal ultrasound might also aid in precise location. This information should be placed in the prenatal record to guide the surgeon if a caesarean delivery is performed. - In post-transplant patients, offer advise on mediations and assisted reproductive technology if reproductive ability is impaired - The American Society of Transplantation consensus criteria for timing of pregnancy : - No rejection in the past year - Adequate and stable graft function - No acute infections that might impact the foetus					

d. Maintenance immunosuppression at stable dosing

7. Review medications for teratogenic risk. Immunosuppressants like mycophenolic acid (eg, mycophenolate mofetil), mammalian target of rapamycin (mTOR) inhibitors (e.g.; sirolimus and everolimus) should not be used during pregnancy given its high risks of adverse foetal effects and should be stopped 6 weeks before conception. Glucocorticoids (e.g. prednisolone), calcineurin inhibitors (e.g.; cyclosporine and tacrolimus) and purine synthesis inhibitors such as azathioprine are considered safe in pregnancy.
8. Vaccination against influenza, pneumococcus, hepatitis B, and tetanus should be administered before conception.

SOP 13: DEPRESSION AND ANXIETY

Assessment	Lab investigations and PE findings	Diagnostic criteria and Differential diagnosis	Care plan		
			Management	Level of personnel	Level of care
• Disease Status: Acute/Remission • Functionality e.g work, interpersonal and activity of daily living • Comorbidities: other psychiatric illness or with medical illnesses • Psychosocial risks e.g unemployment, poor social support • Suicidal risks	TFT (if it has not been done before)	Common Types: Depression • Major depressive disorder • Persistent depressive disorder (dysthymia) Anxiety • Generalised anxiety disorder • Panic disorder • Other anxiety disorder Severity: • Mild • Moderate • Severe with psychosis or suicidality	• Counsel on risk-benefit of treatment options in pregnancy. • For mild-moderate depression/anxiety with low risk of relapse, aim to achieve remission and complete treatment before pregnancy. • For moderate-severe depression and/or high risk of relapse, counsel on risk-benefit of medications. • If medication is indicated but the patient refuses, offer adequate support, refer to FMS or psychiatrist, or offer intensive psychotherapy if available.	MO/ FMS/ Psychiatrist	Health Clinic/ Hospital ± specialist
1. Depression and anxiety are common mental disorders. 2. Depression is the leading cause of disability worldwide and is a major contributor to the overall global burden of disease. 3. Depression can lead to suicide and devastating psychosocial adverse effects.					

4. Suicide is classified as direct maternal mortality in Malaysia.
5. Educate patients on the risk of untreated depression and anxiety on pregnancy and fetus
6. Emphasise on the importance of recognizing symptoms of depression and anxiety
 - a. Feeling depressed or loss of interest
 - b. Disturbed sleep and appetite
 - c. Poor concentration
 - d. Feeling lethargic
 - e. Hopeless, excessive guilt and having suicidal ideation
7. Counsel on the risk-benefit analysis of treatment options to reach a shared decision on a treatment plan.
 - f. Risks of medication i.e. Selective Serotonin Reuptake Inhibitor (SSRI): no risk of teratogenicity, low absolute risks of miscarriage, premature delivery, neonatal adaptation syndrome and primary pulmonary hypertension
 - g. Risk of untreated depression: miscarriage, premature delivery, low birth weight, poor antenatal self-care, interpersonal conflict, functional impairment, mother-baby bonding/attachment difficulties, low breastfeeding initiation, long term behavioural problems in offsprings, suicide and infanticide.
 - h. Assess previous history of depression and anxiety (how many episodes and severity) and medication history
8. Aim to complete treatment before getting pregnant if that is possible (only for mild and moderate depression/anxiety with low risk of relapse). Maintenance treatment of 6-9 months after remission
9. If medication is indicated for severe depression/anxiety but the patient refuses, offer adequate support, refer to FMS or psychiatrist, or offer intensive psychotherapy if available.
10. Arrange for psychological intervention:
 - a. Provide counselling and emotional support in primary care or O&G setting
 - b. Arrange for brief psychological interventions (eg brief cognitive behaviour therapy) by trained personnel in primary care
 - c. Referral to tertiary centre for psychological intervention
11. Baseline psychosocial investigations: corroborative history from social support
12. Early booking when pregnant

SOP 14: SEVERE MENTAL ILLNESS

Assessment	Lab investigations and PE findings	Diagnostic criteria and Differential diagnosis	Care plan		
			Management	Level of personnel	Level of care
• Disease status: in remission or presence of ongoing symptoms • medication side effects • Functionality and capacity to consent • Comorbidities: metabolic syndrome and substance use • Psychosocial risk: Social support and treatment adherence • Risk of self-neglect, aggression and self-harm	• FBS • Lipid profile	Types: • Schizophrenia • Bipolar Disorder • History of postpartum psychosis • Other psychotic disorders	• Refer to FMS or Psychiatrist • Multidisciplinary team involving psychiatrist, O&G, FMS and social worker • Counsel on risk-benefit of treatment options in pregnancy. • Counsel patient and family for compliance, monitoring and social support • Continue medication if patient is stable and medications are not contraindicated in pregnancy • If patient refuses medication or switching is required, refer immediately to FMS or Psychiatrist	MO/ FMS/ Psychiatrist	Health Clinic/ Hospital ± specialist
Remarks 1. Severe mental illness includes psychotic disorders, schizophrenia and bipolar disorder. 2. Schizophrenia and bipolar disorder have a prevalence of around 1 in 100 in the general population. Postpartum psychosis has a prevalence of 1 in 1,000 pregnancies. 3. There is an increased risk of relapse of pre-existing mental disorders during the perinatal period, usually following					

cessation of medication.

4. Preconception planning should start as soon as possible for women with severe mental illness that plan to conceive. Contraception is important if they are not planning to conceive.
5. Detailed counselling regarding the risk-benefit of medication during the pregnancy, the possibility of relapse and effects to the fetus must be given to the woman and her partner. Preconception planning will also involve discussion on types of medication used if she plans to breastfeed.
6. Medication:
 - a. Continue medication if patient is stable and medications are not contra-indicated in pregnancy
 - b. Valproate and carbamazepine should not be used in women who plan to conceive.
 - c. Lithium should not be used in the first trimester of pregnancy.
 - d. Patients on Clozapine should be referred to psychiatrist for management
 - e. Continue depot antipsychotic if patient is responding well and has a previous history of non-adherence with oral medication.
 - f. Avoid benzodiazepine unless for short-term treatment of severe anxiety and agitation
 - g. Choose the drug with the lowest risk profile for the woman, fetus and baby, taking into account previous response to medication.
 - h. Use the lowest effective dose, aiming for a single drug regime but taking into account response to medication.
 - i. Start patient early on folate supplements
7. Educate patients and partner on risk of relapse, risk-benefit of continuing or starting medication
8. Emphasise the importance of follow-up and compliance
9. Optimise patient's mental state
 - a. Optimise medication or switch to low impact medication
 - b. Offer psychological support
 - c. Close monitoring for relapse
10. Assist in substance abstinence
11. Advise on early booking when pregnant

SOP 15: SUBSTANCE USE DISORDER

Assessment	Lab investigations and PE findings	Diagnostic criteria and Differential diagnosis	Care plan		
			Management	Level of personnel	Level of care
- Obtain thorough and adequate history: substance, alcohol, tobacco - Comorbidities: medical and psychiatric conditions - Complications: bio-psycho-social - Psychosocial history: History of partner or spouse taking substances. History of forensic/criminal history	- Urine for drugs (rapid test) - Infectious disease screening	Types: - Alcohol - Nicotine - Cannabis - Amphetamine-type stimulant - Opioid	- Brief psychological intervention e.g Brief Motivational Interviewing - Nicotine replacement therapy - Methadone replacement therapy - Refer to appropriate discipline	MO/ FMS/ Psychiatrist	Health Clinic or Hospital with Psychiatrist/ Psychiatrist with subspeciality addiction
- Preconception planning should start as soon as possible for women with substance use disorder who plan to conceive. - Contraception is important if they are not planning to conceive. - Detailed counselling regarding the risk-benefit of replacement therapy and medication during the pregnancy, the possibility of relapse and effects to the fetus must be given to the woman and her partner. - Preconception planning will also involve discussion on types of replacement therapy and medication used if she plans to breastfeed. - Replacement Therapy and Medication: - Nicotine replacement therapy is safe during pregnancy under supervision of a clinician. A combination of cognitive behavioural therapy and counselling with nicotine replacement therapy is the most effective strategy to achieve smoking cessation during pregnancy. - Methadone Replacement Therapy in pregnancy improves many of the adverse consequences of maternal					

and foetal outcomes associated with untreated opioid use. Infants exposed to methadone during pregnancy typically require treatment for withdrawal symptoms after delivery. Methadone is recommended to be continued during pregnancy.

6. Counsel patients on the risk of substances on pregnancy and fetus.
7. Counsel patients and family on risk-benefit of starting or continuing replacement therapy
8. Emphasise the importance of recognizing symptoms of withdrawal for each substance.
9. Inform patients on the potential risks of substances and the availability of non-pharmacological intervention and pharmacological intervention.
10. Emphasise on the importance of follow-up and adherence
11. Aim of treatment is to reach substance abstinence before pregnancy.
12. Offer counselling and psychological support in primary care or O&G setting
13. Arrange for brief psychological interventions e.g brief motivational interviewing by trained personnel in primary care
14. Referral to tertiary centre for psychological intervention and pharmacological intervention which not available in primary care
15. Close monitoring for relapse
16. Early booking when pregnant

Reference

1. Mental Health Care in the Perinatal Period. Australian Clinical Practice Guideline. October 2017
2. NICE Antenatal and postnatal mental health. December 2014
3. Clinical Practice Guidelines on Major Depressive Disorder, Ministry of Health Malaysia 2020 (work in progress)
4. Guidelines for the identification and management of substance use and substance use disorders in pregnancy (WHO 2014)
5. Mental Health Gap Action Programme Intervention Guide Version 2.0 (WHO 2016)
6. Guidelines for the Psychosocially Assisted Pharmacological Treatment of Opioid Dependence (WHO 2009)
7. The ASSIST Project-Alcohol, Smoking and Substance Involvement Screening Test. (World Health Organization 2009)

PRE PREGNANCY HEALTH EDUCATION

1. Towards a healthy and happy family

A healthy married couple is the basic foundation for a happy family. Factors which influence the health of an individual, family and the community include:

- Lifestyle
- Genetics/familial factors
- Environmental factors

2. Practising a healthy lifestyle

Balanced diet	A diet which contains all the necessary nutrients in the right proportions according to caloric needs and right proportion. Ensure adequate fluid intake.
Social interactions	Husband and wife must be supportive and actively participate in enhancing each other's health. Couples should practice mutual respect and consent for a satisfying and equitable sexual relationship.
Good daily living habit	All men and women in reproductive age should have a healthy lifestyle; avoid unhealthy habits like smoking, consuming alcohol and other types of drug abuse.
Relaxation	Regular exercise decreases stress and lowers the risk of heart disease, stroke and hypertension.
Adequate rest and sleep	Six to eight hours of sleep a day to ensure adequate rest.

3. Genetic factors

Couple, men and women with:-

- Consanguineous marriage (e.g., autosomal recessive disorders)
- Previous child with genetic disorders (e.g., thalassaemia)
- Family history of genetic disorders (e.g., autosomal recessive disorders)
- Women at risk for aneuploidy or chromosomal anomaly (e.g., Down's Syndrome)
- Male disorders (e.g., X-linked disorders – Duchenne Muscular Dystrophy, Haemophilia)
- Unexplained/uninvestigated foetal loss should be counselled for possible genetic problems.

4. Family Planning

It is encouraged for couples to plan their pregnancy in order to contribute positively to the eventual maternal and foetal outcome especially if the women's condition is not yet optimised. Health care providers should be consulted regarding the appropriate and effective contraceptive method.

5. Pregnancy and Birth

- **Physical maturity and age of the mother**

The appropriate age for a woman to get pregnant is at the legal age 18 and above. Women above 40 years are at higher risk of pregnancy complications.

- **Preventing infections**

Men and women in reproductive age group are advised about infections such as sexually transmitted diseases as well as lifestyle diseases which can affect reproductive potential and the unborn child. Hepatitis B, varicella and rubella vaccinations may be advised to all women who are not immune.

- **Antenatal health care**

Couples who are planning to start a family should be in optimal health. A pregnant woman and her partner should attend an antenatal clinic before 12 weeks of gestation.

- **Supplementation**

Folic acid supplementation should be emphasised to all women at least 3 months prior to a pregnancy.

- **Breastfeeding**

Breast milk is the best option for the newborn as it contains all the necessary nutrients, in the right proportions, for the optimum health and growth of the newborn. Exclusive breastfeeding for the first 6 months of the newborn and encouraged to continue for at least 2 years.

- **Childbirth**

Each pregnant woman must be advised on the appropriate place of delivery.

- **Child care**

Every child must be immunised according to the recommended schedule.

6. Screening

- Cervical cancer screening (i.e.; pap smear, HPV DNA test) according to national guideline
- STI (sexually transmitted infection) screening as indicated
- Clinical breast examination
- Diabetes and hypertension screening should be offered at least annually

7. Vaccination

- **Check for rubella status.**

If the mother is not rubella immune, offer rubella vaccination together with contraception, as pregnancy is contraindicated for three months following vaccination. If the mother is unsure of her status, check for Rubella IgG to confirm immunity.

- **Hepatitis B vaccination**

Enquire regarding the patient's Hepatitis B vaccination. If the woman has not been vaccinated, offer Hepatitis B vaccination

- **Varicella (chickenpox)**

Offer and discuss benefits of vaccination pre pregnancy if the mother has never had a varicella infection. If unsure, check for varicella IgG to confirm immunity.

- **COVID-19 vaccination**

If the woman has not received COVID-19 vaccination, offer and discuss the benefit of vaccination.

PRE PREGNANCY COUNSELLING

A recommendation for pre pregnancy counselling should be given to all men and women with risk of pregnancy complications. Such counselling can reduce the incidence of maternal and foetal mortality and morbidity.

A. Objectives of pre pregnancy counselling include:

1. Conducting an initial assessment

- full history (medical, surgical, past obstetric, psychiatric, family, social and substance use history)
- general physical examination
- identification of appropriate screening tests if necessary
- discussion on pregnancy planning

2. Allaying or reducing anxiety

It is necessary to reduce anxiety in women with chronic medical illness. Counselling should include:

- The effect of pre-existing disorder on pregnancy and pregnancy on the disorder.
- The likelihood of possible recurrence of previous complications and how this may possibly be reduced (e.g., intrauterine or neonatal death, hypertension, deep vein thrombosis, miscarriage or preterm labour, mechanical problems of labour or delivery).

3. Determining fitness for pregnancy

Pregnancy should be deferred and contraception should be offered to allow further evaluation and management of known disorders or new findings (e.g., anaemia, heart disease, diabetes and hypertension). Treatment and optimisation of medical and surgical disorders may be required. Reproductive issues should be managed appropriately.

4. Follow up

- Follow-up interval : 6 -12 monthly (based on patient's condition)
- Until patient has no risk for pregnancy

B. Factors Affecting Pregnancy

1. Social behaviour

- Common social behaviours affecting pregnancy:
 - Smoking - Miscarriage, low birth weight, placenta previa, placenta abruption, infant respiratory tract infection, sudden infant death syndrome, impaired fertility
 - Alcohol - Miscarriage, foetal alcohol syndrome, placenta abruption, foetal intrauterine growth restriction (IUGR), low birth weight, central nervous system abnormalities
 - Cocaine - Abortion, premature birth, placental abruption, IUGR, congenital anomalies, neonatal central nervous system (CNS) dysfunction
 - Caffeine - Low birth weight, IUGR
- Any form of substance abuse can affect pregnancy and its outcome.

2. Medication

A potential preventable group of disorders are drug-induced anomalies. Medications during pregnancy should be avoided as far as possible.

Table 1.1: Effects of medications on pregnancy

AGENTS	EFFECTS
Anti-convulsions	Incidence of congenital malformations in children born to epileptic mothers is about 6%. This appears to be largely due to teratogenic effects of anticonvulsant. Combining drugs increases the incidence of congenital defects
Sodium valproate	Increase risk of neural tube defect to about 1/1000 pregnancies. Long term neurological implication to the baby
Lithium carbonate	Increase in cardiovascular abnormality
Warfarin	Various congenital malformations including abnormalities of the central nervous system, nose and bony epiphyses
Alcohol	Low birth weight, microcephaly, congenital heart disease and mental retardation
Androgens	Teratogenesis in first trimester, virilisation of female fetus
Atropine	Foetal tachycardia

AGENTS	EFFECTS
Beta-blockers	IUGR
Diazepam	Respiratory depression
Cyclophosphamide	Teratogenesis in first trimester
Diuretics	IUGR
Diethyl-stilboesterol	Genital anomalies, male infertility, female may develop clear cell carcinoma of the vagina many years later
Methadone	Maternal symptoms of withdrawal inducing foetal compromise and abruption Foetal complications are smaller-than-normal head size, low birth weight, IUGR, preterm delivery, unspecified structural anomalies and foetal withdrawal syndrome
Methotrexate	Neural tube defects
Mycophenolate mofetil (Immunosuppressive drug)	Multiple congenital malformations including facial, heart, renal, ear, eye and tracheo-esophageal malformations
Phenytoin	Embryopathy includes dysmorphic facial features, microcephaly and motor and intellectual retardation
Tetracycline	Tooth enamel hypoplasia and cataract
Terbutaline	Hypoglycaemia
Thalidomide	Phocomelia
Angiotensin converting enzyme inhibitor (ACE-i) and angiotensin receptor blocker (ARB)	Oligohydramnios, bone malformation, prolonged hypotension and renal failure

3. Nutritional status

Nutritional deficiency in women of reproductive age affects not only the general health but also the fertility capacity. Folic acid supplementation is essential to prevent neural tube defects.

4. Medical history

Pre-existing medical conditions may adversely affect mother and fetus. Pre pregnancy intervention is important in counselling regarding risk and in optimising medical management.

CHAPTER 2

PROCESSES AND PROCEDURES OF ANTENATAL CARE

2.1 ANTENATAL CARE

- Antenatal care should address both medical and psychological needs of the woman. Periodic antenatal health check-ups are necessary to establish rapport between the woman and health care providers. Health care providers should deliver individualised health promotional messages.

2.1.1 Antenatal visit

- Early antenatal care (first trimester) is important to screen women for risk factors, identify those with bad obstetric history and manage women with medical complications as these may have bearing on the progress of the pregnancy and its outcome.
- Activities during the antenatal visits should include the spouse/partner or family members as it will provide emotional support to the expectant mother. Their involvement enhances mother's compliance; identify her needs and wants; and discuss the plan for delivery.

2.1.2 Frequency of visits

- Below is the suggested schedule for antenatal contacts including options for virtual consultations for eligible mothers:

Table 2.1: Schedule for Antenatal Visits

Gestation		Mode of consultation
Primigravida	Multigravida	
<12	<12	Physical
16 - 20	16 - 20	Physical
24-26	24-26	Physical or virtual
28 – 30	28 – 30	Physical
32 – 34	32 – 34	Physical or virtual
35 - 36	35 - 36	Physical
37	38	Physical
38	40	Physical
39		Physical
40		Physical

- For high-risk pregnancy and other colour tags, more frequent visits are required. Low-risk mothers can be offered alternative antenatal contacts through virtual consultations if the requirements are fulfilled.
- The reduced physical visits can potentially benefit both mothers and healthcare providers by focusing more on the quality of the visits. This flexibility also provides ease of access for low-risk mothers as they will be at the comfort of their home.

2.1.3 Booking visit

- The first visit is very important and should be done as soon as possible (preferably before 12 weeks of gestation). Even if the first visit may be later in pregnancy, it is still regarded as the booking visit.
- First routine medical examination (RME) by a doctor should be done during the booking visit or within 2 weeks after the booking visit. The following information should be recorded:

a. History

- Detailed menstrual history
 - Last normal menstrual period (LNMP)
 - Regularity of cycles
 - Contraceptive usage

*refer for dating ultrasound scan if mother's period is irregular, stopped contraceptive pills less than 6 months or unsure of LNMP
- Medical history
 - Allergies
 - History of blood transfusion
 - Medical conditions
 - Infections
 - Drug history (traditional medication and other self-prescribed medicines)
- Past surgical history
- Past obstetric history
 - Previous recurrent miscarriage or termination of pregnancy
 - Intrauterine growth restriction and preterm labour
 - Previous LSCS, instrumentation, PPH, anaemia etc

- Intrauterine death
- History of baby with congenital abnormality
- Early or late neonatal death
- Family history
 - Medical disorders such as diabetes mellitus, hypertension
 - Multiple pregnancy
 - Congenital anomalies
- Socioeconomic background
 - Occupation of both the woman and her partner
 - Smoking, drugs and alcohol consumption
 - Education level

b. Physical examination

- Relevant physical examination should be performed :

○ Height ○ Weight ○ Blood pressure ○ Pallor, cyanosis, jaundice ○ Oral hygiene ○ Clubbing ○ Thyroid enlargement & signs of hypo/ hyperthyroidism ○ Cardiovascular system ○ Respiratory system ○ Breast ○ Abdomen	○ Scars of previous operation ○ Palpation: uterine size/ other masses ○ Vaginal examination (when indicated) ○ Oedema ○ Varicose veins ○ The mother's gait: any bony deformity of pelvis ○ Spine: kyphosis/scoliosis
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c. Laboratory Investigations

- Urinalysis: protein (albumin), sugar (glucostix), UFEME (when indicated)
- Blood :
 - Haemoglobin, ABO and Rhesus group
 - Syphilis (VDRL) – if positive, performs TPHA and refers to treatment.
 - HIV (Rapid test) – if reactive proceed with confirmatory test
 - Hepatitis B (HBsAg) antigen (if indicated)

- Thalassaemia screening (if indicated)
- Tuberculosis (if indicated, from TB screening questions)
- Malaria – all mothers who are suspected having signs and symptoms of Malaria must be screened
 - Definition of suspected malaria: Mother with high grade fever with signs & symptoms listed:
 - Headache
 - Malaise & fatigue
 - Abdominal pain
 - Myalgia & joint pain
 - Chills & rigor
 - Sweating
 - Loss of appetite
 - Vomiting

(Refer to The Management Guidelines of Malaria in Malaysia 2013)
 - Priority to the high risk mothers:
 - Orang Asli
 - Those who stay near forest
 - Living in, or returning from, endemic area/countries (Papua New guinea, Pakistan, Indonesia, Bangladesh, India, Myanmar, Nepal and Philippines)
 - Close contact with malarial case
 - Involved in high-risk activities (e.g.; logging, agriculture, recreation, army & hunting within 6 weeks)

d. Ultrasound scan for viability/ dating

- It is recommended to perform an ultrasound examination during the booking visit, if it is available.

e. Management

- Folic acid and iron supplementation to be given at booking if the patient can tolerate it. Combined preparation can be used. Refer WHO recommendations on micronutrient supplement for pregnancy (**Table 15.3**).
- Nutritional advise
- Health education e.g. smoking cessation
- Give information on the antenatal screening test i.e. indications, benefits and limitations

2.1.4 Subsequent physical visits

During the visits:

- Haemoglobin level monthly
- Ask relevant symptoms if problems arise
- Weight and blood pressure
- Urine for protein and glucose
- Symphysio-fundal height
- Assess the lie, presentation of the fetus and head engagement especially after 36 weeks
- Second RME by Medical Officer between 32 to 36 weeks POA

2.1.5 Virtual visits

- Virtual consultation is an option to be given to eligible mothers when all requirements are fulfilled. It is not compulsory.
- Virtual consultation is proposed only for white-coded mothers. Green-coded mothers should be assessed by medical officers for suitability of virtual consultation according to the criteria as in Table 2.2.
- Refer *Garis Panduan Peluasan Pelaksanaan Klinik Virtual (Virtual Clinic)*, *Bahagian Pembangunan Kesihatan Keluarga, 2020* on the details for requirement of virtual consultation.
- Refer **Appendix 2-1** – Consent form for antenatal virtual consultations and **Appendix 2-2** – Checklist for antenatal virtual consultations.

Table 2.2 : Criteria for mothers to be eligible for virtual consultations

For first virtual consultation	For second virtual consultation
• White-coded • Green-coded - Mothers who are coded green due to these factors are allowed for virtual consultations: ○ Primigravida / pseudoprimigravida (only for < 36 years old) ○ Grandmultipara ≥ 5 ○ Age 36-39 (only for multigravida) ○ Pre pregnancy/ booking weight < 45 kg ○ Height < 145cm	• White-coded • Green-coded - Mothers who are coded green due to these factors are allowed for virtual consultations: ○ Rhesus negative ○ Primigravida / pseudoprimigravida (only for < 36 years old) ○ Grandmultipara ≥ 5 ○ Pre pregnancy/ booking weight < 45 kg ○ Height < 145cm

For first virtual consultation	For second virtual consultation
○ History of gynaecological surgery (eg: fibroid, cyst) ○ Unsure of dates ○ History of recurrent miscarriages (≥ 3 consecutive miscarriages) ○ History of baby with birth weight > 4kg ○ History of 3rd and 4th degree perineal tear ○ History of retained placenta ○ History of postpartum haemorrhage ○ History of instrumental delivery ○ Birth spacing <2 years or > 5 years	○ Age 35-39 (only for multigravida) ○ History of gynaecological surgery (eg: fibroid, cyst) ○ Unsure of dates ○ History of recurrent miscarriages (≥ 3 consecutive miscarriages) ○ History of baby with birth weight > 4kg ○ History of 3rd and 4th degree perineal tear ○ History of retained placenta ○ History of postpartum haemorrhage ○ History of instrumental delivery ○ Birth spacing <2 years or > 5 years

Mothers with the condition as below are not eligible for virtual consultation:

- Mothers with social issues (e.g.: poor social support)
- Mothers with previous history of poor compliance to advise or missed appointments

2.2 SCREENING FOR RISK FACTORS

- Checklist should be assessed and documented. The care plan should be based on the protocol given based on *Garis Panduan Senarai Semak bagi Penjagaan Kesihatan Ibu Mengikut Sistem Kod Warna*. Refer **Appendix 2-3** for risk stratification based on colour coding.

2.3 MATERNAL VACCINATION

- Anti-tetanus vaccination (tetanus toxoid, TT)
 - Primigravida – at quickening and second dose 4 weeks later
 - Multigravida – a single dose is given between quickening and before 37 weeks of gestation
- **Tdap may be administered if available in place of ATT
- Optional vaccination can be given if available. Refer **Appendix 2-4**.

- COVID-19 vaccination – if the mother has not received the vaccine, offer and give advice on its importance

2.4 ANTENATAL CARE FOR UNCOMPLICATED AND HIGH RISK PREGNANCY

2.4.1 Routine antenatal care for uncomplicated pregnancies

Visits	Gestation	Content of care
1	Before 12 weeks	• Weight, height, body mass index (BMI) • Blood pressure, pulse rate • Full physical examination including mental health assessment • Ultrasound to confirm the date • Full blood count • Urine protein and sugar • Blood group and rhesus status • HIV screening • Syphilis screening • Hepatitis B screening(if indicated) • GDM screening (i.e.; OGTT) if indicated • VTE risk assessment • Antenatal colour coding based on risk factors
2	16-20 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Urine protein and sugar • Haemoglobin level • Review lab investigation results • Targeted scan if indicated • Placental localization
3	24 – 26 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • Haemoglobin level • GDM screening (i.e.; OGTT) if indicated

Visits	Gestation	Content of care
4	28 – 30 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • Haemoglobin level • Educate on foetal kick chart • Anti-tetanus vaccination
5	32-34 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • Haemoglobin level • Foetal kick chart
6	35-36 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • FBC • Foetal kick chart • Birth plan
7	38 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • Foetal kick chart
8	40 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • Foetal kick chart

Visits	Gestation	Content of care
9	41 weeks	• Weight • Blood pressure, pulse rate • Physical examination • Foetal heart rate auscultation • Urine protein and sugar • Foetal kick chart • Hospital admission for assessment or induction

2.4.2 Routine antenatal care for high risk pregnancies

Please refer to the respective sections for each specific condition. More frequent visits are required for high risk pregnancies. Below are additional cares to basic antenatal management whenever indicated:

- Other investigations related to the conditions (e.g.; renal function, HbA1c, thyroid function, ECG)
- Shared care / combined care / multidisciplinary management – with documented antenatal care plan
- Growth scans
- Detailed scan
- Documented delivery plan

2.5 HEALTH EDUCATION

Health education should be provided for mother and spouse/ family members. The topics should include:

- Diet during pregnancy (Section C: Perinatal Nutrition Care)
- Exercises during and after pregnancy
- Development of the baby
- How to overcome common discomforts in pregnancy
- Preparation for safe delivery – place of delivery, mode of delivery
- Labour process
- Pain relief methods
- Relaxation and breathing techniques
- Basic baby care
- Coping with problems in the first few weeks after delivery
- Education on common disorders in pregnancy (e.g.; hypertensive diseases in pregnancy, gestational diabetes mellitus, anaemia etc)

- Breastfeeding – (Refer Section C: Perinatal Nutrition Care)
- Partners and family/ community role in supporting breastfeeding mothers.
- Importance of vaccination for mother and baby

2.6 HOME VISITS

Home visit during antenatal should be provided for new case, patients who defaulted follow-up and for high-risk mothers as soon as possible. Refer **Appendix 2-5** for protocols on home visit.

- For normal antenatal mothers, one (1) home visit is adequate
- More frequent home visits are required for high-risk mother depending on severity of the risk including defaulter mothers

2.7 ANTEPARTUM FOETAL MONITORING AND SURVEILLANCE

There is a higher incidence of foetal compromise in pregnancy with hypertension, diabetes, heart disorders and other medical disorders. Foetal monitoring during the antepartum period consists of tests for:

- Foetal growth
- Foetal well being

2.7.1 Fetal growth

- Symphysio-fundal height (SFH) tape measurement should be performed routinely from 22 weeks onwards in all pregnancies where the POA is expected to correspond to the centimetres of the SFH.
- If there is a discrepancy between the SFH and POA of $\pm 3\text{cm}$, the patient needs to be re-evaluated with regards to the accuracy of the LNMP and should be referred for an ultrasound. This can be an early indicator of impaired fetal growth.
- Maternal weight gain: The antenatal mothers should be weighed at every antenatal visit.
- There should be a progressive increase in weight throughout pregnancy. For the 1st trimester, generally the weight gain should be about 0.5-2.0 kg in total. In the second and third trimester, weight gains are varied according to women pre pregnancy BMI categories (by using WHO cut-off). Refer to **Table 2.3** for recommended range of gestational weight gain.

Table 2.3: Gestational Weight Gain Range for Single Pregnancy

Pre pregnancy BMI (kg/m ²)	Total Weight Gain in 1 st trimester (kg)	Mean and range of GWG in 2 nd & 3 rd Trimester (kg)	Total Weight Gain (kg)
Underweight < 18.5	0.5 – 2 kg	0.5kg / week (0.44-0.58kg)	12.5 – 18.0
Normal 18.5 – 24.9		0.4kg / week (0.35-0.50kg)	11.5 – 16.0
Overweight 25.0 – 29.9		0.3kg / week (0.23-0.33kg)	7.0 – 11.5
Obese ≥ 30.0		0.2 kg / week (0.17-0.27kg)	5.0 – 9.0

Source: Institute of Medicine, 2009

2.7.2 Ultrasound scanning

- Ultrasound scanning for dating is reliable if the parameters are taken before 20 weeks. Serial scan should be done every 2 – 3 weeks for foetal growth assessment if there is suspicion of IUGR or other disorders.

2.7.3 Fetal monitoring

- Foetal kick chart is an indirect tool for monitoring fetal wellbeing. All mothers should be given the foetal movement chart (Cardiff 'count-to-ten') for recording of foetal movements from 28 weeks gestation onwards and should be told to report to any health facility if movements are less than 10 within 12 hours. This observation should be done at regular intervals every day.
- Foetal heart auscultation: should be routinely practiced from 22 weeks onward. Foetal heart rate should be taken for at least 30 seconds to determine the rate.
- CTG should be performed in cases where there is an abnormal FHR by daptone and high risk of foetal compromise such as poorly controlled hypertension/diabetes, IUGR or postdates.

2.8 ANTENATAL EXERCISE

Refer *Manual Senaman Antenatal & Postnatal Di Klinik Kesihatan, Bahagian Pembangunan Kesihatan Keluarga, Kementerian Kesihatan Malaysia, 2014.*

2.9 DOMESTIC VIOLENCE AND ABUSE

- Antenatal care is a good opportunity to identify and assist women who experience domestic violence. Women who are victims of domestic violence typically experience a phenomenon called 'learned helplessness' and are trapped in a 'cycle of violence' and they may not readily come forward for help. There are also many factors such as fear for their children being taken away that prevents them from disclosing.
- Healthcare providers to be aware of and be vigilant to the presence of cues associated with abuse and prompt further if domestic violence is suspected.
- Recommendations in managing women involved in domestic violence include:
 1. Create an environment for asking about domestic violence
 - Display information in suitable places (e.g.; waiting areas) about the support for those affected by domestic violence and abuse.
 - Ensure frontline staff are familiar with procedure regarding victims of domestic abuse
 2. Ask questions about domestic violence
 - Provide each woman with the opportunity for a one-to-one consultation, without her partner, a family member or a legal guardian present, on at least one occasion. For example during physical examination or ultrasound examination.
 - Be alert to the signs of abuse and raise the issue
 - Use verbal and non-verbal communication skills to develop trust
 - Assure abused women about privacy, safety and confidentiality issues
 - Be compassionate, supportive and respectful towards abused women
 - Do not pressure women to disclose; simply raising the issue can help them

3. Response to women who disclose abuse and provide first line support
 - Listen
 - Inquire about needs
 - Validate women's disclosure
 - Enhance safety and providing support
 - Do not pressure the woman to make immediate decisions about her situation or offer advice or reassurances prematurely.
4. Ensure safety
 - Initial safety assessment
 - Immediate risk: does she and her children require an immediate place of safety
 - Future risk: does she require help to prevent future risk e.g police report
 - Referral to police, social worker, support group, legal advice if indicated
 - Note-taking for legal purposes
 - Mandatory reporting only if children are involved
 - Continuing care
5. Help patient prepare an emergency plan
 - Assist patient make a plan in case she needs to leave
 - Appropriate referral for help e.g social worker
6. Plan for continuation of care
 - Consider your patient's safety as a paramount issue. A woman is usually a good judge of her own safety.
 - Empower her to take control of decision-making; ask what she needs and present choices of actions she may take and services available.
 - Respect the knowledge and coping skills she has developed.
 - Provide emotional support.
 - Ensure confidentiality—the woman may suffer additional abuse if her partner suspects she has disclosed the abuse.
 - High-risk discharge notification to the nearest clinic.
 - Ensure patients remain within the system. Obtain telephone numbers and addresses. Early postnatal home visit is essential.

Reference:

1. WHO Recommendations on Antenatal Care for a Positive Pregnancy Experience. 2016: WHO
2. NICE Clinical guideline [CG110] Pregnancy and complex social factors: a model for service provision for pregnant women with complex social factors. September 2010.
3. NICE Public health guideline [PH50] Domestic violence and abuse: multi-agency working. February 2014.
4. Hegarty K, O'Doherty L. Intimate partner violence - identification and response in general practice. Aust Fam Physician. 2011 Nov;40(11):852-6.

BORANG KEIZINAN KONSULTASI ANTENATAL SECARA MAYA

Konsultasi antenatal secara maya adalah penyampaian perkhidmatan kesihatan secara maya (*virtual*), langsung (*live*) dan interaktif yang merangkumi konsultasi klinikal dan pelan rawatan pelanggan di antara anggota kesihatan dan pelanggan. Konsultasi antenatal secara maya merupakan satu inisiatif yang dilaksanakan bagi memastikan penjarakan sosial dan menyediakan perkhidmatan kesihatan yang berterusan setaraf dengan kemajuan teknologi.

Saya telah diberi penerangan oleh anggota kesihatan tentang pelaksanaan konsultasi antenatal secara maya dan bersetuju:

1. Untuk menyertai sesi konsultasi antenatal secara maya.
2. Maklumat klinikal semasa sesi konsultasi boleh dikongsi dengan anggota kesihatan lain untuk tujuan rujukan sekiranya perlu, bagi kesinambungan perawatan.
3. Sebarang rakaman video atau audio sepanjang sesi konsultasi postnatal secara maya tidak boleh ditular/ dikongsi/ disebar kepada pihak luar oleh kedua-dua pihak
4. Rakaman tidak boleh digunakan sebagai bukti untuk tindakan undang-undang terhadap KKM
5. Anggota kesihatan yang bertugas boleh memberhentikan konsultasi jika:
 - a. berlaku gangguan akses capaian internet atau sebarang masalah teknikal.
 - b. jenis perbualan tiada berkaitan dengan tujuan konsultasi kesihatan saya
6. Saya berhak untuk tidak meneruskan konsultasi antenatal secara maya atas pilihan saya sendiri.
7. Saya akan hadir ke klinik sekiranya terdapat keperluan untuk pemeriksaan lanjutan.
8. Konsultasi secara maya boleh menyebabkan kelewatan dalam rawatan berbanding konsultasi bersemuka.

Ditandatangani:

(Ibu)
Nama :
No. Kad pengenalan:
Tarikh:

(Saksi*)
Nama :
No. Kad pengenalan:
Jawatan :
Tarikh :
Cop jawatan:

** Saksi boleh terdiri dari Pakar Perubatan Keluarga/ Pegawai Perubatan/ Jururawat/ Penolong Pegawai Perubatan

**SENARAI SEMAK PEMANTAUAN IBU ANTENATAL SECARA VIRTUAL UNTUK
ANGGOTA KESIHATAN**

KONSULTASI VIRTUAL PERTAMA PADA 24-26 MINGGU			
No.	Soalan	Jawapan	Tindakan
1.	Untuk meneruskan konsultasi secara <i>virtual</i> , pastikan anggota kesihatan dapat melihat imej ibu sehingga sekurang-kurangnya ke paras dada. Ini bagi membolehkan anggota kesihatan dapat melihat penampilan ibu secara keseluruhan.		
2.	Bertanya khabar ibu?	Sihat / tidak sihat	Jika ibu tidak sihat, nasihat ibu untuk datang klinik
3.	Setuju untuk teruskan konsultasi secara <i>virtual</i> ?	Ya / Tidak	Jika ibu tidak bersetuju untuk teruskan konsultasi secara <i>virtual</i> , beri temujanji klinik
4.	Demam	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada doktor
5.	Masalah pernafasan	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada doktor
6.	Sakit kepala/ pening / loya/ muntah/ mata kabur	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada doktor
7.	Sakit dada /berdebar-debar	Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital
8.	Edema di kaki/ tangan/ muka secara mendadak	Ya / Tidak	Jika edema di kaki, tanya lebih lanjut sehingga tahap mana. Jika sehingga lutut, nasihat ibu untuk datang ke klinik. Jika edema di tangan atau muka, nasihat ibu untuk datang ke klinik
9.	Sakit dada/ susah nafas/ betis bengkak/ sakit / kemerahan	Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital
10.	Pemakanan - selera makan	Normal/ kurang selera	Jika kurang selera makan, beri nasihat mengenai pemakanan atau nasihat untuk datang ke klinik jika perlu
11.	<i>Contractions</i>	Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital

KONSULTASI VIRTUAL PERTAMA PADA 24-26 MINGGU			
No.	Soalan	Jawapan	Tindakan
12.	Pecah air ketuban / <i>leaking</i>	Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital
13.	Pendarahan per-vagina	Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital
14.	Kesihatan mental - Sejak kebelakangan ini, adakah anda sering diganggu perasaan murung atau sedih? - Sejak kebelakangan ini, adakah anda sering kehilangan minat atau keseronokan dalam melakukan kerja harian?	Ya / Tidak	Jika ya kepada salah satu, nasihat ibu untuk datang ke klinik.
15.	Kepatuhan pengambilan ubat / suplemen	Ya / Tidak	Jika tidak patuh, nasihat ibu dan tekankan mengenai kepentingan kepatuhan kepada pengambilan ubat / suplemen tersebut
16.	Soalan umum : Adakah mempunyai masalah lain? – contohnya masalah kencing, buang air besar, corak tidur terganggu	Ya / Tidak	Jururawat nasihat ibu apa yang perlu dilakukan dan jika perlu, berbincang dengan doktor.
17.	Adakah ibu tahu temujanji seterusnya?	Ya / Tidak	Jika tidak, maklumkan tarikh temujanji. Nasihat ibu untuk hubungi klinik/hospital atau datang ke klinik/hospital jika mempunyai sebarang masalah.
Nota: - Jika ibu diminta untuk ke hospital, jururawat / pegawai perubatan perlu beritahu mengenai kes kepada pihak hospital/ PAC dan memaklumkan status ibu jika terdapat sebarang perubahan (Rujuk <i>Garis panduan Perkhidmatan Kecemasan dan Ambulans di Fasilitas Kesihatan Primer</i>). - Jururawat perlu melakukan tindak susul ke atas perancangan pengendalian/ penjagaan ibu.			

KONSULTASI VIRTUAL KEDUA PADA 32-34 MINGGU			
No.	Soalan	Jawapan	Tindakan
1.	Untuk meneruskan konsultasi secara <i>virtual</i> , pastikan anggota kesihatan dapat melihat imej ibu sehingga sekurang-kurangnya ke paras dada. Ini bagi membolehkan anggota kesihatan dapat melihat penampilan ibu secara keseluruhan.		
2.	Bertanya khabar ibu?	Sihat / tidak sihat	Jika ibu tidak sihat, nasihat ibu untuk datang klinik
3.	Setuju untuk teruskan konsultasi secara <i>virtual</i> ?	Ya / Tidak	Jika ibu tidak bersetuju untuk teruskan konsultasi secara <i>virtual</i> , beri temujanji klinik
4.	Demam	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada pegawai perubatan
5.	Masalah pernafasan	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada pegawai perubatan
6.	Sakit kepala/ pening / loya/ muntah/ mata kabur	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik
7.	Sakit dada / berdebar-debar	Ya / Tidak	Jika ya, minta ibu untuk ke hospital dengan segera
8.	Edema di kaki/ tangan/ muka secara mendadak	Ya / Tidak	Jika edema di kaki, tanya lebih lanjut sehingga tahap mana. Jika sehingga lutut, nasihat ibu untuk datang ke klinik. Jika edema di tangan atau muka, nasihat ibu untuk datang ke klinik
9.	Sakit dada/ susah nafas/ betis bengkak/ sakit / kemerahan	Ya / Tidak	Jika ya, minta ibu untuk ke hospital dengan segera
10.	Pemakanan - selera makan	Normal/ kurang selera	Jika kurang selera makan, beri nasihat mengenai pemakanan atau nasihat untuk datang ke klinik jika perlu
11.	Pergerakan janin ● Bilangan - 10 kali dalam tempoh 12 jam ● Kekuatan	● Ya, cukup bilangan dan kekuatan gerakan ● Tidak cukup bilangan dan kekuatan	Jika kurang pergerakan, nasihat ibu untuk datang segera ke klinik

KONSULTASI VIRTUAL KEDUA PADA 32-34 MINGGU			
No.	Soalan	Jawapan	Tindakan
		gerakan	
12.	<i>Contractions</i>	Ya / Tidak	Jika ya, minta ibu untuk ke hospital dengan segera
13.	Pecah air ketuban / <i>leaking</i>	Ya / Tidak	Jika ya, minta ibu untuk ke hospital dengan segera
14.	Pendarahan per-vagina	Ya / Tidak	Jika ya, minta ibu untuk ke hospital dengan segera
15.	Kesihatan mental - Sejak kebelakangan ini, adakah anda sering diganggu perasaan murung atau sedih? - Sejak kebelakangan ini, adakah anda sering kehilangan minat atau keseronokan dalam melakukan kerja harian?	Ya / Tidak	Jika ya kepada salah satu, nasihat ibu untuk datang ke klinik.
16.	Kepatuhan pengambilan ubat / suplemen	Ya / Tidak	Jika tidak patuh, nasihat ibu dan tekankan mengenai kepentingan kepatuhan kepada pengambilan ubat / suplemen tersebut
17.	Adakah terdapat sebarang perubahan kepada perancangan kelahiran yang telah dibincangkan pada temujanji sebelum ini?	Ya / Tidak	Jika ya, jururawat perlu ambil maklum dan ambil tindakan sewajarnya
18.	Soalan umum : Adakah mempunyai masalah lain? – contohnya masalah kencing, buang air besar, corak tidur terganggu	Ya / Tidak	Jururawat nasihat ibu apa yang perlu dilakukan dan jika perlu, berbincang dengan doktor.
19.	Adakah ibu tahu temujanji seterusnya ?	Ya / Tidak	Jika tidak, maklumkan tarikh temujanji. Nasihat ibu untuk hubungi klinik/hospital atau datang ke klinik/hospital jika mempunyai sebarang masalah.
Nota: - Jika ibu diminta untuk ke hospital, jururawat / pegawai perubatan perlu beritahu mengenai kes kepada pihak hospital/ PAC dan memaklumkan status ibu jika terdapat sebarang perubahan (Rujuk <i>Garis panduan Perkhidmatan Kecemasan dan Ambulans di Fasilitas Kesihatan Primer</i>). - Jururawat perlu melakukan tindak susul ke atas perancangan pengendalian/ penjagaan ibu.			

SISTEM KOD WARNA DAN SENARAI SEMAK PENJAGAAN ANTENATAL

Sistem ini menggunakan empat kod warna. Penentuan kod warna ini adalah berdasarkan kepada penilaian faktor risiko ibu semasa sesi konsultasi di klinik/ rumah. Penjagaan ibu adalah berdasarkan kepada kod warna seperti berikut:

KOD WARNA	TAHAP PENJAGAAN
Merah	Rujukan segera ke Hospital dan pengendalian selanjutnya adalah bersama (<i>shared care</i>) Pakar O&G dan/atau Pakar Perubatan Keluarga
Kuning	Rujukan untuk pengendalian oleh Pakar O&G Hospital/Pakar Perubatan Keluarga, dan penjagaan selanjutnya boleh dilakukan bersama (<i>shared care</i>) Pegawai Perubatan dan Jururawat Kesihatan
Hijau	Pengendalian di Klinik Kesihatan oleh Pegawai Perubatan dan Kesihatan dan pengendalian selanjutnya boleh dilakukan bersama (<i>shared care</i>) Jururawat Kesihatan/Jururawat Masyarakat di bawah pengawasan Pegawai Perubatan
Putih	Penjagaan oleh Jururawat Kesihatan/Jururawat Masyarakat di Klinik Kesihatan dan Klinik Desa (sekiranya tiada terdapat faktor risiko yang disenaraikan dalam kod merah, kuning dan hijau, ibu diberi kod warna putih)

Sistem kod warna ini adalah panduan bagi menentukan peringkat penjagaan ibu hamil berdasarkan faktor risiko yang dikenalpasti. Senarai semak risiko menyenaraikan situasi/ faktor risiko yang **KERAP** ditemui dalam kalangan ibu hamil. Sekiranya terdapat situasi/ faktor risiko lain yang tidak tersenarai, bincang dengan Pegawai Perubatan/ Pakar Perubatan Keluarga bagi menentukan kod warna.

Di dalam situasi yang tertentu khususnya di kawasan pedalaman, di mana tidak terdapat Pegawai Perubatan, pengendalian boleh dilakukan oleh Jururawat Kesihatan/ Jururawat Masyarakat dengan pengawasan dari Pegawai Perubatan yang terdekat atau mudah dihubungi.

Pakar O&G/ Pakar Perubatan Keluarga boleh menurunkan kod warna mengikut penilaian tahap risiko semasa ibu hamil. Kod warna terkini, dilekatkan dengan menampakkan sebahagian kecil pelekat warna yang terdahulu.

SENARAI SEMAK PENGENDALIAN IBU HAMIL

KOD MERAH - Rujukan segera ke Hospital dan pengendalian selanjutnya adalah bersama (shared care) Pakar O&G dan/ atau Pakar Perubatan Keluarga.

FAKTOR RISIKO		Tandakan (√) dalam ruangan jika ada faktor														
TRIMESTER		1	2			3						Post - date				
KEKERAPAN PENILAIAN RISIKO		≤12	13-22	23-27		28-32		33-36		37-40			>40			
TARIKH																
Jangkamasa tidak datang haid (POA/POG)																
1	Eclampsia															
2	Pre-eclampsia (tekanan darah tinggi dengan urin protein) - sistolik ≥160mmHg atau diastolik ≥100mmHg tanpa/ dengan urin protein ≥2+															
3	Hypertensive crisis - Tekanan darah tinggi ≥160/110mmHg															
4	Penyakit jantung semasa mengandung, dengan tanda dan gejala (sesak nafas, berdebar-debar)															
5	Sesak nafas dan/ atau kadar pernafasan ≥22/min															
6	Diabetes ketoasidosis (paras glukos dextrostix >11mmol/L dan urin ketone ≥ 2+)															
7	Pendarahan antepartum (termasuk keguguran)															
8	Denyutan jantung janin yang abnormal • FHR <110/min selepas 22/52 • FHR >160/min selepas 32/52															
9	Anemia dengan simptom pada mana-mana gestasi atau Hb ≤7g/dL															
10	Kontraksi rahim pramatang															
11	Keluar air ketuban tanpa kontraksi															
12	Sawan															
13	Demam dengan tanda-tanda sepsis (lesu, dehidrasi, tachycardia)															
14	Kes disyaki denggi (demam berserta sakit kepala, sakit sendi, sakit perut, muntah, cirit birit dan sebagainya)															
15	Gejala deep vein thrombosis (DVT) dan/ atau Pulmonary															

KOD MERAH - Rujukan segera ke Hospital dan pengendalian selanjutnya adalah bersama (shared care) Pakar O&G dan/ atau Pakar Perubatan Keluarga.																		
FAKTOR RISIKO		Tandakan (√) dalam ruangan jika ada faktor																
TRIMESTER		1	2						3								Post - date	
KEKERAPAN PENILAIAN RISIKO		≤12	13-22		23-27		28-32		33-36		37-40				>40			
TARIKH																		
Jangkamasa tidak datang haid (POA/POG)																		
	Embolism																	
16	Keinginan mencederakan/ bunuh diri																	
NAMA & JAWATAN PEMERIKSA																		

KOD KUNING -Rujukan untuk pengendalian oleh Pakar O&G Hospital/ Pakar Perubatan Keluarga, dan penjagaan selanjutnya boleh dilakukan bersama (<i>shared care</i>) Pegawai Perubatan dan Jururawat Kesihatan																	
FAKTOR RISIKO			Tandakan (√) dalam ruangan jika ada faktor														
TRIMESTER			1		2				3								Post-Date
KEKERAPAN PENILAIAN RISIKO			≤12	13-22		23-27		28-32		33-36		37-40				>40	
TARIKH																	
Jangkamasa tidak datang haid (POA/POG)																	
1	Ibu HIV positif																
2	Ibu Hepatitis B positif																
3	Ibu Tuberkulosis/ Malaria/ Syphilis																
4	Tekanan darah >140/90 - <160/100mmHg tanpa/ dengan urin protein																
5	Tekanan darah tinggi (140/90mmHg) dengan urin protein																
6	Ibu diabetes dengan rawatan dan/ atau komplikasi																
7	**Pergerakan janin kurang semasa kandungan ≥32 minggu dengan faktor risiko																
8	Kandungan melebihi 7 hari dari EDD																
9	Ibu dengan masalah perubatan yang memerlukan rawatan bersama di hospital																
10	Ibu tunggal atau ibu remaja (<20 tahun)																
11	Ibu berumur ≥40 tahun																

KOD KUNING -Rujukan untuk pengendalian oleh Pakar O&G Hospital/ Pakar Perubatan Keluarga, dan penjagaan selanjutnya boleh dilakukan bersama (<i>shared care</i>) Pegawai Perubatan dan Jururawat Kesihatan																		
FAKTOR RISIKO		Tandakan (√) dalam ruangan jika ada faktor																
TRIMESTER		1	2					3					Post-Date					
KEKERAPAN PENILAIAN RISIKO		≤12	13-22	23-27			28-32	33-36			37-40			>40				
TARIKH																		
Jangkamasa tidak datang haid (POA/POG)																		
12	Hemoglobin 7 - <9g/dL dan <i>asymptomatic</i>																	
13	<i>Placenta previa</i> tanpa pendarahan																	
14	Maternal pyrexia ≥38°C atau >3 hari																	
15	Penyakit jantung tanpa gejala																	
16	*Ketagihan dadah/ rokok/ alkohol																	
17	Skor risiko antenatal <i>Venous Thromboembolism (VTE)</i> ≥3																	
18	Kandungan kembar																	
19	Masalah kesihatan mental																	
20	BMI ≥ 40 kg/m ²																	
NAMA & JAWATAN PEMERIKSA																		

*Penilaian sekali sahaja.

**Rujukan segera ke hospital untuk penilaian

Nota: Ibu mesti diperiksa oleh Pegawai Perubatan pada hari yang sama tarikh *booking*

KOD HIJAU - Pengendalian di Klinik Kesihatan oleh Pegawai Perubatan dan Kesihatan dan pengendalian selanjutnya boleh dilakukan bersama (sharedcare) Jururawat Kesihatan/ Jururawat Masyarakat dibawah pengawasan Pegawai Perubatan

FAKTOR RISIKO		T n n √ l m u n g n f t															
TRIMESTER	1	2				3				3				Post-Date			
KEKERAPAN PENILAIAN RISIKO	≤	13-22		23-27		28-32		33-36		37-40		>40					
TARIKH																	
Jangkamasa tidak datang haid (POA/POG)																	
1 *Rh negatif																	
2 *Berat badan ibu sebelum mengandung atau ketika booking <45kg																	
3 Obes ($\geq 30.0 \text{ kg/m}^2$)																	
4 Pertambahan berat badan yang mendadak (>2 kg dalam seminggu)																	
5 Berat badan statik atau menurun (dalam tempoh sebulan)																	
6 *Masalah perubatan semasa																	
7 *Masalah ginekologi yang lalu (fibroid, cyst, pembedahan)																	
8 *LNMP yang tidak pasti																	
9 * ≥ 3 kali keguguran yang berturutan																	
10 *Sejarah obstetrik yang lalu:																	
• Pembedahan cesarean																	
• PIH/ eclampsia/ diabetes																	
• Kematian perinatal																	
• Bayi dengan berat lahir <2.5kg atau >4kg																	
• Luka perineum 3 rd degree dan 4 th degree																	
• Retained placenta																	
• Postpartum haemorrhage																	
• Kelahiran instrumental																	
11 **Pergerakan janin kurang semasa kandungan ≥ 32 minggu																	
12 Tekanan darah (140/90mmHg) tanpa urin protein																	
13 Hemoglobin 9 - <11g/dL																	
14 Glukosuria 2 kali																	
15 Air kencing mempunyai protein $\geq 1+$																	
16 Tinggi rahim (SFH) tidak sejajar dengan jangkamasa kandungan +/- 3cm																	

KOD HIJAU - Pengendalian di Klinik Kesihatan oleh Pegawai Perubatan dan Kesihatan dan pengendalian selanjutnya boleh dilakukan bersama (sharedcare) Jururawat Kesihatan/ Jururawat Masyarakat dibawah pengawasan Pegawai Perubatan

FAKTOR RISIKO		T n n √ l m u n g n f t															
TRIMESTER		1	2				3								Post-Date		
KEKERAPAN PENILAIAN RISIKO		≤	13-22		23-27		28-32		33-36		37-40		>40				
TARIKH																	
Jangkamasa tidak datang haid (POA/POG)																	
17	Breech/ oblique/ tranverse dengan tiada tanda sakit bersalin pada 36 minggu kehamilan																
18	Kepala bayi tinggi (<i>head not engaged</i>) semasa kandungan 37 minggu bagi primigravida																
19	Ibu GDM dengan kawalan diet tanpa komplikasi																
20	*Ibu berumur 36 - 39 tahun																
21	*Primigravida/ pseudoprimigravida																
22	*Grandmultipara ≥5																
23	*Jarak kelahiran <2 tahun atau melebihi 5 tahun																
24	*Ketinggian ibu <145cm																
NAMA & JAWATAN PEMERIKSA																	

*Penilaian sekali sahaja.

**Rujukan segera ke hospital untuk penilaian

Nota: Ibu mesti diperiksa oleh Pegawai Perubatan dalam tempoh 2 minggu dari tarikh booking

KOD PUTIH -Penjagaan oleh Jururawat Kesihatan/ Masyarakat di Klinik Kesihatan dan Klinik Desa. Ibu akan hanya diberi kod berwarna putih setelah ia tidak mempunyai sebarang faktor risiko yang tersenarai dalam kod merah/ kuning/ hijau DAN menepati kriteria dibawah.

KRITERIA		TANDAKAN (✓) DALAM RUANG BERKENAAN
TARIKH		
Jangkamasa tidak datang haid (POA/POG)		
1	Gravida 2 – 5 (tidak termasuk pseudoprimigravida)	
2	Tiada masalah obstetrik lalu dan semasa	
3	Tiada masalah perubatan yang lalu dan semasa	
4	Ibu berumur 20 - 35 tahun	
5	Ibu berkahwin dan mempunyai sokongan keluarga	
NAMA & JAWATAN PEMERIKSA		

OPTIONAL VACCINATIONS FOR PREGNANT WOMEN

For pregnant women:

i. Influenza (flu) vaccination

Pregnant mothers are susceptible to severe illness if they acquire influenza (flu) infections, especially in the third trimester.

Hence, all pregnant mothers are recommended to be vaccinated for influenza (flu) in pregnancy, unless they have been vaccinated in the last 12 months.

There are two vaccines which are available, the trivalent and quadrivalent and both are safe in pregnancy. Although there is no teratogenicity associated with the use of the vaccine, it's generally recommended beyond 20 weeks of pregnancy.

Influenza vaccine is contraindicated if patient had severe egg protein allergy.

ii. Tdap vaccination

Although all babies are vaccinated for pertussis (whooping cough), newborn babies aged 2 months and below remain susceptible.

Current recommendations is to vaccinate all pregnant mothers between 28 weeks to 36 weeks of pregnancy for pertussis (Tdap vaccination) which triggers maternal immunoglobulin transfer via the placenta and offers some immunity. These babies still require routine pertussis vaccination at the age of 2 months.

For pregnant women in specific population

i. Asplenia (post splenectomy patients)

These patients will benefit from additional vaccines, which are:

- *Haemophilus influenza* (Hib)
- Meningococcal (2 doses) – 5 yearly booster
- Pneumococcal (2 doses) – 5 yearly booster

These vaccines are not contraindicated in pregnancy

ii. Travelling

Additional vaccines may be essential, depending on the area of travel. Diseases which are endemic in certain countries should be checked with the local authorities prior to making travel arrangements. These includes:

- Yellow fever
- Typhoid
- Meningococcal

iii. Hepatitis A vaccine

This may be beneficial for patients with liver diseases, sewage workers, intravenous drug abusers and haemophilia patients.

For breastfeeding mothers

Life vaccines that are safe during breastfeeding and are not contraindicated, including:

- MMR
- Rubella
- Varicella
- BCG

PROTOCOLS ON HOME VISIT

- Enter the house only after obtaining permission
- Respect the mother and her family
- Communicate well with the mother in order to develop rapport
- Describe clearly the objectives of the visit to the mother
- Avoid making any unfavourable comment or judgement about the patient and family
- Educate the mother and family about personal hygiene and sanitation
- Refer to relevant units, if basic facilities are not available (e.g. environmental sanitation unit, if there is no toilet)
- If the mother prefers home delivery and meets all the criteria, the health worker should check the intended birth site and advise the mother regarding necessary preparation.
- If the mother requires delivery at a hospital or Alternative Birthing Centre, she should be advised with regards to the facility and its locality
- A history and physical examination can be done after you have developed a rapport with the mother. First ascertain the progress of the pregnancy and the well being of the mother. The mother's antenatal book should be updated.

CHAPTER 3

PROCESSES AND PROCEDURES OF INTRAPARTUM CARE

CHAPTER 3: PROCESSES AND PROCEDURES OF INTRAPARTUM CARE

This chapter is a guide for health personnel attending to mothers in labour. The content is developed in such a manner to provide an appropriate assessment of safety and birth outcomes at different levels of health care.

3.1 NORMAL LABOUR AND SAFE DELIVERY

3.1.1 Definition of labour

Labour is a process where there is a presence of regular uterine contractions of increasing intensity and frequency that is associated with progressive dilatation and effacement of the cervix and descent of the presenting part. It may or may not be associated with rupture of membranes and leaking liquor.

3.1.2 Care in Labour

Care of a mother in labour starts with an accurate and legible documentation of the date and time of consultation and signature of the attending doctor or nurse with the name printed.

One must ensure that the mother is managed in an appropriate centre. (Refer to **Appendix 3-1 – Intrapartum Care Flow Chart**). In any situation whereby referral to hospital is required, consider the practical points in **Appendix 3-2**.

A checklist for risk assessment for ABC / LRBC (**Appendix 3-3**) and hospital (**Appendix 3-4**) should be completed by the nurse upon admission of the mother in labour.

- Psychological support
 - As most labour is spontaneous and ends with a normal delivery, the main role of the birth attendant (usually a midwife) is to provide support for the mother and her companion and to monitor the progress of labour.
 - Companionship to the labouring mother should be encouraged. However, all companions are encouraged to undergo an orientation programme when it is available.

- Physical examination
 - A detailed and systematic examination should be carried out on the labouring mother upon admission. All findings must be accurately documented. Vaginal examinations should be done every four hours, unless contraindicated, or sooner if indicated.

3.1.3 Abdominal Examination

Abdominal examination which includes inspection and palpation should be carried out and documented including:

- Inspection
 - Foetal movements
 - Scars
 - Skin changes
 - Linea nigra
 - Striae gravidarum
 - Striae albicans
- Palpation
 - Fundal height
 - Lie
 - Presentation
 - if cephalic, head engagement
 - Liquor volume

3.1.4 Vaginal Assessment

This should be done systematically and with adequate explanation to the mother. The findings should include the nature of:

- Vulva and vagina
- Cervical length (effacement)
- Dilatation of the cervical os
- Presenting part/position
- Station of presenting part
- Membranes (intact/ruptured)
- Blood vessel on the membrane
- Cord (felt/not felt)
- Placenta (felt/not felt)
- Liquor colour and volume drained

3.1.5 Amniotomy

Amniotomy is a process where the amniotic membranes are ruptured either spontaneously or artificially.

Amniotomy is performed in certain conditions:

- i Poor progress of labour
- ii Induction of labour

Artificial rupture of membranes can be performed when all these criteria are fulfilled.

- i. Fetal head is two-fifths palpable per abdomen with regular uterine contractions
- ii. 2 contractions in 10 minutes and the cervical dilatation is more than 4 cm.

Trained staff nurses may do artificial rupture of the membranes if it is still intact in advanced labour.

3.1.6 Analgesia

Choice of appropriate and available analgesia should be offered to all mothers in labour:

- Intramuscular narcotics with anti-emetic.
 - Pethidine 1mg/kg, with Metoclopramide 10 mg or Promethazine 25mg. This can be repeated 4 to 6 hourly.
 - Pethidine should not be given when cervical dilatation is more than 6 cm.
 - Nalbuphine (Nubain) 10 -20mg, repeat 4 to 6 hourly.
- Entonox – Inhalation agent with 50:50 oxygen and nitrous oxide (can be ordered by nurses).
- Epidural analgesia – available in hospital with anaesthetic service
- Non-pharmacological method of pain relief include – companionship, warm bath, music, massage, Transcutaneous Electrical Nerve Stimulation (TENS) etc
- The above methods may not be applicable in birth centres at rural clinics. Companionship, ambulation and family support is important to alleviate pain in the absence of medication

3.2 INTRAPARTUM MONITORING

Intrapartum risk assessment and monitoring of the mother and foetus are essential because complications can arise without warning.

Methods for intrapartum monitoring based on level of care are as below (Table 3.1)

Table 3.1: Methods and appropriate technology for intrapartum monitoring by level of care

Level	Community health clinic	Health clinic and ABC	Low risk birth centre	Hospital without specialist	Hospital with specialist
Person	Community health nurse	Doctor / nurse	Doctor / nurse	Medical team without specialist	Medical team with specialist
Equipment					
Early labour monitoring record/LPC	Yes	Yes	Yes	Yes	Yes
Partograph	Yes	Yes	Yes	Yes	Yes
Foetal stethoscope/daptone	Yes	Yes	Yes	Yes	Yes
Cardiotocography	No	No	Yes	Yes	Yes
Ultrasound	No	Optional	Yes	Yes	Yes

3.2.1 When to document labour observation?

Nurses should commence documentation of contractions and foetal heart rate upon admission by using Labour Progress Chart (LPC) / early labour monitoring record

3.2.2 What is a Partograph?

A partograph is a diagrammatic representation of the progress of labour. It is where all observations of the mother and her foetus are charted in a manner which facilitates monitoring of the progress of labour by the health care worker (Figure 3.2)

Figure 3.2: Partograph

PARTOGRAPH

NAME: _____ PID: _____ ADMISSION DATE: _____
 POA: _____ ARM: _____ BORN: _____

FETAL HEART RATE	180 170 160 150 140 130 120 110 100	
LIQUOR MOULDING		
(cm) CERVIX Plot X Descent of Head Plot O	10 9 8 7 6 5 4 3 2 1 0	LATENT PHASE ACTIVE PHASE
Time	Hours	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
CONTRACTIONS PER 10 MINS	5 4 3 2 1 0	
OXYTOCIN (U/500mL) TITRATION (mL/h)		
DRUGS		
I.V.FLUIDS		
PULSE AND BP	180 170 160 150 140 130 120 110 100 90 80 70 60	
TEMP		
URINE { PROT KET VOL		

The main components that need to be monitored and plotted on the partograph are:

- Foetal condition
- The progress of labour
- Maternal condition

All mothers in labour should be monitored using partograph adapted by Jawatankuasa Pengurusan dan Perkembangan Obstetrik & Ginekologi (JKPPOG)

(The modified WHO partograph commences at 4 cm cervical dilatation and dispenses with the recording of the latent phase of labour)

When to start a partograph?

When a patient is planned for delivery, with cervical dilatation of ≥ 4 cm or contraction > 2 in 10 minute.

The information charted on a partograph includes:

i. **Mother information**

- Name
- Gravida
- Para
- registration number
- Diagnosis/problem list
- Date and time of admission
- Time of ruptured membrane

ii. **Foetal heart rate**

This is recorded every half an hour

iii. **Membranes and amniotic fluid (Liquor)**

The state of the membranes and amniotic fluid (liquor) should be documented as follows:

I : Membranes intact

C : Membranes ruptured, clear liquor

M : Meconium-stained liquor

B : Blood-stained liquor

iv. **Moulding**

Moulding of the foetal skull is recorded as follows:

- No moulding – parietal bones (sagittal suture) are not apposed.
- +1 moulding – parietal bones are touching but not overlapping.
- +2 moulding – parietal bones are overlapped but easily reduced.
- +3 moulding – parietal bones have overlapped and are irreducible.

v. **Cervical dilatation**

This is marked with a cross (X), and begin to plot the partograph when cervical dilatation is 4 cm or more

vi. **Descent of foetal head**

This is assessed as fifths of the head palpable per abdomen, and marked with an (O)

vii. **Alert line**

The Alert Line starts at 4 cm cervical dilatation, and increases to the point of expected full dilatation at a rate of 1 cm per hour. If the progress of labour is normal, this progress line (cervicogram) on the partograph will correspond to the Alert Line or lie to the left of it

viii. **Action line**

The Action Line is parallel and 4 hours to the right of the Alert Line

ix. **Hours**

This charts the time (in hours) elapsed since the onset of the active phase of labour

x. **Time**

The actual time of the clock is recorded

xi. **Contractions**

Uterine contractions are assessed every half an hour and charted as the number of contractions in 10 minutes and duration of contraction in seconds. The duration of contraction reflects the strength of contraction (**Figure 3.1**).

Figure 3.1: Duration of Contraction

Duration of contraction	
Less than 20 seconds (weak)	
20 to 40 seconds (moderate)	
More than 40 seconds (strong)	

xii. **Oxytocin**

The amount of oxytocin added per volume of intravenous fluids and the rate of infusion must be recorded every half an hour

xiii. **Additional drugs**

Any additional drugs given such as Pethidine and Metoclopramide must be recorded at the time of administration

xiv. **Maternal pulse rate**

This is documented every half an hour with a dot (•)

xv. **Maternal blood pressure**

This is recorded every 4 hours (unless more frequently indicated) and marked with arrows (↕)

xvi. **Maternal temperature**

This is recorded every 4 hours

xvii. **Urine protein, ketone and volume**

Each time the mother passes urine or is catheterized, measure her urine volume and record. Test the urine for protein and ketone and record the result

3.2.3 Foetal monitoring methods in labour

a. Intermittent auscultation with a pinard fetoscope/ Doppler Foetal Monitor detector (Daptone)

Auscultation must be done after a contraction. This should be practised every 15 – 30 minutes for mothers who are in labour. In the majority of ABC, this is the only method to detect and monitor foetal heart.

b. Electronic Foetal Monitoring with CTG

CTG should be done on every woman in labour for 20 minutes on admission. If the CTG is suspicious or abnormal, it should be continued and immediate consultation should be done with the specialist.

CTG can be faxed or shared via smartphone to the covering hospital for interpretation and advise if the facilities are available.

- **Admission CTG**

With a suspicious or abnormal admission CTG, there are higher rates of meconium staining of liquor, intrapartum CTG decelerations and other subsequent ominous patterns. A normal admission CTG is reassuring.

- **Indications for close monitoring of intrapartum CTG**

Mothers requiring close monitoring of intrapartum CTG should ideally be managed in the hospital with a specialist. Some of the indications for are as follows:

- Maternal medical conditions :
 - Gestational diabetes mellitus
 - Hypertensive disorders in pregnancy
- Obstetric conditions:
 - Multiple pregnancies
 - Previous caesarean section
 - Intrauterine growth restriction
 - Prelabour rupture of membranes
 - Preterm labour
 - Meconium-stained amniotic fluid
 - Third trimester bleeding
 - Oxytocin induction/augmentation of labour
 - Post-date

Intermittent CTG monitoring every 2 to 4 hourly should be reviewed by a medical officer. A specialist should be consulted if there is any doubt.

Table 3.2: Interpretation of Cardiotocography (CTG)

	Normal	Suspicious	Pathological
Baseline	110-160 bpm	Lacking at least one characteristic of normality, but with no pathological features	< 100 bpm
Variability	5-25 bpm		Reduced variability for > 50 min, increased variability for > 30 min, or sinusoidal pattern for > 30 min
Decelerations	No repetitive decelerations *		Repetitive* late or prolonged decelerations during > 30 min, or 20 min if reduced variability, or one prolonged deceleration with > 5 min
Interpretation	Fetus with no hypoxia/ acidosis	Fetus with a low probability of having hypoxia/ acidosis	Fetus with a high probability of having hypoxia/ acidosis
Clinical Management	No intervention necessary to improve foetal oxygenation state	Action to correct reversible causes if identified, closed monitoring or additional methods to evaluate foetal oxygenation	Immediate action to correct reversible causes, additional methods to evaluate foetal oxygenation, or if this is not possible expedite delivery. In acute situations (cord prolapse, uterine rupture or placental abruption) immediate delivery should be accomplished.

* Decelerations are repetitive if they occur at >50% of uterine contractions

Intrapartum indications for ultrasonography

The role of ultrasonography in the intrapartum period is limited to the following:

- Antepartum haemorrhage – for placental localization and retroplacental clots
- To ascertain lie/ presentation for multiple pregnancies, maternal obesity, malpresentation, and polyhydramnios
- To confirm intrauterine death
- Presence of pelvic masses obstructing labour, when indicated

3.3 NORMAL STAGES OF DELIVERY

3.3.1 Stage 1

It starts from the onset of labour to full dilatation (commonly lasts 8-24 hours in first labour including the latent phase and 3-12 hours in subsequent labour). The first stage is further divided into two phases:

- Latent phase (0 – 4 cm cervical dilatation)
- Active phase (≥ 4 cm of cervical dilatation)

Mother presenting in latent phase should be managed by using Labour Progress Chart (LPC) [Appendix 3-5]:

- Monitor temperature, pulse, blood pressure 4 hourly and urinalysis on admission and when mother passes urine
- Monitor nature of contractions (length, strength and frequency) 4 hourly
- Abdominal examination finding – fundal height, lie, presentation and engagement on admission
- Vaginal examination finding – vulva, vagina, cervical effacement and dilation, station, position, membrane (if absent nature of the liquor) and to rule out cord presentation.
- Pain assessment and offer pain relief if possible
- Auscultate/ listen for FHR for a minimum of 1 minute immediately after a contraction and every four hourly

- Encourage frequent drinks and eating light meals to maintain hydration and energy
- Encourage mobilisation and mother should adopt whatever position they find most comfortable
- 2 - 6 hourly passing of urine
- If the labour progresses, transfer patient to labour room

Mother presenting in active phase of labour:

- Partograph should be started.
- Drinking water and eating light meals may be given if the labour is progressing normally in low risk mothers. This might be contraindicated if they have received opioids or they develop risk factors that make a general anaesthetic more likely.
- Mother may become more comfortable by changing position in bed or by ambulation.

3.3.2

Stage 2

It starts from full dilatation of the cervix to delivery of the baby (commonly ends within 1 hour). The start of the second stage is not clear but a vaginal examination is indicated when the mother has a sensation of bearing down.

- During the second stage mother should be informed that they should be guided by their own urge to push.
- Strategies to assist birth with effective pushing;
 - Change of mother's position
 - Empty her bladder
 - Support and encouragement

Intrapartum intervention to reduce perineal trauma include:

- a. Either the 'hands-on' (guarding the perineum and flexing the baby's head) or the 'hands poised' (with hands off the perineum and baby's head but in readiness) technique can be used to facilitate spontaneous birth.
- b. Episiotomy may be considered at this time if indicated. Episiotomy is an incision performed medio-laterally in the perineum during crowning of the presenting part in order to prevent extensive perineal tearing. It should be performed selectively and not routinely.

- Episiotomy should be considered only in the case of complicated vaginal deliveries (breech, shoulder dystocia, forceps and vacuum) and for previous third or fourth degree tears.
- Episiotomy should not be performed too early as excessive bleeding will result.
- Local anaesthesia should be provided to the mother before episiotomy repair.
- A rectal examination should be done on completion of episiotomy repair.
- If a third or fourth degree tear is suspected, the mother should be referred to a hospital with a specialist. Haemostasis should be secured before referral. This may be either by:
 - i. Suturing the bleeder
 - ii. Pack the wound to ensure pressure
 Antibiotic therapy should be initiated at the earliest opportunity
- The episiotomy rate should ideally not exceed 30% in any centre (according to the Director General of Health's circular 1/2008).

3.3.3 Stage 3

It starts from delivery of the baby to delivery of the placenta (usually lasts 15 – 30 minutes). Active management of the third stage helps to prevent postpartum haemorrhage. Active management of the third stage of labour includes:

- Immediate oxytocin
- Controlled cord traction, and
- Uterine massage

a) Oxytocin

- Within 1 minute of delivery of the baby, give IM oxytocin 10 units or IM Syntometrine (5 units oxytocin plus 0.5mg ergometrine) or IM/IV Carbetocin 100 mcg (oxytocin analogue) after palpating the abdomen to rule out the presence of an additional fetus.
- Oxytocin/oxytocin analogue are drugs of choice because they are effective 2 to 3 minutes after injection, has minimal side effects and can be used in all mothers.

- Do not give syntometrine to mother with hypertension and heart disease because it increases the risk of convulsions and cerebrovascular accidents.

b) Controlled cord traction

- Clamp the cord close to the perineum using Spencer Wells artery forceps. Hold the clamped cord at the end of forceps with one hand.
- Place the other hand just above the woman's pubic bone and stabilise the uterus by applying counter traction during controlled cord traction. This helps prevent inversion of the uterus.
- Wait for signs of placental separation i.e.
 - Lengthening of the cord
 - Gushing of blood
 - Uterus raised and hard
- Gently pull downward the cord to deliver the placenta.
- Continue to apply counter traction to the uterus with the other hand.
- If the placenta does not descend during 30–40 seconds of controlled cord traction (i.e. there are no signs of placental separation), do not continue to pull on the cord.
- After delivery of the placenta, hold the placenta in two hands and gently turn it until the membranes are twisted.
- Slowly pull to complete the delivery of placenta.
- Examine the placenta to be sure it is complete. If suspected to be incomplete, refer to the hospital.
- If the cord is pulled off (cord snapped), manual removal of the placenta may be necessary.
- If uterine inversion occurs, replace immediately (Refer to Chapter 9 – Obstetric Emergencies).

c) Uterine Massage

- Immediately massage the fundus of the uterus through the woman's abdomen until the uterus is contracted.
- If the uterus is not contracted and ongoing bleeding, actions in **Table 3.3** can be followed.

Table 3.3: Actions during uterine atony and ongoing bleeding

Actions	Health Clinics / ABC / District Hospital without Specialist / Low-risk Birthing Centre	Hospital With Specialist
1. Initiate oxytocic drugs and bleeding stops	Yes	Yes
2. Uterine massage every 15 minutes for the first 1 hour and bleeding stops	Yes	Yes
3. Other uterotonic drugs and bleeding still continues and uterus remain not contracted	Refer to hospital with specialist *If uterine tamponade technique is available, (e.g.; Bakri balloon), to insert before transfer patient	● Uterine tamponade ● B-Lynch suture } Uterine conservation ● Hysterectomy

3.4 LEVEL OF CARE FOR INTRAPARTUM MANAGEMENT

Apart from hospitals, other public health facilities which provide intrapartum services are Alternative Birthing Centre (ABC) and Low-Risk Birthing Centre (LRBC). The descriptions are as below:-

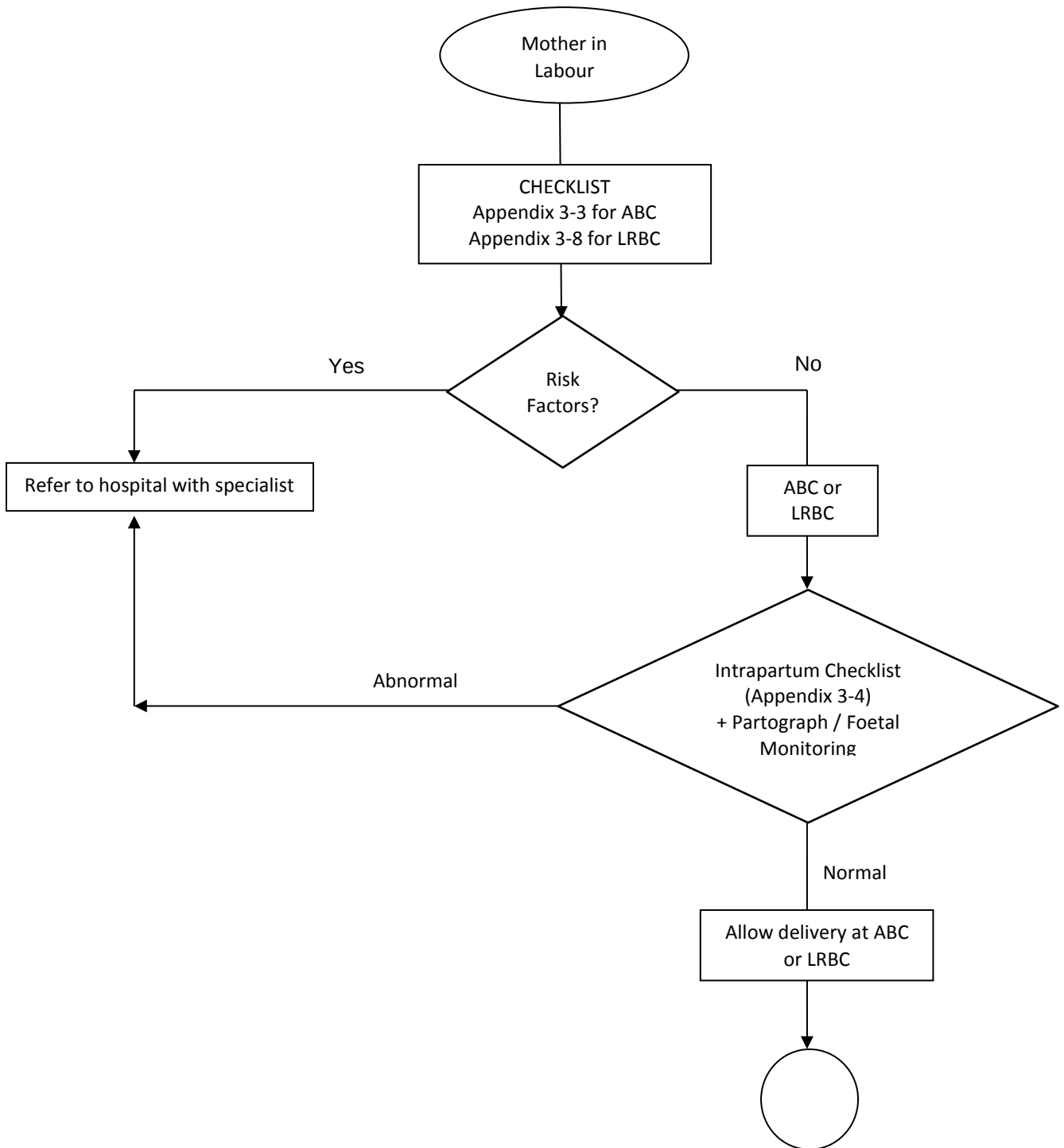
	HOME DELIVERY	ALTERNATIVE BIRTHING CENTRE (ABC)	LOW-RISK BIRTHING CENTRE (LRBC)
DEFINITION	Mothers delivering at home, conducted by trained medical personnel	A centre run by trained staff nurses where mothers can deliver in a safe medical setting	A centre run by trained staff nurses where mothers can deliver under supervision of medical officers / specialists (if required) in a safe medical setting.
CRITERIA	White-coded	White-coded	White-code Green-coded (after assessment by specialist)
FACILITY	Delivery bag (List of items in Appendix 3-6) Current recommendation does not favour usage of ravin enema before delivery process To bring along the eclampsia kit if patient has symptoms of impending pre-eclampsia Reference: <i>Manual Perkhidmatan Kesihatan Ibu dan Anak Bagi Anggota</i>	Following current specification for ABC centre Reference: <i>Garis panduan Pusat Bersalin Alternatif di Klinik Kesihatan, KKM, 2013</i>	Following current specification for LRBC Refer Appendix 3-7 and Appendix 3.8

	HOME DELIVERY	ALTERNATIVE BIRTHING CENTRE (ABC)	LOW-RISK BIRTHING CENTRE (LRBC)
	<i>Kejururawatan di Perkhidmatan Kesihatan Awam, KKM, 2016</i>		
MONITORING	Minimum 1 hour observation by attending medical personnel after delivery. Monitoring of BP, PR and uterus contraction to be done every 15 minutes.	Mothers will be admitted and observed in the ABC for 6 hours before being discharged home. Vital signs should be monitored every 15 minutes for the first 1 hour. Then 2 – 4 hourly depending on mother's condition	Mothers will be admitted and observed in the postnatal ward for 6 hours before allowing discharged home. Vital signs should be monitored every 15 minutes for the first 1 hour. Then 2 – 4 hourly depending on the mother's condition. Discharge only after assessment by medical officer

References:

- *FIGO Classification of Intrapartum Cardiotocography, 2015*
- *NICE Guideline Intrapartum care for healthy women and babies, 2017*

INTRAPARTUM CARE FLOWCHART (ABC / LRBC)



PRACTICAL POINTS DURING TRANSFER OF MOTHERS

Practical points to be noted during transfer of mothers:

Pre-transfer:

- Meticulous planning and coordination to identify personnel and mode of transport
- Resuscitate and stabilise the mother
- Coordinate safe embarkation on to vehicle

During transfer:

- Maintain stability of mother
- Constant monitoring and documentation of vital signs, treatment and incidents during transfer
- If acute problems arise, stop vehicle to carry out resuscitative measures or divert to nearest health facility

On arrival:

- Hand over to appropriate person
- Ensure safe disembarkation

Steps to be taken prior to transfer to hospital in certain conditions :

RISK FACTORS	PLAN OF ACTION
Leaking liquor or rupture of membrane > 6 hours	i. Check foetal heart rate and give mother oxygen if there is sign of foetal distress ii. Rule out cord prolapse
Cord Prolapse	i. Elevate the perineum by putting two pillows under the mother's buttock ii. Give oxygen to the mother iii. If the cord is protruding through the vagina, cover it with a pad or gauze which has been soaked with warm water and push it back into the vagina if possible. iv. Distend bladder with 500-750 mls of saline/water v. If cord is within the vagina, push the presenting part away from the cord
Gestation less than 36 weeks	IM Dexamethasone 12 mg stat. Can be given by trained Staff Nurse according to <i>Garis panduan Pemberian Suntikan Kortikosteroid (IM Dexamethasone) bagi Ibu Mengandung Berisiko Kelahiran Prematur di Klinik Kesihatan 2013</i>

SENARAI SEMAK BAGI KELAYAKAN IBU UNTUK BERSALIN DI ABC/RUMAH

KRITERIA UNTUK BERSALIN DI ABC/ RUMAH		
IBU DIBENARKAN BERSALIN DI PUSAT BERSALIN ALTERNATIF (ABC) / RUMAH SEKIRANYA MEMENUHI SYARAT BERIKUT :-		
KRITERIA		TANDAKAN ✓ DALAM RUANG BERKENAAN
TARIKH		
Jangkamasa tidak datang haid (POA/POG)		
1	Gravida 2 – 5 (tidak termasuk pseudoprimigravida)	
2	Tiada masalah obstetrik lalu dan semasa	
3	Tiada masalah perubatan yang lalu dan semasa	
4	Ibu berumur 20 - 35 tahun	
5	Ibu berkahwin dan mempunyai sokongan keluarga	
6	POA 37- 40 minggu	
7	Anggaran berat bayi 2.5-3.5kg	
8	Persekitaran rumah ibu sesuai (jika ingin bersalin di rumah)	
NAMA & JAWATAN PEMERIKSA		

*Ibu dibenarkan bersalin di ABC/ rumah sekiranya semua kriteria di atas dipenuhi.

SENARAI SEMAK JAGAAN INTRAPARTUM DI ABC/ RUMAH

Nota: Jika terdapat faktor risiko, ibu perlu dirujuk kepada Pegawai Perubatan di klinik/hospital berhampiran.

Tarikh :
 Nama :
 No.K/P :

No	Faktor risiko	Tandakan (✓) pada ruang berkenaan
PERINGKAT PERTAMA KELAHIRAN		
1.	Demam $\geq 37.5^{\circ}\text{C}$	
2.	Proteinuria (1+ atau lebih)	
3.	Tekanan darah tinggi ($\geq 140/90$ mmHg)	
4.	Kedudukan janin yang abnormal (menyongsang, menyerong, melintang)	
5.	Masalah perubatan semasa seperti sakit jantung, asma, diabetes, hipertensi, sawan, anemia, TB, HIV positif, Hepatitis B positif dan syphilis.	
6.	Kandungan lebih 40 minggu	
7.	Kandungan kurang 37 minggu	
8.	Keluar air ketuban lebih 6 jam dan masih dalam peringkat <i>latent phase</i>	
9.	Air ketuban mengandungi najis janin (<i>meconium</i>)	
10.	Kadar denyutan jantung bayi $<110/\text{min}$ atau $>160/\text{min}$	
11.	Sakit bersalin >12 jam untuk primigravida, >8 jam untuk multigravida	
12.	Dilatasi serviks yang statik selama 4 jam	
13.	Tali pusat terkeluar	
14.	Pendarahan (<i>intrapartum haemorrhage</i>)	
15.	Kontraksi rahim tidak tetap (<i>irregular/incoordinated</i>) > 4 jam	
PERINGKAT KEDUA KELAHIRAN		
1.	Peringkat kedua >1 jam untuk primigravida, >30 minit untuk multigravida	
2.	Ibu mengalami pendarahan, sesak nafas, denyutan nadi $>100/\text{min}$ atau cyanosis	

No	Faktor risiko	Tandakan (✓) pada ruang berkenaan
3.	Kadar denyutan jantung bayi <110/min atau >160/min	
4.	Shoulder dystocia	
PERINGKAT KETIGA KELAHIRAN		
1.	<i>Retained placenta</i>	
2.	Cebisan uri / <i>placenta</i> / membrane (POC) tertinggal di dalam rahim	
3.	Luka perineum (<i>tear</i>) tahap kedua atau ketiga	
4.	Pendarahan postpartum (PPH) ≥ 500ml, contohnya disebabkan oleh <i>uterine atony</i> , <i>uterine inversion</i> atau masalah pembekuan darah (<i>clotting defect</i>).	
5.	Ibu mengalami sesak nafas, denyutan nadi >100/min atau <i>cyanosis</i>	

Nama anggota :

Jawatan :

Klinik Kesihatan/ Klinik Desa :

Nota: Sila kepilkan Senarai Semak Jagaan Intrapartum ini jika ibu dirujuk ke klinik/hospital

APPENDIX 3-5

MINISTRY OF HEALTH MALAYSIA LABOUR PROGRESS CHART

NAME:

RN:

WARD:

BED NO:

HEIGHT:

WEIGHT:

Date	Time	Blood Pressure	Pulse Rate	Respiratory Rate	Foetal Heart Rate	SpO ₂	Uterine Contraction: strength/ frequency/ duration	Pad Chart	Remarks	Name of Staff

SENARAI KEPERLUAN PERALATAN UNTUK MENYAMBUT KELAHIRAN

A	Delivery Set	Kuantiti
1.	<i>Kidney dish</i> (10 inci)	2
2.	<i>Kidney dish</i> (6 inci)	1
3.	<i>Gallipot</i>	2-3
4.	<i>Artery forceps</i> (<i>Spencer Wells forcep</i>)	2
5.	<i>Dissecting forceps</i>	1
6.	<i>Cord scissor</i>	1

B	Losen dan Krim	Kuantiti
1.	<i>Hibitane in spirit</i> 1:200 (<i>Chlorhexidine</i> 0.5% <i>Alcohol</i>) untuk <i>cord dressing</i>	1 botol
2.	<i>Hibitane in spirit</i> 1:2000	1 botol
3.	<i>Flavin</i> 0.1% (<i>Acriflavine lotion</i> 0.1%) untuk ubat luka	1 botol
4.	<i>Alcohol</i> 70%	1 botol
5.	<i>Swab hibitane</i> (<i>Swab soaked in hibitane in spirit</i>) untuk membersihkan peralatan	1 botol

C	VE Set	Kuantiti
1.	<i>Bowl</i>	1
2.	<i>Gallipot</i>	2

D	Peralatan Sterile Loose Pack	Kuantiti
1.	<i>Dressing towel</i>	1
2.	<i>Stitch scissor</i>	1
3.	<i>Needle holder</i>	1
4.	<i>Artery forcep</i>	2
5.	<i>Jug</i>	1
6.	<i>Swab</i>	
7.	<i>Pad</i>	
8.	<i>Gauze</i>	
9.	<i>Paper hand towel</i>	
10.	<i>Linen paper</i>	
11.	<i>Syringe</i> 2ml	
12.	<i>Needle</i>	
13.	<i>Cord clamp</i>	
14.	<i>Cat gut</i>	

E	Peralatan Mengambil Pemeriksaan Vital	Kuantiti
1.	BP Set	1
2.	Stethoscope	1
3.	Daptone / Fetoscope	1
4.	Penimbang bayi	1
5.	Pita pengukur	1
6.	<i>Clinical thermometer</i>	1
7.	<i>Disposable lancet dan glass slide – untuk kes yang mengalami demam*</i>	

F	Set Infusi Intravena
1.	<i>IV Fluid</i>
2.	<i>IV Drip set</i>
3.	<i>Branula size 16G, 18G, 20G</i>
4.	Plaster
5.	Gunting

G	Ubatan
1.	<i>Syntometrine (4 ampoules)</i>
2.	<i>Syntocinon (2 ampoules)</i>
3.	<i>Carboprost</i>
4.	<i>Tranexamic acid</i>
5.	Vitamin K
6.	Hepatitis B
7.	Tablet Paracetamol

H	Lain-lain Peralatan
1.	<i>Mucus extractor/sucker</i>
2.	Kertas turas saringan G6PD
3.	Tiub botol untuk saringan <i>congenital hypothyroidism</i>
4.	Beg plastik kecil
5.	<i>Glove</i>
6.	Apron
7.	<i>Mask</i>
8.	<i>Paddle pad soap/hand rub dan paper hand towel</i>

I	Kad dan Borang-borang
1.	Buku Rekod Kesihatan Bayi dan Kanak-kanak 0-6 tahun
2.	Borang Partogram
3.	Senarai Semak Intrapartum
4.	Stiker Kod Warna
5.	Borang Makmal
6.	Borang Rujukan Intrapartum (IP-1)

LOW-RISK BIRTHING CENTRE (LRBC)

Low-risk birthing centre is a mother friendly facility. The objective of LRBC is to conduct low-risk deliveries. The choice of low risk deliveries will include patients tagged as white and green. For green-coded mothers, O&G specialists need to assess and decide the suitability for delivery in LRBC.

Objective of Low-risk Birth Centre (LRBC)

- a. To provide an alternative high-quality care and safe delivery to low risk mothers.
- b. To enable mothers to experience positive and satisfactory birth with minimal medical and surgical intervention
- c. To reduce congestion in the main labour room and obstetric ward, therefore reducing overcrowding and optimising patient care.

Services Provided

1. Observation wards (antenatal observation and postnatal observation)
 - Admission
 - History taking and examination
 - Maternal vital sign monitoring/ postpartum care
 - Foetal monitoring/ care of newborn
2. Delivery bay
 - Admission CTG
 - Partogram
 - Active management of labour
 - Mother Friendly Care Policy during active labour will be practiced in LRBC, promoting breastfeeding
 - Vaccination
3. CTG assessment
 - Admission CTG
 - When indicated during labour
4. Ultrasound assessment
 - If indicated

Kriteria Pemilihan untuk bersalin di Pusat Bersalin Berisiko Rendah

BIL	KRITERIA KEMASUKAN
1.	Warganegara Malaysia
2.	Berumur 18 – 40 tahun
3.	Perkahwinan yang sah
4.	Ketinggian ibu melebihi 145 cm
5.	BMI < 40 kg/m ² pada usia kandungan 36 minggu
6.	Gravida 2 hingga 6 atau jumlah kelahiran tidak melebihi 5 kali
7.	Telah dipastikan tarikh jangka bersalin
8.	Kehamilan tunggal (<i>singleton</i>)
9.	Usia kandungan ≥37 minggu dan ≤41 minggu
10.	Kepala bayi berkedudukan di bawah (<i>cephalic</i>)
11.	Anggaran berat bayi 2.2- 3.5 kg pada usia kandungan 36 minggu
12.	Bersetuju mematuhi Program Imunisasi Kebangsaan
13.	TIADA sejarah lampau seperti berikut:
	i. Lahir mati atau kecederaan bayi ketika bersalin
	ii. Anak dengan masalah perkembangan otak/ mengidap penyakit metabolik
	iii. Kelahiran secara pembedahan atau pembedahan pada rahim
	iv. Uri melekat selepas kelahiran bayi
	v. Tumpah darah selepas kelahiran bayi (<i>postpartum haemorrhage</i>)
	vi. Koyakan perineum tahap 3 atau 4 selepas kelahiran bayi
	vii. Kelahiran instrumental
14.	TIADA masalah kandungan semasa seperti berikut:
	i. Tumpah darah sebelum kelahiran (<i>antepartum haemorrhage</i>)
	ii. Kekurangan atau terlebih air ketuban (<i>oligohydramnios/ polyhydramnios</i>)
	iii. Darah tinggi ketika mengandung
	iv. Kencing manis ketika mengandung
	v. Masalah perubatan, kardiovaskular
	vi. Pecah air ketuban melebihi 12 jam
	vii. Jangkitan GBS
	viii. Ibu berstatus <i>Rhesus</i> negatif
	ix. Hepatitis/ HIV/ penyakit kelamin

BIL	KRITERIA KEMASUKAN
	x. Pengambilan dadah/ merokok
	xi. Anaemia (paras haemoglobin <9g/dL pada usia kandungan 36 minggu)
	xii. Masalah psikiatri
	xiii. Kecacatan fizikal yang mengganggu proses kelahiran
	xiv. Ketidaksuburan > 5 tahun (<i>involuntary</i>)

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CHAPTER 4

PROCESSES AND PROCEDURES OF POSTNATAL CARE

CHAPTER 4: PROCESSES AND PROCEDURES OF POSTNATAL CARE

4.1 INTRODUCTION

- A significant number of maternal deaths as well as morbidity occur during the postpartum period. About two-thirds of maternal deaths occur during this phase.
- Postpartum complications can be grouped into acute life threatening, mid- and long-term chronic conditions. Increased awareness of warning signs and appropriate intervention is needed at all levels.
- Skilled care and early identification of problems can reduce the incidence of death and disability.
- Postpartum period (puerperium) is from the end of labour until the reproductive organs return to their original non-pregnant condition. It lasts for 42 days. The postpartum period is a critical transitional time for a woman, her newborn and her family, on a physiological, emotional and social level. Inadequate postnatal care can reduce opportunities for early detection and management of problems and disease.
- The aims of care in the postpartum period are:
 1. Support of the mother and her family in the transition period for the new family member and respond to their needs
 2. Prevention, early diagnosis and treatment of complications of mother and neonate, including the prevention of vertical transmission of diseases from mother to neonate
 3. Referral of mother and neonate for specialist care when necessary
 4. Counselling on baby care and immunisation of the infant
 5. Support for breastfeeding
 6. Counselling on maternal nutrition and exercise
 7. Counselling and service provision for contraception and the resumption of sexual activity
 8. Provision of emotional and psychological support in particularly in special circumstances

4.2 POSTNATAL CONTACTS

- Postnatal care starts immediately after the mother gives birth. Optimal postnatal care should be carried out soon after delivery both in hospital and health clinic. For continuous medical care, all deliveries must be notified to the respective health clinic. The staff in the health clinic must take action within 24 hours of reception of notification.

- According to WHO Recommendations on Antenatal Care for a Positive Pregnancy Experience (2016), 'contact' implies an active encounter between the women and the healthcare provider that is not implicit with the word 'visit'. 'Contact' can be adapted to local contexts either through home visits, clinic visits, phone calls or mobile phone-based contacts. All mothers should be informed about the importance of postnatal contacts, service and schedule from the antenatal period.
- In the event of a baby being admitted in the hospital, but the mother has been discharged and continues to accompany the baby, postnatal care should be provided in accordance with KKM policy as outlined in Garispanduan Perawatan Ibu Postnatal di Hospital 2015. The provision of this care must be documented in the *Buku Rekod Kesihatan Ibu*.
- In the event of a mother being admitted in the hospital but the baby has been discharged, and continues to stay with the mother, neonatal care should be provided. This care should be provided by the nursing staff either from the obstetrics or paediatric department in the hospital. The provision of this care must be documented in the *Buku Rekod Kesihatan Bayi dan Kanak-kanak*.

4.3 FREQUENCY OF CONTACTS

- The schedule of postnatal home visits is determined by the condition of mother and baby upon discharge and assessment of their health status throughout postnatal contacts. For mothers with uneventful pregnancy and delivery, it is recommended that home visits should be at least 5 times (Refer **Table 4.1**). As for high risk mothers and babies or any abnormalities detected, the number of contacts should be more frequent. Please refer to the postnatal checklist and flowchart. Any abnormality observed during these visits may require more visits with appropriate referral to be made.
- Postnatal mothers who give birth at the alternative birthing centre (ABC) or at home and attended by trained health personnel must have postnatal care via home visit within 24 hours after delivery (Day 1 of baby's life).
- WHO recommends at least four (4) postnatal contacts as below (WHO Recommendations on Postnatal care of the Mother and Newborn 2013):
 - If birth is in a health facility, mothers and newborns should receive postnatal care in the facility for at least 24 hours after birth.

- If birth is at home, the first postnatal contact should be as early as possible within 24 hours of birth.
- At least three additional postnatal contacts are recommended for all mothers and newborns, on day 3 (48–72 hours), between days 7–14 after birth, and six weeks after birth.
- For Malaysia, the minimum postnatal home visits are 5 times with 1 clinic visit. The additional postnatal home visits on day 5 and between day 14–16 are based on consensus from the expert committee.
- Postnatal contacts through virtual consultation can also be offered to low-risk mothers and newborns if virtual requirement is fulfilled as outlined in *Garis panduan Peluasan Pelaksanaan Klinik Virtual (Virtual Clinic), Bahagian Pembangunan Kesihatan Keluarga 2020*.

Table 4.1: Schedule for minimum postnatal contacts

Days (according to age of baby)	Contacts (home visit / virtual consultation / clinic visit)
Day 1	Home visit for mother who had safe delivery at home
Day 2	Home visit
Day 3	Home visit
Day 5	Home visit *assessment for suitability of virtual consultation
Day 8 to 10	Home visit or Virtual (1 visit)
Day 14 to 16	Home visit (1 visit)
Day 30	Clinic visit

Note:

1. Weigh baby on day 5 and day 14–16. For babies with jaundice weigh babies according to *Clinical Practice Guidelines Management of Neonatal Jaundice, MOH, 2014*.
2. Consent for virtual consultation will be taken during postnatal contacts on day 5 by nurses during home visit.

High-risk postnatal mother and baby

- The postnatal contacts for high-risk mother and/or baby should be carried out more frequently. The urgency, frequency, mode and number of contacts are determined by the condition of mother and baby upon discharge and assessment of their health status throughout postnatal contacts.
- Upon discharge of a high-risk postnatal mother and/or baby, the medical personnel / physician must identify and notify the health clinic immediately by phone / email / fax or existing online system. This is to ensure timely postnatal home visit is done. A documented management plan should be attached to the mother's copy of Buku Rekod Kesihatan Ibu.
- Below are the criteria for high-risk postnatal mother and baby (**Table 4.2**). Mothers and babies under this category are not suitable for virtual consultation.

Table 4.2: Postnatal mother and baby in the high risk criteria

High risk postnatal mother	High risk baby
1. Maternal collapse	1. History of prolonged labour
2. Moderate/ severe PIH / pre-eclampsia/ eclampsia / HELLP Syndrome	2. Baby to an obese mother
3. Diabetes on treatment	3. Small for gestation (SGA)
4. Maternal obesity, BMI ≥ 40 kg/m ²	4. Large for gestation (LGA)
5. Medical disorders in pregnancy (i.e; cardiac disease, renal disease, autoimmune diseases, epilepsy, any chronic diseases eg: active tuberculosis, HIV, hepatitis B/C etc)	5. Baby to GDM mother
6. Malignancy	6. At risk of severe NNJ
7. Mental health	7. Preterm
a. Pre-existing : eg: suicidal ideation, major depression disorder, bipolar , schizophrenia, previous history of postpartum depression	8. Under the management and follow-up for neonatal jaundice, poor weight gain /weight loss > 7%, or other conditions.
b. At risk of perinatal mental health issues including adverse foetal outcome/ traumatic birth	

High risk postnatal mother	High risk baby
8. Sepsis / infection 9. Current VTE 10. Massive postpartum haemorrhage and obstetric anal sphincter injury (OASIS) 11. Social issue: teenage pregnancy, substance abuse, domestic violence, low socioeconomic status, lack of family support (including single mother). 12. Mothers of vulnerable group (eg: Orang Asli, mothers under custodial care, immigrants) 13. Mothers with postpartum VTE score ≥ 3 14. AOR discharge	

Note:

- This risk stratification is to be done before discharge.
- Any postnatal readmission should be re-triaged prior to discharge.
- This list is also applicable following management of miscarriage or termination of pregnancy.
- This list is not exhaustive, if any mother or newborn out of this list requires a higher level of care, tag as high-risk

4.4 COMPONENTS OF POSTNATAL CARE

4.4.1 Home visits

- For detailed procedure and checklist of postnatal home visit, refer *Manual Perkhidmatan Kesihatan Ibu & Anak bagi Anggota Kejururawatan di Perkhidmatan Kesihatan Awam 2016*, *Buku Rekod Kesihatan Ibu* (KIK/1(a)/96 Pind. 2020) and *Buku Rekod Kesihatan Kanak-kanak*.
- At each postnatal contact, enquiries should be made about general well-being and assessments of mother's vital signs, lochia, urinary and bowel function, healing of wound, fatigue, perineal pain and perineal hygiene, breast pain, abdominal tenderness, calf pain and other relevant signs and symptoms. Mothers should also be given information about the physiological process of recovery after birth, nutrition, hygiene and contraceptives.
- Basic care for all newborns should include promoting and supporting early and exclusive breastfeeding, keeping the baby warm, providing

hygienic umbilical cord and skin care, identifying conditions requiring additional care (e.g: jaundice) and counselling on when to take a newborn to a health facility.

- Breastfeeding progress should be assessed at each postnatal contact. At each postnatal contact, women should be asked about their emotional wellbeing, what family and social support they have and their usual coping strategies for dealing with day-to-day matters.
- All mothers and their families/partners should be encouraged to tell their health care professional about any changes in mood, emotional state and behaviour that are outside of the woman's normal pattern.
- After day 10 postnatal, mothers should be asked about resolution or symptoms of baby blues. If symptoms have not resolved the health staff should arrange further support and communicate with the medical officer regarding assessment for postnatal depression and further evaluation.
- If any abnormality is detected in mothers or babies, referral to the hospital or clinic must be made. Mothers should be advised of the signs and symptoms of potentially life-threatening conditions in themselves or babies, to contact health staff immediately or call for emergency help.

4.4.2 Clinic visits

- Indications for clinic visits for postnatal mothers can be either a scheduled visit at 1 month or various conditions requiring medical review at clinic.
- For deliveries at home, both mother and baby need to be reviewed by a medical officer at least within 24 hours after delivery.
- Among common situations which require clinic visits during the postnatal phase are blood pressure review, laboratory investigations, referral to medical officers or mother having complaints related to herself/baby.
- During the scheduled visit at 1 month, enquiries are made about general well-being of the mother and other relevant signs and symptoms including pervaginal bleeding/discharge, urinary and bowel function and healing of the wound.
- Assessment of vital signs, physical examinations and laboratory investigations i.e.; haemoglobin level and urine analysis for sugar and albumin, are conducted and managed accordingly.
- They should be counselled on birth spacing and family planning ideally before discharge following delivery. Contraceptive options should be discussed and provided if agreeable. Sexual intercourse may be resumed after the mother's vaginal bleeding has stopped and perineal

wound stitches has healed. Usually, this would have recovered within four to six weeks following delivery.

- For babies, enquiries about feeding, urinary and bowel activities and general conditions are made in addition to anthropometry measurements, physical examinations and developmental assessments. Immunisation is given as scheduled.
- For detailed procedure and checklist of postnatal home visit, refer Manual Perkhidmatan Kesihatan Ibu & Anak bagi Anggota Kejururawatan di Perkhidmatan Kesihatan Awam KKM 2016, Buku Rekod Kesihatan Ibu (KIK/1(a)/96 Pind 2020) and Buku Rekod Kesihatan Kanak-kanak.
- Specific conditions for clinic visits:
 - For mothers with gestational diabetes (GDM), OGTT should be repeated during the postnatal phase between 42 days until 3 months.
 - Mothers with medical conditions, appropriate management and follow-up must be arranged.

4.4.3 Virtual Consultations

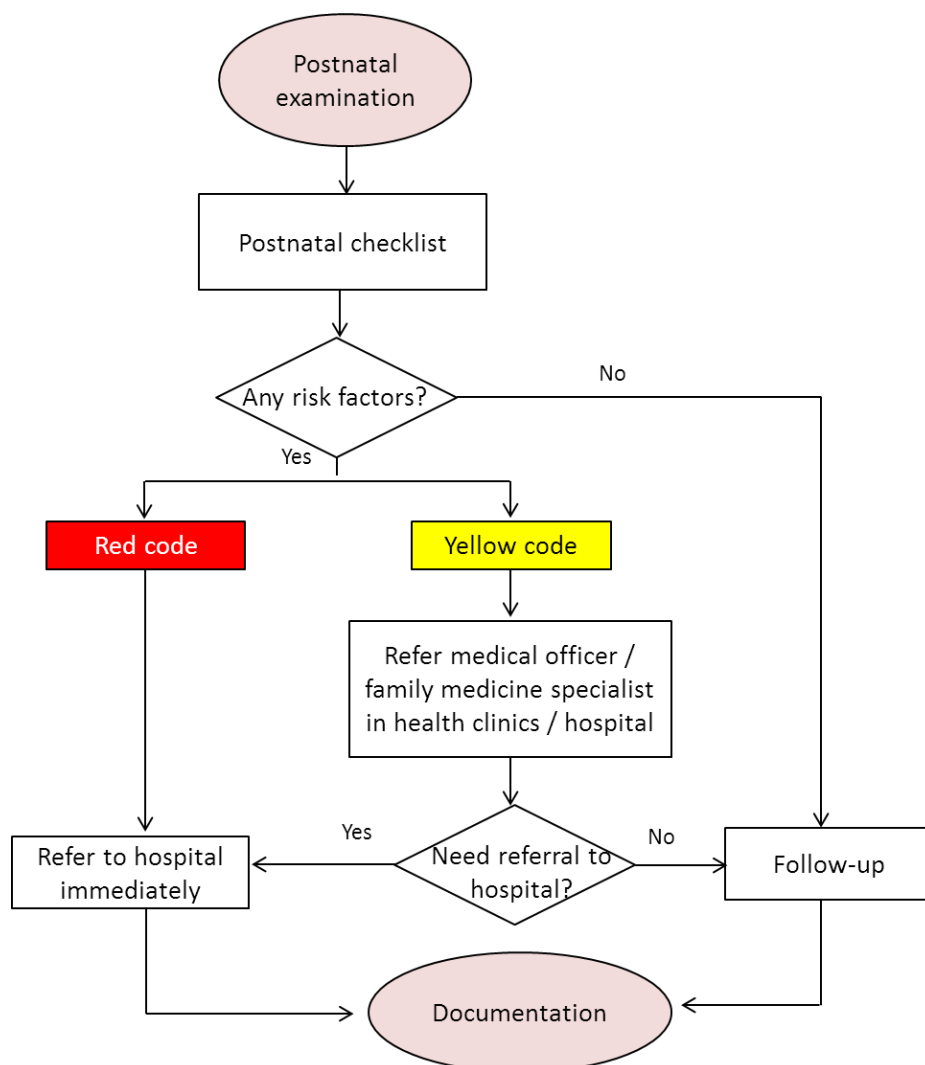
- Virtual consultation is an option for eligible postnatal mothers when all requirements are fulfilled.
- Mothers and newborns should be assessed for suitability of virtual consultations on day 5 by nurses during home visit. **Mothers with the condition as below are not eligible for virtual consultation:**
 - High risk mothers and babies as in **Table 4.2**. If either mother or newborns is high-risk, they are not suitable for virtual consultations.
 - Mothers with yellow or red postnatal coding based on *Garis panduan Senarai Semak bagi Penjagaan Kesihatan Ibu Mengikut Sistem Kod Warna (2020)*.
 - Mothers with social issues e.g.: poor social support
 - Mothers with previous history of poor compliance to advise or missed appointments
- Refer *Garis panduan Peluasan Pelaksanaan Klinik Virtual (Virtual Clinic)*, *Bahagian Pembangunan Kesihatan Keluarga, 2020* on the details for requirement of virtual consultation.
- Refer **Appendix 4-1** – Consent form for postnatal virtual consultations and **Appendix 4-2** – Checklist for postnatal virtual consultations (mother and baby).

4.5

RISK STRATIFICATION IN POSTNATAL PERIOD

- During the postnatal contacts, any abnormalities detected should be referred for further management in health clinic or hospital according to the risk checklist (Refer *Garispanduan Senarai Semak bagi Penjagaan Kesihatan Ibu mengikut Sistem Kod Warna, KKM 2020*). Mothers with risk factors are stratified to either red code or yellow code which determines the promptness of management. **Figure 4.1** illustrates the flow for action during postnatal contacts.

Figure 4.1: Flow of action during postnatal contacts



Post-miscarriage home visit is not a routine care. It is only carried out at the discretion of the attending doctor. Post-miscarriage women need to be advised on:

- **Next menses**

Most women will resume menses within 4 to 6 weeks post-miscarriage/abortion. However it may not be of the normal volume or duration of her previous cycles. It is advisable for the patient to record her menses to ensure the regularity of the menstrual cycle before planning to conceive.

- **Physical activities**

There is no restriction to resume normal physical activities following a miscarriage. However it is advisable to avoid strenuous activities such as jogging and lifting heavy weight during the immediate post miscarriage period.

- **Diet**

A well-balanced diet with appropriate amount of fluid intake will assist the body to return to its normal form. There is no particular restriction with regards to the types of food that may be or may not be consumed during the post miscarriage period.

- **Sexual relationship**

It may be appropriate for the couple to resume sexual intercourse only after vaginal bleeding has stopped.

- **Contraception**

It is advisable for a woman to avoid a pregnancy soon after the miscarriage. This is likely to happen in the event of unprotected intercourse. Contraception advise should be offered in order to space her pregnancy. This advise should be based on WHO Medical Eligibility Criteria for Contraceptive Use 2015 or *Garis panduan Kriteria Kelayakan Perubatan & Soal Jawab Amalan Penggunaan Kaedah Kontraseptif*, MOH 2017.

- **Emotional support**

Following miscarriage, a proportion of women may experience various levels of emotional changes. At times, these changes may be similar to that of a woman who has lost a baby at term. These reactions may be attributed to abrupt changes in hormonal levels or due to the loss of a wanted pregnancy. Counselling in the form of emotional support should be offered to women who experience these changes.

- **Advise on next pregnancy**

Next pregnancy can take place as long as the mother has no medical illness or constraint. Mothers are encouraged to embark on their next pregnancy once they are ready.

4.7

POSTPARTUM PRE PREGNANCY CARE

Identification of patients who require referral to the pre pregnancy service should be done at postnatal visit at 1 month (Refer to pre pregnancy risk factors – **Chapter 1**).

a. Women who should avoid future pregnancy:

This refers to a mother with serious medical conditions which may be detrimental to her life if she embarks on another pregnancy. The woman and partner should be counselled and provided with appropriate contraceptives.

b. Women who are likely to be high-risk in future pregnancies:

They should be advised on the risk during pregnancy in relation to her current health conditions. Other important information is the need to attend pre pregnancy care and use of contraceptives. They must be enrolled to the pre pregnancy care either in hospital or health clinic at the earliest possible after postnatal period.

4.8

CONTRACEPTIVES

- Discussions on birth spacing and family planning should be initiated early from the antenatal phase. Contraceptive options should be discussed and provided if agreeable.
- Postpartum mothers can become pregnant again even before they have their first menses. Breastfeeding is not a reliable form of birth control.
- Refer *Garispanduan Kriteria Kelayakan Perubatan & Soal Jawab Amalan Penggunaan Kaedah Kontraseptif*, BPKK, KKM 2017, *Buku Panduan Latihan Pemasangan dan Pengeluaran Alat Dalam Rahim*, BPKK, KKM 2019, and WHO's Four Cornerstones of Family Planning Guidance, WHO.

4.9

RESUMING SEXUAL INTIMACY

- Among the needs of women in the postpartum period are information and counselling on sexual life.
- Sexual intercourse may be resumed after the mother's vaginal bleeding has stopped and perineal wound stitches has healed. Usually, recovery is within four to six weeks following delivery.

- The couple should decide together, with the advise of their health care provider, when to resume sexual intimacy. Initially, sex following birth may be painful. Advise to use a lubricant or trying different positions that allow the woman to be in control of penetration may help.

4.10

POSTNATAL EXERCISES

- All women should be encouraged to mobilize as soon as possible following the birth. They should be encouraged to take gentle exercise and make time to rest during the postnatal period.

Refer to *Manual Senaman Antenatal dan Postnatal di Klinik Kesihatan*, BPKK, KKM 2014.

BORANG KEIZINAN KONSULTASI POSTNATAL SECARA MAYA

Konsultasi postnatal secara maya adalah penyampaian perkhidmatan kesihatan secara maya (*virtual*), langsung (*live*) dan interaktif yang merangkumi konsultasi klinikal dan pelan rawatan pelanggan di antara anggota kesihatan dan pelanggan. Konsultasi postnatal secara maya merupakan satu inisiatif yang dilaksanakan bagi memastikan penjarakan sosial dan menyediakan perkhidmatan kesihatan yang berterusan setaraf dengan kemajuan teknologi.

Saya telah diberi penerangan oleh anggota kesihatan tentang pelaksanaan konsultasi postnatal secara maya dan bersetuju:

1. Untuk menyertai sesi konsultasi postnatal secara maya.
2. Maklumat klinikal semasa sesi konsultasi boleh dikongsi dengan anggota kesihatan lain untuk tujuan rujukan sekiranya perlu, bagi kesinambungan perawatan.
3. Sebarang rakaman video atau audio sepanjang sesi konsultasi postnatal secara maya tidak boleh ditular/ dikongsi/ disebar kepada pihak luar oleh kedua-dua pihak
4. Rakaman tidak boleh digunakan sebagai bukti untuk tindakan undang-undang terhadap KKM
5. Anggota kesihatan yang bertugas boleh memberhentikan konsultasi jika:
 - a. berlaku gangguan akses capaian internet atau sebarang masalah teknikal.
 - b. jenis perbualan tiada berkaitan dengan tujuan konsultasi kesihatan saya
6. Saya berhak untuk tidak meneruskan konsultasi postnatal secara maya atas pilihan saya sendiri.
7. Saya akan hadir ke klinik sekiranya terdapat keperluan untuk pemeriksaan lanjutan.
8. Konsultasi secara maya boleh menyebabkan kelewatan dalam rawatan berbanding konsultasi bersemuka.

Ditandatangani:

(Ibu)
Nama :
No. Kad pengenalan:
Tarikh:

(Saksi*)
Nama :
No. Kad pengenalan:
Jawatan :
Tarikh :
Cop jawatan:

** Saksi boleh terdiri dari Pakar Perubatan Keluarga/ Pegawai Perubatan/ Jururawat/ Penolong Pegawai Perubatan

**SENARAI SEMAK PEMANTAUAN IBU POSTNATAL SECARA VIRTUAL UNTUK
ANGGOTA KESIHATAN**

KONSULTASI POSTNATAL SECARA VIRTUAL (HARI 8-10)			
No.	Soalan	Jawapan	Tindakan
1.	Untuk meneruskan konsultasi secara virtual, pastikan anggota kesihatan dapat melihat imej ibu sehingga sekurang-kurangnya ke paras dada. Ini bagi membolehkan anggota kesihatan dapat melihat penampilan ibu secara keseluruhan.		
2.	Bertanya khabar ibu?	Sihat / tidak sihat	Jika ibu tidak sihat, nasihat ibu untuk datang klinik
3.	Setuju untuk teruskan konsultasi secara <i>virtual</i>	Ya / Tidak	Jika ibu tidak bersetuju untuk teruskan konsultasi secara virtual, beri temujanji klinik
4.	Demam	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada doktor
5.	Masalah pernafasan	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada doktor
6.	Pening/ pitam/ letih / lesu / berdebar-debar	Ya / Tidak	Jika ya, nasihat ibu untuk datang ke klinik dan maklumkan kepada doktor
7.	Sakit dada Sesak nafas Kesukaran bernafas ketika baring Cepat penat	Ya / Tidak Ya / Tidak Ya / Tidak Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital
8.	Sakit kepala/ pening / loya/ muntah/ mata kabur	Ya / Tidak	Jika ya kepada salah satu, nasihat ibu untuk datang ke klinik.
9.	Pendarahan / lochia yang banyak /	Normal / tidak normal	Jika tidak normal, nasihat ibu untuk

KONSULTASI POSTNATAL SECARA VIRTUAL (HARI 8-10)			
No.	Soalan	Jawapan	Tindakan
	berbau / discaj vagina / sakit perut		datang ke klinik
10.	Payudara dan penyusuan	○ Bengkak payudara ○ Sakit payudara ○ Masalah penyusuan	Jika terdapat masalah, nasihat ibu untuk datang ke klinik
11.	Bagaimana keadaan luka perineum/ pembedahan? ○ Sangat sakit ○ Kemerahan ○ Bengkak ○ Jahitan terbuka ○ Lelehan/ discaj	Ya / Tidak Ya / Tidak Ya / Tidak Ya / Tidak Ya / Tidak	Jika ya kepada salah satu, nasihat ibu untuk datang ke klinik.
12.	Edema di kaki/ tangan/ muka secara mendadak	Ya / Tidak	Jika edema di kaki, tanya lebih lanjut sehingga tahap mana. Jika sehingga lutut, nasihat ibu untuk datang ke klinik. Jika edema di tangan atau muka, nasihat ibu untuk datang ke klinik
13.	Betis bengkak/ sakit / kemerahan	Ya / Tidak	Jika ya, minta ibu untuk segera ke hospital
14.	Adakah ibu mempunyai tanda-tanda dehidrasi/kurang minum air: - kering bibir - kurang kencing, atau kencing berwarna gelap	Ya / Tidak	Jika tidak, nasihat ibu kepentingan hidrasi dan minum air yang cukup
15.	Kesihatan mental ○ Sejak kebelakangan ini, adakah anda sering diganggui perasaan murung atau sedih?	Ya / Tidak	Jika ya kepada salah satu, nasihat ibu untuk datang ke klinik.

KONSULTASI POSTNATAL SECARA VIRTUAL (HARI 8-10)			
No.	Soalan	Jawapan	Tindakan
	Sejak kebelakangan ini, adakah anda sering kehilangan minat atau keseronokan dalam melakukan kerja harian?		
16.	Kepatuhan pengambilan ubat / suplemen	Ya / Tidak	Jika tidak patuh, nasihat ibu dan tekankan mengenai kepentingan kepatuhan kepada pengambilan ubat / suplemen tersebut
17.	Soalan umum : Adakah mempunyai masalah lain? – contohnya masalah kencing, buang air besar, corak tidur terganggu	Ya / Tidak	Jururawat nasihat ibu apa yang perlu dilakukan dan jika perlu, berbincang dengan doktor.
18.	Adakah ibu tahu temujanji ke klinik pada 1 bulan selepas bersalin?	Ya / Tidak	Jika tidak, maklumkan tarikh temujanji. Nasihat ibu untuk hubungi klinik/hospital atau datang ke klinik/hospital jika mempunyai sebarang masalah.
Nota: 1. Jika ibu diminta untuk ke hospital, jururawat / pegawai perubatan perlu beritahu mengenai kes kepada pihak hospital/ PAC dan memaklumkan status ibu jika terdapat sebarang perubahan (Rujuk <i>Garis panduan Perkhidmatan Kecemasan dan Ambulans di Fasiliti Kesihatan Primer</i>). 2. Jururawat perlu melakukan tindak susul ke atas perancangan pengendalian/ penjagaan ibu.			

SENARAI SEMAK PEMANTAUAN BAYI SECARA *VIRTUAL* UNTUK ANGGOTA KESIHATAN

KONSULTASI POSTNATAL SECARA <i>VIRTUAL</i> UNTUK BAYI (HARI 8-10)					
No.	Perkara	Soalan	Penerangan	Jawapan	Tindakan
1.	Tindakbalas bayi	Adakah bayi aktif?	Tanda-tanda tidak aktif: - Tidur lama - Susah dikejutkan - Kurang pergerakan - Kurang menyusu	Ya/Tidak	Sekiranya tidak aktif, sila segera bawa ke klinik atau hospital terdekat
2.	Demam	Adakah bayi demam?	Sentuhan yang panas atau direkodkan > 37.5°C	Ya/Tidak	Sekiranya rasa demam, sila segera bawa ke klinik atau hospital terdekat
3.	Jaundis	Adakah bayi kelihatan kuning?	<i>Blanching test</i> sekiranya pernah diajar.	Ya/Tidak	Sekiranya jaundis yang pernah menerima rawatan, sila semak TSB yang terakhir dan rujuk kepada doktor untuk tindakan seterusnya. Sekiranya jaundis kes baru, jaundis bertambah, ataupun ada keraguan, nasihat ibu untuk bawa ke klinik hari yang sama.

KONSULTASI POSTNATAL SECARA VIRTUAL UNTUK BAYI (HARI 8-10)					
No.	Perkara	Soalan	Penerangan	Jawapan	Tindakan
4.	Kekerapan Penyusuan	Berapa kali bayi menyusu?	Sekurang-kurangnya 8-12 kali dalam tempoh 24 jam	Ya/Tidak	Sekiranya Tidak, lawatan rumah/ klinik perlu dilakukan.
5.	Kencing	Berapa kali bayi kencing?	Menukar lampin basah sekurang-kurangnya 5-6 kali dalam tempoh 24 jam	Ya/Tidak	Sekiranya Tidak, lawatan rumah/klinik perlu dilakukan.
6.	Membuang air besar	Berapa kali bayi membuang air besar?	Sekurang-kurangnya 2 kali dalam tempoh 24 jam	Ya/Tidak	Sekiranya Tidak, lawatan rumah/klinik perlu dilakukan.
7.	Pernafasan	Adakah bayi bernafas seperti biasa?	Tanda-tanda pernafasan yang tidak normal: -Sesak nafas -Nafas laju -Lekuk di dada	Ya/Tidak	Sekiranya tidak normal, sila segera bawa ke klinik atau hospital terdekat
8.	Abdomen	Adakah perut dan tali pusat bayi kelihatan biasa?	Tanda-tanda yang membimbangkan: -Perut kembung -Tali pusat merah ataupun bernanah	Ya/Tidak	Sekiranya tidak normal, sila segera bawa ke klinik atau hospital terdekat
9.	Lain-lain	Adakah mempunyai kebimbangan yang lain?	Nyatakan: _____	Ya/Tidak	Tindakan bergantung kepada penilaian. Jika perlu, sila rujuk doktor

KONSULTASI POSTNATAL SECARA VIRTUAL UNTUK BAYI (HARI 8-10)					
No.	Perkara	Soalan	Penerangan	Jawapan	Tindakan
Tanda-tanda bahaya (<i>warning signs</i>) yang menandakan bayi memerlukan tindakan segera: • kelihatan pucat dan kebiruan • muntah kerap/hijau • sawan • tidak aktif • demam • tidak menyusu dengan baik					
Nota: 1. Jika dinasihatkan untuk ke hospital, jururawat / pegawai perubatan perlu beritahu mengenai kes kepada pihak hospital dan memaklumkan status bayi jika terdapat sebarang perubahan (Rujuk <i>Garis panduan Perkhidmatan Kecemasan dan Ambulans di Fasiliti Kesihatan Primer</i>). 2. Jururawat perlu melakukan tindaksusul ke atas perancangan pengendalian/ penjagaan bayi.					

CHAPTER 5

MEDICAL COMPLICATIONS IN PREGNANCY

CHAPTER 5: MEDICAL COMPLICATIONS IN PREGNANCY

5.1 HAEMATOLOGICAL DISORDER IN PREGNANCY

5.1.1 Anaemia in Pregnancy

PHASE	PLAN OF ACTION
Pre pregnancy	• Women with known anaemia need to have her anaemia corrected before embarking on pregnancy • Effective communication with women about diet and healthy eating is essential.
At diagnosis in pregnancy	• FBC must be done at booking • Anaemia : Hb <11g/dL – investigate for the cause. • The most common cause is iron-deficiency anaemia (IDA). Hemoglobinopathy should be screened especially after IDA has been ruled out. IDA & hemoglobinopathy may co-exist. • Hematinics should be continued while investigation is on-going. An improvement after a course of oral iron supplementation indicates IDA. • Anaemia workout includes: ○ Serum ferritin ○ Transferrin saturation (if serum ferritin is not conclusive) - send for TIBC & Serum Iron ○ FBP ○ BFMP ○ Stool for ova & cyst • Role of serum ferritin: ○ Routine antenatal serum ferritin testing along with FBC at booking would detect iron depletion before a patient becomes anaemic. Therefore allowing better opportunity to replenish iron store to prevent maternal or neonatal complications. ○ If routine antenatal serum ferritin is not feasible (e.g. cost constraint), it should then be performed in patients at increased risk of iron deficiency (ID) or bleeding (e.g. multiparity, previous IDA, multiple pregnancy, placenta previa, previous scars, history of PPH, etc). ○ Iron deficiency is already advanced by the time anaemia is detected. The deficiency has consequences even when no anaemia is clinically apparent. ○ Point-of-care test (POCT) ferritin would potentially lower the cost of the test and facilitate earlier management of ID/ IDA. • Oral Iron therapy: ○ For prevention of anaemia (normal Hb):

PHASE	PLAN OF ACTION
	▪ Daily oral iron with 30-60 mg of elemental iron (higher dose of 60mg elemental iron is preferred in view of high prevalence of anaemia in pregnancy such as in Malaysia) ▪ Every other day (EOD) dosing of 30-60 mg elemental iron (e.g. Mon/Wed/Fri) to reduce gastrointestinal side effects ▪ Weekly oral iron with 120 mg elemental iron if daily iron is not acceptable due to side effects ○ For treatment of anaemia (Hb <11.0 g/dL): ▪ Daily oral iron with 120 mg elemental iron ▪ Resume the standard daily antenatal iron dose once her anaemia has been corrected. ▪ Include Folic Acid 400 µg (0.4 mg) with iron supplement (daily or every other day dosing) or 2800 µg (2.8 mg) with weekly iron supplement. Folic acid should be commenced as early as possible, ideally before conception to prevent neural tube defects. ● IV Iron therapy: ○ IV Iron should be administered if oral iron is not tolerated, not-compliant or if response is poor (i.e. <1 g/L in 2 weeks or <2 g/L in 4 weeks) ○ Do not use IM route ○ DO NOT administer IV Iron in first trimester ○ Use of Erythropoietin Stimulating Agent (ESA) may be considered if poor response after IV Iron, after consultation with haematologist. ● Transfusion is not a treatment for anaemia. It is only for bleeding, unstable patients. Use non-blood alternatives instead of transfusion whenever possible. ● Asymptomatic anaemia ○ Hb 7-11 g/dL irrespective of gestational age – follow up at health clinic. ○ Try a course of oral iron. If there is poor / no response after 1 or 2 types of oral iron, switch to IV Iron unless contraindicated. If gestational age is advanced, use IV Iron sooner. The response time to IV Iron is the same, though faster response may be seen, hence timing of treatment is important. ○ Hb <7 g/dL and POA <34weeks – refer FMS ○ Hb <7 g/dL and >34weeks – refer O&G ○ Asymptomatic patients may be referred to FMS for IV Iron in health clinics if available. ● Symptomatic anaemia – irrespective of Hb level or gestational age, refer to O&G specialist.

PHASE	PLAN OF ACTION
Subsequent antenatal follow-up	• Monitor monthly Hb level at health clinic • There may be a need to repeat serum ferritin especially after a course of IV Iron. • Foetal growth monitoring at health clinic
Delivery plan	• Hospital delivery for anaemic patients, with GSH on standby. • PPH prophylaxis • Generally may allow postdates, unless specified otherwise. • There is no indication to transfuse just to top up a patient's Hb level as long as the patient is stable and asymptomatic.
Postpartum	• Discuss options of contraception with patient / couple (Refer Medical Eligibility Criteria for Contraceptive) • Continue iron supplementation for 3 to 6 months postpartum • Use iron supplementation to improve Hb if the patient had PPH – IV Iron if a faster increment in Hb and better replenishment of iron stores are needed (e.g. after massive PPH, severe postpartum anaemia). • Blood transfusion is required if evidence of active bleeding or sign and symptoms of heart failure
Upon discharge from hospital	• Routine discharge procedure • High risk postnatal notification if a patient had PPH and awaiting for Hb to increase with iron supplementation (oral or IV).

REMARKS:

1. WHO defines anaemia in pregnancy as Hb < 11 g/dL. Trimester-specific Hb thresholds for diagnosing anaemia are: • 1st & 3rd trimester: Hb 11.0 g/dL • 2nd trimester: Hb 10.5 g/dL • Postpartum: Hb 10 g/dL WHO further defines the severity of anaemia in pregnancy as below: • Mild: Hb 10.0-10.9 g/dL • Moderate: 7.0-9.9 g/dL • Severe: <7.0 g/dL Mild anaemia is a misnomer: iron deficiency is already advanced by the time anaemia is detected. The deficiency has consequences even when no anaemia is clinically apparent. Prevalence of anaemia among pregnant women ≥40% denotes severe public health significance.
2. If anaemia (Hb <11 g/dL) – investigate for the cause • Iron deficiency anaemia (IDA): ☐ If FBC shows hypochromic microcytic anaemia, to send for Serum Ferritin. ☐ Serum Ferritin <30µg/l indicates iron depletion (sensitivity of 90%, specificity

of 85%) ☐ If Serum Ferritin between 30-100µg/l, proceed to calculate Transferrin Saturation (TSat) by sending for TIBC and Serum Iron. TSat <20% indicates iron depletion. • Hemoglobinopathy ☐ If IDA is ruled out, send for Hb analysis to screen for hemoglobinopathy.
3. Indications for assessment of Serum Ferritin: • Anaemic women when estimation of iron stores is necessary: ☐ Known hemoglobinopathy (to rule out concomitant IDA) • Non-anaemic women with high risk of iron depletion: ☐ Previous anaemia ☐ Multiparity ☐ Short pregnancy interval of <1yr ☐ Vegetarians ☐ Teenage pregnancies ☐ Recent history of bleeding • Non-anaemic women where estimation of iron stores is necessary: ☐ High risk of bleeding ☐ Jehovah's witness / women who refuse blood or blood product
4. Referral to hospital when: • Significant symptoms • Severe anaemia (Hb <7.0 g/dL) • Anaemia in late gestation (>34weeks) • Failure to respond to oral iron (this can be managed in health clinic if IV Iron is available) Asymptomatic patients may be referred to FMS for IV Iron in health clinic if available.
5. Oral Iron preparation containing 60 mg elemental iron: • Ferrous Gluconate 500 mg • Ferrous Sulphate 300 mg • Ferrous Fumarate 180 mg
6. IV Iron – examples: • Iron sucrose such as Iron (III) hydroxide sucrose complex (Venofer) • Low Molecular Weight Iron Dextran such as Iron (III) hydroxide dextran complex (Cosmofer) • Iron (III) carboxymaltose (Ferinject) • Iron (III) isomaltoside (Monofer) The latter two are examples of single high-dose preparation. • Do not use Iron Sucrose Similar (ISS) as they are associated with more side effects and are less efficacious.

- Do not use IM route for parenteral iron administration.
- Avoid parenteral Iron during 1st trimester.

7. Patient Blood Management (PBM) has three pillars:

- 1st Pillar – optimising erythropoiesis
- 2nd Pillar – minimise bleeding and blood loss
- 3rd Pillar – harness and optimise physiologic tolerance of anaemia

Early identification and treatment of iron deficiency and iron deficiency anaemia is an important component of PBM which includes diagnosis and use of iron therapy including IV Iron. IV Iron should be considered for stable postpartum anaemia patients.

Reference:

1. WHO Recommendations on Antenatal Care for a Positive Pregnancy Experience, 2016.
2. UK Guidelines on the Management of Iron Deficiency in Pregnancy, British Committee for Standards in Haematology, 2012
3. Flores CJ, Sethna F, Stephens B, et al. Improving Patient Blood Management in Obstetrics: snapshots of a practice improvement partnership. BMJ Quality Improvement Reports 2017; 6:e000009. DOI:10.1136/bmjquality-2017-000009
4. Moretti D, Goede JS, Seder C, et al. Oral iron supplements increase hepcidin and decrease iron absorption from daily or twice-daily doses in iron-depleted young women. Blood, 22 October 2015
5. Holm C, Thomsen LL, Norgaard A, Langhoff-Roos J. Single-dose intravenous iron infusion versus red blood cell transfusion for the treatment of severe postpartum anaemia: a randomised controlled pilot study. The International Journal of Transfusion Medicine 2016; DOI: 10.1111/vox.12475
6. Auerbach M. Commentary: Iron deficiency of pregnancy – a new approach involving intravenous iron. Reproductive Health 2018, 15 (Suppl 1):96; <https://doi.org/10.1186/s12978-018-0536-1>

5.1.2 Thalassaemia in Pregnancy

PHASE	PLAN OF ACTION
Pre pregnancy	• If either partner or the couple is a thalassaemia carrier, refer to FMS / O&G for counselling, including information regarding prenatal diagnosis • Screen untested partner • Advise for early booking before 12 weeks of gestation
At diagnosis	• All antenatal women should be offered screening if they have a family history of haemoglobinopathy • All women who have MCV $\leq$ 80fL and MCH $\leq$ 27pg and a normal ferritin level (e.g. $> 30\mu\text{g/L}$) should be offered Hb analysis • If confirmed carrier, screen partner for thalassaemia • If the couple is carrier, refer to O&G Clinic for counselling $\pm$ prenatal diagnosis
Subsequent antenatal follow-up	• Monitor Hb level. If anaemic, check serum ferritin and for iron supplement if serum ferritin $< 30\text{ ng/ml}$ • Folic acid should be given throughout pregnancy • Referral to O&G if Hb $< 7\text{ g/dL}$ or symptomatic of anaemia
Delivery plan	• Keep Hb $> 7.0\text{ g/dL}$ • Generally may allow post date, unless specified otherwise • If moderate or severe anaemia, deliver at specialist hospital and give PPH prophylaxis
Postpartum	• If both parents are carriers – refer baby to paediatrics at delivery • If only the mother is a carrier – baby to be seen at 6 months in health clinics • Discuss contraception options with couple
Upon discharge from hospital	• Pre pregnancy counselling appointment for newly diagnosed thalassaemic couple

5.1.3 Thrombocytopenia in Pregnancy

PHASE	PLAN OF ACTION
Pre pregnancy (if known pre-existing case) or at booking / diagnosis	- Known thrombocytopenia in pregnancy need to be seen in pre pregnancy clinic and under specialist care - Screen for thrombocytopenia at booking by performing FBC - Repeat FBC is required if FBC is thrombocytopenic - Referral to medical / O&G if the repeated FBC remain thrombocytopenic - Look for bleeding tendencies, immediate referral is required if present clinically - Tests to be done: ☐ FBP ☐ LFT ☐ RP ☐ Viral screening (HIV, HCV, HBV) ☐ ANA - Refer patient with asymptomatic thrombocytopenia in pregnancy with platelet $<100 \times 10^9/L$ or symptomatic thrombocytopenia to O&G clinic / Medical clinic - Patient with mild asymptomatic thrombocytopenia ($100-150 \times 10^9/L$) can be monitored at Health Clinic
Subsequent antenatal follow-up	- Follow plan laid out by O&G / medical / haematology clinic - Follow up of mild asymptomatic thrombocytopenia in pregnancy at health clinic with monthly FBC monitoring
Delivery plan	- For hospital delivery - Give PPH prophylaxis - Consider anaesthetic referral for moderate to severe thrombocytopenia - Medical / haematology input if severe thrombocytopenia for possible need of intervention (steroid / azathioprine / IV IgG or platelet transfusion) - Avoid traumatic delivery / foetal scalp sampling / foetal scalp electrode - Refer baby to Paediatrics to rule out neonatal thrombocytopenia - Avoid Vit K injection for the baby until thrombocytopenia is ruled out (to discuss with Paediatric team)
Postpartum	Discuss options of contraception with patient / couple (Refer WHO Medical Eligibility Criteria for Contraceptive Use, 2015)
Upon discharge from hospital	- Repeat FBC at 6 weeks postpartum for gestational thrombocytopenia - For other causes of thrombocytopenia, follow medical/ haematology plan

REMARKS:

● Definition of thrombocytopenia in pregnancy: Platelet $<150 \times 10^9/L$ ☐ Mild: $100-150 \times 10^9/L$ ☐ Moderate: $50-100 \times 10^9/L$ ☐ Severe: $<50 \times 10^9/L$
● Thrombocytopenia occurs in 7-8% of all pregnancy: ☐ 70-80% are gestational thrombocytopenia ☐ 15-20% are severe pre-eclampsia ☐ $<1\%$ are HELLP syndrome and APS
● Differential diagnosis: ☐ Hereditary ☐ Autoimmune disease (SLE, APS) ☐ Pre-eclampsia ☐ HELLP syndrome ☐ DVC ☐ Drug-induced ☐ Viral Infection (HIV, Dengue, HCV) ☐ Hypersplenism due to chronic liver disease ☐ Haematological malignancy ☐ Gestational thrombocytopenia ☐ Spurious-platelet clumping
● In gestational thrombocytopenia, platelet should have normalised by 6 weeks postpartum

Reference:

1. Myers B. Thrombocytopenia in pregnancy, Royal College of Obstetrics and Gynaecology guidelines, 2009
2. Clinical Practice Guidelines on Management of Immune Thrombocytopenic Purpura, MOH 2006

5.1.4 Rhesus Isoimmunisation in Pregnancy

PHASE	PLAN OF ACTION
At booking / diagnosis	• Blood group & Rh status at first antenatal visit • Check husband's blood group & Rh status if patient is Rh Negative • Indirect Coomb's test / indirect agglutination test are required to detect prior sensitization ☐ At first antenatal visit for known case of Rhesus negative ☐ After diagnosis of new cases
Subsequent antenatal follow-up	• If Indirect Coomb's test / indirect agglutination test is positive, refer to MFM or O&G • If initial Indirect Coomb's test / indirect agglutination is negative, repeat test is required at 24-26 weeks • Routine Antenatal Anti-D Prophylaxis (RAADP) for non-sensitised patient, either one of these regimes: ☐ 2-dose regime: IM Anti-D Immunoglobulin 500iu at 28 weeks and 34 weeks ☐ Single dose regime: IM Anti-D Immunoglobulin 1500iu between 28-34 weeks • RAADP may be administered in a health clinic if Anti-D immunoglobulin is available.
Delivery plan	• Hospital delivery • Mode and timing of delivery as per obstetric indications
Postpartum	• Refer baby to Paediatrics • Administer IM Anti-D immunoglobulin 500iu within 72 hours if baby is Rh positive • IM Anti-D immunoglobulin 500iu is also required for IUD >20 weeks at diagnosis and again after delivery, unless delivery is soon after IUD is diagnosed • If baby's direct Coomb's test / direct agglutination test is positive, baby is at risk of haemolytic disease of newborn (HDN) and Kleihauer test for the mother should be performed to quantify the foetal maternal haemorrhage (FMH). Further Anti-D required if FMH >4ml.
Upon discharge from hospital	• Contraceptive advise

Remarks:

Direct Coomb's/ agglutination test is performed to detect anti-globulins attached to red blood cell surfaces and their interactions in-vivo. Direct Coomb's test is important to diagnose autoimmune haemolytic anaemia.

Indirect Coomb's / agglutination test is performed to detect the presence of antiglobulins in the serum in an unbound state and to detect their interactions in-vitro with Coombs' anti-human globulins. Indirect Coomb's test is important for prenatal testing for pregnant women prior to blood transfusion.

Anti-D Prophylaxis:

- 1) Routine – Routine Antenatal Anti-D Prophylaxis (RAADP) & postnatal prophylaxis
- 2) Any potentially sensitising events

GESTATIONAL WEEK	POTENTIALLY SENSITISING EVENTS	ANTI-D PROPHYLAXIS
<12 weeks	Vaginal bleeding + severe pain ERPOC/ instrumentation of uterus Medical / Surgical TOP Ectopic / Molar pregnancy	250iu Anti-D Immunoglobulin within 72 hours of event
12-20 weeks	Amniocentesis, CVS, cordocentesis PV bleeding in pregnancy Abdominal trauma (sharp/blunt, open/closed) Ectopic pregnancy Miscarriages, threatened miscarriage ERPOC / TOP In-utero therapeutic interventions	250iu Anti-D Immunoglobulin within 72 hours of event
20weeks to term	APH ECV Abdominal trauma (sharp/blunt, open/closed) Delivery Intraoperative cell salvage IUD	Regardless of whether RAADP has been given, 500iu Anti-D Immunoglobulin to be given within 72 hours of event

5.2 HYPERTENSIVE DISORDER IN PREGNANCY

5.2.1 Gestational Hypertension / Pre-eclampsia

PHASE	DEFINITION	PLAN OF ACTION
At booking	Gestational Hypertension is detected after 20 week. Hypertension in pregnancy is defined as BP 140/90 on 2 occasions 4 to 6 hours apart. Pre-eclampsia is defined as gestational hypertension with proteinuria.	- Identify patient at risk - Offer women with high risk of pre-eclampsia for screening (i.e.; prior pre-eclampsia, chronic hypertension, pregestational diabetes, BMI >30, antiphospholipid syndrome, receipt of assisted reproduction) - If fulfil criteria as high risk for pre-eclampsia: - To start Cardipirin 100mg ON or Aspirin 150mg ON once foetal heart activity is seen (not recommended after 20 weeks). - Recommended dose of calcium is 1.5 – 2g elemental calcium after 20 weeks until delivery. (Calcium Carbonate 1g BD or Calcium Lactate 600mg TDS. Calcium Carbonate is preferred due to better absorption).
At diagnosis of gestational hypertension	Mild: - SBP 140 – 149 mmHg - DBP 90 – 99 mmHg - without proteinuria Moderate: - SBP 150 – 159 mmHg - DBP 100 – 109 mmHg Severe: - SBP ≥160mmHg - DBP ≥110 mmHg	- Assess severity: - Blood pressure - Urine albumin - SFH - Hb and platelet (baseline) - Renal function (baseline) - Liver function test (baseline) - Uric acid (as a screening for referral) Results should be reviewed by medical officer within 1 week - Choice of anti-HPT: - Methyldopa - Labetalol - Nifedipine - Treatment target: - SBP: 140-149 mmHg - DBP: 90-99 mmHg - Consider reducing anti-HPT if BP < 140/90 mmHg
At diagnosis of pre-eclampsia		Refer for admission and specialist assessment

PHASE	DEFINITION	PLAN OF ACTION
At diagnosis of severe pre-eclampsia / eclampsia		• Management □ Inform O&G specialist on-call □ If BP is not controlled, consider IV Labetolol 20mg/ IV Hydralazine 5mg or oral Nifedipine 10mg. □ Give loading dose of IM MgSO₄ 5g (with Lignocaine 2%) before transfer (after discussing with O&G specialist on-call /FMS). □ Maintenance dose of 5g every 4 hours (at alternate buttocks) as per <i>Garis panduan Pemberian Suntikan Intramuscular MgSO₄ di Peringkat Penjagaan Kesihatan Primer bagi Kes Severe Pre-eclampsia / Eclampsia, 2014</i> • May request for Obstetrics Emergency Retrieval Team (OERT) if available
Subsequent antenatal follow-up		• Educate and advise mother to return immediately if develop symptoms of impending eclampsia • Mild gestational hypertension - manage at health clinic - weekly BP monitoring • If BP well controlled, case can be managed by MO and refer to FMS for assessment at 20-24 weeks and 32-34 week • Moderate gestational hypertension - refer to FMS/O&G - biweekly BP monitoring • Any hypertension with proteinuria or severe gestational hypertension - to refer for admission • Foetal surveillance with SFH, FKC, foetal growth monitoring by serial ultrasound starting at 28 weeks, at 4-weekly interval
Delivery plan		• Delivery plan to be outlined by O&G specialist at about 36 weeks • Hospital delivery
Postpartum		• Continue to monitor BP after delivery until 6 weeks postpartum (the frequency should be individualised) • Wean down treatment dose of anti-HPT and do not stop abruptly • Discuss options of contraception with couple • Please note that persistent hypertension in young women need to be investigated for the

PHASE	DEFINITION	PLAN OF ACTION
		secondary causes and managed accordingly
Upon discharge from hospital		• Notification of high-risk discharge from hospital to respective health clinics as per guideline. • Respective health clinic will continue with follow-up care (unless specified otherwise on high risk discharge summary). • Home visit: BP monitoring, signs & symptoms of pre-eclampsia (frequency individualised). • Two weeks review at health clinic by MO.

5.2.2 Chronic Hypertension In Pregnancy

PHASE	PLAN OF ACTION
Pre pregnancy	• Refer FMS for assessment & counselling • Screen for secondary causes of hypertension • Need to rule out secondary complications of hypertension • Review safety of medication • Discuss pre-eclampsia prophylaxis
At booking / diagnosis	• Dating scan • Start aspirin once foetal heart activity is seen and calcium after 20 weeks • Recommended monitoring: ☐ Blood pressure ☐ Body weight ☐ Urine albumin ☐ SFH ☐ Hb and platelet (baseline) ☐ Renal function (baseline) ☐ Liver function test (baseline) ☐ Uric acid (as a screening for referral) ☐ ECG. If it is suggestive of left ventricular hypertrophy (LVH), for echocardiography. - Results should be reviewed by medical officer within 1 week • Choice of anti HPT: ☐ <20 weeks: Methyldopa is preferred. ☐ >20 weeks: Methyldopa, Labetalol and Nifedipine can be used • Treatment target: ☐ SBP: 140-149 mmHg ☐ DBP: 90-99 mmHg • Consider reducing anti-HPT if BP < 140/90 mmHg
Subsequent antenatal follow-up	• Educate and advise mother to return immediately if develop symptoms of impending eclampsia • Nutritional intervention or advise by dietitian/ nurse • Mild hypertension – manage at a health clinic. Weekly BP monitoring • If BP is well-controlled, the case can be managed by MO. • First assessment by FMS at around 20 weeks of gestation, refer to O&G for further management if required. Subsequent assessment to be decided by FMS based on patient's condition • Moderate hypertension refers to FMS/O&G. Biweekly BP monitoring. • Any hypertension with proteinuria or severe hypertension to refer for admission • Foetal surveillance with SFH, FKC, foetal growth monitoring by serial ultrasound, starting at 28 weeks, at 4-weekly interval
Delivery plan	• Delivery plan to be outlined by O&G specialist at about 36 weeks • Hospital delivery
Postpartum	• Continue antenatal anti-HPT after delivery

PHASE	PLAN OF ACTION
	□ Aim for BP <140/90 mmHg. ● Continue to monitor BP after delivery (frequency of monitoring should be individualised) ● Discuss options of contraception with couples and reinforce the importance of contraception and planned pregnancy. ● Restart old medication, as most hypertensive medication is safe for breastfeeding
Upon discharge from hospital	● Notification of high-risk discharge from hospital to respective health clinics as per guideline ● Respective health clinics will continue with follow-up care (unless specified otherwise on high-risk discharge summary). ● Home visit: BP monitoring, signs & symptoms of pre-eclampsia (frequency of visit and monitoring is individualised) ● Review by MO at a health clinic in two weeks.

5.3 DIABETES IN PREGNANCY

5.3.1 Gestational Diabetes Mellitus

*Refer Clinical Practice Guidelines – Management of Diabetes in Pregnancy,
Ministry of Health, 2017*

5.3.2 Pre-existing Diabetes in Pregnancy

*Refer Clinical Practice Guidelines – Management of Diabetes in Pregnancy,
Ministry of Health, 2017*

5.4 THYROID DISORDER IN PREGNANCY

5.4.1 Hyperthyroidism in Pregnancy

PHASE	DEFINITION	PLAN OF ACTION												
Pre pregnancy	Clinical hyperthyroidism is confirmed in the presence of a suppressed ($<0.1\text{mU/L}$) or undetected ($<0.01\text{mU/L}$) TSH and an elevated FT4 <table border="1"> <thead> <tr> <th>Tri - mester</th><th>TSH (mU/L)</th><th>FT4 (pmol/L)</th></tr> </thead> <tbody> <tr> <td>First</td><td>0-5.5</td><td>10 -16</td></tr> <tr> <td>Second</td><td>0.5-3.5</td><td>9-15.5</td></tr> <tr> <td>Third</td><td>0.5-4</td><td>8-14.5</td></tr> </tbody> </table> (Reference: <i>Oxford Handbook of Obstetrics and Gynaecology, 2013</i>)	Tri - mester	TSH (mU/L)	FT4 (pmol/L)	First	0-5.5	10 -16	Second	0.5-3.5	9-15.5	Third	0.5-4	8-14.5	- Refer for counselling by FMS / O&G (pre pregnancy clinic) - Thyrotoxic women should be rendered euthyroid before attempting pregnancy - The treatment of thyrotoxicosis should be optimised with the lowest dose of effective treatment
Tri - mester	TSH (mU/L)	FT4 (pmol/L)												
First	0-5.5	10 -16												
Second	0.5-3.5	9-15.5												
Third	0.5-4	8-14.5												
At booking / diagnosis		- Refer FMS for assessment / consult O&G specialists for clinics without FMS. - Combined Clinic appointment after seen by FMS / after commencement of anti-thyroid drugs (ATD). - Monitor: ☐ FT4 and TSH level to be done 4 weekly after initiation of therapy and then 4-6 weekly after achieving the target value ☐ Aim of treatment is to maintain the FT4 levels at, or just above upper limit of normal ☐ Check FBC, LFT and TFT 4 weeks after commencement of ATD Treatment of hyperthyroidism in pregnancy: - PTU is the preferred ATD during first trimester and can be continued up to 16 weeks of gestation - Patients, who are already on carbimazole pre pregnancy can continue with the current regime without switching to PTU. - After first trimester, carbimazole is recommended - Propranolol is for symptomatic control usually only for 2-4 weeks and should not be continued without review.												

PHASE	DEFINITION	PLAN OF ACTION
		Treatment of subclinical hyperthyroidism is not recommended as it is usually transient in first trimester & gradually improves in later gestation
Subsequent antenatal follow-up		- Shared care between FMS and Combined Clinic team every trimester. - Detail scan by fetomaternal team at 24 weeks. - Urgent consultation with endocrinologist if hyperthyroidism is difficult to control
Delivery plan		- May allow postdate, unless specified otherwise. - Hospital delivery
Postpartum		- Refer all babies born to mothers with hyperthyroidism to Paediatric team - It is important to trace and review TFT for the babies - Discuss options of contraception
Upon discharge from hospital		- Arrange FMS appointment for postnatal review - Pre pregnancy Clinic appointment postnatal (if future pregnancy possible)

REMARKS:

- **Important to distinguish Grave's disease from gestational transient thyrotoxicosis (GTT)**
- GTT is defined as transient hyperthyroidism, limited to the first half of pregnancy, characterised by elevated serum FT4 and suppressed or undetectable serum TSH, in the absence of thyroid autoimmunity.
- The usual presentation is hyperemesis gravidarum (due to high levels of HCG).
- The presence of autoimmunity, goitre, ophthalmopathy, family history, would suggest Grave's, therefore recommended to treat with ATD
- **Management of GTT:** supportive treatment, treat dehydration, ATDs are not recommended, low-dose short-term beta-blockers may be considered
- **TSH receptor autoantibodies (TRAb):** Measuring TRAb can help to assess foetal risk
 - **TRAb –ve** : No antithyroid drugs needed as very low risk of foetal / neonatal thyrotoxicosis
 - **TRAb +ve** : Follow up foetal growth and monitor TFTs

- 1-5% of neonates born to women with Graves' disease have foetal hyperthyroidism/thyrotoxicosis due to transplacental transfer of TRAb

- **Uncontrolled hyperthyroidism can cause:**

Mother	Foetal / Neonatal
☐ Miscarriage ☐ Pre-eclampsia ☐ Preterm delivery ☐ Congestive heart failure (CHF) ☐ Thyroid storm ☐ Placental abruption	☐ Prematurity ☐ Small size for gestational age ☐ Intrauterine foetal death ☐ Goitre ☐ Thyrotoxicosis ☐ Transient hyperthyroidism (neonates) ☐ Hydrops fetalis

- **Adverse effects of drugs used in hyperthyroidism:**

Propylthiouracil (PTU)	Carbimazole	Propranolol
Rash, fever, agranulocytosis, risk of liver toxicity	Rash, fever, agranulocytosis, aplasia cutis, methimazole embryopathy	Bronchospasm, intrauterine growth restriction, neonatal hypoglycaemia

- If patient is allergic to carbimazole, change to PTU
- Maintaining FT4 levels in the upper normal of non-pregnant reference range usually protects the fetus from hypothyroidism
- Carbimazole is safe for lactating mothers and infants up to 20mg/day.
- Administer ATDs following a feeding, in divided doses to reduce excretion in breast milk.
- Subtotal thyroidectomy may be indicated if:
 - Patient has severe reaction to ATD
 - Persistent high doses of ATD are required (Carbimazole > 30mg or PTU > 450mg/day)
 - Non-adherence or uncontrolled hyperthyroidism
- The optimal timing of surgery is in the second trimester
- Refer Medical Eligibility Criteria for Contraception Use, WHO 2015 for contraception counselling
- Women with history of Postpartum Thyroiditis (PPT) have a markedly increased risk of developing permanent primary hypothyroidism in the five-to-ten year's period after the episode of PPT. An annual TSH level should be performed in these women

Reference:

1. The American Thyroid Association Taskforce on Thyroid Disease during Pregnancy and Postpartum, Stagnaro-Green, A., Abalovich, M., Alexander, E., Azizi, F., Mestman, J. Wiersinga, W. (2011). Guidelines of the American Thyroid Association for the Diagnosis and Management of Thyroid Disease during Pregnancy and Postpartum. *Thyroid*, 21(10), 1081–1125. <http://doi.org/10.1089/thy.2011.0087>
2. 2017 Guidelines of the American Thyroid Association for the Diagnosis and Management of Thyroid Disease during Pregnancy and the Postpartum. Erik K. Alexander, Elizabeth N. Pearce, Gregory A. Brent, Rosalind S. Brown, Herbert Chen, Chrysoula Dosiou, William A. Grobman, Peter Laurberg, John H. Lazarus, Susan J. Mandel, Robin P. Peeters, and Scott Sullivan. *THYROID*, Volume 27, Number 3, 2017. American Thyroid Association. Mary Ann Liebert, Inc. DOI: 10.1089/thy.2016.0457
3. 2015 American Thyroid Association Management Guidelines for Adult Patients with Thyroid Nodules and Differentiated Thyroid Cancer. Bryan R. Haugen, Erik K. Alexander, Keith C. Bible, et al. The American Thyroid Association Guidelines Task Force on Thyroid Nodules and Differentiated Thyroid Cancer.
4. Oxford Handbook of Obstetric and Gynaecology 3rd edition (2013)

5.4.2 Hypothyroidism in Pregnancy

PHASE	DEFINITION	PLAN OF ACTION
Pre pregnancy	Overt hypothyroidism :	- Refer FMS for counselling- pre pregnancy care - If already on L-Thyroxine, maintain TSH < 2.5 mIU/L
Booking	- An elevated TSH (>2.5 mIU/L) in conjunction with a decreased FT4 concentration or - Women with TSH level >10 mIU/L, irrespective with their FT4 level Subclinical hypothyroidism - Serum TSH above the upper limit of the trimester-specific reference range with a normal FT4	- Overt and subclinical hypothyroidism case should be referred to FMS/ combined clinic as it should be treated - As soon as pregnancy is confirmed, the dose of L-Thyroxine should be increased by 25%-30% - Target TSH differs according to each trimester as stated below - To screen for hypothyroidism in pregnant women with risk factors including history of thyroid disease, type 1 diabetes mellitus, or other autoimmune diseases; current or past use of thyroid therapy; or a family history of autoimmune thyroid disease (American Academy of Family Physician – Thyroid disease in Pregnancy, 2016). - Take blood for serum TSH in women with: ☐ History of thyroid dysfunction or thyroid surgery ☐ Symptoms of thyroid dysfunction or presence of goitre ☐ History of head or neck radiation ☐ Use of amiodarone or lithium, or recent administration of iodinated radiologic contrast
Subsequent antenatal follow up		- Dose of L-Thyroxine should be adjusted according to target TSH as below - Target TSH ☐ 1st trimester: 0.1-2.5 mIU/L ☐ 2nd trimester: 0.2-3.0 mIU/L ☐ 3rd trimester: 0.3-3.0 mIU/L - TFT should be measured every 4- 6 weeks - Shared care between FMS and Combined Clinic team
Delivery		- May allow post-date unless there is other obstetric indications for earlier delivery - For hospital delivery
Post partum		- After deliver, most patients need to reduce the dose of L-Thyroxine to the dose before pregnant - L-Thyroxine can be continued in breastfeeding mother - Repeat serum TSH at 6 weeks postpartum - The newborn babies should be referred to

PHASE	DEFINITION	PLAN OF ACTION
		paediatric team for assessment • Discuss options of contraception (Refer Medical Eligibility Criteria for Contraception Use, WHO 2015)
Upon discharge		• Refer FMS/MO at 2 month postpartum to review TSH result and contraception advise

5.5

BRONCHIAL ASTHMA IN PREGNANCY

PHASE	DEFINITION	PLAN OF ACTION
Pre pregnancy	Asthma diagnosed before pregnancy	• Establish diagnosis of asthma accurately • Preferably every asthma patient need to have spirometry test • Check the best baseline peak flow • Assess level of asthma control as per CPG Management of Asthma in Adults 2017 or GINA guideline • Optimise treatment before getting pregnant • Address trigger factors • Check inhaler techniques • Educate patient on the stepwise approach • Refer to FMS if asthma is partially controlled • Refer to Medication Therapy Adherence Clinic (MTAC) to optimise treatment • Advise on smoking cessation in women who smokes
At booking		• Assess level of asthma control • If asthma is well-controlled, mothers can be followed up by MO • If asthma is partially controlled (Step 2-3) follow-up by FMS • Refer to Combined Clinic if indicated • Educate patients on stepwise approach • Patient should have a written asthma action plan
Subsequent antenatal follow-up		• Assess level of asthma control • Monitor PEF at every visit • Optimise treatment • 2-4 weekly review depending on level of asthma control • Foetal surveillance with SFH, FKC, foetal growth monitoring by serial ultrasound, starting at 28 weeks till 36 weeks. • Admit if indicated during exacerbations • Refer to O&G specialist for further management if required
Delivery plan		• For hospital delivery • May allow post-date unless there is other obstetric indications for delivery
Postpartum		• Continue asthma medications • Discharge patient once stable • Emphasise on contraception • Continue stepwise approach

PHASE	DEFINITION	PLAN OF ACTION
Upon discharge from hospital		• Notify to health clinics prior to discharge • In mild asthma, mothers should be reviewed by MO • For moderate asthma, follow-up by FMS

Reference:

1. Global Strategy for Asthma Management and Prevention (2017 update) by Global Initiative for Asthma (GINA)
2. Clinical Practice Guidelines Management of Asthma in Adults 2017, Ministry of Health Malaysia

5.6 CARDIAC DISEASE IN PREGNANCY

5.6.1 Heart Disease in Pregnancy

Refer Clinical Practice Guidelines – Heart Disease in Pregnancy, Ministry of Health, 2nd Edition 2016

5.6.2 Peripartum Cardiomyopathy

Refer Clinical Practice Guidelines – Heart Disease in Pregnancy, Ministry of Health, 2nd Edition 2016

5.7 INFECTIOUS DISEASE IN PREGNANCY

5.7.1 Retroviral Disease

a. HIV in pregnancy

Refer Garis Panduan Pengukuhan Program Pencegahan Jangkitan HIV dan Sifilis dari Ibu-ke-Anak, Kementerian Kesihatan Malaysia, 2021

b. Serodiscordant Couple (Positive Male, Negative Female)

PHASE	PLAN OF ACTION
Pre pregnancy	• Refer couple for pre pregnancy counselling • Pregnancy must be well-planned • Husband should be on HAART and is virally suppressed before attempting conception • Plan for timed unprotected intercourse during fertile period • Advise on protected sexual intercourse other than that timed period • Advise for early antenatal booking • Recommend wife to have pre-exposure prophylaxis (PrEP) if husband is not on treatment/ non-compliant /unknown viral load/ unsuppressed viral load • Wife does not have to be on PrEP if husband is virally suppressed, but need proper counselling and close monitoring
At diagnosis	• Early booking if UPT positive • To do HIV screening each trimester • Emphasize on safe sex or abstinence
Subsequent antenatal follow-up	• Mother must be closely monitored • If HIV screening is reactive, to refer to FMS
Delivery plan	• Mode of delivery as per obstetric indication
Postpartum	• Closely monitor mother and child • Discuss options for contraception. BTL can be offered if patient/ couple had completed family
Upon discharge from hospital	• Routine discharge procedure • Repeat rapid test

Reference:

1. Management of HIV Infection in Pregnant Women, Ministry of Health, 2008
2. Malaysian Consensus Guidelines on Antiretroviral Therapy, Ministry of Health, 2017

5.7.2 Syphilis in Pregnancy

Refer Garis Panduan Pengukuhan Program Pencegahan Jangkitan HIV dan Sifilis dari Ibu-ke-Anak, Kementerian Kesihatan Malaysia, 2021

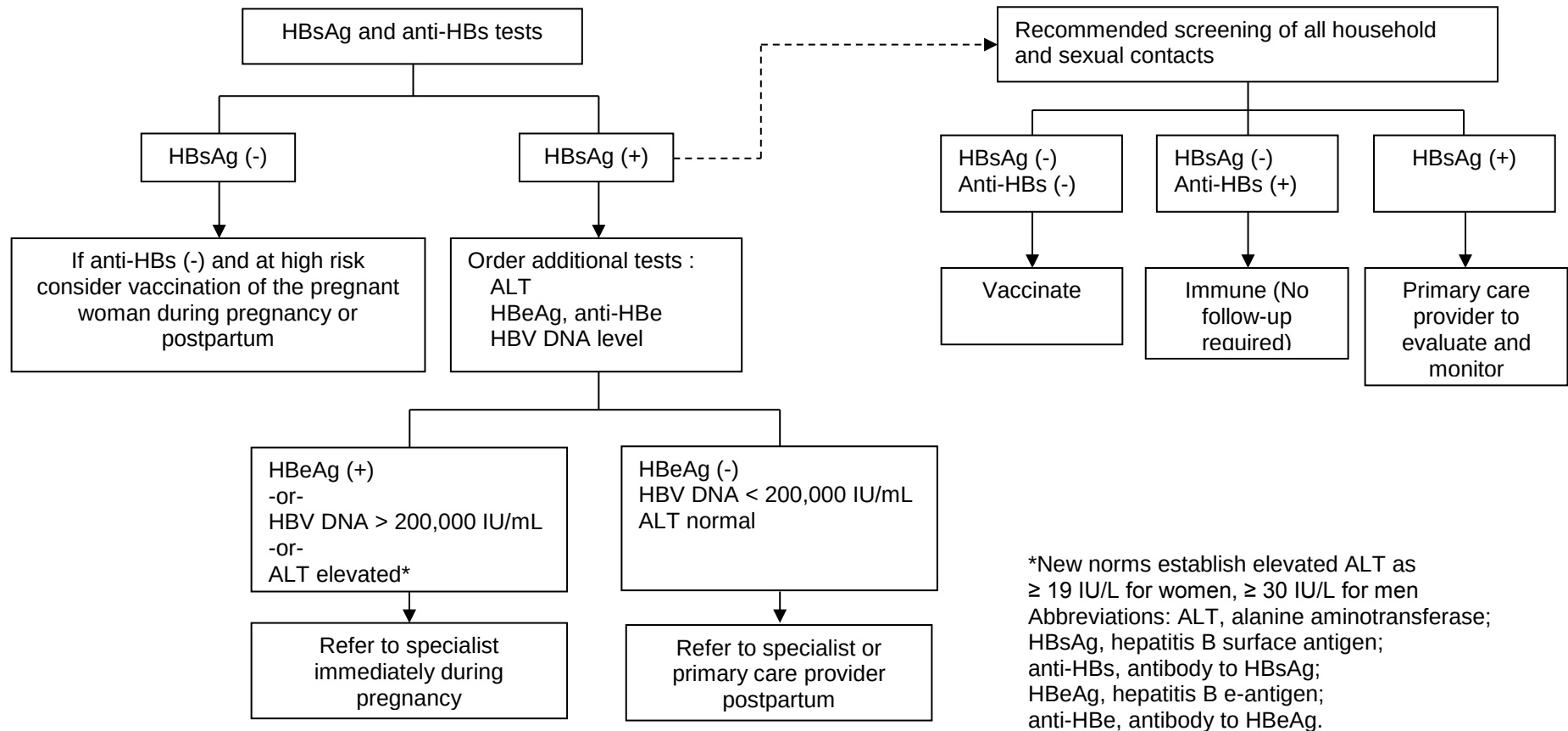
5.7.3 Hepatitis B in Pregnancy

PHASE	PLAN OF ACTION
At booking/ diagnosis	• Screen for Hepatitis C co-infection if not done previously • Screen husband and family • Counsel on high-risk behaviours • Determine infectivity • Notify for newly diagnosed as required under the Infectious Disease Prevention and Control Act 1988
Subsequent antenatal follow-up	• To do HBe antigen and Hbe antibody • Monitor liver functions every trimester • Look for hepatic flares and admit if indicated • Refer to combined clinic / O&G if indicated • If positive HBe antigen, arrange for ultrasound of hepatobiliary system • Admit if patient is jaundiced • Look for other comorbidities • If low infectivity (antibody Hbe reactive), for MO follow up • Stable high infectivity can be followed up by FMS • If available, measure HBV viral load at 28 weeks. If Hep B DNA is high in third trimester (>200,000 IU/ml, refer gastroenterologist or MFM for antiviral therapy)
Delivery plan	• Refer to O&G & decide mode of delivery
Delivery	• For hospital delivery • Universal precaution • Hep B immunoglobulin and Hepatitis B vaccination (1st dose) to be given at birth or no later than 12 hours after delivery • Refer baby to paediatric team • Allow breastfeeding
Postpartum	• Advise on contraception (Refer Medical Eligibility Criteria for Contraception Use, WHO 2015)
Upon discharge from hospital	• Arrange for pre pregnancy care • Ensure mother has follow up with gastroenterology/medical team • Advise patient on importance of early booking in next pregnancy

Reference:

1. Asian-Pacific Clinical Practice Guidelines on the Management of Hepatitis B: a 2015 update
2. Management of Hepatitis B in Pregnancy: The Royal Australian and New Zealand College of Obstetricians and Gynaecologists July 2016
3. South Australian Perinatal Practice Guidelines: Hepatitis B in Pregnancy
4. Chronic Hepatitis B in Pregnancy A Workshop Consensus Statement on Screening, Evaluation, and Management
5. Sabah Obstetric Shared Care Guidelines 2018

FLOW CHART FOR CHRONIC HEPATITIS B IN PREGNANCY MANAGEMENT APPROACH



Recommended Approach for Hepatitis B Virus (HBV) Screening, Evaluation, Vaccination, and Referral of Pregnant Women

5.7.4 Tuberculosis in Pregnancy

PHASE	PLAN OF ACTION
Pre pregnancy	Women with tuberculosis (TB) on treatment must be advised for contraception until she has completed treatment
At booking / diagnosis	- To screen all pregnant mothers according to the guideline by asking whether she has prolonged cough more than 2 weeks - For patient with suspected tuberculosis based on history and physical examination, investigations including: ☐ Sputum AFB direct smear X 3 (Induce sputum if unable to produce sputum) ☐ CXR with abdominal shield (preferably after 12 weeks gestation) ☐ Biopsy if indicated (extrapulmonary tuberculosis) - For newly diagnosed tuberculosis: ☐ Refer Pusat Rawatan 1 (PR 1) and FMS once for follow up ☐ Notify and contact tracing as per protocol ☐ Baseline investigations (FBC, RBS, BUSE, creatinine, LFT, sputum MTB C&S, HIV rapid test) ☐ Refer combined clinic/ visiting O&G specialist in district hospital - For relapsed pulmonary TB (PTB): ☐ may start re-treatment and refer to respiratory team if suspected MDR TB - Treatment regime: - For pulmonary tuberculosis – 2 EHRZ / 4HR (Pyridoxine 30 mg OD to be given with Isoniazid) - For extrapulmonary tuberculosis – refer to respective team - For drug-resistance TB (DR-TB) – refer to chest physician Streptomycin can cause foetal ototoxicity and should not be used during pregnancy
Subsequent antenatal follow-up	- Check on directly observed therapy (DOT) - Continue anti-TB treatment - Monitor patient as per guidelines. - Ultrasound monthly after 24 weeks to look for IUGR
Delivery	For hospital delivery
Postpartum	- Refer baby to Paediatric team to initiate isoniazid prophylaxis - Defer giving BCG if baby is symptomatic - Inform health clinic upon discharge to ensure continuity of TB treatment - Allow breastfeeding - Advise on contraception (Refer Medical Eligibility Criteria for Contraception Use, WHO 2015). Rifamycin drugs such as rifampicin or rifabutin reduce contraceptive efficacy of both OCP and POP.

Remarks:

1. Diagnosing TB in pregnancy can be difficult as it is masked by pregnancy changes.
2. TB in pregnancy has been associated with increased risk of maternal and perinatal morbidity namely premature delivery, SGA and LBW
3. Mantoux test is considered safe and valid for use in pregnancy
4. Chest X-ray must not be delayed in diagnosis of PTB even before 12 weeks because the risk of ionizing radiation is so low for one with an abdominal shield
5. First-line anti-TB drugs are safe in pregnancy and breastfeeding
6. Breastfeeding should be continued – use 3-ply surgical mask if the mother is still infectious

Reference:

1. Garis Panduan Kawalan Dan Pencegahan TBC Dalam Kalangan Ibu Mengandung, Programme Kawalan TBC Kebangsaan, Kementerian Kesihatan Malaysia, 2018
2. Clinical Practice Guidelines Management of Tuberculosis (3rd edition), Ministry of Health, 2012

5.7.5 Chickenpox in Pregnancy

PHASE	PLAN OF ACTION
At diagnosis	● Pregnancy with significant exposure: □ Administer VZIg (ideal to be administered within 96 hours to 10 days of exposure) ▪ If VZIg is given – pregnant women should be managed as potentially infectious from 8 to 28 days (8 to 21 days if no VZIg is given) □ A second dose of VZIg may be required if further exposure is reported and 3 weeks have elapsed since the last dose ● Assess for complications (i.e., pneumonia, hepatitis and encephalitis) □ If complications developed, to refer O&G and medical team □ In the absence of complication but the mother has risk factors[†] or co morbidities, to refer O&G for assessment □ If no complications, to manage as outpatient ● Onset of symptoms: □ If onset within 72 hours and >20 weeks POG, to treat with oral Acyclovir 800mg 5 times daily for 7 days. Intravenous acyclovir may be required for severe disease □ if <20 weeks POG, need to discuss risk and benefit of acyclovir ● Refer immediately to hospital for severe presentation (refer O&G and medical team) ● Isolate either at home or ward (if admission is required) ● Educate mothers on prevention of spread of disease – infectious state until the lesions have crusted (usually about 5 days after the onset of rash) ● Symptomatic treatment and emphasise on hygiene to prevent secondary bacterial infection of the lesions
Subsequent antenatal follow-up	● If infection occurred ≤ 20 weeks POG, refer to O&G for detail scan at 24 weeks POG
Delivery plan	● Preferably to delay delivery 7 days after onset of symptoms
Delivery	● Hospital delivery ● Timing of delivery as per obstetric indications
Postpartum	● Breastfeeding is encouraged ● Baby should be referred to paediatric team after delivery, irrespective of the gestation when maternal varicella zoster infection developed ● Advise on contraception (Refer Medical Eligibility Criteria for Contraception Use, WHO 2015).
Pre pregnancy & Postpartum	● Varicella vaccination can be considered for seronegative patient □ To avoid pregnancy in the next 4 weeks after the completing the vaccine schedule □ To continue breastfeeding after vaccination

† Risk factor and comorbid that requires the mother to be assessed by specialist even though without complications:

- ☐ Smoker
- ☐ Chronic lung disease
- ☐ Corticosteroids use in the preceding 3 months
- ☐ In the second half of pregnancy

REMARKS:

● Causative agent: Varicella zoster virus ● Mode of transmission: Direct contact with vesicle fluid & respiratory droplets (2 days before appearance of rash up to the healing of active rash). ● Incidence: uncommon for primary varicella infection.
● Maternal complication** ☐ Pneumonia (40% risk of death) ☐ Hepatitis ☐ Encephalitis ● Foetal complications ☐ Congenital Varicella Syndrome (0.4 to 2%) early 2nd trimester (< 20 week) ▪ Skin scarring in dermatomal distribution ▪ Eye defects (microphthalmia, chorioretinitis, cataracts) ▪ Limb hypoplasia ▪ Neurological abnormalities (microcephaly, cerebral cortical atrophy, mental retardation or dysfunction of bowel and bladder sphincters) ● Neonatal complication ☐ Neonatal varicella (within first ten days of life) – if mother developed rash 5 days before or 2 days after delivery ▪ If maternal infection occurs 1-4 weeks before delivery, up to 50% of infants are infected and up to 23% develop clinical varicella ▪ Severe chickenpox may occur if infants are born within 7 days of onset of mother's rash or if mother develops rash up to 7 days after delivery ☐ Risk of death – 30%
● Severe presentations that need immediate referral including: ☐ Dense rash ☐ Respiratory symptoms ☐ Neurological symptoms ☐ Haemorrhagic rashes ☐ Immunosuppressive drugs use
● Aim of treatment is to reduce maternal complication ☐ Significant varicella infection such as pneumonitis should be treated ● Side effects of acyclovir – cardiovascular effect, polydactyly, limb reduction & hypospadias
● Varicella Zoster vaccination(attenuated live-virus vaccine) in pregnant women is NOT recommended

Reference:

1. Chickenpox in Pregnancy (Green-top Guideline No. 13), Royal College of Obstetricians & Gynaecologists (RCOG), 2015
2. Clinical Practice Guideline No. 274-Management of Varicella Infection (Chickenpox) in Pregnancy, Society of Obstetricians & Gynaecologists of Canada (SOGC), 2012

5.7.6 Group B Streptococcus in Pregnancy

PHASE	PLAN OF ACTION
At booking	• Routine universal screening for GBS is currently not recommended. • High risk screening by identifying clinical risk factors: □ History of previously affected baby (GBS sepsis, both early & late onset) ▪ intrapartum antimicrobial prophylaxis (IAP) will be required, repeat test is not necessary □ History of maternal GBS carriage but baby did not contract GBS sepsis ▪ to screen for GBS at 35-37 weeks (singleton) or 32-34 weeks (twins) • HVS (high vaginal swab) C&S is the usual test, especially for symptomatic woman (diagnostic). • LVS (low vaginal swab) and/ or rectal swab or VRS (vaginal-rectal swab) may be used for screening of asymptomatic women. However, confirmation with the laboratory is needed for LVS / rectal swab / VRS (regarding transport media). If LVS / rectal swab / VRS cannot be sent, then resort to HVS.
Antenatal Care	• In mothers with history of maternal GBS carriage, but baby did not contract GBS sepsis: □ Screen for GBS C&S at 35-37 weeks (singleton) or 32-34 weeks (twins): ▪ If positive – for IAP ▪ If negative – IAP is not required • In mothers with proven GBS during this pregnancy □ GBS bacteriuria / UTI → treat and will require IAP even if GBS has been eradicated from urine as evidenced from repeat urine C&S 2 weeks after treatment. □ GBS cultured from vaginal / rectal swab for symptomatic patients → treat and repeat swab 2 weeks after treatment. If it has been eradicated, IAP would not be required □ GBS cultured from vaginal / rectal swab (incidentally and asymptomatic) → no need antenatal treatment but IAP would be required.

PHASE	PLAN OF ACTION
Intrapartum care	● Clinical factors that are associated with increased risk of baby contracting GBS sepsis: ☐ Previous baby affected with GBS sepsis ☐ Maternal GBS carriage from bacteriological investigation e.g. urine or vaginal C&S ☐ Preterm birth ☐ Prolonged rupture of membranes ☐ Suspected intrapartum maternal infection including chorioamnionitis ● Therefore, IAP would be required for: ☐ Mother with history of previously affected baby with GBS sepsis ☐ Mother with history of GBS UTI, even after it has been treated / eradicated ☐ Mother whose latest vaginal / rectal swabs was GBS positive (intentional swabs e.g. 35-37 weeks screening, or incidental swabs) ☐ Mother with history of GBS carriage but no repeat swab was done – counsel patient that the likelihood of maternal GBS carriage in this pregnancy is 50% and offer IAP ☐ Preterm labour (PTL) ☐ Preterm prelabour rupture of membranes (PPROM): ▪ PPRM >34+0 weeks: start IAP and expedite delivery ▪ PPRM <34+0 weeks: start antibiotic and conservative management with close monitoring as perinatal risks associated with prematurity will likely outweigh perinatal infection. Deliver when indicated. ☐ Prelabour rupture of membranes (PROM): ▪ If known GBS carrier – start immediate IAP and induce for delivery ▪ If not a GBS carrier – start IAP at 18hr of PROM and induce for delivery as per protocol (not later than 24hr of PROM) ● In suspected intrapartum maternal infection (maternal pyrexia >38°C) – start broad spectrum antibiotic which covers for GBS. ● IAP is not required for women undergoing planned caesarean section in the absence of labour and with intact membranes. The routine prophylactic antibiotics for surgery is necessary and adequate. ● IAP regimens: ☐ IV Penicillin G 5million units, followed by 2.5 million units 4-hourly until delivery (after test dose) ☐ OR

PHASE	PLAN OF ACTION
	☐ IV Ampicillin 2g followed by 1g 4-hourly until delivery ☐ Consider Vancomycin for patients with suspected penicillin allergy
Postpartum care	● Refer baby to Paediatric team ● Counsel patient for all future pregnancies / deliveries: ☐ IAP if baby has GBS sepsis ☐ Screen for GBS 35-37 weeks if she has GBS this / past pregnancy and baby is not affected ● Document the above instruction in patient's antenatal (red) book ● Advise on contraception

Reference:

1. Hughes RG, Brocklehurst P, Steer PJ, Heath P, Stenson BM on behalf of the Royal College of Obstetricians and Gynaecologists. Prevention of early-onset neonatal group B streptococcal disease. Green-top Guideline No.36. BJOG 2017;124:e280-e305

5.7.7 Dengue in Pregnancy

Refer Clinical Practice Guidelines – Management of Dengue Infection in Adults, Ministry of Health, 3rd Edition 2015

5.7.8 Malaria in Pregnancy

Refer to Management Guidelines of Malaria in Malaysia 2013

5.8 KIDNEY DISEASE IN PREGNANCY

5.8.1 Urinary Tract Infection

Refer Consensus Guidelines on the Management of Urinary Tract Infections in Pregnancy, MOH 2021

5.8.2 Chronic Kidney Disease

PHASE	PLAN OF ACTION
Pre pregnancy	• All women in the reproductive age group with renal impairment should be referred for pre pregnancy counselling by O&G and nephrologist (multidisciplinary team) • Pregnancy may be considered in women having mild renal impairment (serum creatinine < 125µmol/L) and well-controlled blood pressure • Women with moderate to severe renal impairment should be counselled to avoid pregnancy due to greater adverse maternal and foetal outcomes • Rule out relative contraindication to pregnancy: ☐ ESRF ☐ Renal replacement therapy / haemodialysis ☐ Recent transplant < 1-2 years ☐ Transplant with recent rejection • Baseline investigations: Pre-eclampsia profile
Antenatal	• Shared care between health clinic and O&G team • First trimester dating scan • For Aspirin 150mg OD and Calcium Carbonate 1g BD at booking • At every visit, women should be screened for complications, hypertension, proteinuria and pre-eclampsia • UTI screening for asymptomatic bacteriuria at each visit • Renal function monitoring at each trimester or more frequent if needed. • Anomaly scan at 24 weeks (Indication: drug exposure in pregnancy) • Ultrasound for foetal growth every 4 weeks starting from 24 weeks POA till delivery
Delivery	• Hospital delivery • Timing of delivery as per obstetric indications
Postpartum	• Encourage breastfeeding • Advise on contraception (Refer Medical Eligibility Criteria for Contraception Use, WHO 2015).

REMARKS:

● Severity grading in chronic kidney disease follows serum creatinine level	
Severity	Serum Creatinine Level (μmol/L)
Mild	90-124
Moderate	125-225
Severe	>225
● Pregnancy in women with moderate to severe renal impairment (serum creatinine > 125 μmol/L) results in increased risk of adverse maternal and foetal outcomes	
● Common maternal complications include accelerated decline in renal function, hypertension, proteinuria and pre-eclampsia	
● Common foetal complications include spontaneous abortion/ neonatal death, prematurity, low birth weight/ SGA baby	

Reference:

1. Clinical Practice Guidelines, Management of Chronic Kidney Disease (2nd edition), Ministry of Health, Malaysia 2018

5.9 CONNECTIVE TISSUE DISEASE IN PREGNANCY

5.9.1 Rheumatoid Arthritis

PHASE	PLAN OF ACTION
Pre pregnancy	• Refer to pre pregnancy clinic (FMS/ O&G/ Rheumatologist/ Physician) • Avoid unplanned pregnancy • Pregnancy should be deferred until disease is under good control on medications compatible with pregnancy • Folic acid 5mg daily • Majority will achieve disease control in pregnancy • NSAIDs should be discontinued during periconception period and used sparingly during the first trimester • Medications that can generally be used throughout pregnancy are hydroxychloroquine (HCQ), sulfasalazine (SSZ) and azathioprine (AZA) • Pregnancy is contraindicated for patients on methotrexate, leflunomide, JAK-2 inhibitors and biological agents due to its teratogenicity
At booking/ diagnosis	• Continue calcium supplementation during pregnancy • For patients already on medical / rheumatology follow-up → refer Combined Clinic • For patients not on medical / rheumatology follow-up → refer medical / rheumatology AND Combined Clinic • OGTT for patients on prednisolone • Foetal detail scan
Delivery plan	• Timing and mode of delivery as per obstetric indications • Hospital delivery
Postpartum	• Monitor for flares. • Refer rheumatologist for follow-up • Advise on contraception (Refer Medical Eligibility Criteria for Contraception Use, WHO 2015).
Lactation	• Safe to continue: NSAIDS (but aspirin should be avoided), corticosteroids, HCQ, SSZ, TNF inhibitors, AZA • Inadequate data: JAK inhibitors (tofacitinib) • Contraindicated: methotrexate, leflunomide, cyclosporine, cyclophosphamide, chlorambucil and other biologics

5.9.2 Systemic Lupus Erythematosus

PHASE	PLAN OF ACTION
Pre pregnancy	• Refer to pre pregnancy clinic (FMS/ O&G/Rheumatologist/Physician) • Pregnancy should be deferred until disease is under good control on medications compatible with pregnancy, whenever possible • Folic acid 5mg daily • Pregnancy may increase the risk of disease flares • Assess disease activity, major organ involvement, hypercoagulability and concurrent medical conditions • Pregnancy is allowed if: ☐ Disease in remission for ≥ 6 months ☐ BP is well-controlled ☐ eGFR >60ml/min ☐ Proteinuria <1g/day (proteinuria 2+) • Test for anti-Ro and anti-La is recommended if available, as determining the status of these autoantibodies improves counselling regarding pregnancy and foetal risk
At booking/ diagnosis	• Refer for rheumatology clinic assessment • Early combined clinic appointment after assessment by rheumatologist • Assessment to detect disease flares: ☐ Symptoms: ▪ Joint pain ▪ Cutaneous manifestations ▪ Serositis • Blood pressure should be monitor closely • Investigations: ▪ FBC, RP, LFT, ESR, uric acid ▪ Red cell cast in urine (UFEME) ▪ Urine 24hour protein (if proteinuria present) ▪ Reduction in complement (C3, C4) values ▪ Increase in dsDNA antibody titer
Subsequent antenatal follow-up	• T. Aspirin 150mg daily from 12 weeks until delivery • Calcium supplementation starts at 20 weeks until delivery • During antenatal follow up, look out for signs and symptoms of disease flares, anaemia, pre-eclampsia, foetal growth restriction and other disease complications
Delivery plan	• Hospital delivery • Timing and mode of delivery as per obstetric indications
Post-partum	• Monitor for flares • If antiphospholipid syndrome is present, continue pharmacological thromboprophylaxis for 6 weeks after delivery • Advise patient regarding contraception (Refer Medical Eligibility Criteria for Contraceptive Use, WHO 2015)
Lactation	• Generally safe

5.9.3 Antiphospholipid Syndrome (APS)

PHASE	PLAN OF ACTION
Pre pregnancy	• Assessment of past thrombotic event (arterial or venous) • Assessment of any associated condition (e.g. SLE) and thrombotic risk (e.g. obesity, smoking) • APS patients on oral anticoagulation should be informed about the potential teratogenic effects of warfarin • Oral anticoagulation should be switched to a therapeutic dose of low-molecular-weight heparin (LMWH) either before or very shortly after conception. • Defer pregnancy in order to improve general health and reduce risk, such as the need for weight loss, following an acute thrombotic event or treatment of active SLE • Pregnancy should be actively discouraged in certain condition e.g; in pulmonary hypertension, the risk of maternal death is estimated at 35%
At booking/ diagnosis	• Refer specialist for assessment / initiation of VTE prophylaxis • Start low-dose aspirin (150mg OD) on confirmation of pregnancy
Subsequent antenatal follow-up	• Refer hospital for combined care • Foetal growth should be monitored monthly from 24 weeks of gestation
Delivery plan	• Timing and mode of delivery as per obstetric indication • Delivery should be planned for thrombotic APS • There is no contraindication for vaginal delivery for patients on LMWH, but vigilance for PPH is recommended • LMWH should be discontinued at least 12 hours prior to planned delivery / IOL
Postpartum	• Continue VTE prophylaxis for 6 weeks for those with history of thrombosis • Avoid oestrogen-containing contraception pills. Refer Medical Eligibility Criteria for Contraceptive Use by WHO 2015

5.10

NEUROLOGICAL DISORDER IN PREGNANCY – EPILEPSY

PHASE	PLAN OF ACTION
Pre pregnancy	• Refer all women with epilepsy in the reproductive age and planning to conceive to the pre pregnancy clinic • Emphasise on compliance to antiepileptic drug (AED) • Counsel patient on the risk of treatment (teratogenicity) vs risk of defaulting treatment (seizure risk to mother and fetus) • Advise to take folic acid 5mg daily • Aim for seizure control at least 6 months before conception • Aim to use the lowest effective dose of a single anticonvulsant whenever possible • Advise early booking once pregnant
At diagnosis in pregnancy	• Perform first trimester dating scan and anomaly scan (i.e.NTD) • Refer combined clinic • Folic acid 5mg daily till delivery • Maintain the pre-existing AED if seizure is well-controlled
Subsequent antenatal follow-up	• Shared care between FMS and combined clinic • For detailed scan at 18 to 24 weeks gestation • Watch out for signs of depression, anxiety and neuropsychiatric symptoms in all mothers on AED • Advise compliance to AED • Advise to avoid triggers of seizures such as sleep deprivation and stress • Perform serial growth scans monthly from 24 weeks of gestation • Routine serum AED level monitoring during pregnancy is not recommended
Delivery Plan	• Generally may allow postdate unless specified otherwise/compounded by other factors
Delivery	• Timing and mode of delivery as per obstetric indications • Provide adequate analgesia • Continue AED in labour
Postpartum	• Reinforce the importance of contraception and planned pregnancy. Advise on contraception according to Medical Eligibility Criteria for Contraceptive Use, WHO 2015 • Adjustment of AED dose (if required) need further discussion with treating physician/neurologist • Breastfeeding is encouraged • Neonates need to be monitored for side effects of AEDs e.g.: drowsiness, jitteriness and hypotonia • Monitor the mother for at least 24 hours postpartum • Screen mothers for depression • Advise on safety issues and strategies to prevent accidental injury while caring of newborns with involvement of family members •

PHASE	PLAN OF ACTION
Upon discharge from hospital	• Notification of high-risk discharge from hospital to respective health clinics as per guideline. • Arrange appointment with FMS at 1 month postpartum • Provide neurology / medical clinic appointment • To give pre pregnancy clinic appointment with FMS at 3 months postpartum (if future pregnancy is possible / anticipated)

REMARKS:

• Seizure frequency in pregnant women with epilepsy: 60% no change, 30% increase, 10% decrease
• Pregnancies in women with epilepsy should be planned, when possible
• Give women a clear understanding of the risks of uncontrolled seizures and the possible teratogenicity of AED. Where possible, avoid sodium valproate and AED polytherapy. Phenytoin, carbamazepine, sodium valproate, lamotrigine and levetiracetam can cross the placenta • In-utero exposure to carbamazepine, lamotrigine, levetiracetam and phenytoin does not appear to adversely affect neurodevelopment of the child • Major malformation: NTD (valproate), orofacial clefts (phenobarbitone), heart disease (phenytoin and valproate). • Minor malformation: Foetal anticonvulsant syndrome - also need to be ruled out
Safety strategies to prevent accidental injury while caring of newborns: • Nurse the baby on the floor • Use very shallow baby baths and do not bathe the baby unaccompanied • Lay the baby down if there is a warning aura • Avoid sleep deprivation and alcohol
Contraception: • IUCD is a preferred choice of contraception • If opting for hormonal contraception, higher dose is required

Reference:

1. Epilepsy in Pregnancy (Green-top Guideline No. 68) , Royal College of Obstetricians & Gynecologists (RCOG), 2016
2. Diagnosis and management of epilepsy in adults (SIGN 143), Scottish Intercollegiate Guidelines Network (SIGN) 2015
3. Consensus Guidelines on the Management of Epilepsy, Malaysian Society of Neurosciences, 2017
4. Women and Epilepsy, Edmonton Epilepsy Association, 2011
5. Clinical Guideline Epilepsy and Pregnancy Management, South Australian Maternal and Neonatal Clinical Network, 2014

5.11 THROMBOEMBOLISM IN PREGNANCY

Refer Training Manual on Prevention and Treatment of Thromboembolism in Pregnancy and Puerperium, Ministry of Health 2018

5.12 OBESITY IN PREGNANCY

Refer Consensus Clinical Guideline on Obesity in Pregnancy, Ministry of Health 2022

5.13

SMOKING IN PREGNANCY

PHASE	PLAN OF ACTION
Pre pregnancy	• Offer quit smoking services and pre pregnancy care to all reproductive women who are smoking
At booking / diagnosis	• Smoking history should be elicited in all pregnant women • If the woman is an active smoker, assess: □ Stage of her willingness to quit ▪ Pre-contemplation - No quit plan in next 6 months ▪ Contemplation - Quit plan in next 6 months. Seriously considering ▪ Preparation - Quit plan for next month. ▪ Action - Quit. Day 1-6 months tobacco free ▪ Maintenance - Quit. More than 6 months □ Level of addiction using Fagerstrom's score for nicotine dependence (Table 5.1) □ Analyse level of carbon monoxide (CO) in blood by breath test ▪ Level >3 ppm (suspect smoking), (usually smoker >7ppm) ▪ Test is available at Quit Smoking Clinic
Subsequent antenatal follow-up	• No safe smoking level in pregnancy • Stopping smoking at any time during pregnancy is beneficial to mother and baby • Advise all pregnant women to avoid cigarette smokes during pregnancy including secondhand smokes • Refer O&G if complications develop • Refer quit smoking programme (need support to cope with withdrawal and craving) • Treatment: □ Non-pharmacological - Psychosocial intervention is effective in supporting pregnant smokers to quit : ▪ Advise regarding smoking-related risks ▪ Pregnancy-specific self-help materials counselling session with a health educator. ▪ Counsellor provided counselling session and telephone follow up calls during pregnancy and after delivery ▪ Quit plan : set date for quit smoking ▪ Counselling • Motivation – 5R ◆ Relevance – Why quitting is important to them ◆ Risks – Negative consequences of ongoing habit ◆ Rewards – Benefits of tobacco cessation ◆ Roadblock – Identify impediments to quitting (eg: withdrawal symptom, fear, weight gain) ◆ Repetition – Repeat every time the patient comes to the clinic

PHASE	PLAN OF ACTION
	☐ Pharmacological ▪ Indicated when non-pharmacological treatment failed ▪ Nicotine replacement therapy (NRT) – Safer than smoking (no tar and carbon monoxide) ▪ NRT helps to manage craving ▪ Risks & benefits should be discussed ▪ Only prescribe NRT once quit date is set ◆ Prescribe NRT for 2 weeks then reassess ◆ Continue NRT if they have quit smoking ▪ Be cautious of NRT use in CVD • Bupropion and Varenicline should not be offered to pregnant or breastfeeding women. • Electronic cigarettes are not recommended in pregnancy because long term risk to baby is unknown
Delivery plan	• Mode and timing of delivery as per obstetric indications • Hospital delivery • Refer baby to paediatric team • Advise on contraception. Refer Medical Eligibility Criteria for Contraceptive Use, WHO 2015
Postpartum	• If still smoking – advise for quit smoking for better quality of breast milk • Refer pre pregnancy care
Upon discharge from hospital	• Routine discharge procedure

REMARKS:

- Risk of active smoking in pregnancy:
 - ☐ Miscarriage
 - ☐ Ectopic pregnancy
 - ☐ Stillbirth (one-third of stillbirth are related to smoking)
 - ☐ Congenital abnormalities (cleft lips & palate)
 - ☐ IUGR
 - ☐ Abruptio placenta
 - ☐ Premature birth
 - ☐ Risk of sudden infant death syndrome (SIDS)
 - ☐ Asthma, chest infection & ear infection
 - ☐ Risk of ADHD
 - ☐ Poor performance at school in children
- Risk of passive smoking in pregnancy:
 - ☐ Stillbirth
 - ☐ Premature birth
 - ☐ IUGR

Table 5.1: Fagerstrom' Test for Nicotine Dependence

PLEASE TICK (✓) ONE BOX FOR EACH QUESTION		
How soon after waking do you smoke your first cigarette?	Within 5 minutes 5-30 minutes 3-60 minutes	☐ 3 ☐ 2 ☐ 1
Do you find it difficult to refrain from smoking in places where it is forbidden? E.g. church, library, etc	Yes No	☐ 1 ☐ 0
Which cigarette would you hate to give up?	The first in the morning Any other	☐ 1 ☐ 0
How many cigarettes a day do you smoke?	10 or less 11-20 21-30 31 or more	☐ 0 ☐ 1 ☐ 2 ☐ 3
Do you smoke more frequently in the morning?	Yes No	☐ 1 ☐ 0
Do you smoke even if you are sick in bed most of the day?	Yes No	☐ 1 ☐ 0
Total score		
SCORE	1- 2 = low dependance 5-7 = moderate dependance 3 – 4 = low to moderate dependance 8+ = high dependance	

Reference:

1. MIMS Stop Smoking Cessation Guidelines 2014.
2. Smoking: Stopping in Pregnancy and After Childbirth, National Institute for Health and Care Excellence (PH 26) June 2010.
3. Health & Care Information, Royal College of Obstetricians & Gynaecologists, 2015
4. Clinical Practice Guideline, Treatment of Tobacco Use Disorder, Ministry of Health Malaysia, 2016

5.14

TEENAGE PREGNANCY

PHASE	DEFINITION	PLAN OF ACTION
Pre pregnancy		• Refer FMS for assessment & counselling • Folic acid pre-conception • Rule out anaemia and pre-existing medical condition
At booking / diagnosis	Teenage mothers: aged 10–19 years old (WHO)	• Use HEADSS framework to establish rapport and assess psycho-social- economic status • Implement Verbal Confidentiality Contract (VCC) and apply CRET (Confidentiality, Rapport, Empathy, Trust) • Identify risk factors for ☐ Infection (UTI, STIs, pneumonia etc.) ☐ Medical condition (DM, HPT, heart disease, asthma, anaemia etc.) ☐ Malnutrition (underweight, overweight, obese, micronutrient deficiencies) Dating scan • Screening for HIV, VDRL, Hepatitis B, blood group • Routine monitoring of : ☐ Blood pressure ☐ Body weight ☐ Urine albumin ☐ SFH ☐ Haemoglobin • Assess mental health • Explore condition/ situation surrounding conception : ☐ Marital status/ teenager's age at conception ☐ Wanted/ unwanted pregnancy ☐ Victim of abuse/ rape (statutory /coercion) ☐ Alcohol/ substance abuse ☐ Domestic violence ☐ Spouse/ partner's history ▪ Substance use/abuse ▪ Occupation ▪ Sexual history (STI/ multiple partner) ☐ Family/ social support • Consider involving the police/ medical social worker/ child protector/ other disciplines in accordance to relevant act eg. the Child Act 2001 / Penal Code (Act 574) / Sexual Offences Against Children Act (Act 792)/ Control of Infectious Diseases Act 2020 (Act 342)/ etc (Refer <i>Garis Panduan Pengendalian Masalah Kesehatan Seksual dan Reproduksi Remaja di Klinik Kesehatan</i> or other relevant acts and guidelines)

PHASE	DEFINITION	PLAN OF ACTION
Subsequent antenatal follow-up		● Routine monitoring of : ▪ Blood pressure ▪ Body weight ▪ Urine protein ▪ SFH ▪ Hb ▪ Foetal kick chart ● Routine and repeated screening of mood disorders, alcohol/ substance abuse ● Hematinics to reduce anaemia ● Aspirin and calcium to reduce pre-eclampsia risk ● Advise on balanced diet ● Ultrasound for foetal growth to look for SGA or IUGR ● High Vaginal Swab if STIs suspected ● Screening for diabetes mellitus / other common medical conditions in pregnancy as per guideline ● Educate patients and caregivers on ☐ signs and symptoms of preterm labour ☐ physiological changes of pregnancy ☐ early warning symptoms and signs of ▪ fever ▪ headache/ blurring of vision ▪ abnormal PV discharge and/or foul smelling discharge ▪ abdominal pain ▪ dysuria ▪ palpitation/dyspnoea/reduce effort tolerance ☐ basic personal hygiene ☐ sexual history and safe sex /contraception ☐ unmarried teenagers (in particular) ▪ afterbirth childcare (adoption/ keep baby/ abortion in special cases of harm to mother's life or physical or mental health) ▪ education/ life skills & employment
Delivery plan		● Hospital delivery ● Legal guardian should accompany patients to delivery suites for teenage <18 years old
Postpartum		● Refer to medical social worker/ social worker as per hospital policy or to facilitate adoption process (if indicated) ● Refer school counsellor / educator for continuation of education if applicable ● Contact police if indicated / there is an on-going investigations

PHASE	DEFINITION	PLAN OF ACTION
		• Discuss contraception and spacing • Facilitate breastfeeding • Assess for postpartum blues / depression • Reinforce on personal hygiene, wound care, perineal care and early warning symptoms of puerperal sepsis (fever, pelvic pain, abnormal PV discharge and/or foul smelling)
Upon discharge from hospital		• Inform nearest health clinic for home visit • Physical and mental well-being of patient • Eat balanced diet and do appropriate exercise/ physical activity • Ensure well-being of baby • Breastfeeding support • Pre pregnancy care before next conception • Further reinforce personal hygiene, perineal care and early warning symptoms • TCA early if any complication arises (e.g. fever, abnormal PV bleeding and/or abnormal foul smelling PV discharge)

5.15

MENTAL DISORDER IN PREGNANCY

PHASE	PLAN OF ACTION
Pre pregnancy	• If known case of mental illnesses, refer to FMS/ psychiatrist for assessment and preparation for pregnancy
At booking/ diagnosis	• Screening for depression may be done for high-risk group, e.g those with health problems causing disability, past history of depression, a family history of depression and those with other mental health problems. • Screening tool to be used is the Two-Question Case-Finding Instrument ☐ Patient Health Questionnaire-4 (PHQ-4) ☐ Edinburgh Postnatal Depression Scale (EPDS) • Consult psychiatrist for assessment • Assess: ☐ symptoms related to mental health problem ☐ social support (family, housing, employment) ☐ preparedness towards the pregnancy and acceptance of pregnancy (planned / unplanned pregnancy) ☐ mental health status during previous pregnancy ☐ possibility of self-harm or suicide • Screen for substance abuse, e.g. alcohol, smoking, drugs
Subsequent antenatal follow-up	• Shared care between FMS / psychiatrist / O&G • Detailed scan at 24 weeks if indicated • OGTT for patient on antipsychotic medication • Monitor for excessive weight gain in women taking antipsychotic • Monitor regularly for symptoms of relapse throughout pregnancy • Provide ongoing social support with referral to social workers if needed. • Advise for sleep hygiene.
Delivery plan	• Generally may allow postdate unless specified otherwise • Hospital delivery
Postpartum	• Look for symptoms of postpartum psychosis / depression • Look for symptoms of relapse • Advise on contraception according to <i>Medical Eligibility Criteria for Contraceptive Use, WHO 2015</i> • Encourage breastfeeding. Advise to take psychotropic medication just after breastfeeding • Monitor the baby for adverse effects such as drowsiness, hypotonia, rigidity, tremor and withdrawal symptoms • Review family support
Upon discharge from hospital	• Notification of high-risk discharge from hospital to respective health clinics as per guideline.

PHASE	PLAN OF ACTION
	• Family planning counselling before discharged • Follow-up within two weeks in health clinic for possible symptoms of postpartum psychosis/ depression after childbirth • Follow up in psychiatry clinic • Pre pregnancy clinic appointment

REMARKS:

1. Depression and Anxiety in Pregnancy

- Perinatal depression and anxiety are common. In the perinatal period, 1 in 10 women develop depression. The prevalence of anxiety disorders is 10-15%.
- Untreated depression and anxiety lead to adverse effects to fetus/child, mother and family e.g. low birth weight, poor antenatal care, interpersonal difficulties and impaired children's neurobehavioural development.
- Depression and anxiety are underrecognized and undertreated due to multiple barriers eg lack of awareness or knowledge, stigma, time constraints.
- If patients are screened positive for depression or anxiety:
 - ☐ Proceed for further assessment to establish diagnosis
 - ☐ Assess psychosocial risks
 - socioeconomic status
 - unintended pregnancy
 - unmarried
 - intimate partner violence
 - insufficient emotional and practical support
 - ☐ Assess suicidal risk
- Diagnosis of depression or anxiety is based on the criteria from Diagnostic and Statistical Manual 5th Edition (DSM-5) or International Statistical Classification of Diseases and Related Health Problems (ICD-10)
- Counsel on risk-benefit analysis of treatment options (psychological intervention vs medication vs combination).
- Take note that access to psychological interventions depends on availability of the services and require logistic arrangement to attend several sessions.
- For mild to moderate depression or anxiety:
 - ☐ Provide brief psychosocial intervention in primary care or O&G setting (e.g. supportive psychotherapy, counselling, problem solving therapy or relaxation therapy)
 - ☐ Arrange for brief psychological intervention by trained personnel in primary care e.g. brief cognitive behaviour therapy (CBT)
 - ☐ Refer to tertiary centre for intensive psychological interventions e.g.CBT, interpersonal psychotherapy (IPT).
- For moderate to severe depression or anxiety:
 - ☐ Counsel on risk-benefit of medication.

- Risks of medication, i.e.; Selective Serotonin Reuptake Inhibitors (SSRI): no risk of teratogenicity, low absolute risks of miscarriage, premature delivery, neonatal adaptation syndrome and primary pulmonary hypertension.
- Risk of untreated depression: miscarriage, premature delivery, low birth weight, poor antenatal self-care, interpersonal conflict, functional impairment, mother-baby bonding/attachment difficulties, low breastfeeding initiation, long-term behavioural problems in offsprings, suicide and infanticide.
- Based on risk-benefit ratio, start SSRI (e.g.; Sertraline 50-100 mg OD)
- Avoid benzodiazepines in pregnancy.
- Offer psychological intervention when medications have taken effect.
- If medication is indicated but the patient refused, offer adequate support, refer to FMS or psychiatrist, and expedite intensive psychotherapy (CBT or IPT)

2. Severe Mental Illness in Pregnancy

- Manage under a multi-disciplinary team of family medicine, psychiatry and obstetrics.
- Medication:
 - If a patient is stable on antipsychotic, to continue on the same medication. Refer as soon as possible for patients on Clozapine
 - Patients on mood stabilizers (sodium valproate, carbamazepine, lamotrigine and lithium) need to have their medication withheld immediately. Consult a psychiatrist immediately for switching.
- Monitoring for early signs of relapse.
- If a patient has chosen not to be on medication or has defaulted, close monitoring is required. Structured individual, group and family therapy can be offered to patients to prevent relapse.
- If patient has become pregnant while on a known teratogenic medication,
 - explain that stopping or switching the medication after pregnancy is confirmed may not remove the risk of foetal malformations
 - refer for foetal abnormality screening and neonatology consultation
 - explain the need for additional monitoring and the risks to the fetus if medication is continued
 - sodium valproate is associated with major malformation, cardiac malformation and adverse cognitive outcomes

5.16

ALCOHOL ABUSE IN PREGNANCY

PHASE	PLAN OF ACTION																									
Pre pregnancy	● Counsel about the adverse effects of alcohol on pregnancy outcomes ● Promote alcohol abstinence if planning for pregnancy																									
At booking/ diagnosis	● SCREENING – Need to ask all women for alcohol use □ If positive, need to screen with ‘AUDIT-10’ questionnaire <table><tr><th>Score</th><th>Risk</th><th>Zone</th></tr><tr><td>0 – 7</td><td>Low risk</td><td>1</td></tr><tr><td>8 – 15</td><td>Hazardous drinker</td><td>2</td></tr><tr><td>16 – 19</td><td>Harmful & dependency</td><td>3</td></tr><tr><td>0 – 40</td><td>High risk for alcohol related harm</td><td>4</td></tr></table> ● ASSESSMENT during first contact; □ History should include alcohol pattern, complication, comorbid, MSE & social support □ Physical examination including general signs of chronic alcohol, gastrointestinal/abdomen & CNS. □ Investigation including; FBC-Hb, MCV, LFT (GGT, AST, ALT), (Ultrasound hepatobiliary system, OGDS, x-ray & EEG if indicated) ● COUNSEL those who screened positive women regarding risks of alcohol use □ Advise maternal cessation of alcohol intake to reduce complications to mother and foetus ● INTERVENTION involving multidisciplinary team approach, depending on zone as per Malaysian guideline.<table><tr><th>Zone</th><th>Intervention</th></tr><tr><td>1</td><td>Health education</td></tr><tr><td>2</td><td>Simple advise</td></tr><tr><td>3</td><td>Extended intervention (MO / FMS)</td></tr><tr><td>4</td><td>Refer specialist in addiction (Psychiatrist or FMS with subspecialty in addiction)</td></tr></table>	Score	Risk	Zone	0 – 7	Low risk	1	8 – 15	Hazardous drinker	2	16 – 19	Harmful & dependency	3	0 – 40	High risk for alcohol related harm	4	Zone	Intervention	1	Health education	2	Simple advise	3	Extended intervention (MO / FMS)	4	Refer specialist in addiction (Psychiatrist or FMS with subspecialty in addiction)
Score	Risk	Zone																								
0 – 7	Low risk	1																								
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Zone	Intervention																									
1	Health education																									
2	Simple advise																									
3	Extended intervention (MO / FMS)																									
4	Refer specialist in addiction (Psychiatrist or FMS with subspecialty in addiction)																									
Subsequent antenatal follow-up	● Frequent antenatal follow up to monitor maternal and foetal status (maternal alcohol consumption habits and complications such as withdrawals, foetal surveillance for pregnancy complications) ● Emphasise on maternal cessation of alcohol intake ● Refer O&G for detailed scan (at 24 weeks) for women with alcohol exposure in first trimester																									
Delivery plan	● Similar with normal pregnancy ● Hospital delivery																									

PHASE	PLAN OF ACTION
Post-partum	- Refer baby of women with heavy drinking to paediatric team (possibility of neonatal withdrawal) - Encourage breastfeeding and advise to avoid alcohol - Advise on contraception. Refer Medical Eligibility Criteria for Contraceptive Use, WHO 2015
Upon discharge from hospital	Notification of high-risk discharge from hospital to respective health clinics as per guideline.

REMARKS:

- Alcohol – contains ethanol (depressant drug) 1 unit = 10 ml @ 10 gm ethanol (Malaysia) 13.5% aged 13 years and above ever consumed alcoholic beverages in year 2019 in Malaysia. In general, most of the current drinkers aged 18 years and above, were categorised as low risk drinkers with a proportion of 81.5% (95% CI: 75.32, 86.36). The proportion of risky drinkers was 17.6% (95% CI: 12.79, 23.71) while the proportion of drinkers with probable alcohol dependence was 0.9% (NHMS 2019)
- There is NO safe level of alcohol consumption during pregnancy. It is best to abstain from alcohol during pregnancy & breastfeeding - Recommendations on alcohol consumption: - Pregnant women / women planning a pregnancy should be advised to avoid alcohol in the first 3 months in pregnancy - If women choose to drink alcohol during pregnancy, they should be advised to drink no more than 2 units and not more than twice per week (after 3 months of gestation) - Avoid binge drinking (>6 unit per occasion)
- Complications of alcohol to the mother & baby: - Spontaneous miscarriage - Stillbirth - IUGR - Low birth weight - Foetal alcohol spectrum disorder (FASD), which includes foetal alcohol syndrome (FAS), partial foetal alcohol syndrome, alcohol-related neurodevelopmental disorder, alcohol-related birth defects ** Risk is progressively increased with greater alcohol consumption

Reference:

1. Royal College of Obstetricians & Gynaecologist – Health & Care Information (February 2015)
2. NICE Antenatal Care- Routine Care for the Healthy Pregnant Woman (March 2008)
3. Nykjaer.C, et al. J.Epidemiol Community Health 2014;0:1-8, doi:10.1136/jech-2013-202934
4. Garis Panduan Penilaian Risiko dan Intervensi Primer Kemudaran Alkohol, Cawangan Penyakit Tidak Berjangkit, Bahagian Kawalan Penyakit, KKM 2010
5. Maklumat Kesihatan- Intervensi, Pencegahan dan Pengurangan Kemudaran Alkohol, Unit Alkohol & Substans, NCD, KKM (2013)
6. Alcohol Use Disorder: A Comparison Between DSM–IV and DSM–5, National Institute on Alcohol Abuse and Alcoholism, www.niaaa.nih.gov • 301.443.3860
7. SOGC Alcohol Use and Pregnancy Consensus Clinical Guidelines 2010

Table 5.2 Alcohol Use Disorders Identification Test (AUDIT)

The Alcohol Use Disorders Identification Test : Interview Version Read questions as written. Record answers carefully. Begin the AUDIT by saying 'Now I am going to ask you some questions about your alcoholic beverages during this past year'. Explain what is meant by 'alcoholic beverages' by using local examples of beer, wine, vodka etc. Code answers in terms of 'standard drinks'. Place the correct answer number in the box at the right.	
1. How often do you have a drink containing alcohol? (1) Never (skip to Questions 9-10) (2) Monthly or less (3) 2 to 4 times a month (4) 2 to 3 times a week (5) 4 or more times a week	6. How often during the last year have you needed a first drink in the morning to get yourself going after a heavy drinking session? (1) Never (2) Less than monthly (3) Monthly (4) Weekly (5) Daily or almost daily
2. How many drinks containing alcohol do you have on a typical day when you are drinking? (1) 1 or 2 (2) 3 or 4 (3) 5 or 6 (4) 7, 8 or 9 (5) 10 or more	7. How often during the last year have you had a feeling of guilt or remorse after drinking? (1) Never (2) Less than monthly (3) Monthly (4) Weekly (5) Daily or almost daily
3. How often do you have six or more drinks on one occasion? (1) Never (2) Less than monthly (3) Monthly (4) Weekly (5) Daily or almost daily	8. How often during the last year have you been unable to remember what happened the night before because you had been drinking? (1) Never (2) Less than monthly (3) Monthly (4) Weekly (5) Daily or almost daily
Skip to Question 9 if Total Score for Questions 2 and 3 is 0	
4. How often during the last year have you found that you were not able to stop drinking once you had started? (1) Never (2) Less than monthly (3) Monthly (4) Weekly (5) Daily or almost daily	9. Have you or someone else been injured as a result of your drinking? (1) No (2) Yes, but not in the last year (3) Yes, during the last year
5. How often during the last year have you failed to do what was normally expected from you because of drinking? (1) Never (2) Less than monthly (3) Monthly (4) Weekly (5) Daily or almost daily	10. Has a relative or friend or a doctor or another health worker been concerned about your drinking or suggested you cut down? (1) No (2) Yes, but not in the last year (3) Yes, during the last year
RECORD TOTAL OF SPECIFIC ITEMS HERE	

Table 5.3 : A comparison between DSM-IV and DSM-5

DSM-IV		DSM-5	
In the past year, have you:		In the past year, have you:	
Any 1 = ALCOHOL ABUSE	Found that drinking – or being sick from drinking – often interfered with taking care of your home or family? Or caused job troubles? Or school problems?	1	Had times when you ended up drinking more, or longer, than you intended?
	More than once gotten into situations while or after drinking that increased your chances of getting hurt (such as driving, swimming, using machinery, walking in a dangerous area, or having unsafe sex?)	2	More than once wanted to cut down or stop drinking, or tried to, but couldn't?
	More than once gotten arrested, been held at a police station, or had other legal problems because of your drinking? <i>**This is not included in DSM-5</i>	3	Spent a lot of time drinking? Or being sick or getting over other aftereffects?
	Continued to drink even though it was causing trouble with your family or friends?	4	Wanted a drink so badly you couldn't think of anything else? <i>**This is new to DSM-5</i>
Any 3 = ALCOHOL DEPENDENCE	Had to drink much more than you once did to get the effect you want? Or found that your usual number of drinks had much less effect than before?	5	Found that drinking – or being sick from drinking – often interfered with taking care of your home or family? Or caused job troubles? Or school problems?
	Found that when effects of alcohol were wearing off, you had withdrawal symptoms, such as trouble sleeping, shakiness, restlessness, nausea, sweating, a racing heart, or a seizure? Or sensed things that were not there?	6	Continued to drink even though it was causing trouble with your family or friends?
	Had times when you ended drinking more, or longer, than you intended?	7	Given up or cut back on activities that were important or interesting to you, or gave you pleasure, in order to drink?
	More than once wanted to cut down or stop drinking, or tried to, but couldn't?	8	More than once gotten into situations while or after drinking that increased your chances of getting hurt (such as driving, swimming, using machinery, walking in a dangerous area, or having unsafe sex)?
	Spent a lot of time drinking? Or being sick or getting over other aftereffects?	9	Continued to drink even though it was making you feel depressed or anxious or adding to another health problem? Or after having had a memory blackout?
	Given up or cut back on activities that were important or interesting to you, or gave you pleasure, in order to drink?	10	Had to drink much more than you once did to get the effect you want? Or found that your usual number of drinks had much less effect than before?
	Continued to drink even though it was making you feel depressed or anxious or adding to another health problem? Or after having had a memory blackout?	11	Found that when the effects of alcohol were wearing off, you had withdrawal symptoms, such as trouble sleeping, shakiness, restlessness, nausea, sweating, a racing heart, or a seizure? Or sensed things that were not there?

The presence of at least 2 of these symptoms indicates an Alcohol Use Disorder (AUD).

The severity of AUD is defined as:

Mild:
The presence of 2 to 3 symptoms

Moderate:
The presence of 4 to 5 symptoms

Severe:
The presence of 6 or more symptoms

5.17 SUBSTANCE ABUSE IN PREGNANCY

	PHASE	PLAN OF ACTION
1	Pre pregnancy	● Assess: ➢ types of substance abuse ➢ duration of substance abuse ➢ status – withdrawal, intoxication ➢ psychosocial support ➢ associated infectious diseases ● Inform National Anti-Drugs Agency (AADK) ● Psychiatric referral for management ● Defer pregnancy until remission with contraception
2	At booking/ diagnosis	● Assess: ➢ Status of substance use disorder ➢ Screening for STIs ● If patient on medication-assisted therapy, continuation of therapy is advised ● Refer FMS ● Combined care with psychiatrist, FMS and O&G team for pregnancy care
3	Subsequent antenatal follow-up	● Detailed scan at 24 weeks ● Monthly growth scan after 28 weeks ● Admission is required for following cases: ➢ Develop withdrawal symptoms during pregnancy ➢ Psychological implication
4	Delivery plan	● May consider to taper down opioid-agonist ● Pain management during intrapartum ● Hospital delivery
5	Postpartum	● Neonatal assessment by paediatrics team for Neonatal Abstinence Syndrome ● Breastfeeding is not contraindicated ● Psychiatric assessment before discharge ● Advise on contraception. Refer Medical Eligibility Criteria for Contraceptive Use, WHO 2015
6	Upon discharge from hospital	● Notification of high-risk discharge from hospital to respective health clinics as per guideline.

REMARKS:

1. Assessment during first contact:

a) History

- Patterns of substance use (The ASSIST [Alcohol, Smoking and Substance Involvement Screening Test] helps identify current or potential problems resulting from substance use and motivate those at risk to change their substance use behaviour).
- Medical or psychiatric comorbidity
- Blood-borne and other infectious diseases
- Psychosocial problems such as relationship with a partner/ other people living in the same household, homelessness, poverty and violence

b) Physical examination

- Including general, sign of chronic substance use (Difficulty caring for self, poor dentition, parasitic skin infections such as lice or scabies, malnutrition), injection marks, GI/abdomen & CNS

c) Investigation

- Urine drug screen: whenever intoxication, withdrawal, or overdose is suspected
- HIV, Hepatitis B and C screening if the person has been injecting drugs
- Testing for sexually transmitted infections, including HIV, syphilis, chlamydia, gonorrhoea, and human papilloma virus (HPV)
- Obtain a tuberculosis test, sputum sample, and a chest x-ray if tuberculosis is suspected

2. Substance use disorders during the perinatal period have been identified as critical to the health of mothers and babies.

3. Substance use contributes to obstetrics, paediatrics and mental health complications.

4. Women with perinatal substance use disorder presented with extremely complicated issues:

- unplanned pregnancy
- late or no antenatal booking
- low socio-economic background
- poor living condition complicated with issues of estrangement from family
- involvement with other high-risk behaviors.

5. They presented with a high rate of mental health comorbidities and risks of self-neglect, self-harm as well as posing a risk of neglect and abuse to the babies.

6. In addition, they might be victims of interpersonal violence. Frequently there were histories of child abuse, neglect or sexual abuse.

7. There were lots of gaps in providing intervention which include lack of multi-agency coordination, unavailability of case management, lack of skills to provide specialised care and lack of human resources.

8. To some women, the prospect of becoming a mother turned out to be a golden opportunity for a life changing decision of abstinence but they need intensive support.

9. Outlining a comprehensive individual care plan covering issues in pregnancy, childbirth, postpartum and mother-baby relationship was a daunting task, given the complexity of issues faced by these women.
10. For professionals in perinatal mental health, addiction, obstetrics, paediatrics, social welfare and the National Anti-Drug Agency; the ability to coordinate responsibilities and share information in monitoring and delivering interventions required lots of effort.
11. Early identification in antenatal care may be the first step towards engaging women with substance use disorder into treatment but the high prevalence of unintended pregnancy made this a big challenge.
12. Fine tuning the integrated care is crucial and the way forward; where a policy, guidelines, standard operational procedures and training will ensure an effective and coordinated service delivery.
13. Case-management approach with regular multidisciplinary discussion is postulated to be a good strategy to start with; although working within lots of limitations, this will build up the experience and collaboration among the team member

Reference:

1. Guidelines for the identification and management of substance use and substance use disorders in pregnancy (WHO 2014)
2. Mental Health Gap Action Programme Intervention Guide Version 2.0 (WHO 2016)
3. Guidelines for the Psychosocially Assisted Pharmacological Treatment of Opioid Dependence (WHO 2009)
4. The ASSIST Project-Alcohol, Smoking and Substance Involvement Screening Test (World Health Organization 2009)
5. Antenatal and Perinatal Mental Health: Clinical Management and Service Guidance (NICE)
6. Laura P .M, Madeleine B., Nehama D. Guidelines for management of pregnant women with substance use disorder. The Journal of Consultation-Liaison Psychiatry (Psychosomatic) Dec 3, 2015
7. Grella CE. Background and overview of mental health and substance abuse treatment system: meeting the needs of women who are pregnant or parenting. J Psychoactive Drugs 1996 Oct-Dec
8. Grella CE. Services for perinatal women with substance abuse and mental health disorders: the unmet need. J Psychoactive Drugs. 1997 Jan-Mar

Table 5.4: Alcohol, Smoking, and Substance Involvement Screening Test (ASSIST)

1. In your life, which of the following substances have you ever tried? (non-medical use only)			
a. Tobacco products	☐ Yes ☐ No	f. Inhalants	☐ Yes ☐ No
b. Alcoholic beverages	☐ Yes ☐ No	g. Sedatives or sleeping pills	☐ Yes ☐ No
c. Cannabis	☐ Yes ☐ No	h. Hallucinogens	☐ Yes ☐ No
d. Cocaine	☐ Yes ☐ No	i. Opioids	☐ Yes ☐ No
e. Amphetamine-type stimulants	☐ Yes ☐ No	j. Others	☐ Yes ☐ No
2. During the past 3 months, how often have you used the substances you mentioned (first drug, second drug, etc)?			
☐ Never (0) ☐ Once/ twice (2) ☐ Monthly (3) ☐ Weekly (4) ☐ Daily/almost daily (6)			
3. During the past 3 months, how often have you had a strong desire or urge to use (first drug, second drug, etc)?			
☐ Never (0) ☐ Once/ twice (3) ☐ Monthly (4) ☐ Weekly (5) ☐ Daily/almost daily (6)			
4. During the past 3 months, how often has your use of (first drug, second drug, etc) led to health, social, legal or financial problems?			
☐ Never (0) ☐ Once/ twice (4) ☐ Monthly (5) ☐ Weekly (6) ☐ Daily/almost daily (7)			
5. During the past 3 months, how often have you failed to do what was normally expected of you because of you use of (first drug, second drug, etc)?			
☐ Never (0) ☐ Once/ twice (5) ☐ Monthly (6) ☐ Weekly (7) ☐ Daily/almost daily (8)			
6. Has a friend or relative or anyone else ever expressed concern about your use of first drug, second drug, etc)?			
☐ No, never (0) ☐ Yes, in the past 3 months (6) ☐ Yes, but not in the past 3 months (3)			
7. Have you ever tried and failed to control, cut down, or stop using (first drug, second drug, etc)?			
☐ No, never (0) ☐ Yes, in the past 3 months (6) ☐ Yes, but not in the past 3 months (3)			
8. Have you ever used any drug by injection? (non-medical use only)			
☐ No, never (0) ☐ Yes, in the past 3 months (6) ☐ Yes, but not in the past 3 months (3)			

ASSIST risk score and associated risk level and intervention			
Alcohol	All other substances	Risk level	Intervention
0 – 10	0 – 3	Lower risk	General health advise
11 – 26	4 – 26	Moderate risk	- Brief intervention - Take home booklet & information
27 +	27 +	High risk	- Brief intervention - Take home booklet & information - Referral to specialist assessment and treatment
Injected drugs in last 3 months		Moderate and high risk	- Risk of injecting card - Brief intervention - Take home booklet & information - Referral for testing for blood-borne virus (BBV) - Referral to specialist assessment and treatment

CHAPTER 6

ANTENATAL COMPLICATIONS

CHAPTER 6: ANTENATAL COMPLICATIONS

6.1 UNSURE OF DATE

Signs & Symptoms

Symptoms :

- Asymptomatic

Signs:

- Uterus larger or smaller than date

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
● Perform ultrasound scan for dating as soon as possible (within 1 week)	-	● History: ○ Detail menstrual history ○ Date of UPT ○ Early scan ● Measure SFH ● Foetal growth by scan and plot foetal parameters chart ● If foetal parameters from scan < 22 weeks ○ REDD from the scan can be used ● If parameters measure >22 weeks, ○ Do not rely on the given REDD. ○ Scan must be repeated every 3-4 weeks later to support the working gestational age ○ Given concern that a suboptimally-dated pregnancy could actually be weeks further along than it is believed to be, initiate foetal surveillance at 39-40 weeks of gestation	FMS/MO	HC
			FMS/MO	HC
			FMS/MO	HC
			O&G	HC/ Hospital
			O&G	HC/ Hospital
		● The timing of delivery		

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
		should be based on the best clinical estimate of gestational age • If foetal parameters and SFH are not corresponding, to refer O&G		

6.2 PRETERM LABOUR

Signs & Symptoms

Symptoms:

- Contraction pain before 37 completed weeks
- PV bleeding

Signs:

- Contractions felt
- Cervical / os changes on digital vaginal examination

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• FBC • UFEME • HVS C+S • Predictors of preterm labour if available (e.g.; Actim Partus, foetal Fibronectin etc)	• UTI • Abruptio Placenta • Braxton hicks • Contraction	• Refer to the nearest hospital • IM Dexamethasone 12mg, 2 doses 12 hours apart, if POA between 24 to 36 weeks (1st dose can be given in the clinic) • Tocolysis if indicated. • If delivery is imminent, prepare for delivery	O&G doctor MO/MS	Hospital HC/ Hospital

6.3 PRETERM PRELABOUR RUPTURE OF MEMBRANE (PPROM)

Signs & Symptoms

Symptoms:

- Leaking without contraction before 37 completed weeks
- vaginal discharge may be present

Signs:

- Fever
- Uterus < dates
- Leakage of fluid seen in speculum examination

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• FBC • UFEME • HVS C&S • Amnicator or litmus paper showing alkali or other point-of-care test if available, e.g.; Actim PROM	• Vaginal discharge secondary to vaginal infection • Urinary incontinence	• Refer to the nearest hospital • IM Dexamethasone 12 mg, 2 doses 12 hours apart, if POA between 24 to 36 weeks. First dose can be given in the clinic after discussion with specialist (FMS or O&G) • Erythromycin 400mg BD for 10 days	O&G doctor	Hospital
			MO/FMS	HC/ hospital

6.4 PRELABOUR RUPTURE OF MEMBRANES (PROM)

Signs & Symptoms

Symptoms:

- Leaking without contraction after 37 completed weeks
- vaginal discharge may be present

Signs:

- Fever
- Uterus < dates
- Leakage of fluid seen in speculum examination

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• FBC • UFEME • HVS C&S	• Vaginal discharge secondary to	• Refer to the nearest hospital	MO / O&G doctor	HC/ Hospital

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
Amniculator test or litmus paper indicate alkali reaction or other point-of-care test if available, e.g.; Actim PROM	vaginal infections Urinary incontinence			

6.5 UTERUS LARGER THAN DATES

Signs & Symptoms

Symptoms:

- Distended abdomen
- Compressive symptoms
- Asymptomatic

Signs:

- Uterus > dates (≥ 3 cm discrepancy between the SFH and POA)
- Shifting dullness
- Abnormal lie
- Multiple foetal pole
- Excessive maternal weight gain

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
- Plot growth chart - Ultrasound scan: - Amniotic fluid index (AFI) - Estimated foetal weight (EFW) - Multiple	- Multiple pregnancy - Pelvic tumour - Polyhydramnios - Wrong dates - Foetal anomaly - Placenta previa	Refer hospital for further management	FMS/MO/O&G	HC / Hospital

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
- pregnancy ○ Pelvic tumour ○ Foetal anomaly ● OGTT if indicated				

6.6 UTERUS SMALLER THAN DATES

Signs & Symptoms

Symptoms:

- Small abdomen
- Unsure of dates
- Leaking liquor

Signs:

- Uterus < dates (≤ 3 cm discrepancy between the SFH and POA)
- Clinically reduced liquor
- Easily felt parts
- Poor maternal weight gain

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
● Plot growth chart ● Ultrasound scan: ○ AFI ○ Foetal parameters ○ Foetal anomaly ● Serial ultrasound if corresponding to dates and AFI is normal	● Oligohydramnios ● IUGR ● Intrauterine death ● Wrong dates ● Foetal abnormality ● Normal fetus	Refer hospital for further management	FMS / MO/ O&G	Hospital

6.7 BREECH AT TERM

Signs & Symptoms

Symptoms:

- Asymptomatic

Signs:

- Breech presentation from palpation

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• Ultrasound scan: ○ Parameters ○ AFI ○ Placental localization ○ Foetal anomalies ○ Pelvic mass	• Foetal anomalies • Wrong dates • Polyhydramnios • Presence of pelvic mass • Placenta Praevia	Refer to the nearest hospital at about 36 weeks for further management. Possible options: • ECV • Elective LSCS • Vaginal breech delivery	MO/FMS O&G/MO	HC Hospital

6.8 MALPRESENTATION (AT 36 WEEKS AND BEYOND)

Signs & Symptoms

Symptoms:

- Asymptomatic

Signs:

- Transverse/ oblique lie

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• Ultrasound scan: ○ Parameters ○ AFI ○ Placental localization ○ Foetal anomalies ○ Pelvic mass	• Foetal anomalies • Wrong dates • Polyhydramnios • Presence of pelvic mass • Placenta Praevia	Refer hospital for further management	O&G	Hospital

6.9 MULTIPLE PREGNANCY

Signs & Symptoms

Symptoms:

- excessive nausea and vomiting
- uterus larger than date

Signs:

- SFH > POA/POG
- multiple foetal pole

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• First 1st ultrasound is recommended , (best at 14 weeks) to determine chorionicity ○ If mono-chorionic, screen for Twin to twin transfusion syndrome (TTTS) • serial ultrasound	• MCDA • DCDA • Triplet • Molar pregnancy with normal ongoing pregnancy • Pelvic tumour • Polyhydramnios • Wrong dates	• When to refer: ○ KD to refer immediately to KK ○ KK to be seen by MO/ FMS • Refer O&G within 1 week to determine chorionicity, counselling and outline of antenatal follow-up plan • Refer fetomaternal specialist as soon as chorionicity is determined for further antenatal care plan • URGENT REFERRAL IF: ○ Monoamniotic ○ Suspected twin-to-twin transfusion syndrome (TTTS) ○ Foetal structural abnormality ○ Suspected discordance in weight >18% (weight) ○ High order multiple	FMS/ MO/ O&G	HC/ Hospital

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
		- gestation (≥ 3) ○ Single foetal demise ● If monochorionic (MC) twins or high order multiple pregnancy (≥ 3), to follow up in fetomaternal clinic/ general O&G clinic of hospital (high risk pregnancy) ● If dichorionic (DC) twins, patient will be followed-up both at health clinic and hospital (general O&G clinic) ● All multiple pregnancies require monthly growth scan ● Delivery plan to be outlined by O&G: ○ High order multiple pregnancy: soon after diagnosis is confirmed ○ Twins: depend on chorionicity, presentation and any other associated factors. Plan by third trimester ○ Uncomplicated monochorionic twins: to deliver by 36 weeks ○ Uncomplicated dichorionic		

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
		twins: deliver by 38 weeks • Mode of delivery should be individualised - based on gestational age, co morbidity, availability of expertise in management of vaginal twin birth and mothers' preference.		

6.10 PREVIOUS CAESAREAN SECTION

Signs & Symptoms

Symptoms:

- Asymptomatic

Signs:

- Scar at the lower abdomen (suprapubic / sub-umbilical)

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• Ultrasound for placental localization ○ if upper segment ○ if placenta praevia		• Review indication & complications of the previous caesarean section • Refer hospital immediately if pain • Refer hospital at 32 - 34 weeks • Refer hospital immediately if placenta	MO/FMS MO/FMS/hospital O&G MO/specialist	HC/hospital

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
		praevia with previous scar	O&G MO/ specialist	

6.11 REDUCED FOETAL MOVEMENT

Signs & Symptoms

Symptoms:

- Reduced foetal movement
- <10 movements within 12 hours
- Progressively longer in a day to reach 10 kicks
- Any subjective feeling of reduced foetal movement (frequent ± intensity)

Signs:

- Foetal heart rate :
 - Normal
 - Bradycardia
 - Tachycardia
 - Irregular
 - Absent

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
● CTG (CTG prior to 28 weeks is difficult to interpret in view of foetal immaturity) ● Ultrasound scan: ○ AFI ○ foetal biometry ± morphology ● Doppler studies	● Intrauterine Death ● IUGR ● Foetal anomaly ● Normal fetus	● Refer to hospital for further assessment and management ● Repeat CTG & USG if required	FMS / MO	HC / Hospital

6.12 POSTDATES (EDD + 7 DAYS)

Signs & Symptoms Asymptomatic

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
- Reassess the dates - Ultrasound - CTG	Wrong dates with a healthy fetus	- If wrong dates - Correct the date - Continue follow up	FMS/ MO	HC / Hospital
		- If postdates (EDD+ 7 days) - Refer to hospital for further management - KIV IOL (depending on individual hospital protocol)	FMS/ MO	HC/ Hospital

6.13 HYPEREMESIS GRAVIDARUM

Signs & Symptoms

Symptoms:

- intractable vomiting
- unable to tolerate orally
- hypersalivation and spitting
- retching

Signs:

- dehydration
- weight loss greater than 5% of body weight
- signs of muscle wasting

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
- Urinalysis - Urea & electrolytes - FBC - Ultrasound to rule out multiple pregnancy and molar pregnancy	- UTI - hyperthyroidism - multiple pregnancy - molar pregnancy - peptic ulcer - cholecystitis - pyelonephritis	- PUQE score - Refer hospital for assessment and inpatient management based on severity - Aims of treatment: - Alleviate symptoms of nausea/vomiting - Rehydration - Correction of	FMS/MO/&G	Hospital

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
If indicated: • thyroid function test • liver function test • blood glucose	• hepatitis • pancreatitis	electrolyte imbalance ○ Prevention of complications		

6.14 RECURRENT MISCARRIAGES (LOSS OF 3 OR MORE CONSECUTIVE PREGNANCIES)

Signs & Symptoms Asymptomatic

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• Lupus anticoagulant and anticardiolipin antibodies 6 weeks apart • OGTT If indicated: • Parental test for peripheral karyotyping • Pelvic ultrasound (2D or 3D)/ hysteroscopy/ hysterosalpingogram (HSG) / MRI • Serial transvaginal sonography (TVS) in early trimester to detect cervical shortening / funnelling • Thyroid Function Test / Thyroid antibodies	• Anti-phospholipid syndrome	• Low dose aspirin once pregnancy is confirmed and low molecular weight heparin (LMWH) after 1 st trimester	• FMS/ Physician/ O&G Specialist	HC / Hospital
	• Diabetes Mellitus			
	• Genetic Factor	• Refer CPG Diabetes in Pregnancy (2018)	• FMS / MO	HC / Hospital
	• Uterine anomaly	• Genetic counselling/ referral to geneticist if available	• FMS/ O&G / MFM / Geneticist	HC / Hospital
	• Cervical incompetence	• Offer early prenatal diagnostic test	• MO / O&G Specialist	HC / Hospital
	• Thyroid Disorder	• Hysteroscopic resection of uterine septa	• MO / O&G Specialist	HC /

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• Serum prolactin • Thrombophilic screening including Factor V Leiden, Prothrombin, Protein S & C • Vaginal swab C&S • STD workout	• Hyperprolactinemia • Inherited Thrombophilia • Infection (eg TORCHES, Bacterial Vaginosis)	• could be performed • Cervical cerclage / cervical pessary / transabdominal cerclage • Anti-thyroxine for hyperthyroidism / L-thyroxine for hypothyroidism • Treatment with bromocriptine before pregnancy • LMWH throughout antenatal period • Treatment accordingly to known and treatable organism	• MO / FMS/ Physician / Endocrinologist/ O&G Specialist • MO / Physician/ O&G • MO/ FMS/ O&G / Genitourinary medicine specialist (if available)	- Hospital - HC / Hospital - HC / Hospital - HC / Hospital

6.15 PREVIOUS HISTORY OF UNEXPLAINED INTRAUTERINE DEATH (IUD)

Signs & Symptoms Asymptomatic

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• OGTT • Growth chart • Detail scan at 18-24 weeks if available	• Poorly controlled DM • Genetic disorder • Undiagnosed infection	• Refer O&G for shared care	FMS/ MO/O&G	HC/ Hospital

6.16 HISTORY OF FOETAL ABNORMALITY

Signs & Symptoms Symptoms:
• Asymptomatic

Signs:
• Uterus may be smaller or larger than dates

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• Ultrasound scan: ○ for dating ○ anomaly	-	Refer O&G or MFM for assessment that could include blood investigations, cell-free foetal DNA (cffDNA) (if available) and detailed scan	FMS/ MO / O&G	HC/ Hospital

6.17 SYMPTOMATIC VAGINAL DISCHARGE

Signs & Symptoms Symptoms:
• vaginal discharge more than normal
• with or without:

- itchiness
- dysuria
- lower abdominal discomfort
- dyspareunia

- fever

Signs:

- abdominal tenderness
- discharge seen in the vagina
- redness and swelling of the cervix, vagina and vulva

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• If cervicitis: ○ endocervical swab i. gram stain for pus cells, intracellular gram-negative diplococci ii. culture for <i>Neisseria gonorrhoea</i> (Thayer-Martin culture medium) iii. Antigen/ NAAT test for <i>Chlamydia trachomatis</i> iv. Pap smear • If vaginitis: ○ Vaginal Swab i. Wet mount from posterior fornix for trichomonas vaginalis ii. gram stain for pus cells, clue cells and yeast iii. candida culture - swab from lateral fornix	• <i>Neisseria gonorrhoea</i> – gram stain shows pus cells, intracellular gram negative diplococci • <i>Chlamydia trachomatis</i> - antigen/ NAAT positive • <i>Trichomonas vaginalis</i> - wet mount slide microscopy shows motile flagellates, oval or pear shaped organism with jerky movement	• Vaginal candidiasis – clotrimazole pessary 500mg ON stat dose or clotrimazole pessary 200mg ON for 3 days or Nystatin pessary 100,000 unit daily for 14 days • Bacterial vaginosis - oral metronidazole 400mg OD for 5-7 days • <i>Trichomonas vaginalis</i> – oral metronidazole 400mg OD for 5-7 days • <i>Neisseria gonorrhoea</i> - IM ceftriaxone 500mg stat and T.azithromycin 1g stat (single dose) or IM spectinomycin 2g as a single dose • <i>Chlamydia trachomatis</i> - erythromycin stearate 500mg QID for 7 days or erythromycin ethyl succinate 800mg QID for 7 days or amoxycillin 500mg tds for 7 days or azithromycin 1g PO stat • In areas where laboratory facilities and investigations are	MO	MO/FMS

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
		limited, a modified syndromic approach may be used. ○ if cervicitis is noted on speculum examination: ▪ oral azithromycin 1 g stat dose and IM ceftriaxone 500mg stat ▪ OR IM ceftriaxone 500mg stat and oral erythromycin ethyl succinate 800mg for 10-14 days ○ if vaginitis is noted in speculum examination, ▪ oral metronidazole 2g stat dose and nystatin pessary 100,000 unit dose for 14 days ▪ OR clotrimazole pessary 500mg stat ▪ OR clotrimazole pessary 200mg ON for 3 nights		

6.18 ADVANCED MATERNAL AGE (≥ 40 YRS OLD)

Signs & Symptoms

Symptoms:

- Asymptomatic

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
• First trimester ultrasound • Offer prenatal screening for chromosomal abnormality (if available) • OGTT	-	Refer FMS and/or O&G if suspected foetal anomaly	MO/ FMS/ O&G	HC/ Hospital

6.19 MATERNAL SEPSIS

Signs & Symptoms

Sepsis

- Infection plus systemic manifestations of infection

Clinical signs:

- Temp $\geq 38^{\circ}\text{C}$ or $< 36^{\circ}\text{C}$
- HR $>100\text{bpm}$
- RR $>20/\text{min}$ or PaCO₂ $<32\text{mmHg}$
- Leukophilia $>12 \times 10^9/\text{L}$ or Leucopenia $<4 \times 10^9/\text{L}$

Systemic Inflammatory Response Syndrome (SIRS) =

Presence of ≥ 2 of above signs

Severe Sepsis

- Sepsis is associated with organ dysfunction or tissue hypoperfusion (hypotension, arterial hypoxemia, lactic acidosis, renal failure, liver dysfunction, coagulation abnormalities and mental status changes).

Septic Shock

- Sepsis associated with hypotension despite IV fluid resuscitation leading to cell dysfunction and, if prolonged, cell death

Investigations	Differential Diagnosis	Care Plan		
		Management	Level of Personnel	Level of Care
● FBC ● Blood C&S ● Other cultures as guided by clinical suspicion of the focus of infection e.g. throat swabs, mid-stream urine, high vaginal swab, CSF, sputum, wound swab ● UFEME ● Serum lactate – to be taken within 6 hours of suspicion of severe sepsis to guide management. Serum lactate >4mmol/l is indicative of tissue hypoperfusion ● Any relevant imaging studies	● UTI ● Pyelonephritis ● Pneumonia ● Chorioamnionitis ● Wound infection ● Endometritis ● Puerperal sepsis ● Dengue ● Malaria ● Influenza-like Illness (ILI)* * ILI is an acute respiratory infection with fever $\geq 38^{\circ}\text{C}$ and cough. Pregnant women are one of the high risk categories.	Hospital care according to causes, consider: ● Antibiotic – IV broad-spectrum antibiotics are recommended within 1 hour of suspicion of severe sepsis, with or without septic shock. ● Wound care ● Supportive management ● Manage according to causes ○ Dengue – as per Management of Dengue Infection in Adults CPG ○ Malaria – as per WHO Guidelines for the Treatment of Malaria ○ ILI – start Tamiflu as per guidelines, dose depends on severity.	Specialist	Hospital

CHAPTER 7

INTRAPARTUM COMPLICATIONS

7.1 FALSE LABOUR

In false labour, the cervix remains undilated, and uterine contractions remain impalpable or infrequent. No further action needs to be taken in the absence of other complications.

Misdiagnosis of false labour or prolonged latent phase leads to unnecessary induction of labour or augmentation, which may fail. This may lead to unnecessary caesarean section or chorioamnionitis.

7.2 ABNORMAL LABOUR PROGRESS CHART (LPC) / EARLY LABOUR MONITORING RECORD

- Abnormal latent phase
 - Cervical dilatation remains less than 4 cm despite 8 hours of regular contractions
 - The duration may be longer for primigravidae
- Any abnormal LPC/ early labour monitoring record at the hospital without specialist or at lower levels should be referred and transferred to a hospital with specialist for further action.

7.3 ABNORMAL PARTOGRAPH

The following features in a partograph indicate poor progress of labour:

- Cervical dilatation to the right of Alert Line
- Cervical dilatation at or beyond the Action Line

7.3.1 Diagnosis of poor progress of labour**a. Primary dysfunctional labour**

The rate of cervical dilatation is less than 1 cm/hour in the active phase of labour due to ineffective uterine contractions of less than 3 in 10 minutes, each lasting less than 40 seconds.

b. Cephalopelvic Disproportion (CPD)

Secondary arrest of cervical dilatation and descent of the presenting part occurs despite good uterine contractions. This can be either:

- Absolute – due to big fetus or small pelvis
- Relative – due to foetal malposition

7.3.2 Management of abnormal partograph

Management of abnormal partograph at the hospital without specialist or at lower level

a. Moving to the right of the alert line

In the active phase of labour, plotting of cervical dilatation will normally remain on, or to the left of the alert line. However, some will cross to the right of the alert line and this warns that labour may be prolonged.

When this occurs in the absence of adequate facilities for obstetric emergencies and operative delivery, the woman must be transferred to a hospital where such facilities are available after consultation with the Specialist

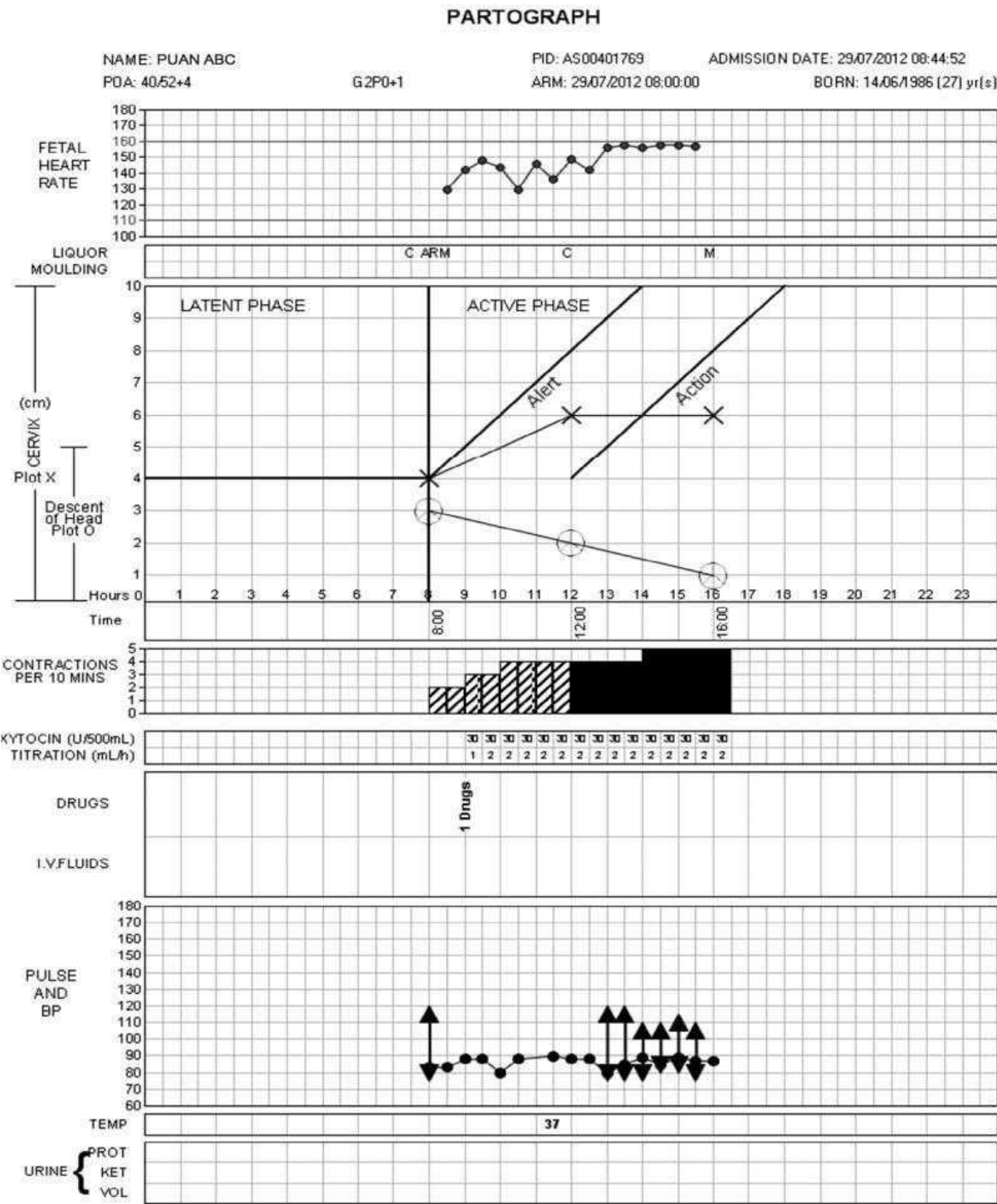
b. At or beyond the action line

Mothers who are at or beyond the action line should ideally be managed in a hospital with a specialist. If a woman's labour reaches or crosses this line, a decision must be made about the cause of poor progress, and appropriate action taken.

This decision and action must be taken in a hospital with facilities to deal with obstetric emergencies and operative delivery

**Inefficient contractions are less common in multigravida than primigravidae.
Hence, every effort should be made to rule out CPD in multigravidae before augmenting with oxytocin.**

Figure 7.1: Abnormal Partograph



7.4 OBSTRUCTED LABOUR

Obstructed labour means that, in spite of strong contractions of the uterus, the fetus cannot descend through the pelvis because there is an insurmountable barrier preventing its descent. Obstruction usually occurs at pelvic brim, but occasionally it may occur in the cavity or at the outlet of the pelvis.

7.4.1 Evidence of obstructed labour

- Secondary arrest of cervical dilatation and descent of presenting part
- Large caput
- Third degree moulding
- Oedematous cervix
- Maternal/foetal distress

7.4.2 Management of obstructed labour

- Rehydrate the mother
- Give supportive care
- Refer mother to the nearest higher level of care or for a caesarean section in hospital with a specialist.

7.5 MANAGEMENT OF ABNORMAL FOETAL HEART RATE (FHR) PATTERNS

- Prop up and turn patient to the left lateral position to alleviate vena caval compression
- Discontinue intravenous oxytocin if any evidence of hyperstimulation
- Perform vaginal examination to rule out cord presentation/prolapse
- Transfer mother immediately to a hospital with facilities for operative delivery

STANDARD OPERATING PROCEDURES

SOP	Condition	Signs & Symptoms	Laboratory Investigation & Findings	Diagnostic Criteria and Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
1	Meconium-stained liquor	Greenish/yellowish discolouration of liquor	• FBC • GXM • CTG	Breech presentation	• Initiate continuous electronic foetal heart activity monitoring • May be necessary to expedite delivery • May need operative delivery • MO/ Paediatrician on standby	MO/Specialist • O&G • Anaesthetist • Paediatrician	Hospital with Specialist
2	Abnormal foetal heart rate	Refer to Foetal Monitoring section	• FBC • GXM • CTG	Refer to Foetal Monitoring section	• Initial management: ○ Left lateral position ○ Stop oxytocin ○ VE to rule out cord presentati	MO/Specialist • O&G • Anaesthetist • Paediatrician	Hospital with Specialist

SOP	Condition	Signs & Symptoms	Laboratory Investigation & Findings	Diagnostic Criteria and Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
					- on/ cord prolapse ○ IV infusion • Expedite delivery as appropriate • MO/ Paediatrician on standby • May need operative delivery		
3	Prolonged first stage of labour	• Latent phase > 8 hours • Crossed alert line on partograph	• FBC • GXM • CTG		• Exclude cephalopelvic disproportion • Augmentation if appropriate • May need caesarean section • Augmentation if appropriate	MO/Specialist • O&G • Anaesthetist • Paediatrician	Hospital with specialist
4	Prolonged second stage	• Primigravida > 60 mins • Multigravida > 30 mins	• FBC • GXM • CTG		• Instrumental delivery • Caesarean section	MO/Specialist • O&G • Anaesthetist • Paediatrician	Hospital with specialist

GUIDELINES FOR PERFORMING LOWER SEGMENT CAESAREAN SECTION (LSCS) AT DISTRICT HOSPITALS WITHOUT SPECIALIST

Caesarean section in district hospitals can only be performed by medical officers who have been credentialed and certified competent.

General criteria for LSCS that can be performed in district

All caesarean sections to be performed in district hospitals must be decided by the specialist who provides the coverage for the hospital, having taken into consideration the availability of appropriate staff and resources.

1. Maternal:
 - i. BMI < 30
 - ii. Parity ≤ 4
 - iii. No midline or Pfannenstiel scar on abdomen. Laparoscopy or Lanz incision are permissible
 - iv. Upper segment placenta
 - v. No significant uterine fibroids or ovarian cyst
 - vi. Blood available
2. Fetal:
 - i. Able to be supported by district paediatrics team
 - ii. No known abnormality that needs tertiary paediatrics support
3. Anaesthetic factors:
 - i. ASA 1 or 2
 - ii. BMI < 30
 - iii. Non-difficult airway
 - iv. No history of CVS diseases
 - v. No history of bleeding disorders
 - vi. No spine abnormalities
 - vii. No eventful anaesthetic history

Potentially difficult LSCS that has to be performed in the district will need the most experienced MO to perform, after discussing with specialist

- Second stage LSCS
- Failed instrumental delivery
- Obstructed labour that cannot be transferred to a specialist hospital in time
- Presence of foetal compromise
- Abruptio / APH

Elective LSCS at district hospital

- For cases that fulfil the above criteria, to arrange LSCS for 38-39 weeks gestation.
- If LSCS is done before 39 weeks, offer IM Dexamethasone 12 mg x 2 doses at 12 or 24 hours apart.

Drugs

1. Premedication
 - IV Ranitidine 50mg
 - Sodium citrate 15mls
2. Prophylactic antibiotics up to 1 hour before incision
 - IV Unasyn 1.5g
 - alternatively IV Cefuroxime 1.5g and IV Metronidazole 500mg
3. PPH prophylaxis/ treatment medications
 - Oxytocin 5 IU by slow intravenous injection (may repeat dose)
 - Oxytocin infusion (40 IU in 500 ml isotonic crystalloids at 125 ml/hour) unless fluid restriction is necessary
 - Cases of PPH should be referred to a specialist on phone cover for advise for plan of management
 - Carboprost 0.25 mg by intramuscular injection repeated at intervals of not less than 15 minutes to a maximum of eight doses (use with caution in women with asthma)
 - IV Tranexamic acid 2g and followed by 1g/hour for 6 hours
4. Thromboprophylaxis
 - Clexane or heparin as per protocol (Prevention & Treatment of Thromboembolism in Pregnancy and Puerperium 2018)

Staffing for LSCS

- 2 O&G MO for each LSCS where possible
- 1 Paediatric MO on standby
- 1 Anaesthetic medical officer and or medical assistant
- 1 Anaesthetic assistant (medical assistant or nurse)
- 1 scrub nurse
- 1 circulating nurse
- 1 recovery nurse
- 1 counter nurse
- 1 Pembantu Perubatan Kesihatan

Instruments and Equipment for LSCS

Ensure that the LSCS set is complete as listed below:

No.	Instrument	Unit
1.	Instrument tray 540x325x64mm	1
2.	Kidney dish 12" (SH533)	2
3.	Kidney dish 10"(JG50R)	1
4.	Bowl 6" (JG524R)	1
5.	Gallipot 10oz,6oz	2
6.	Standard tooth dissecting forceps (BD560R)	1
7.	Standard plain dissecting forceps (BD050R)	1
8.	Mcindoe dissecting forceps (BD236R)	1
9.	Gillies dissecting forceps (BD 660R)	1
10.	Waugh plain dissecting forceps (BD049R)	1
11.	Waugh plain dissecting forceps (BD559R)	1
12.	B/P scalpel handle no.3 (BB073R)	1
13.	B/P scalpel handle no.4 (BB084R)	2
14.	Mayo scissors (C) 6 3/4" (BC557R)	2
15.	Mayo scissors (STR) 6" (BC545R)	1
16.	Metzenbaum scissors (C) 7" (BC606R)	1
17.	Mayo needle holder 7 ½" golden handle (BN065R)	1
18.	Mayo needle holder 7 ½" golden handle (BN066R)	1
19.	Mayo needle holder 7 ½" golden handle (BN067R)	1
20.	Babcock tissue forceps (EA031R)	2
21.	Allis tissue forceps (EA015R)	2
22.	Kocher artery forceps (S) (BH642)	2
23.	Spencer wells artery forceps (str) 7 ½" (BH336R)	4
24.	Halstead artery forceps (C) 7 ½" (BH203R)	6

No.	Instrument	Unit
25.	Spencer wells artery forceps (C) 7 ½ " (BH337R)	4
26.	Allis Thomas toothed (EA020R)	2
27.	Green Armytage forceps (FT269R)	4
28.	Towel clips (BF432R)	6
29.	Sponge holders (BF122R)	6
30.	Doyen retractor 48x90mm (BT723R)	1
31.	Sinus forceps (C) (BF006R)	1
32.	Canny Ryall retractor Medium (BT048R)	2
33.	Canny Ryall retractor Small (BT047R)	2
34.	Wrigley's obstetric forceps	1 pair
35.	Yankauer sucker	1
36.	Instrument pins	2
37.	Silicone tubing	3m

Ensure other essential medical and non-medical equipment needed to perform LSCS is available and functioning well e.g. anaesthetic machine, diathermy machine, resuscitation trolley and defibrillator.

General Anaesthesia Service in the Operating Theatre

1. All patients shall be assessed pre-operatively by the anaesthetising doctor.
2. Pre-operative fasting practice shall be in accordance with the *Guidelines on Preoperative Fasting, College of Anaesthesiologists, Academy of Medicine Malaysia, revised 2008*.
3. In OT, the correct identification of the patients are verified by the nurse and doctor by using SSSL checklist which also includes valid informed consent for surgery and anaesthesia (refer to *Consent for Treatment of Patients by Registered Medical Practitioner: Malaysian Medical Council, 2013*).
4. Separation of children or intellectually challenged patients from parents or guardians prior to anaesthesia is to be discouraged.
5. The minimum standards for the safe conduct of anaesthesia in the OT shall be strictly adhered to (refer to *Recommendations on Minimum Facilities for Safe Anaesthesia Practice in Operating Suites and Other Anaesthetising Locations: Australia and New Zealand College of Anaesthesiologists. P255. 2012*).
6. A skilled assistant shall be available in every operating room to assist in the administration of anaesthesia.
7. Formal hand-over of patient information shall take place whenever there is a change of caregivers during anaesthesia even temporarily e.g. during relief for breaks or permanently.
8. The assistant medical officer (AMO) shall be responsible for the regular preventive maintenance of the equipment in the unit through the hospital's concession company.
9. Monitoring of patients under anaesthesia shall comply with the recommended standards (refer to *Recommendations for Safety Standards and Monitoring during Anaesthesia and Recovery: College of Anaesthesiologists Malaysia, 2008*).
10. All anaesthetic locations shall be equipped with anaesthetic gas scavenging system.
11. Post-anaesthesia patients shall be monitored in the recovery room according to the level of care determined by the physiologic status of the patient.

Anaesthetic Equipments in OT

1. Anaesthesia machine, ventilator, physiologic monitor
2. Emergency trolley with cardiac pacing and defibrillation device
3. Airway adjuncts: Guedel airways, LMA, endotracheal tubes, bougies
4. Laryngoscope with various blades sizes (+/- McCoy Blades)
5. Suction apparatus, Yankauers, suction catheters
6. Infusion syringe pump
7. Warming blanket/mattress
8. Patient transport trolley
9. Patient transfer device
10. Blood warming device
11. Blood refrigerator
12. Drugs refrigerator
13. Equipment drying cabinet
14. Anaesthesia trolley

Recovery room

1. Recovery room monitor
2. Oxygen
3. Suction apparatus
4. Patient recovery trolley

List of medications

No.	Types of medication	Medications
1.	IV induction agents	▪ Propofol ▪ Thiopentone ▪ Ketamine ▪ Midazolam
2.	Muscle relaxants	▪ Atracurium ▪ Rocuronium ▪ Suxamethonium
3.	Reversal drugs	▪ Atropine ▪ Neostigmine
4.	DDA drugs	▪ Morphine ▪ Fentanyl ▪ Ketamine
5.	Volatile agents	▪ Sevoflurane
6.	Emergency drugs	▪ Adrenaline ▪ Atropine ▪ Calcium gluconate ▪ Dextrose 50% ▪ Ephedrine ▪ Hydrocortisone ▪ Chlorpheniramine ▪ Dexamethasone ▪ Dopamine ▪ Flumazenil ▪ Frusemide ▪ Metoclopramide ▪ Naloxone
6.	Local anaesthetics	▪ Lignocaine ▪ 0.5% Hyperbaric bupivacaine ▪ Plain Marcaine
7.	Lubricants	▪ K-Y jelly
8.	Suppositories	▪ Diclofenac sodium ▪ Paracetamol
9.	Inhalational drugs	▪ Salbutamol metered dose inhaler

LOGBOOK FOR LSCS FOR DISTRICT MEDICAL OFFICER

No.	Name of patient	NRIC.	Indication of LSCS	Assist/ Performed	Date	Initial of supervisor

COMPETENCY ASSESSMENT : LOWER SEGMENT CAESAREAN SECTION

Target	Expected competence level					Supervisor signs when competence level achieved	
	1	2	3	4	5	Signature	Date
Uncomplicated lower segment caesarean sections including those without previous scars							
Signature to confirm completion: Name of the Supervisor/ Specialist: Date: Hospital:							

General Information:

The level of competence ranges from observation (Level 1) to independent practice (Level 4 or 5). The officer should achieve at least level 4 to be credentialed to be able to perform uncomplicated LSCS independently. The officer should keep a logbook of the cases of LSCS assisted or performed. Minimum of 5 LSCS needs to be performed before a competency assessment is carried out. When you feel ready for competency assessment, it is your responsibility to organise with your supervisor.

SCORING SYSTEM:

- 1: Passive attendance, assistance
- 2: Needs close supervision
- 3: Able to carry out procedure under some supervision
- 4: Able to carry out procedure without supervision
- 5: Able to supervise and teach the procedure

*The general aim is to get at least mark 4.

CHAPTER 8

POSTNATAL COMPLICATIONS

8.1 MENTAL HEALTH IN POSTNATAL PERIOD**8.1.1 Postnatal blues**

- Postnatal blues is common, affecting around 8 in 10 postnatal mothers and considered as a normal situation after childbirth.
- It is a mild and transient condition where mothers may experience a range of feelings: being tearful, overwhelmed, irritable and emotionally fragile with sadness, loneliness, anxiety and insomnia
- It may occur immediately after childbirth, commonly within the first few days of delivery, peak around one week and resolve by the end of the second week of postpartum.
- It may be associated with sudden hormonal changes, discomfort from breast engorgement and birth pain, stress of parenthood and childcare, isolation, sleep deprivation and exhaustion.
- It normally resolves as mothers learn to adjust to her new life and with understanding and support.

8.1.2 Postnatal depression

- Postnatal depression is a major depressive disorder with onset in the postnatal period, up to 1 year. It is considered as a common psychiatric disorder affecting 1 in 10 postnatal women.
- In postnatal depression, women experience symptoms of depression for more than 2 weeks:
 - depressed mood
 - loss of interest and pleasure in activities they usually enjoy
 - loss of appetite or eating much more than usual
 - inability to sleep or sleeping too much
 - fatigue
 - diminished ability to concentrate or make decisions
 - restlessness or becoming slow
 - feeling worthless
 - recurrent thoughts of death and suicide
- It may also be associated with
 - irritability
 - anxiety and panic attacks
 - difficulty in bonding with baby
 - feeling like they are not a good mother
 - thoughts of harming baby

- It is commonly underrecognized and underdiagnosed due to multiple barriers such as ignorance or not having awareness and fear of stigma on women's side; time constraint and unwillingness to explore on health providers' side.
- It may not be easily diagnosed as there is overlap between depressive symptoms and what women commonly experience in perinatal period such as feeling fatigue, having changes in sleep and appetite.
- Screening for postnatal depression will be useful and can be done from 6 to 12 weeks postnatal and be repeated once in later postnatal years. Useful tools include: Whooley Two-Question Screen, Patient Health Questionnaire-2 (PHQ-2) or Edinburgh Postnatal Depression Scale (EPDS).
- Management:
 - Mild depression: provide psychosocial intervention (e.g. counselling, peer-support) and provide or arrange psychological intervention (e.g. cognitive behavioural therapy, interpersonal psychotherapy).
 - Moderate depression: consider risk-benefit of antidepressant in combination of psychosocial and psychological intervention as above.
 - Severe depression: antidepressant is most likely indicated. Consider SSRI, e.g. Sertraline 50-100 mg daily.
 - Consult/refer to FMS or psychiatrist particularly for severe functional impairment, high risk of suicide or depression with psychosis.
 - Assess mother-baby interaction and address difficulties in parenting and childcare.

8.1.3 Severe mental illness (for postpartum psychosis refer to obstetric emergency)

- Mental disorders have a high risk of relapse during the postpartum period.
- Management during the postnatal period will involve multidisciplinary teams eg: O&G, paediatrics, psychiatry and social worker.
- Close monitoring is crucial for the first six weeks and may be required for up to one year postnatal in women with severe mental illness. Home visit by maternal child health teams and community psychiatry services may be required.
- Ask about change in behaviour, agitation, suspiciousness, confusion and hallucinations during the postnatal follow-up. Obtain collaborative history

from family and partners regarding emotional lability and abnormal behaviour.

- Observe difficulties in mother-baby interaction as well as parenting and child-care abilities and offer appropriate support.
- Address psychosocial problems such as poor support, financial difficulties, housing problems and unemployment
- Breastfeeding
 - Breastfeeding is encouraged for most patients on psychotropics.
 - Aim for antipsychotic that has an infant plasma level of less than 10%
 - Olanzapine is antipsychotic of choice (infant plasma level 1.6%)
 - Patients on Lithium and Clozapine are not recommended to breastfeed.
 - Mothers who are too heavily sedated should not sleep with the baby
 - Breastfeeding might need to cease if patient is too unwell, to avoid sleep disruption or requires night time sedation
- Sleep preservation is important for prevention of relapse.
- Admission during a relapse in the postpartum period
 - Admission is necessary if there is danger to patient and infant.
 - As much as possible, mother and infant should be kept together by mobilising social support or activating community mental health care.

8.1.4

Substance abuse in postpartum period

- Women with perinatal substance use disorder presented with extremely complicated issues and they may present in labour with no antenatal check-up.
- Ask about history of substance use, withdrawal symptoms, psychosocial issues as well as comorbid psychiatric and medical conditions.
- Management during the postnatal period must involve a multidisciplinary team eg: O&G, Paediatrics, Psychiatry and Social worker.
- Address the needs of the care of the baby that may involve parenting capacity issues, social support or adoption.
- Provide support for women to meet needs such as shelter, food and safety plan
- Arrange follow-up (collaboration of hospital and health clinic).
- For women on methadone replacement therapy,
 - relevant information includes confirmation of identification, last dose and current prescription
 - observe signs and symptoms of withdrawal

- Arrangements for short-course of directly observed treatment (DOTS)
- The client's next of kin is allowed to take on behalf of the client during confinement period (around 2 weeks) then the client needs to attend methadone clinic on her own
- Arrangements for take away depend on the client's drug history and recent opioid use as well as the client's stability - urine drug test result (negative in 1-2 years)
- Breastfeeding:
 - For individual risk-benefit analysis
 - Breast-feeding is not contraindicated in patients on methadone replacement therapy
 - For patients still taking substance, generally breastfeeding is not contraindicated unless the woman is a polysubstance user. She may require advice regarding time from substance use to breastfeeding or expressing breast milk for baby.
- Do not breastfeed for 72 hours after using amphetamines. To express and discard milk after drug use.
- To limit alcohol to two standard drinks in a day. Not to consume immediately before feeding. Consider expressing breast milk in advance
- Short acting benzodiazepines may be used for a limited time but long acting should be avoided. Advise not to breastfeed immediately after taking short acting benzodiazepines

8.2 COPING WITH DEATHS

8.2.1 Grief and bereavement

- Grief is the process of experiencing psychological, behavioural, social and physical reactions to loss that may evolve over time. It is a normal reaction, and its absence may be abnormal and indicative of pathology.
- Bereavement is the entire experience of family members and friends in the anticipation of death and subsequent adjustment to living following the death of a loved one.
- The emotional and somatic responses to death differ from person to person. The grief response will be more intense if the death occurs in a person who is closely related. The process of grief involves a few stages.

Table 8.1: Normal grief reaction

Stage 1 Hours to days	• Denial and disbelief • Numbness
Stage 2 Weeks to 6 months	• Sadness, weeping, waves of grief • Somatic symptoms of anxiety • Restlessness • Poor sleep • Diminished appetite • Guilt, blame of others
Stage 3 Weeks to months	• Symptoms resolve • Social activities resumed • Memories of good times • Symptoms may recur at anniversaries

- Abnormal or pathological grief:
 - Symptoms are more intense than usual
 - Symptoms prolonged beyond 6 months
 - Symptoms delayed in onset
- a. Abnormally intense grief:
 - Up to 35% of bereaved people meet the criteria for a depressive disorder at some time during grieving.
 - Most of these depressive disorders resolve within six months but about 20 % persist for longer periods.
 - These persons are more likely:
 - to have poor social adjustment
 - visit doctors frequently
 - to use alcohol
 - Suicidal thoughts may occur when grief is intense. The rate of suicides is increased most in the year after bereavement, but continues to be high for five years after the death of a spouse or parent.
 - Elderly widowers are at higher risk than other bereaved people.

- The presence of suicidal ideas should prompt appropriate assessment of suicide risk.
- b. Prolonged grief:
- Defined as grief lasting for more than 6 months. However, it is difficult to set such a defined limit to normal grief and complete resolution may take much longer.
 - Instead of the normal progression, symptoms of the first and second stage persist.
 - Such prolongation may be associated with a depressive disorder but can occur without such a disorder.
- c. Delayed grief:
- It is said to occur when the first stage of grief does not appear until more than two weeks after the death.
 - It is said to be more frequent after sudden traumatic or unexpected deaths.
- d. Inhibited and distorted grief:
- Absence of grief is a pathological variant of grieving.
 - Inhibited grief refers to a reaction that lacks some normal features.
 - Distorted grief refers to features (other than depressive symptoms) that are either unusual in degree, for example marked hostility, over – activity, and extreme social withdrawal, or else unusual in kind, for example expression of physical symptoms that were part of the last illness of the deceased.
- The mortality of bereavement
 - Several studies have shown an increased rate of mortality among bereaved spouses and other closed relatives, with the greatest increase being in the first 6 months after bereavement.
 - Abnormal grief reactions are more likely in the following circumstances:
 - When the death was sudden and unexpected,
 - When the bereaved person had a very close, or dependent, or ambivalent relationship with the deceased
 - When the survivor is insecure, or has difficulty in expressing, feelings, or has suffered a previous psychiatric disorder

- When the survivor has to care for dependent children and so cannot show grief easily.

8.2.2 Management of grief and bereavement

- In planning the management it is important to take into account the individual circumstances of the patient as well as the general guidelines outlined below:

i. Counselling

The bereaved person needs:

- To talk about the loss
- To express feelings of sadness, guilt or anger
- To understand the normal course of grieving.

It is helpful to forewarn a bereaved person about unusual experience such as feeling as if the dead person were present, illusions, and hallucinations, otherwise these experiences may be alarming.

Help may be needed :

- to accept that loss is real
- to work through stages of grief
- to adjust to life without the deceased

Viewing the dead body and putting away the dead person's belonging help this transition, and a bereaved person should be encouraged to perform the actions.

Practical problems may need to be discussed, including funeral arrangements and financial difficulties.

As time passes, the bereaved person should be encouraged to resume social contacts, to talk to other people about the loss, to remember happy and fulfilling experiences that were shared with the deceased, and to consider positive activities that the latter would have wanted survivors to undertake.

ii. Drug treatment

Cannot remove the distress of normal grief, but it can relieve severe anxiety.

In the second stage, antidepressant drugs may be beneficial if the criteria for depressive disorder are met, though such usage has not been evaluated in this special group.

iii. Support groups

Can be useful, however, it may be difficult to sustain effective functioning.

iv. Psychotherapy

It is not practical, nor is there evidence that it is helpful, to provide psychotherapy for all bereaved persons

Best Practice Guide:

- Provide an environment and circumstances for feeling hurt, guilty, angry or other strongly negative feelings.
- Allow the spouse and relatives to ventilate
- Validate the extent of grief
- Facilitate procedures for removal of the body to the home for last rites.
- Be sensitive for the need for postmortem in cases of sudden death
- Do home visit to explore feelings of guilt or blame and explain /reinforce circumstances of death.
- Encouraging the spouse to build a support network of family, friends and professional-bereavement clinics are useful.
- Consider the needs of the children and refer to the appropriate welfare authorities where appropriate.
- Be alert for suicidal intention or behaviour
- Remember that grief takes time
- Stages of grief are not always predictable

For health workers handling death among their patients:

Do

- Direct expression of sympathy
- Talk about deceased by name
- Elicit question about circumstances of the death
- Elicit question about feeling and about how the death has affected the person

Don't

- Have a casual or passive attitude
- Give statements that death is for the best
- Assume that the bereaved is strong and will get through this
- Avoid discussing the death

To help remember grief work: Remember TEAR

- **T** - To accept the reality of the loss
- **E** - Experience the pain of the loss
- **A** - Adjust to the new environment without the lost object
- **R** - Reinvest in the new reality

8.3

MATERNAL DEATH

- The death of a pregnant woman in the antepartum, intrapartum or postpartum period up to 42 days after the delivery.
- Special characteristics of a maternal death:
 - Sudden, may be totally unexpected
 - May occur in a healthy woman in the prime of her life
 - Sudden end to hope of going home with a healthy wife and infant after a pregnancy that was full of great expectations
- A maternal death is a great loss as the mother has multiple roles :
 - Mother-newborn child, previous children
 - Wife
 - Wage earner
 - Role model
 - Member of the family
 - Member of the community.
- After a maternal death:
 1. Provide an accurate diagnosis. (Medical Officers/ Specialists)
 - International Statistical Classification of Diseases and Related Health Problems 10th Revision (ICD-10) is used.
 - The cause of death should be discussed with the obstetrician covering the district before release of the body.
 - All maternal deaths where the cause of death is not known including home deaths should be reported to the police. Once a postmortem order has been issued, a detailed post-mortem should be carried out. In the event where a post-mortem order cannot be obtained, consent for a clinical post-mortem examination should be requested. This request could be for a limited postmortem examination or permission to carry out needle biopsies.
 2. Documentation and report:
 - Events leading to the death must be completely documented.
 - All investigations done must be documented and the results traced.

- The burial permit and death certificate should be completed by the medical officer.
 - The maternal death coordinator (Health Matron/Sister in the case of a home/private sector death or labour ward sister for a hospital death) should be notified within 24 hours using KIK/KI – 1 form.
 - The antenatal record should be summarised in the Maternal Mortality Report. All maternal death cases need to be reported and notified using the standard Maternal Mortality Report form. The report should be filled and sent to the District Health Officer for further action.
 - All health facilities providing antenatal care should be stated in the report.
 - The Maternal Mortality Report (KIK/KI-2) should be promptly filled within two weeks and sent to the state health department for further action. This format should bear no indication of the place or the persons involved in the death.
3. Provision of spousal support and bereavement counselling (Nurse/ Medical Officers/Specialist)

8.4 PERINATAL LOSS

- Perinatal loss is all types of loss which include early neonatal death, intrauterine death, stillbirth, spontaneous miscarriages and medical terminations.
- Refer Guideline for Stillbirth and Under Five Mortality Reporting System, Ministry of Health 2018 for detailed procedures on the notification, investigation of all stillbirths and Under -5 mortality.

Reference:

1. David M. Taylor, Thomas R. E. Barnes, Allan H. Young. (2018). The Maudsley Prescribing Guidelines, 13th Edition
2. Royal Australian and New Zealand College of Obstetricians and Gynaecologists. Substance use in pregnancy. College Statement- C-Obs 55. 2013.
3. Queensland Clinical Guidelines. Perinatal substance use: neonatal. Guideline No. MN16.38-V1-R21. Queensland Health. 2016.
4. World Health Organisation. Guidelines for the identification and management of substance use and substance use disorders in pregnancy. 2014.
5. Commonwealth of Australia. National clinical guidelines for the management of drug use during pregnancy, birth and the early development years of the newborn. 2006.
6. Guideline for Stillbirth and Under Five Mortality Reporting System, Ministry of Health 2018

STANDARD OPERATING PROCEDURES

SOP	Conditions	Signs & Symptoms	Laboratory Investigations & Findings	Diagnostic Criteria & Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
1	Perineal wound problem	• Pain at wound site • Swelling around the site • Bleeding from wound site • Abnormal discharge • Wound breakdown	• Swab C&S • FBC	N/A	• Wound care • Antibiotics When required to consider:- • Incision & drainage • Secondary suturing	Nurses/ MO/ Specialist	HC/ Hospital
2	Post caesarean care: • Wound break down	• Gaping wound • serous discharge ± foul smell	Wound swab C&S	a. Superficial and small size < 4cm and involve skin and fat layer b. Deep and size >4cm involved rectus sheath and below c. Result of examination and investigation	• Dressing & Toilet • Antibiotics oral cloxacillin 500mg QID for 5 days • Review C&S result and treat accordingly • Refer to hospital • Wound toilet • Antibiotic • Suturing after wound is clean	MO MO / Specialist	HC Hospital with or without specialist

SOP	Conditions	Signs & Symptoms	Laboratory Investigations & Findings	Diagnostic Criteria & Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
	• Infected wound • Wound haematoma	• Pus • Redness • Pain • Fever $\pm$ foul smell • Swelling • Redness • Pain	Wound swab C&S FBC if indicated FBC	Result of examination and investigation Result of examination	• Refer to hospital • Wound toilet • Antibiotic • Suturing after wound is clean • Conservative if small (<4 cm), not increasing in size and minimal pain • Require evacuation if increasing in size or expanding or with pain	MO/ O&G Specialist MO / Specialist	Hospital with or without specialist Hospital with or without specialist
3	Heart diseases	• Dyspnoea • Cyanosis • Cardiac murmur • Signs of cardiac failure	Any or all of the following investigation as required: - FBC - ECG - CXR - ECHO	• Pulmonary oedema • CCF • Embolism	• More frequent postnatal visit • Assessment of cardiac status during postnatal visit • Any worsening symptoms or inter-current symptoms to refer to hospital • Ensure cardiology appointment and follow up • Continue medication • Avoid aggravating factors • Advise contraceptives according to MEC Pre pregnancy clinic (PPC) referral: • Patients who are not fit for pregnancy, to be	MO/ FMS/ Specialist	Hospitals with specialist

SOP	Conditions	Signs & Symptoms	Laboratory Investigations & Findings	Diagnostic Criteria & Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
					referred to PPC during the postpartum period • Patient to be informed to attend PPC when plan for next pregnancy in view of might need cardiac assessment prior to conception		
4	Urinary retention	• Unable to pass urine • Distended bladder • Lower abdominal discomfort • Fever	• Urine FEME • Urine C&S • Ultrasound may be considered	• UTI • Neurogenic Bladder	• Intermittent or continuous bladder catheterisation • Antibiotic for infection	Nurses / MO / Specialist	HC/ Hospital
5	Urinary incontinence	Intermittent or continuous incontinence	• Urine FEME • Urine C&S	Need to rule out any urinary fistula	Management according to cause • Treatment of UTI • Pelvic floor exercise • Referral to gynaecologist/ uro-gynaecology	MO/Specialist	HC/ Hospital
6	Subinvolution of uterus	• Uterus does not involute as expected • Severe pain • Abnormal bleeding	Ultrasound to rule out: • retained POC • uterine or ovarian mass	Need to rule out retained placenta or POC	• Reassurance if ultrasound normal and asymptomatic • If abnormal USG or severe pain to refer hospital	MO/ FMS/ O&G Specialist	HC/ Hospital
7	Secondary postpartum haemorrhage	Excessive bleeding from the genital tract after 24 hours post delivery	• FBC • GXM • Coagulation profile • Ultrasound	• Uterine atony • Retained POC • Coagulations disorder • Endometritis • Pelvic arteriovenous	Refer to training manual on management of PPH	MO/ FMS/ O&G Specialist	Hospital

SOP	Conditions	Signs & Symptoms	Laboratory Investigations & Findings	Diagnostic Criteria & Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
			d	malformation (AVM) – if all of the above have been ruled out			
8	Puerperal pyrexia	- Temp > 38°C - Abnormal vagina discharge	- FBC - Blood C&S - Urine FEME - Urine C&S - HVS C&S - Wound swab C&S - Lower limb ultrasound doppler	- Genital tract infection - Wound infection – episiotomy & LSCS - UTI - Other intercurrent infections such as URTI, malaria, pneumonia, dengue - DVT	According to causes, consider: - Antibiotic - Wound care - Supportive management - Admission to hospital	MO/ FMS/ O&G Specialist	Hospital
9	Breast engorgement	- Pain - Swelling		- Mastitis - Abscess	- Cold and warm compression - Express breast milk - Antibiotics and analgesics - If unresolved, for incision & drainage	Nurses / MO/ Specialist	All level
10	Deep Vein Thrombosis	- Unilateral calf swelling - Tenderness - Pain on walking - Low grade fever	Lower limb ultrasound doppler	- Redness of calf - Tenderness on palpation - Decrease peripheral pulsation	- Inform medical officer/ FMS - Refer to hospital urgently	Specialist	Hospital with specialist
11	Pulmonary embolism	- Difficulty in breathing - Sudden onset	- Chest X ray - Abnormal		- Resuscitation - Inform medical officer/FMS	Specialist and multidisciplinary team	Hospital with specialist

SOP	Conditions	Signs & Symptoms	Laboratory Investigations & Findings	Diagnostic Criteria & Differential Diagnosis	Care of Plan		
					Management	Level of Personnel	Level of Care
		chest pain • Non- productive cough • Tachypnoea • Tachycardia • Hypotension • Low O ₂ saturation • Maternal collapse	ECG changes • Arterial blood gases • CTPA • V/Q scan		• Refer to hospital urgently		

CHAPTER 9

OBSTETRIC EMERGENCIES

9.1 RED ALERT SYSTEM

The Red Alert System is described in the Report on Confidential Enquiries into Maternal Mortality in Malaysia for the year 1992. It should be operational in all hospitals dealing with obstetric cases where it helps to improve emergency response time and reduces maternal morbidity and mortality. This system should also be implemented in other health facilities, but amended to suit the facility concerned.

9.1.1 How the Red Alert System Functions

- Red Alert code is triggered in the presence of an obstetric emergency case in any health facility.
- The nurse or doctor in charge of the area activates the system by calling the telephone operator of the hospital or in the case of health clinics to call their colleagues for assistance by saying “Red Alert”
- The operator will immediately initiate a ‘call system’ to get staff involved to attend to this emergency.
- Each health facility is to establish its own system
- The doctors involved (see below) will go to the area of concern

9.1.2 Staff involved in Red Alert (hospital)

- Medical officers on call
 - Specialist on call
 - Consultant on call
 - Blood bank
 - Sister on-call
- } From O&G, Medical and Anaesthesiology Departments

9.1.3 Staff involved in Red Alert (health clinic)

- Staff attending the patient to trigger red alert (nurse or doctor)
- To call FMS / Medical officer in charge
- Ambulance driver if available
- Medical assistant
- The person in charge (nurse or doctor) must consult the specialist on call in the nearest specialist hospital
- Preparation for transfer of patients by clinic ambulance or by an obstetrics retrieval team should be prompted
- The aim is to get as many staff as possible to assist simultaneously to stabilise the patient and facilitate the transfer to a specialist hospital

9.1.4 Indications to activate Red Alert (but not limited only to these conditions)

- Severe antepartum haemorrhage
- Postpartum haemorrhage
- Maternal collapse
- Eclampsia
- Uterine inversion
- Cord prolapse
- Shoulder dystocia

9.2 REFERRAL SYSTEMS AND RETRIEVAL & RESUSCITATION TEAM

9.2.1 When should a nurse or medical officer refer or consult a higher level of care?

- Abnormalities of the foetal heart rate
- Delay in the first or second stage of labour.
- Any meconium-stained liquor
- Obstetric emergency – antepartum haemorrhage, cord presentation or prolapse, postpartum haemorrhage, uterine inversion, shoulder dystocia, eclampsia, maternal collapse or a need for advanced neonatal resuscitation is anticipated.
- Retained placenta
- Maternal pyrexia in labour (38°C once or 37.5°C on two occasions 2 hours apart)
- Malpresentation or breech presentation diagnosed for the first time at the onset of labour
- Either raised diastolic blood pressure (over 90 mmHg) and or systolic blood pressure (over 140 mmHg) on two consecutive readings taken 30 minutes apart
- Uncertainty about the presence of a foetal heart activity
- Third or fourth degree tears or other complicated perineal trauma requiring repair.
- Any other medical conditions in which the medical officer is unable to manage

9.2.2 Referral System

Inter-hospital / inter-centre transfer should be considered if the necessary resources or personnel for optimal maternal care are not available at the facility providing the care. The resources available at the referring and the receiving centres/hospitals should be considered. The risks and benefits of transporting or not transporting the mother, should be assessed.

9.2.3 In-utero transfer

All conditions potentially requiring specialised care for the neonate (medical/ surgical) e.g. preterm labour, foetal growth restriction and congenital anomaly requiring surgical intervention may benefit from in-utero transfer. This has proven to result in a better neonatal outcome compared to neonatal transfer after delivery.

9.2.4 Retrieval and Resuscitation Team

A system should be available to transport trained medical personnel from higher level centres to provide assistance in the referring centre/home. In some hospitals, the team is called the “Flying Squad”.

Team members

- Medical officer / specialist
- Assistant medical officer (optional)
- Staff nurse from labour ward
- Ambulance driver
- Male attendant from Emergency Department

Indications for mobilisation of the Retrieval Team

- Antepartum haemorrhage
- Postpartum haemorrhage
- Eclampsia
- Severe pre-eclampsia
- Maternal collapse
- Mother in labour
- Cord prolapse
- Uterine inversion
- Shoulder dystocia

How the team functions

- A telephone call is received from a peripheral or private maternity centre
- Staff in Labour Ward triggers the operation of the team by calling the person in-charge from the Emergency Department
- The team assembles at the Emergency Department and sets out to the referring centre as soon as possible preferably within 10 minutes of the initial call
- The team retrieves the mother at the referring centre and brings her to the hospital
- The referring centre needs to perform initial resuscitation

Resuscitative equipment

It is dependent on the nature of the cases being retrieved. A complete set of medical equipment should be brought in the ambulance:

- Defibrillator
- Mobile vital sign monitors including pulse oximetry
- Non-pneumatic anti-shock garments (if available)
- Infusion pumps
- IV fluids/volume expanders
- Matched or unmatched blood and blood products

9.2.5

Referral letter

- A standard Intrapartum Referral Form (IP-1) should be used to refer a mother in labour.
- The intrapartum checklist of the mother (**Appendix 3-4**) should be updated and attached to the referral form.
- The receiving hospital should likewise reply using a standard reply form (IP-2) when the mother is discharged from their care.
- Effective communication between the centres involved will help maintain good working relationships and understanding and ensure continuity of care for the mother.
- As suggested under “Level of Care” – (SOP – Intrapartum Management), apart from mother in normal labour, mother with all other conditions as stated in the Checklist of Intrapartum Risk Factors (**Appendix 3-4**) should be managed in a hospital and not in a health centre/ community health clinic or home.

9.3

CORD PROLAPSE

Cord prolapse has been defined as the descent of the umbilical cord through the cervix alongside (occult) or past the presenting part (overt) in the presence of ruptured membranes.

Cord presentation is the presence of the umbilical cord between the foetal presenting part and the cervix, with or without rupture of membranes.

Risk factors for cord prolapse include:

- Multiparity
- Prematurity less than 37 weeks
- Artificial rupture of membranes

- External cephalic version (during procedure)
- Low birth weight, less than 2.5 kg
- Foetal congenital anomalies
- Vaginal manipulation of the foetus with ruptured membranes
- Internal podalic version
- Breech presentation
- Stabilising induction of labour
- Second twin
- Insertion of uterine pressure transducer
- Unengaged presenting part
- Polyhydramnios
- Transverse, oblique and unstable lie (when the longitudinal axis of the foetus is changing repeatedly)
- Low-lying placenta or other abnormal placentation

Note: Refer to **Appendix 3-2** for measures to be taken before transfer to the tertiary centre.

9.4 UTERINE RUPTURE

Uterine rupture is defined as a disruption of the uterine muscle extending to and involving the uterine serosa or disruption of the uterine muscle with extension to the bladder or broad ligament.

Signs and symptoms of uterine rupture:

- Shock
- Rapid maternal pulses
- Abdominal distension
- Free fluid present in the abdominal cavity
- Abnormal uterine contour
- Tender abdomen
- Easily palpable foetal parts
- Absent foetal movements and foetal heart sounds
- Abnormal CTG

What actions need to be taken?

- If a patient is unstable, call for help. Start resuscitation immediately. Refer to O&G specialist on call at the nearest hospital and arrange for transfer.

- If uterine rupture is suspected and the patient is stable, for the medical officer to refer to O&G specialist on call in the nearest hospital and arrange for transfer.

9.5 SHOULDER DYSTOCIA

Shoulder dystocia is defined as a delivery that requires additional obstetric manoeuvres to release the shoulders after gentle downward traction to deliver the fetus has failed. Shoulder dystocia occurs when either the anterior or less commonly, the posterior foetal shoulder impacts on the maternal symphysis or sacral promontory.

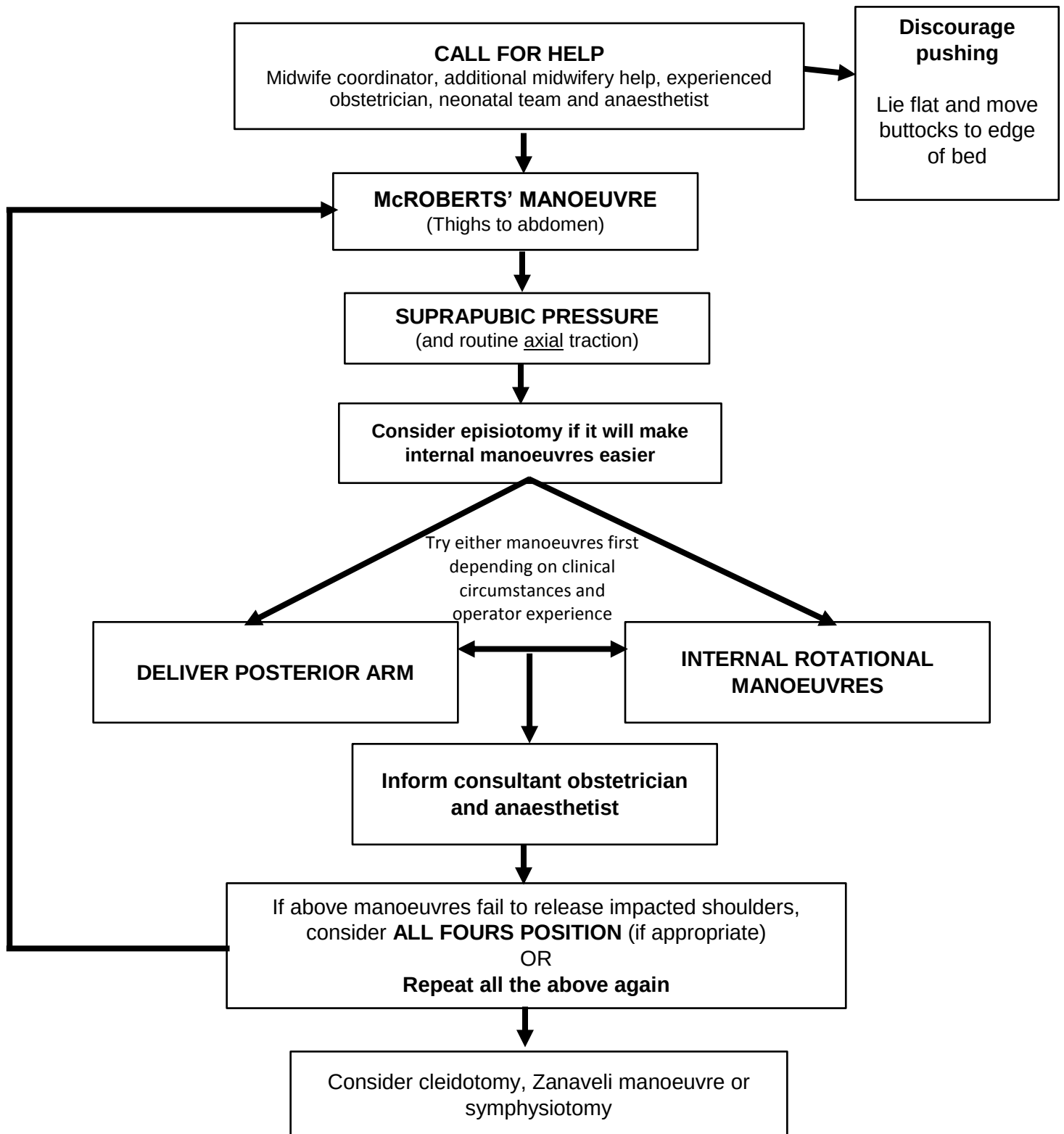
Table 9.1 : Factors Associated with Shoulder Dystocia

Pre-labour	Intrapartum
• Short stature • Previous shoulder dystocia • Macrosomia • Diabetes mellitus/ GDM • Maternal BMI > 30 kg/m² • Induction of labour	• Prolonged first stage of labour • Secondary arrest • Prolonged second stage of labour • Oxytocin augmentation • Assisted vaginal delivery

Note:

- Health care providers in clinics should undergo obstetric emergency drills such as for shoulder dystocia on a regular basis.
- One should refer immediately to a tertiary centre if shoulder dystocia is anticipated. However, if delivery occurs at your centre refer to Standard Operating Procedure for Cord Prolapse and **Figure 9.1** for measures to overcome this complication.

Figure 9.1 : Algorithm for Management of Shoulder Dystocia



Baby to be reviewed by neonate team after birth and referred for specialist review if any concerns

DOCUMENT ALL ACTIONS AND COMPLETE INCIDENT REPORTING FORM

*Adapted from Royal College of Obstetricians and Gynaecologists Green-top Guideline No.42,
2nd edition, 2012*

9.6

MATERNAL COLLAPSE

Maternal collapse is defined as an acute event involving the cardiorespiratory systems and/or brain, resulting in a reduced or absent conscious level (and potentially death), at any stage in pregnancy and up to six weeks after delivery. The common reversible causes of collapse in any woman can be remembered using the well known 4Ts and the 4Hs employed by the Resuscitation Council (UK).

Table 9.2 : Causes of maternal collapse

Reversible cause		Cause in Pregnancy
4H's	Hypovolaemia	Bleeding (may be concealed) (obstetric/ other) or relative hypovolaemia of dense spinal block; septic or neurogenic shock.
	Hypoxia	Pregnant mother can become hypoxic more quickly Cardiac events: peripartum cardiomyopathy, myocardial infarction, aortic dissection, large-vessel aneurysms
	Hypo/ hyperkalaemia and other electrolyte disturbances	No more likely
	Hypothermia	No more likely
4T's	Thromboembolism	Amniotic fluid embolus, pulmonary embolus, air embolus, myocardial infarction.
	Tension pneumothorax	Following trauma/suicide attempt
	Toxicity	Local anaesthetic, magnesium, other
	Tamponade (cardiac)	Following trauma/suicide attempt
Eclampsia and pre-eclampsia		Includes intracranial haemorrhage

9.7

UTERINE INVERSION

- This is an obstetric emergency (Red Alert)
- Sign and symptoms of uterine inversion include:
 - Profuse bleeding
 - Absence of uterine fundus on abdominal examination
 - An obvious defect of the fundus on abdominal examination
 - Visible uterus at the perineum in second and third degree inversion
 - Evidence of shock
- Refer immediately to an O&G specialist in the nearest hospital, the patient should ideally be transferred and managed in specialist hospital
- Any attempt to reduce the inversion should only be done by a trained person and after getting approval from an O&G specialist.
- Replace the inverted uterus immediately (with placenta if still attached) by slowly and steadily pushing upwards.
- The last part to come out should be the first part to go in.
- Do not attempt to remove the placenta as this can lead to severe postpartum haemorrhage.
- With the passage of time, the constriction ring around the inverted uterus becomes more rigid and the uterus more engorged with blood.
- If the attempt fails, refer to the nearest hospital after stabilising the mother.
- In hospital if manual reduction failed, can proceed to the O'Sullivan Method.
- If all the above fails, consider surgical reduction.

Risk factors for uterine inversion include:

- Short umbilical cord
- Excessive traction on the umbilical cord
- Excessive fundal pressure
- Fundal implantation of the placenta
- Retained placenta and abnormal adherence of the placenta
- Chronic endometritis
- Vaginal births after previous caesarean section
- Precipitous or prolonged labour
- Previous uterine inversion

9.8

SEVERE PRE-ECLAMPSIA

Patients who presents with the following signs and symptoms:

- Headache

- Blurring of vision
- Epigastric pain
- Nausea and vomiting
- Hyperreflexia (ankle clonus)

Potential complications from severe pre-eclampsia :

- Eclampsia
- Shortness of breath – may indicate acute pulmonary oedema
- Per vaginal bleeding and abdominal pain to suggest placenta abruption

Management of severe pre-eclampsia:

- At KK level, refer to FMS or medical officer to attend urgently
- To set IV access
- To control blood pressure if there is hypertensive crisis by giving oral Nifedipine 10mg stat
- To consult O&G specialist on-call in nearest specialist hospital
- To give bolus IM Magnesium sulphate injection + 2% lignocaine
- Transfer patient to a specialist hospital once blood pressure has been stabilised but there should not be a delay to transfer the patient if blood pressure is difficult to stabilise
- Obstetric retrieval team should be considered if available
- Refer to Training Manual on Hypertensive Disorders in Pregnancy (2018)

9.9 ECLAMPSIA

- Eclampsia is an obstetric emergency.
- Any seizure occurring in pregnant women NOT due to other causes is considered eclampsia until proven otherwise.

9.9.1 Management of Eclampsia

Management of eclampsia is essentially similar to that of severe pre-eclampsia. In addition, resuscitation and control of seizure should be of priority.

1. Call for help! (INITIATE RED ALERT)
2. Resuscitation
 - a. Ictal phase: left lateral position and oxygen supplementation.
 - b. Post-ictal: assessment of response, protect airway, assess breathing and circulation (may need maternal

cardiorespiratory resuscitation if there is absence of normal breathing).

3. Secure at least 2 intravenous lines.
4. Prevent recurrent seizure with loading dose of IV MgSO_4 4g slow bolus over 10-15 mins or IM MgSO_4 5g + 1ml 2% lignocaine in each buttock followed by maintenance dose.
5. For maintenance dose, IM MgSO_4 5g every 4 hours in alternate buttocks (without lignocaine)
6. If the woman develops another fit after 1 hour of loading dose or maintenance dose, administer another 5g of IM MgSO_4 . *Refer Garis panduan Pemberian Suntikan Intramuscular Magnesium Sulphate di Peringkat Penjagaan Kesihatan Primer bagi Kes Severe Pre-eclampsia/Eclampsia, MOH, 2014.*
7. Start parenteral anti-hypertensive if systolic BP $\geq$ 160 mmHg or diastolic BP $\geq$ 110mmHg.
8. Monitor closely according to protocol while awaiting transfer to specialist hospital.
9. Monitor the fetus accordingly.
10. Do not give too much fluid as the patient is at risk of acute pulmonary oedema, total fluids should not exceed 2L in 24 hours.
11. Refer to Training Manual on Hypertensive Disorders in Pregnancy (2018)

Table 9.3: Intravenous Magnesium Sulphate dosage

Loading dose		
Dose	4g	
Concentration	1 ampoule = 2.47g / 5 ml	
Preparation	Withdraw 8 ml (4g) of MgSO ₄ + 12 ml of normal saline = 20 ml	
Administration	To give 4g MgSO ₄ in slow bolus for 15 – 20 minutes.	
In cases where seizure occurs after administration of MgSO ₄ , a further bolus of 2 g MgSO ₄ can be given with close monitoring of Mg level. Serum magnesium can be taken before the repeated MgSO ₄ bolus if the situation allows.		
Maintenance dose		
	Infusion pump	Syringe pump
Dose	1 g/hour	
Concentration	1 ampoule = 2.47g / 5 ml	
Preparation	Withdraw 10 ampoules (50 ml) of MgSO ₄ + 450 ml of normal saline = 25g/500ml	Withdraw 2 ampoules (5 g/10 ml) of MgSO ₄ + 40 ml of normal saline = 5 g/50 ml (1 g/10 ml)
Infusion	Infusion rate: 21 ml/hour (1 g/ hour) Continue infusion of MgSO ₄ for 24 hours after delivery or last seizure, whichever occurs later.	Infusion rate: 10 ml/hour (1 g/ hour) Continue infusion of MgSO ₄ for 24 hours after delivery or last seizure, whichever occurs later.

Table 9.4: Intramuscular Magnesium Sulphate dosage

Loading dose	
Dose	Total of 10 g (5 g each buttock)
Concentration	1 ampoule = 2.47 g/5 ml
Preparation	Withdraw 2 ampoules (5 g/10 ml) of MgSO ₄ + 1ml of local anaesthesia (2% lignocaine)
Administration	Deep intramuscular injections into each buttock.
In cases where seizure occurs after administration of MgSO ₄ , a further bolus of 5 g intramuscular MgSO ₄ can be given with close monitoring of Mg level. Serum magnesium can be taken before the repeated MgSO ₄ bolus if the situation allows.	
Maintenance Doses	
Dose	5 g every 4 hours
Concentration	1 ampoule = 2.47 g/5 ml
Preparation	Withdraw 2 ampoules (5 g/10 ml) of MgSO ₄ + 1ml of local anaesthesia
Administration	Deep intramuscular injection into alternate buttock every 4 hourly until 24 hours after delivery or last seizure, whichever occurs later.

9.9.2 Maternal and Foetal Monitoring in Eclampsia

Magnesium sulphate can potentially cause cardiorespiratory depression. Therefore, the woman needs to be monitored for signs and symptoms of MgSO₄ toxicity.

Table 9.5: Maternal monitoring for Magnesium Sulphate toxicity

Parameters	Interval of monitoring	Target
Blood pressure	Every 15 minutes	140 – 150/ 90 – 109 mmHg
Pulse rate	Every 15 minutes	> 60 bpm
Respiratory rate	Hourly	> 16 breaths per minute
Urine output	Hourly	< 30 mls/H or < 100 mls/4 hours or < 0.5 ml/kg/H
Deep tendon reflex of knee	Hourly	Presence of reflexes

Obstetric haemorrhage can be divided into bleeding while the fetus is still in-utero which is antepartum haemorrhage (APH), or bleeding after delivery of the fetus which is defined as postpartum haemorrhage (PPH). The causes for APH and PPH are very different. However, the principles of management and good communication remain the same.

Principles of Managing Obstetric Haemorrhage

1. Recognition of obstetric haemorrhage and assessment of severity
2. Communication and call for help
3. Resuscitation
4. Identification and treatment of the cause of haemorrhage
5. Monitoring and documentation

Communication, Resuscitation, Treatment & Monitoring should occur **simultaneously**.

- Timing is critical in the management of obstetric haemorrhage
- Early involvement & intervention by specialist/consultant can be lifesaving

Recognition and Assessment of Haemorrhage

Ability to recognise APH/PPH early and to appreciate the severity is the most crucial aspect of management.

The three commonest shortfalls identified in maternal deaths due to massive PPH are delay to diagnose, failure to appreciate severity and the delay, inadequate and/or inappropriate therapy.

A systemic assessment is essential in obstetric haemorrhage to establish:

- The severity of her condition based on:
 - Visual estimation of blood loss
 - Clinical signs and symptoms of hypovolaemic shock
 - Obstetric Shock Index
 - Cause of bleeding

Table 9.6 : Clinical Signs and Symptoms of Hypovolaemic Shock

Classification of shock	Shock Index	Blood loss (ml) (% of blood volume) **	Pulse Rate (beats per min)	Blood Pressure	Respiratory Rate	Mental Status	Urine output (ml/hour)
Class I (No shock)	<0.6	Up to 750 ml (< 15%)	< 100	Normal	14 – 20	Normal	> 30
Class II (Mild shock)	≥ 0.6 to < 1	750 – 1500 ml (15 – 30%)	> 120	Normal but may have narrow pulse pressure	20 – 30	Anxious	20 – 30
Class III (Moderate shock)	≥ 1 to < 1.4	1500 – 2000 ml (30 – 40%)	> 120	Reduced	30 – 40	Confused	5 – 15
Class IV (Severe shock)	≥ 1.4	> 2000 ml (> 40%)	> 140	Very low, can be unrecordable	> 35	Lethargic	Nil

** based on body weight of 50kg

Source: Gutierrez G, Wulf-Gutierrez M & Reines D, *Clinical review: Hemorrhagic shock; Critical Care* (2004)

Obstetric Shock Index (SI)

Visual estimation of blood loss is usually inaccurate and blood loss is often underestimated. Physiological compensatory mechanisms of pregnancy and the puerperium may also mask decompensation until late in hypovolaemic shock. Abnormality in vital signs is a late sign of hypovolaemic shock. This will lead to late recognition of hypovolaemic shock, and delay in resuscitation.

Obstetric Shock Index (SI) is defined as HR/SBP. It is a useful tool as an early marker of compromise and thus should be practiced to improve the management of all obstetric haemorrhages. For the pregnant population, normal SI ranges from 0.7- 0.9; SI > 1 predicts adverse clinical outcome.

OBSTETRIC SHOCK INDEX = HR/SBP
(NORMAL = 0.7 – 0.9)

Minor Obstetric Haemorrhage

1. Identify early and call for help
2. Set 1 large bore intravenous cannula of at least 18 G or 16 G.
3. Fluid resuscitation with crystalloids
4. Monitor vital signs every 15 minutes until bleeding stops
5. Continue to assess blood loss and Shock Index periodically if there is ongoing loss.
6. Monitor foetal heart if it is antepartum haemorrhage
7. Refer early to O&G specialist on-call from the nearest hospital and transfer patient

Major Obstetric Haemorrhage

1. Activate RED ALERT
2. Refer stat to O&G specialist on-call
3. Place the woman flat and keep her warm.
4. Supplementation of oxygen up to the rate of 15 L/min via facemask.
5. Set 2 large bore intravenous cannulas – 16G or 18G.
6. Fluid resuscitation with crystalloids first and then colloids at the recommended ratio while waiting for transfer to a hospital or for an obstetric retrieval team to arrive.
7. Monitor vital signs every 15 minutes or more often until patient is transferred
8. Continue to assess blood loss and Shock Index periodically until the patient arrives in a specialist hospital
9. Monitor foetal heart if it is antepartum haemorrhage

9.10.1 ANTEPARTUM HAEMORRHAGE

Antepartum haemorrhage (APH) is defined as any pervaginal bleeding occurring after 22 weeks of pregnancy. Prior to 22 weeks, any pervaginal bleeding is categorized under threatened miscarriage.

Two main causes of APH are:

1. Abruptio placenta
2. Placenta praevia

Abruptio placenta

Defined as the premature separation of a normally located placenta from the uterus prior to the delivery of the fetus.

- Bleeding in placenta abruptio is usually accompanied by severe abdominal pain

- Bleeding in placenta abruptio may be REVEALED / CONCEALED / MIXED
- Do not treat patient based on the amount of bleeding seen. Include Obstetric Shock Index in assessment.

Placenta praevia

Placenta praevia is defined as a placenta which is implanted partially or entirely in the lower segment of the uterus. Bleeding is usually painless but can be life-threatening.

Table 9.7: Classification of Antepartum Haemorrhage

Classification	Definition
Spotting	Staining, streaking or blood spotting noted on underwear or sanitary protection
Minor haemorrhage	Blood loss is less than 500ml and has stopped
Major haemorrhage	Blood loss of 500-1000ml, with no signs of clinical shock
Massive haemorrhage	Blood loss is more than 1000ml and/or signs of clinical shock

9.10.2 POSTPARTUM HAEMORRHAGE

Postpartum haemorrhage (PPH) is the second commonest cause of direct maternal deaths in Malaysia.

Table 9.8: Classifications of Postpartum Haemorrhage

Classification	Definitions
Primary	Blood loss of 500 ml or more from the genital tract within 24 hours after vaginal delivery and 1000ml following operative delivery
Secondary	Any abnormal or excessive bleeding from the genital tract after 24 hours to 6 weeks postpartum

Table 9.9: Severity of Postpartum Haemorrhage

Severity	Amount of blood loss (ml)
Minor	500 – 1000
Major	> 1000
Massive	> 1500 *RCOG defined severe PPH as blood loss of ≥ 2000 ml; however, with the logistic issues, we have lowered the threshold to prevent potential delays in resuscitation.

Risk Identification and Prevention

Assessment of risk for PPH should be done for all pregnant women throughout their antenatal care. Risk identification allows a safe delivery plan to be outlined and documented in the woman's antenatal card.

Care should be taken to optimise high-risk women before delivery. A safe delivery plan includes delivery in hospital with specialists and support from blood banks.

Risk Factors Associated with PPH (Antenatal)

- Grandmultipara
- Anaemia in pregnancy
- Previous history of PPH
- Previous history of retained placenta
- Previous history of caesarean section
- Previous history of myomectomy
- Multiple pregnancies
- Macrosomia
- Placenta praevia
- Any bleeding disorder
- Women on anticoagulant
- Presence of uterine fibroid
- Overdistended uterus

Prevention of PPH (Antenatal)

- Optimization of haemoglobin level by 36 weeks
- Identification of high risk women
- Women who are high risk for PPH from remote area may need to stay near a hospital with O&G specialist from 36 weeks onwards (e.g.: transit home, relatives' house)
- Women with placenta praevia should be managed accordingly
- Women with previous caesarean section and placenta praevia should be referred to rule out morbidly adherent placenta

General Principles in Managing of PPH

- Recognition and assessment of severity
- Communication and call for help
- Resuscitation as necessary
- Treatment to arrest the bleeding
- Monitoring and documentation
- Early referral and prompt transfer
- Refer to the Training Manual on Management of Postpartum Haemorrhage (PPH), 2016

Causes of Bleeding

The causes of bleeding – 4 T's

- Tone (70%) – uterine atony
- Trauma (20%) – genital tract trauma including uterine rupture
- Tissue (10%) – retained placenta or product of conception
- Thrombin (<1%) – pre-existing or acquired coagulopathy (DIC from placenta abruption or severe pre-eclampsia)

Management of common causes of PPH

a. UTERINE ATONY

1. Uterine massage
2. Empty urinary bladder with continuous bladder drainage
3. Oxytocics – oxytocin or syntometrine
 - Oxytocin (Pitocin) – IM Pitocin bolus 10 units/IV Pitocin bolus 5 units slow bolus over 1 – 2 minutes.
 - Dose may be repeated after 5 minutes – up to a total dose of 10 units.
 - Start IV infusion of oxytocin infusion 40 units in 1 pint normal saline for 4 hours (125 mls/H).
 - Syntometrine – IM 1 ampule stat (5 units oxytocin and 0.5 mg ergometrine)
 - Contraindicated in hypertension and cardiac disease.

- If the mother is unbooked or unsure of medical or antenatal history, use oxytocin.
- IM Carboprost 0.25 microgram stat. This can be repeated every 15 minutes up to 8 doses
- IV Tranexamic acid 1g stat

b. GENITAL TRACT TRAUMA

Vulval/Vaginal/Cervical Tears:

1. Attempt repair if possible but do not delay transfer.
2. If repair is not feasible or bleeding continues, control the bleeding temporarily with vaginal packing (either using a roller gauze or 2-3 rolled sanitary pads) while awaiting transfer (if it happens in the district hospital) or definitive management.

c. RETAINED PLACENTA

1. Retained placenta is diagnosed after 30 minutes if active management of the third stage is practiced (controlled cord traction). Otherwise, it is diagnosed after 1 hour.
2. MRP should not be attempted in a health clinic unless in certain circumstances and only with the approval of an O&G specialist
3. IV oxytocin infusion 40 units should be started while awaiting transfer.

SECONDARY PPH

Defined as abnormal or excessive bleeding from the genital tract after 24 hours and up to 6 weeks post-delivery.

• Causes of Secondary PPH

- Infection (most common)
- Retained product of conception
- Unrecognised lower genital tract trauma
- Bleeding disorder
- Persistent trophoblastic disease (uncommon)
- Others – chronic sub-involution of uterus (uncommon), uterine AV malformation (rare)

- **Management of Secondary PPH**

General principles of management are essentially similar to management of primary PPH.

1. Recognition and assessment of the severity
2. Resuscitation if required
3. Initiate treatment based on the cause:
 - Intravenous broad-spectrum antibiotic should be initiated
 - Uterotonics to be given if indicated
4. Refer and transfer patient accordingly
5. Refer to the Training Manual on Management of Postpartum Haemorrhage (2016)

9.11

PUERPERAL SEPSIS

World Health Organisation (WHO) defined puerperal sepsis as “an infection of the genital tract is occurring at any time between the rupture of membranes or labour and the 42nd day postpartum”, in which, two or more of the following are present:

- pelvic pain
- fever (oral temperature 38.5°C or higher on any occasion)
- abnormal vaginal discharge, for example, presence of pus
- abnormal smell/foul odour of discharge
- delay in the rate of reduction of the size of the uterus

Diagnosis of sepsis in pregnant women has been made difficult as the signs and symptoms may not be present or less distinctive compared to the non-pregnant population due to physiological changes in pregnancy. This causes a delay in diagnosis of puerperal sepsis, often until it is too late for intervention. Disease progression may be rapid, hence early recognition leading to diagnosis of puerperal sepsis and commencement of appropriate treatment play crucial roles in improving the outcome.

Common pathogens causing sepsis in the puerperium are:

- Group A streptococcus (e.g.; *Streptococcus pyogenes*), Group B streptococcus (*Streptococcus agalactiae*)
- *Escherichia coli*
- *Staphylococcus aureus*
- Methicillin resistant *Staphylococcus aureus*, *clostridium septicum* and *morganella morganii*

Puerperal sepsis carries a significant preventable maternal morbidity and mortality risk. The diagnosis of sepsis is clinical. Early diagnosis and

treatment of sepsis may improve outcomes of those unfortunate maternal deaths secondary to sepsis.

The National CEMD report commonly identify the following shortfalls

- Delay and suboptimal treatment
- Delay in referring to specialist hospitals
- Failure to appreciate severity

Management plan:

- Early identification during clinic or home visits
- Patients need to be assessed by medical officer
- To initiate IV or oral antibiotics
 - Broad spectrum antibiotics (cephalosporins, clavulanic acid, sulbactam)
 - Metronidazole
- Patients suspected of puerperal sepsis should be referred to an O&G specialist and should be transferred

9.12

POSTPARTUM PSYCHOSIS

- Postpartum psychosis is a psychiatric emergency that affects about one in every 1000 new mothers.
- Postpartum psychosis presents as rapid onset of:
 - frank psychosis such as hallucinations or delusions
 - cognitive impairment and confusion
 - mood swings
 - disorganised behaviours
- Postpartum psychosis carries the risk of maternal harm and infanticide and must be intervened urgently. It has been estimated that untreated postnatal psychosis carries a four percent risk of infanticide and a five percent risk of suicide
- Factors that increase risk for postpartum psychosis includes:
 - primiparous mothers who are single
 - women who are older
 - women with a past psychiatric history and family history of schizophrenia or bipolar disorder
 - prenatal depression
 - past history of postpartum psychosis (recurrence 25 % to 75%)
- Acute management for a patient who present with postpartum psychosis
 - Ensure safety of mother and infant
 - Locate the patient and infant if the family came by proxy. If mother came alone, locate the infant

- Urgent referral to tertiary centre for review by psychiatrist
- Admission will be decided by the psychiatry team
- There is no clear prevention for postnatal psychosis but identification of risk factors and lowering of aggravating factors may help. Encourage social support, sleep hygiene and mobilise help for new mothers.
- The overall prognosis is positive, especially when symptoms emerge less than 1 month after delivery. Among patients who develop postpartum psychosis immediately after childbirth, 72%–88% have bipolar illness or schizoaffective disorder and 12% have schizophrenia.

9.13

PERINATAL SUICIDE AND PARASUICIDE

- Perinatal suicides are suicides that occur during pregnancy and up to six weeks postpartum, but often expanded to the entire first postpartum year (as maternal psychiatric illness most often persists beyond six weeks postpartum).
- Suicide accounted for about 5–20% of maternal deaths during pregnancy and the first postnatal year in high-income countries and 1–5% in low-income and middle-income countries.
- Parasuicide (self-harm) is self-poisoning or self-injury, irrespective of the apparent purpose of the act. Self-harm ideation is more common than suicide attempts or deaths, with the prevalence of 5 to 14%. These are often associated with Borderline Personality Disorder or trauma syndromes.
- Risk factors for perinatal suicide:
 - Younger age
 - Being unmarried
 - Previous history of psychiatric disorders (e.g. mood disorders, substance use disorder)
 - Previous history of suicidal attempt or suicidal ideation
 - Psychiatric comorbidity
 - Shorter illness duration
 - Family history of psychiatric disorders
 - Family history of suicidal attempt or suicidal ideation
 - Family conflict
 - Exposure to domestic violence
 - Loneliness and lack of support
 - Partner who rejected paternity
 - Unintended pregnancy
- Protective factors
 - Support and connectedness
 - Engagement to health services

- Life skills (eg coping strategies)
- Sense of purpose or meaning of life
- Faith, cultural and religious beliefs that discourage suicide

9.13.1 Assessment

- Assess suicidal thought
 - *Do you have any suicidal thoughts?'*
 - *How frequent and persistent are they?*
- Assess suicidal plan
 - *Do you have a plan?*
- Assess mental state, e.g.
 - *Are you depressed?*
 - *Are you feeling hopeless?*
 - Commanding hallucinations
- Assess risk factors
- Assess protective factors
 - *What is stopping you?*
 - *Do you have people to support you?*

9.13.2 Triaging of patients with suicide risks

- Low Risk of Suicide
 - Some mild or passive suicide ideation, with no intent or plan
 - No history of suicide attempt
 - Available social support
- Moderate Risks
 - Suicidal ideation with some level of suicide intent, but who have taken no action on the plan
 - No other acute risk factors
 - History of psychiatric illness & receiving treatment
- High risks
 - Made a serious or nearly lethal suicide attempt
 - Persistent suicide ideation or intermittent ideation with intent and/or planning
 - Psychosis, including command hallucinations
 - Other signs of acute risk
 - Recent onset of major psychiatric syndromes, especially depression
 - Been recently discharged from a psychiatric inpatient unit
 - History of acts/threats of aggression or impulsivity

9.13.3

Management

- Risk of suicide can be reduced by recognising past history and taking proactive management
- Screen during antenatal care for a personal or family history of mental illness. If there is history of mental illness:
 - Document clearly in the antenatal book and clinical case notes
 - Counsel the patient regarding risks and plan of care
- Activate a plan for multidisciplinary monitoring and support:
 - maternal and child health care service
 - high-risk pregnancy care
 - psychiatry and mental health service
 - social worker
- At each visit, assess for suicide risk especially if there are mood changes
- Manage suicide risks by mitigating risk factors and strengthening protective factors.
- Prepare plan of care during antenatal, intrapartum, discharge and postnatal.
- Admit when there is high suicidal risk.

9.14

POSTPARTUM MOTHERS WITH INFANTICIDE RISK

- Thoughts of harming one's child is not uncommon but are considered so taboo that most mothers will not voice it out.
- There are many factors that lead to infanticide such as unintended pregnancies, substance use, domestic violence, poverty, marital conflict and mental illness.
- Untreated postpartum psychosis carries a four percent risk of infanticide, usually resulting from a psychotic belief about the infant or negative maternal reaction to separation.
- Infanticide may also occur in severe postpartum depression causing neglect and abuse
- Empathetic questioning of at risk patients regarding such thoughts is mandatory. If risk to the safety of the infant is present, actions must be taken. Alternative caregivers such as family members must be identified or admission of the infant while the mother undergoes treatment.
- Referral to social worker and child protector is mandatory if the infant is in immediate risk

STANDARD OPERATING PROCEDURES

SOP	Condition	Symptoms/ signs	Laboratory Investigation & findings	Diagnostic criteria and differential diagnosis	Care of plan		
					Management	Level of personnel	Level of care
1	Maternal Pyrexia	Temperature $\geq$ 38°C	• TWBC • Septic work up, including Blood C&S if maternal temperature $\geq$ 38°C • UFEME • Urine C&S • HVS C&S • CXR (if indicated)	• Chorio-amnionitis • Intercurrent infection; UTI, URTI • DVT	• Broad spectrum antibiotic • Expedite delivery if chorioamnionitis • Assessment of baby at delivery	MO/Specialist (Medical/ Surgical/ Padiatric)	Hospital with Specialist

SOP	Condition	Symptoms/ signs	Laboratory Investigation & findings	Diagnostic criteria and differential diagnosis	Care of plan		
					Management	Level of personnel	Level of care
2	Cord Prolapse	• Presence of cord outside the cervix • Membranes absent	• FBC • GXM		Initial management: • Elevate mother's buttocks • Oxygen to mother • Replace cord into the vagina with warm gauze/pad • Inflate bladder with normal saline • Expedite delivery as appropriate	MO/Specialist (O&G / Anaesthesia / Paediatric)	Hospital with Specialist
3	Shoulder Dystocia	Delay in delivery of shoulder	• FBC • GXM		• Call for the most senior staff available at the centre. • IV line • The mother must be in lithotomy position, legs up in stirrups with buttock at the edge of the bed. • Empty the bladder • Extend episiotomy • McRobert manoeuvre: Hyperflex hips and knees and abducts hips. • Suprapubic	All levels	All levels

SOP	Condition	Symptoms/ signs	Laboratory Investigation & findings	Diagnostic criteria and differential diagnosis	Care of plan		
					Management	Level of personnel	Level of care
					pressure to dislodge anterior shoulder. • Downward axial traction on fetus. • Failing the above, deliver the posterior shoulder followed by the anterior shoulder. • Failing the above, activate referral/ retrieval system		
4	Postpartum Haemorrhage	Bleeding from the genital tract >500mls in vaginal delivery and > 1000mls in Caesarean section Or enough blood loss to cause hypotension or shock	• FBC • GXM • Coagulation profile	• Uterine atony • Retained placenta • Trauma: Cervical tear, vaginal wall tear/ haemato ma • Uterine inversion • Coagulati on	• TRIGGER RED ALERT • IV line with 16-18G cannula • Resuscitation • Oxytocics/ Prostaglandins Refer Training Manual on Management of PPH	All levels	All levels

SOP	Condition	Symptoms/ signs	Laboratory Investigation & findings	Diagnostic criteria and differential diagnosis	Care of plan		
					Management	Level of personnel	Level of care
				disorder			
5	Antepartum Haemorrhage	PV bleed during antepartum period	• FBC • GXM • Coagulation profile • CTG • Ultrasound	• Bleeding Placenta Praevia • Abruptio placenta • Uterine rupture	• IV line • Resuscitation • Refer hospital with specialist • Expedite delivery	MO/Specialist (O&G / Anesthesia / Peadiatric)	Hospital with Specialist

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2. Susan Hatters Friedman, Renée Sorrentino. Commentary: Postpartum Psychosis, Infanticide, and Insanity—Implications for Forensic Psychiatry *Journal of the American Academy of Psychiatry and the Law* Online Sep 2012, 40 (3) 326-332
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4. Khalifeh, H., Hunt, I. M., Appleby, L., & Howard, L. M. (2016). Suicide in perinatal and non-perinatal women in contact with psychiatric services: 15 year findings from a UK national inquiry. *The lancet. Psychiatry*, 3(3), 233–242.

SECTION B: NEONATAL CARE

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INTRODUCTION TO NEONATAL CARE SECTION

This neonatal section outlines the care plans and work processes for a baby at birth, the immediate period after birth and thereafter at home. Routine care for most babies who are healthy is as laid out in the flow charts and complications necessitating other interventions and management will be discussed in the relevant chapters. Existing Ministry of Health documents e.g. Integrated Plan for the Detection and Management of Neonatal Jaundice (2017), National Screening Programme for Congenital Hypothyroidism (2018), Paediatric Protocols for Malaysian Hospital 4th edition (2018), Garispanduan Sistem Kawalan Keselamatan Bayi (2007) are intended to be used in conjunction with this manual and will be referenced in the relevant sections. Common neonatal health problems such as skin rashes and feeding problems will be addressed but specific management of serious neonatal medical conditions are not included in the manual except for highlighting the recognition of signs of the seriously ill child and how he/ she should be referred and or transported. We encourage reference to other resources where information is lacking in the manual.

CHAPTER 10

OVERVIEW OF NEWBORN CARE

10.1 AT BIRTH

- a. Identify high-risk factors and request for Paediatric Unit doctor's standby if necessary (**Chapter 11**)
- b. Resuscitate if necessary (**Chapter 11**)
- c. Facilitate skin-to-skin contact immediately and initiate breastfeeding within first hour of birth (**Chapter 11**)
- d. Check vital signs (**Chapter 12**) for range of normal neonatal vital signs and ensure thermal protection (**Chapter 11**)
- e. Perform physical examination to exclude congenital anomalies such as abnormal facies, cleft palate and lips, abdominal wall defects, neural tube defects, imperforate anus etc (**Chapter 12**)
- f. Ensure cord blood is sent for G6PD and TSH screening (**Chapter 12**)
- g. Administer Vitamin K and Hepatitis B vaccination (**Chapter 12**)
- h. Transfer the baby to relevant level of care as necessary (**Chapter 11**)

10.2 AFTER BIRTH

- a. Perform physical examination, and chart findings on newborn physical examination findings in Child Health Record Book (**Appendix 10-1**)
- b. Ensure thermal protection (**Chapter 11**)
- c. If baby is nursed in postnatal or neonatal wards, regular observations and vital signs must be taken
- d. Check for hypoglycaemia in high risk cases (**Chapter 12**)
- e. Check for jaundice and monitor severity (**Chapter 12**)
- f. Encourage breastfeeding and bonding (**Chapter 13**)
- g. Administer BCG vaccination (**Chapter 12**)
- h. Document results of G6PD, TSH screening, universal hearing screening (if available), and pulse oximetry screening (if available) in the child health card. If abnormal, appropriate action is to be taken
- i. Promote education on bathing, cleanliness, skin and cord care, postnatal follow-up, immunisation, and warning signs (**Chapter 10**)

Purpose

To ensure newborn babies are safely discharged, they should meet basic criteria and have appropriate arrangements for continuous care.

The baby should be healthy in the clinical judgement of the health care provider and the mother should have demonstrated a reasonable ability to care for the baby.

Criteria for Discharge of Term Baby**i. Assessment of baby**

- Feeding well - at least two successful feedings
- Size appropriate for gestational age
- If small-for-gestational age
 - no (further) hypoglycaemia
 - and has been discharged by the paediatric doctor
- If large-for-gestational age
 - no (further) hypoglycaemia
 - and has been discharged by the paediatric doctor
- Passed checklist (**Chapter 13**) for breastfeeding technique, and adequacy (observation and health education)
- Normal body temperature (axillary temperature of 36.5°C to 37.5°C)
- Pink and has no breathing difficulties
- No evidence of sepsis. If there is risk of sepsis -observe for at least 24-48 hours
- Minimal neonatal jaundice (except for Day 1 jaundice)
- Passed urine
- Passed meconium
- Cord is dry and clean
- No significant eye discharge
- Physical examination done by doctor and baby discharged from additional observation and treatment

ii. Immunisation and others

- Received BCG, Hepatitis B & Vitamin K
- G6PD and TSH, results documented in Home Based Child Health Record.
- Document results of G6PD, TSH screening, universal newborn hearing screening (if available), and pulse oximetry screening (if

available) in the child health card. If abnormal, appropriate action is to be taken.

- If results are not available yet, arrangements **MUST** be made to trace results and document in the Home Based Child Health Record.
- Health education on bathing, cleanliness, skin and cord care, postnatal follow-up, immunisation, and warning signs.
- Mother is able to provide routine baby care and recognise signs of illness and other problems related to newborn.

iii. Follow-up appointments

- Home Based Child Health Record to be filled and given to mother
- Notify the health clinic once the baby is discharged so that arrangements for follow-up by community health nurse can be done (preferably within 24 hours of discharge).
- Outpatient appointment to see doctor if necessary.
- Follow-up appointment for baby in Paediatric clinic, if required.

10.4

DISCHARGE OF NEWBORN WITH SPECIAL NEEDS

Five critical components must be looked into when providing post-discharge care:

i. Parent education

- General care plans and health education on bathing, cleanliness, skin and cord care, postnatal follow-up, immunisation, and warning signs.
- Specific care plans as below.

ii. Primary care implementation

- Timely immunisation
- Regular hearing and vision tests
- Nutritional support and monitoring
- Growth and development assessment

iii. Evaluate current medical problems

- Physical handicap
- Psychomotor development

iv. Develop home care plan

- Identify parent/caregiver
- Identify equipments needed
- Identify community support
- Have emergency plan

v. Follow up care – coordination between the specialists

Specific Care Plans:

i. Preterm

- care with thermal protection
- more patience with feeding – ensure appropriate weight gain
- avoid overcrowded places and URTI contact – to reduce risk of respiratory tract infection
- follow immunisation schedule according to chronological age
- to nurse in a supine position to reduce risk of SIDS (Sudden Infant Death Syndrome).
- avoid cigarette smoke
- educate on hand hygiene
- close supervision at all times
- follow up for weight at health clinic
- follow up for myopia, hearing and neurodevelopment
- Basic Life Support (BLS) training for parents/caregivers.

ii. At risk families due to family issues

e.g. single or young parents, marital problems, lack of social support, poverty, domestic violence or substance abuse

- review family's coping skills – advise on handling crying baby
- if any financial assistance required – refer Jabatan Kebajikan Masyarakat (JKM)
- mobilise extended family support and supervision especially in drug addiction
- more frequent home/health clinic visits
- alternative care placement of baby may be required – refer JKM

iii. Baby with special health care needs

- Cleft palate – feeding technique, growth, care with aspiration, hearing tests

- Nasogastric feeding – regular change of nasogastric tube, mother to know
- Oxygen dependence – avoid cigarette smoke, avoid URTI contact
- Tracheostomy – parents to learn suctioning and tracheal care
- Colostomy care
- Basic Life Support (BLS) training for parents/caregivers.

iv. Babies with multiple problems

- Multiple congenital abnormalities – multi-disciplinary assessment and follow-up, early intervention programme
- Motor/ sensory disability - multi-disciplinary assessment and follow-up, early intervention programme

For further management of children with special needs, health care providers are required to refer to the following documents developed by the Division of Family Health Development, MOH :

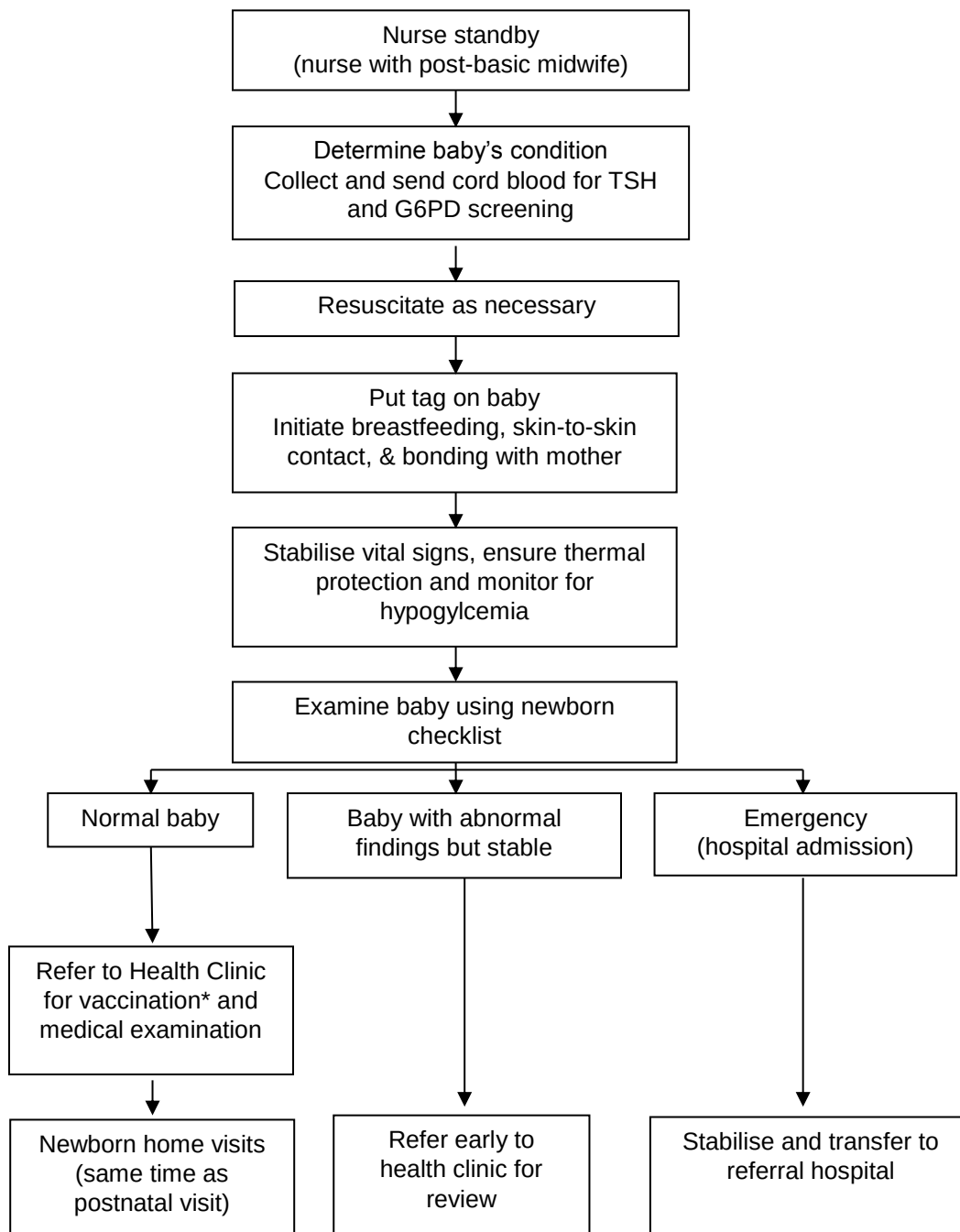
- A series of six manuals on Management of Children with Disabilities
- Care of Children with Special Needs: Manual on Management of Children with:
 - Fine Motor Delay
 - Visual Impairment
 - Communication problems
 - Personal & Social problems

10.5 HOME VISITS

Postnatal home visits are done for mothers and babies according to schedule and whenever necessary (**Chapter 4**)

- Re-examine baby and chart findings on in Child Health Record Book (both mother's and clinic's copy)
- Ensure thermal protection.
- Weigh baby and check for normal weight gain pattern (**Chapter 13**)
- Assess adequacy of feeding and technique (**Chapter 13**)
- Check for jaundice and monitor severity
- Promote education on bathing, cleanliness, skin and cord care, postnatal follow-up, immunisation and warning signs.
- Educate on the role of traditional practice after delivery (mother and newborn) (**Appendix 10-5**)

10.6 WORK PROCESS FOR HOME/LOW-RISK BIRTH CENTRE (LRBC) DELIVERIES



Hepatitis B and Vitamin K injections:

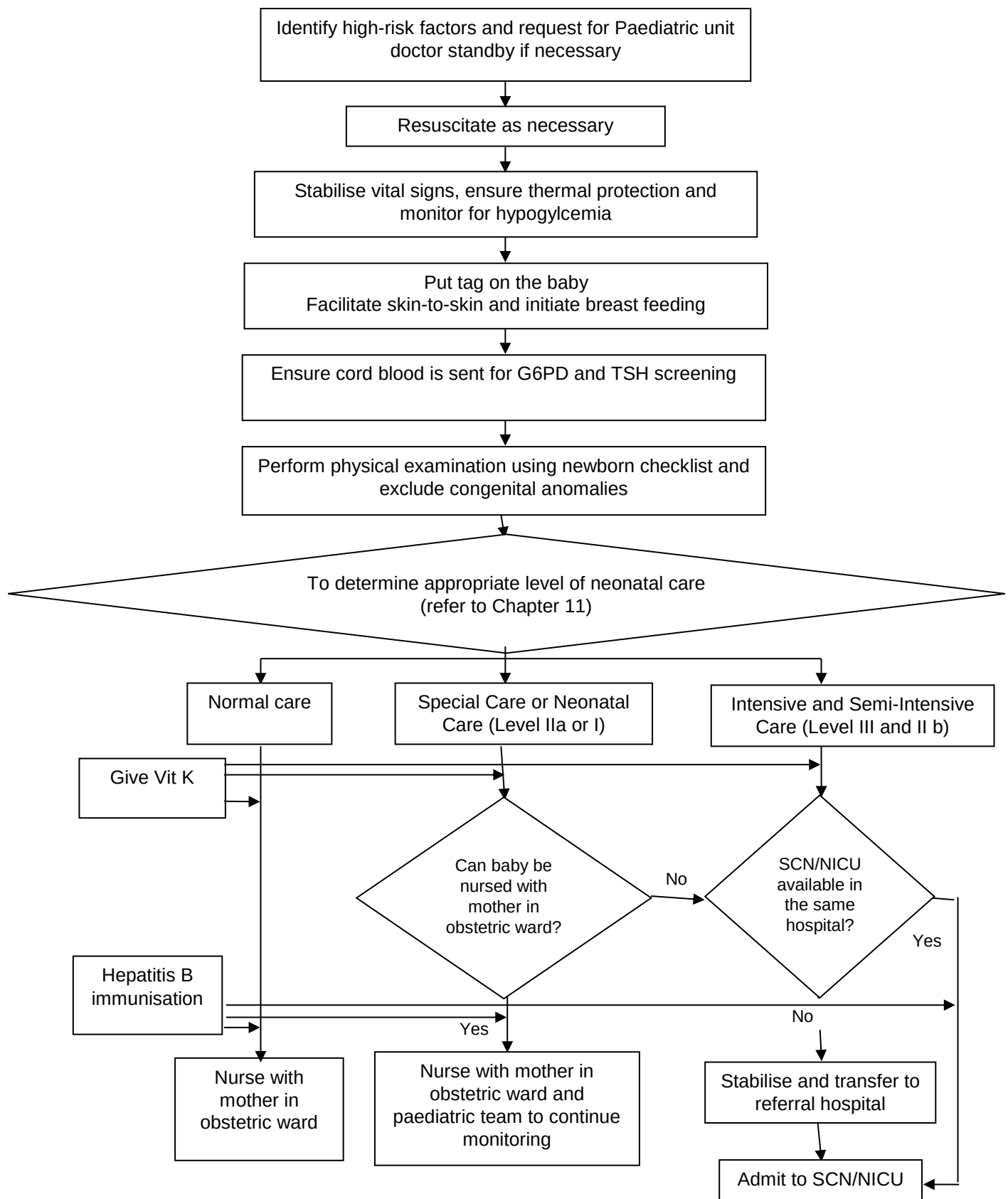
Home delivery- Refer to Health Clinic for Hepatitis B 1st Dose

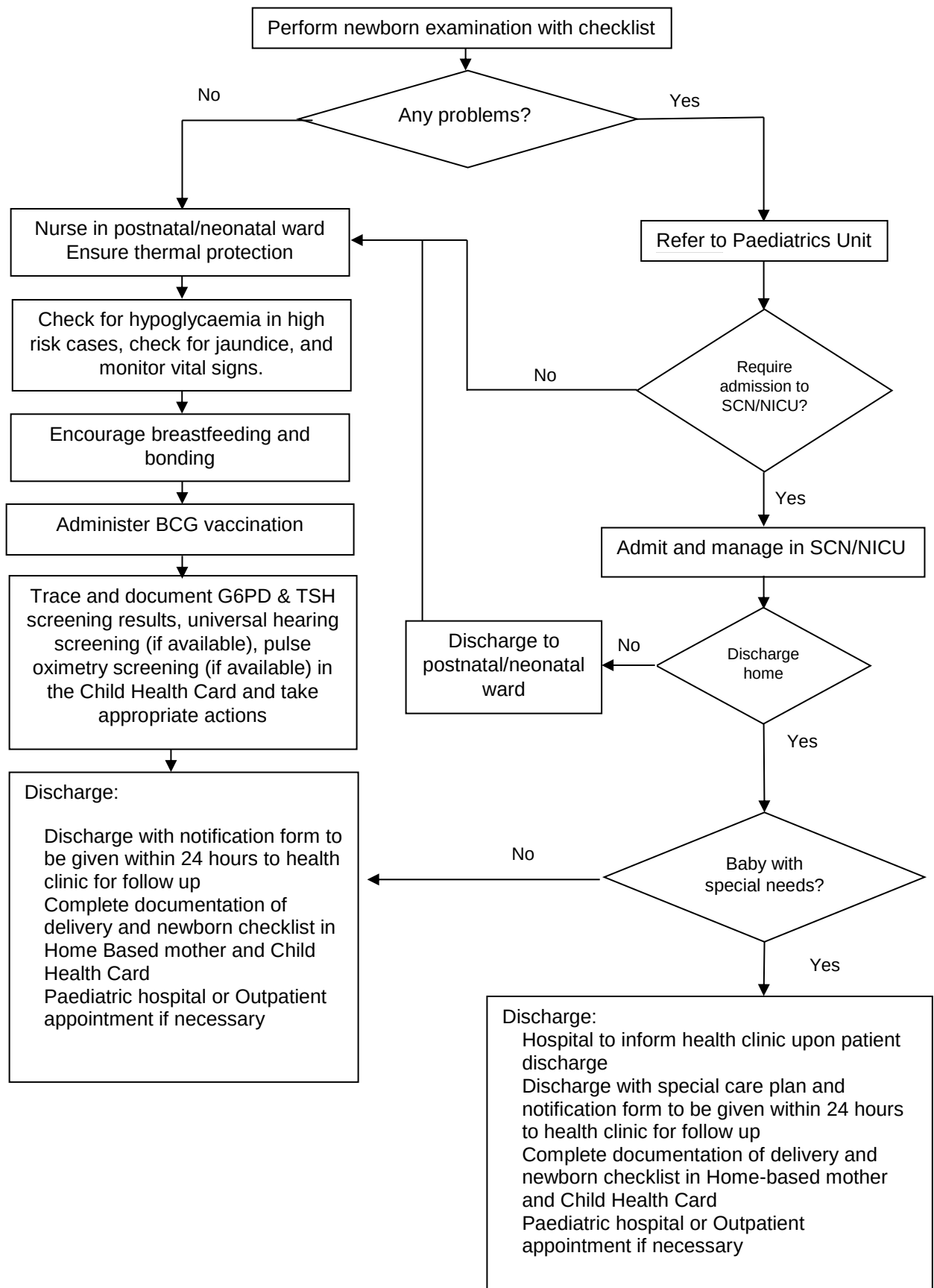
LRBC Delivery – Give Hepatitis B 1st Dose and IM Vitamin K after birth

BCG:

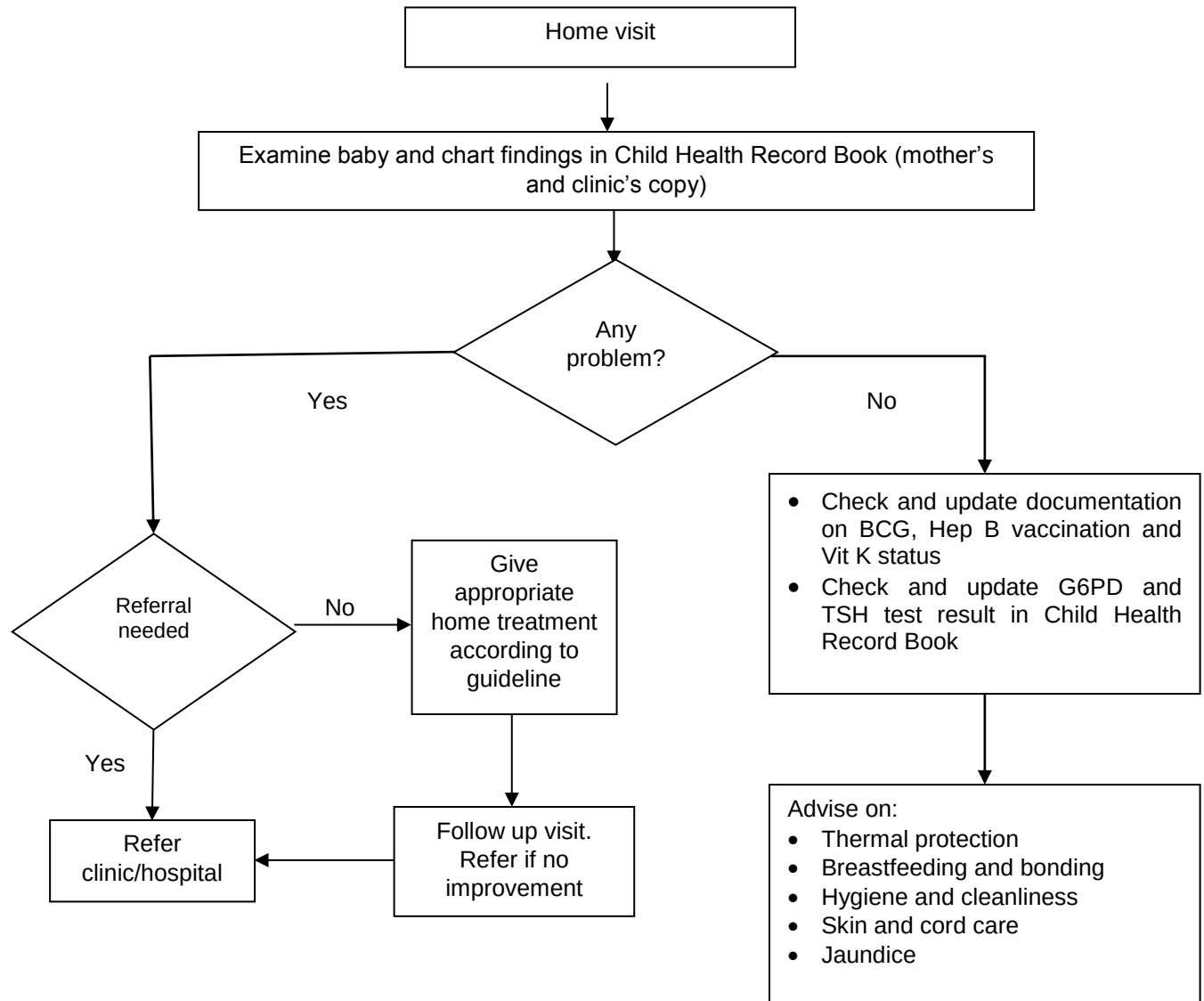
Home delivery- Refer to Health Clinic/hospital for BCG

LRBC Delivery – Give BCG in Health Clinic/hospital





WORK PROCESS DURING HOME VISIT



NEWBORN PHYSICAL EXAMINATION AS IN *BUKU REKOD KESIHATAN BAYI DAN KANAK-KANAK (0- 6 TAHUN)*

SARINGAN BAYI BARU LAHIR

PEMERIKSAAN FIZIKAL: (Diisi oleh Pegawai Perubatan)

Vital sign:

Suhu Badan:..... C Kadar respiratori:..... min

Kadar denyutan jantung: min

Buang air besar dalam masa 24 jam : ☐ Ya ☐ Tidak

Buang air kecil dalam masa 24 jam : ☐ Ya ☐ Tidak

Memulakan penyusuan susu ibu : ☐ Ya ☐ Tidak

Pemeriksaan Am : Tanda (☒) di ruangan berkenaan

Pemeriksaan	Normal	Abnormal	Catatan
Pemeriksaan Am			
Wajah bayi			
Keadaan Kulit			
Kepala Kulit			
Telinga Hidung			
Mata (termasuk <i>red light reflex</i>)			
Mulut			
Gusi			
Lelangit			
Dada			
Jantung			
Abdomen			
Spine			
Anus			
Genitalia			
Nadi Femoral			
Pinggul (Hips)			
Tangan			
Kaki			
<i>Reflexes</i>			

Maklumat Tambahan (jika ada):

GARIS PANDUAN PEMERIKSAAN BAYI BARU LAHIR MENGIKUT BUKU REKOD KESIHATAN BAYI DAN KANAK-KANAK (0 – 6 TAHUN)

Kelahiran di hospital:

- Pemeriksaan dijalankan oleh doktor di wad postnatal sebelum discaj.
- Tanda (✓) pada ruang berkaitan.
- Tulis di ruangan catatan jika berkenaan

Kelahiran di rumah:

- Pemeriksaan dijalankan oleh doktor di klinik kesihatan berhampiran semasa lawatan pertama ke klinik.
- Tanda (✓) pada ruang berkaitan.
- Tulis di ruangan catatan jika berkenaan.

Pemeriksaan	Penemuan
Vital signs	• Ambil suhu badan, kadar respiratori dan kadar denyutan jantung dan rekodkan. • Kadar pernafasan normal: 40-60/min • Kadar denyutan jantung normal: 120-160/min • Rujuk kepada pegawai perubatan sekiranya kadar pernafasan/denyutan jantung luar daripada julat normal
Buang air besar dalam masa 24 jam	• Tanya ibu atau penjaga sama ada bayi ada buang air besar atau tidak dalam masa 24 jam yang lalu. • Tanda (✓) di petak berkenaan.
Buang air kecil dalam masa 24 jam	• Tanya ibu atau penjaga sama ada bayi ada buang air kecil atau tidak dalam masa 24 jam yang lalu. • Tanda (✓) di petak berkenaan.
Penyusuan susu ibu di mulakan	• Tanda (✓) di petak berkenaan.

1. Pemerhatian Am

Perhatikan secara am keadaan dan kecergasan bayi. Bayi yang tidak aktif/kurang cergas, tangisan lemah atau '*irritable*' perlu dirujuk

2. Wajah Bayi

Rupa bayi yang luar biasa sama ada *asymmetry* atau mempunyai ciri-ciri *dysmorphic* seperti *Down Syndrome*.

3. Keadaan Kulit

Warna Kulit: perhatikan warna kulit bayi sama ada pucat, *cyanosis* atau jaundis

Keadaan Kulit: periksa sama ada terdapat sebarang masalah seperti ruam, septic spot yang meluas (*extensive*), *petechiae* dan lain-lain. Periksa status hidrasi.

4. Kepala/Leher

Periksa kepala untuk:

- Bonjol atau lekuk pada ubun-ubun (*bulging or depressed fontanelle*).
- *Caput succedaneum* – ialah benjolan yang bukan hematoma. Akan susut selepas beberapa hari.
- *Cephalhaematoma* – keadaan di mana hematoma tidak merentasi garis *sutures* di kepala.
- *Subaponeurotic haemorrhage* – hematoma yang merentasi garis *sutures* yang boleh menyebabkan renjatan hipovolemik. Ini perlu dirujuk segera

Periksa leher untuk:

- Sternomastoid tumor yang boleh menyebabkan teleng (*torticollis*)
- Pembengkakan seperti *cystic hygroma*

5. Telinga/ Hidung

Perhatikan posisi dan rupa bentuk telinga seperti *low-set ear* menunjukkan ciri *Down Syndrome*.

Perhatikan untuk *nasal flaring* jika ada.

6. Mata

Jika terdapat keadaan seperti merah, bertahi, bernanah, *congenital cataract*, pendarahan bahagian sclera (*subconjunctival haemorrhage*), *congenital ptosis*, atau juling.

Gunakan *ophthalmoscope* untuk memeriksa *red light reflex*.

7. Mulut/ Gusi/ Lelangit

Periksa untuk sumbing bibir (*cleft lip*), sumbing lelangit (*cleft palate*), *tongue tie*, *macroglossia* atau terdapat *natal teeth* (berisiko mengalami aspirasi jika gigi longgar). Periksa juga untuk *oral thrush* (tompokan putih pada lidah atau gusi)

8. Dada

Periksa untuk bentuk dada yang tidak normal, kadar dan cara pernafasan. Kadar pernafasan yang normal adalah 40 – 60 / minit dan tiada *grunting* atau *stridor*

9. Jantung

Periksa kadar dan bunyi denyutan jantung. Kadar denyutan normal adalah 120 – 160/ minit.

10. Abdomen

Jika terdapat keadaan abdomen yang kembung berserta dengan muntah, cirit atau tidak membuang air besar, perlu dirujuk segera. Periksa juga keadaan tali pusat

11. Spine

Perhatikan untuk skoliosis, *Spina Bifida* atau tanda kulit seperti lipoma, *haemangioma* atau *tuft of hair*.

12. Anus

Periksa untuk patensi dan kehadiran fistula.

13. Genitalia

Lelaki:

- Periksa kedudukan pembukaan urethra. Keadaan seperti *hypospadias*, *epispadias* adalah luar biasa.
- Keadaan *undescended testes* perlu dirujuk.
- Pembesaran pada kerandut zakar mungkin disebabkan *hydrocele*, *inguinal hernia*, *tumour*

Perempuan:

- Perhati untuk labia minora dan labia majora, clitoris, urethral dan *vaginal orifice*.
- Jika terdapat discaj dari vagina berwarna putih atau sedikit perdarahan dalam minggu pertama adalah normal.

Sekiranya jantina tidak dapat dikenalpasti, perlu rujukan segera untuk memastikan bukan disebabkan oleh gejala merbahaya seperti *Congenital Adrenal Hyperplasia*

14. Nadi Femoral

Periksa nadi femoral, jika tiada atau lemah perlu dirujuk (kemungkinan *coartation of aorta*)

15. Pinggul (Hips)

Perhati pergerakan di kedua belah sendi pinggul adakah seimbang (*symmetrical*). Rujuk kepada pakar ortopedik sekiranya Ortolani atau Barlow test adalah positif (*clunk sensation*)

16. Kaki

Periksa untuk *congenital talipes equinovarus* (CTEV), panjang kedua belah kaki dan tapak kaki adalah sama

17. Tangan

Boleh menggerakkan tangan dengan bebas. Periksa untuk *webbed fingers*, *polydactyly*, *syndactyly* dan warna kuku jari.

18. Reflexes

Grasp dan *moro reflex*.

Unilateral moro reflex menunjukkan kecederaan *brachial plexus*, *Erb's palsy* atau *fracture clavicle/humerus*.

Sucking reflex tidak perlu dilakukan jika bayi telah menyusu dengan baik.

Catatan:

1. Sila rujuk bayi kepada Pegawai Perubatan/Pakar Perubatan Keluarga/Pakar Pediatrik di klinik atau hospital berdekatan dengan segera jika perlu.
2. Sila rujuk *Perinatal Care Manual: Neonatal Care, Chapter 12, Ministry of Health, 4th Edition, 2020*.

Rawatan Postnatal (Home Visit)

AKTIVITI	PENEMUAN		
Tarikh			
Umur semasa lawatan			
Berat badan (kg), Panjang (cm), Lilitan kepada (cm)			
Suhu (Suhu normal ketiak 36.5 - 37.0°C)			
Tindakbalas (Aktif/Tidak Aktif)			
Keadaan kulit (warna, hidrasi)			
Penilaian penyusuan			
Buang air kecil			
Buang air besar			
Sistem Kardiovaskular (Kadar denyutan jantung normal: 120- 160/min)			
Sistem Pernafasan (kadar pernafasan normal: 40-60/min)			
Pemeriksaan fizikal lain: kepala, leher, mata, mulut, telinga, abdomen, sistem tulang			
Alat kelamin			
Sistem neurologi (<i>reflex</i> dan <i>muscle tone</i>)			
Bimbingan awal ibu bapa			
Catatan			
Tandatangan & Nama			

GARIS PANDUAN PEMERIKSAAN RAWATAN POSTNATAL MENGIKUT BUKU REKOD KESIHATAN BAYI DAN KANAK-KANAK (0 – 6 TAHUN)

RAWATAN POSTNATAL (Hari ke 2, 3, 5, 8-10 dan 14-16)

Aktiviti	Penemuan
Tarikh	Tulis tarikh dan masa lawatan ke rumah dibuat.
Umur semasa lawatan	Umur bayi semasa lawatan.
Parameter	Ukur berat badan (kg), panjang (cm) dan lilitan kepala.
Suhu Badan	Ambil suhu badan bayi dan catat bacaan (°C)
Tindakbalas	- Lihat secara am keadaan bayi sama ada aktif atau tidak. - Tanya ibu keadaan bayi.
Warna dan keadaan kulit bayi	- Perhatikan warna bayi sama ada normal, biru (<i>cyanose</i>) atau pucat (<i>pale</i>) dan catat di ruang penemuan. - Ambil tindakan yang sewajarnya dan rujuk jika perlu. - Periksa sama ada kulit baik atau mempunyai masalah seperti ruam, <i>septic spot</i> dan lain-lain. - Periksa untuk jaundis. Sekiranya jaundis ambil tindakan seterusnya mengikut garis panduan pengendalian jaundis dan di isi di dalam borang yang telah disediakan (MMN/NNJ2016)
Penilaian penyusuan	Tanya ibu sama ada mempunyai masalah penyusuan seperti: - Bengkak buah dada. - Puting buah dada pecah (<i>cracked nipples</i>). - Bayi tak mahu menyusu. - Tiada susu ibu /tidak cukup susu ibu dan lain-lain. - Beri nasihat mengenai penyusuan susu ibu. - Pantau cara ibu menyusukan anak
Buang air besar	Tanya ibu sama ada bayi buang air besar dengan baik (kekerapan dan warna). Sekurang-kurangnya 2 kali sehari.
Buang air kecil	Tanya ibu sama ada bayi buang air kecil dengan baik (kekerapan dan warna). Sekurang-kurangnya 5-6 kali sehari.
Sistem Kardiovaskular (Kadar denyutan jantung normal: 120-160/min)	- Kira pernafasan dalam satu minit, - Ulangi perkiraan dan rujuk ke klinik atau hospital berdekatan jika kadar di luar julat normal

Aktiviti	Penemuan
Sistem Pernafasan (kadar pernafasan normal: 40-60/min)	- Kira pernafasan dalam satu minit - Ulangi perkiraan dan rujuk ke klinik atau hospital berdekatan jika kadar denyutan jantung di luar julat normal
Pemeriksaan fizikal lain: kepala, leher, mata, mulut, telinga, abdomen, sistem tulang	Rujuk kepada pemeriksaan fizikal bayi baru lahir
Alat kelamin	- Periksa rupa bentuk genitalia dan catit penemuan - Periksa keadaan testis, '<i>undescended testes</i>' perlu dirujuk. - Rujuk kepada pemeriksaan fizikal bayi baru lahir
Sistem neurologi (<i>reflex</i> dan <i>muscle tone</i>)	Periksa <i>grasp</i> dan <i>moro reflex</i> bayi (rujuk pemeriksaan bayi baru lahir)
Bimbingan awal ibu bapa	- Baca dan terangkan kepada ibubapa/penjaga mengenai Panduan Ibubapa - Tanya ibu jika ada masalah dan catat di ruang yang disediakan. - Beri nasihat yang bersesuaian dan catitkan
Catatan	Catitkan sebarang maklumat tambahan sekiranya ada
Tandatangan & Nama	- Tandatangan dan tulis nama (cop) pegawai yang melakukan pemeriksaan kesihatan bayi setiap kali lawatan ke rumah dijalankan. - Pada lawatan antara hari ke 10, 15 dan ke 20, ingatkan ibubapa/penjaga lawatan ke klinik pada umur 1 bulan.

Catatan:

1. Sila rujuk bayi kepada Pegawai Perubatan/Pakar Perubatan Keluarga/Pakar Pediatrik di klinik atau hospital berdekatan dengan segera jika perlu.
2. Sila rujuk *Perinatal Care Manual : Neonatal Care, Section 14* dan *Appendix 10-2, Ministry of Health, 2nd Edition, 2010*.

ROLE OF TRADITIONAL PRACTICE AMONG MOTHERS AND THE NEWBORN

Good Practice	Harmful Practice	Unsure Benefit
• Breastfeeding exclusively • Using coconut oil to remove cradle cap • Frequent breastfeeding to encourage production of milk • Baby sleeping with mother in same room	• Too early weaning e.g. mashed bananas, rice porridge, glucose water etc • Goat's milk for jaundiced baby • Bathing jaundiced babies with herbs. This may mask jaundice and sometimes trigger an acute haemolysis in G6PD deficient babies • Applying ash, 'celak mata' to the umbilical stump • Using knitted cap when there is cradle cap (aggravates the cradle cap) • Restriction of fluids in postnatal mothers (especially among Malays), certain food, vegetables, fruits, chickens, egg and seafood • Discarding colostrum • Application of irritant on baby's skin • Application of heated object to abdomen and scrotum after bath • Using breastmilk to wash eyes • Baby sleeping with mother on the same bed • Baby sleeping prone (on	• Feeding of gripe water for relief of abdominal distension/ colic • Use of herbs and <i>jamu</i> by mother during postnatal period

Good Practice	Harmful Practice	Unsure Benefit
	his stomach) • Application of medicated oil/ ointment on the baby's stomach ie Yu Yee oil • Exposing jaundiced babies to direct sunlight • Feeding plain water or glucose water to jaundiced babies • Feeding pearl powder to babies	

CHAPTER 11

RESUSCITATION AND STABILISATION

11.1 RESUSCITATION OF NEWBORN:

- Should be done by trained health care personnel
- Resuscitation equipment should be in working order
- Standard precautions should be observed

11.1.1 Perinatal risk factors increasing the likelihood for neonatal resuscitation

Antepartum factors	Intrapartum Factors
○ Maternal age >35 years ○ Maternal diabetes ○ Pregnancy induced hypertension ○ Chronic hypertension ○ Maternal cardiac, renal, pulmonary, thyroid or neurologic disease ○ Maternal infection ○ Maternal substance abuse ○ Drug therapy (e.g.magnesium,adrenergic agonists) ○ No prenatal care ○ Foetal Anaemia or Rhesus-isoimmunisation ○ Previous foetal or neonatal death ○ Bleeding in second or third trimester ○ Oligo/polyhydramnios ○ Premature rupture of membranes ○ Foetal hydrops ○ Post-term gestation ○ Multiple gestation ○ Size-dates discrepancy ○ Foetal Malformation or anomalies ○ Diminished foetal activity	○ Emergency Caesarean section ○ Breech or other abnormal presentation ○ Forceps or vacuum assisted delivery ○ Premature labour ○ Precipitous labour ○ Shoulder dystocia ○ Chorioamnionitis ○ Prolonged rupture of membrane (>18 hours before delivery) ○ Prolonged labour (>24 hours) ○ Macrosomia ○ Non-reassuring foetal heart rate pattern or persistent foetal bradycardia ○ Use of general anaesthesia ○ Maternal magnesium therapy ○ Uterine hyperstimulation/ uterine tachysystole with foetal heart rate changes ○ Narcotics to mother within 4 hours of delivery ○ Meconium stained liquor ○ Prolapsed cord ○ Abruptio placenta ○ Placenta praevia/ Vasa praevia ○ Significant intrapartum bleeding

Stand-by for High-Risk Deliveries

A medical officer trained in neonatal resuscitation should ideally be present at all high-risk deliveries. It is recommended that the following situations warrant to standby:

- Preterm infants < 35 weeks
- Severe IUGR (weight $\leq$ 1.8kg)
- Meconium stained liquor
- Abnormal CTG or scalp pH < 7.20 or other indications of foetal distress
- Cord prolapse
- Antepartum haemorrhage
- Multiple births with anticipated problems
- Breech or other abnormal presentations
- Shoulder dystocia
- Instrumental delivery (not for uncomplicated low forceps or vacuum lift- out)
- Caesarean section under general anaesthesia
- Emergency caesarean section
- Infants with significant congenital malformations diagnosed antenatally

11.1.2 Neonatal resuscitation equipments

1. Suction equipment
 - Mechanical suction and tubing
 - Suction catheters 5F, or 6F, 8F, 10F, 12F or 14F
2. Bag and mask equipment/ T-Piece resuscitator and mask
 - Device for delivering positive pressure ventilation
 - Face masks : newborn & premature sizes, cushioned rim mask preferred
 - Wall oxygen and air with flowmeters & tubing
3. Intubation equipment
 - Laryngoscope with straight blades, No. 00, 0 and 1
 - Endotracheal tubes, 2.5, 3.0 and 3.5mm internal diameter and stylets
4. Cardiorespiratory monitor
 - Stethoscope
 - Pulse oximeter
 - Electronic cardiac monitor

5. Medications
 - Adrenaline 1:10 000 (0.1mg/ml)
 - Normal Saline
 - Dextrose 10%
6. Catheters
 - Umbilical catheters 3.5F and 5F
7. Miscellaneous
 - Radiant warmer or other heat source
 - Pre warmed linens
 - Plastic wrap or cling film
8. Home delivery (uncommon) for resuscitation for babies
 - Manual suction apparatus

11.1.3 Thermal protection and ABC's of resuscitation

1. Provide **Warmth**
 - Preheat radiant warmer, warm the linens, dry baby.
 - Plastic wrap or cling film if indicated.
2. Establish an open **Airway**
 - Position the head, suction mouth and nose if necessary
3. Initiate **Breathing**
 - Tactile stimulation
 - Positive pressure ventilation with bag and mask or ET if needed
4. Maintain **Circulation**
 - Chest compressions
5. Administer **Drugs**
 - Adrenaline as you continue PPV and chest compression
 - Normal saline for volume expansion as indicated

Note :

- Refer **Figure 11.1** for Overview of Resuscitation in the Delivery Room
- The neonatal resuscitation process will follow the current NRP guideline in use in each local facility and is dependent on the resources available.

Table 11.1 : Guidelines for ETT size (NRP 7th edition)

Infant weight	Gestational age (weeks)	Tube size (mm)
<1000g	< 28	2.5
1000-2000g	28-34	3.0
>2000g	> 34	3.5

Table 11.2: Guidelines for ETT position (NRP 6th edition)

Infant weight	Oral intubation – weight in kg + 6 cm (Tip to lip distance)
1 kg	7 cm
2 kg	8 cm
3 kg	9 cm

11.1.4 Apgar scoring

- Mechanism for documenting newborn's condition at specific intervals after birth
- Should not be used to determine need for resuscitation.
- Resuscitative efforts should be initiated promptly after birth, if required

Table 11.3: Apgar Score

Sign	0	1	2
Heart rate	Absent	< 100/min	>100/min
Respiration	Absent	Slow, irregular	Good, crying
Muscle tone	Limp	Some flexion	Active
Reflex irritability	No response	Grimace	Crying
Colour	Blue	Pink centrally Extremities blue	Pink all over

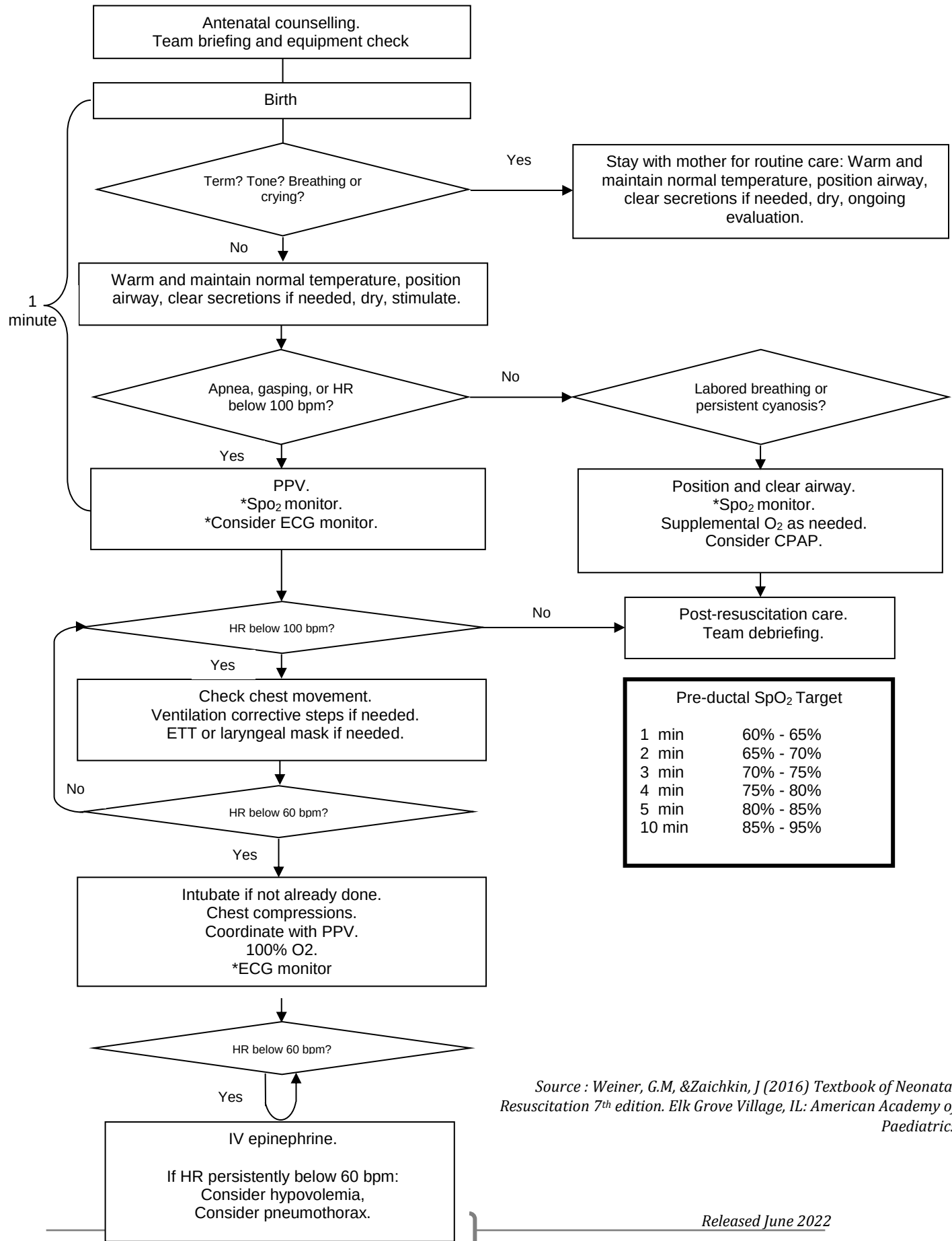
Cessation of Cardiopulmonary Resuscitation

- Decision should be based on cause of arrest, response to resuscitation and remedial factors
- Death or severe neurological abnormality is predicted by a failure to obtain a heart rate by 10 minutes despite adequate resuscitation and failure to respond to adrenaline

*The adherence to the above guideline shall depend on the local facilities and the availability of resources.

Reference: "Textbook of Neonatal Resuscitation" by American Heart Association & American Academy of Paediatrics, 6th and 7th Edition.

Figure 11.1 : Neonatal Resuscitation Flow Diagram



11.2 THERMAL PROTECTION

Thermal protection of the newborn is a series of measures taken at birth and in the first days of life to ensure that the newborn maintains a normal body temperature.

How the newborn loses heat

The temperature inside the mother's womb is 38°C. Leaving the warmth of the womb at birth, the wet newborn finds itself in a much colder environment and immediately starts losing heat.

The newborn loses heat in 4 different ways:

- Evaporation (amniotic fluid from the baby's body)
- Conduction (naked baby on a cold surface)
- Convection (naked baby exposed to cooler surrounding, draught)
- Radiation (baby to cooler objects)

**Table 11.4 : The 10 steps of “Warm Chain” to minimise heat loss in newborn
(Adapted from WHO 1997, for well, term babies)**

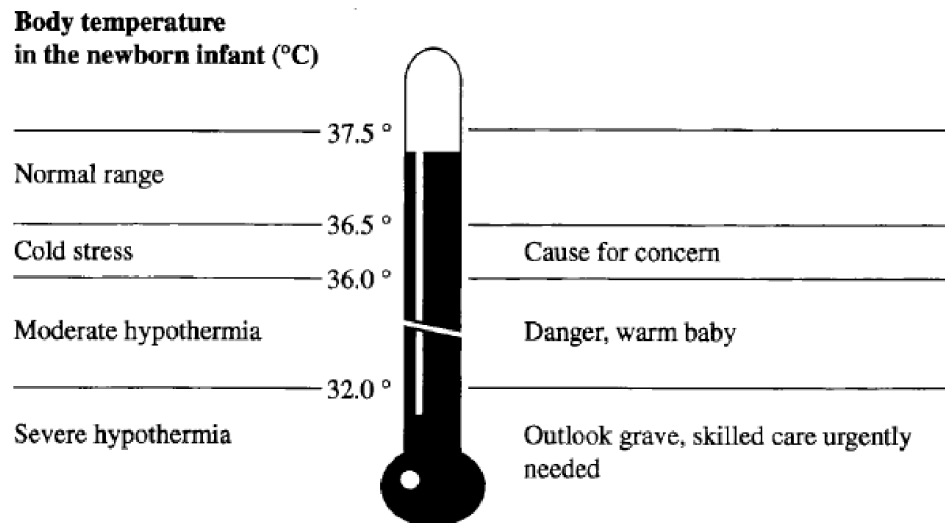
	Steps	Procedure
1.	Warm delivery room	• The delivery room temperature should be at least 25°C, free from the draughts from open windows & doors, or fans.
2.	Immediate drying	• Immediately dry the newborn after birth with a warm towel. • Discard the wet towel and wrap the newborn with another dry towel and cap
3.	Warm resuscitation	• Newborn that needs resuscitation, should be kept warm by putting the newborn under an additional source of heat ie radiant warmer • Preterm $\leq$ 32 weeks should be wrapped in polyethylene plastic bag
4.	Skin-to-skin contact	• The stable newborn can be placed on the mother's chest or abdomen while being dried. • Newborns can be maintained in skin-to-skin contact with the mother: ○ while she is being attended to (placenta delivery, suturing) ○ during transfer to the postnatal unit, recovery room ○ during assessments and initial interventions ○ for the first hours after birth

	Steps	Procedure
5.	Breastfeeding	• Breastfeeding should begin as soon as possible after delivery, preferably within an hour • Baby should breastfeed on demand
6.	Bathing and weighing postponed	• Bathing should not be done before 6 hours after birth, preferably on the 2nd or 3rd day of life so long as the newborn is healthy and the temperature is normal • Bathe newborn in a warm room, using warm water. After bathing, immediately wrap the newborn in a dry warm towel, dry thoroughly, dress quickly and place near the mother
7.	Appropriate clothing and bathing	• Newborn's clothing should be appropriate for the environmental temperature
8.	Mother and newborn rooming in together	• Keep mother and newborn together 24 hours a day (rooming-in), in a warm room
9.	Warm transportation	• Keep newborn warm during transport • Use a transport incubator, where available
10.	Training and awareness raising	• All healthcare personnel involved in the process on birth and care of the newborn should be aware of the risks of hypothermia and hyperthermia • They should be taught the principle of thermal protection of the newborn.

11.2.1 Hypothermia in The Newborn

Hypothermia is defined as body temperature below 36.5°C

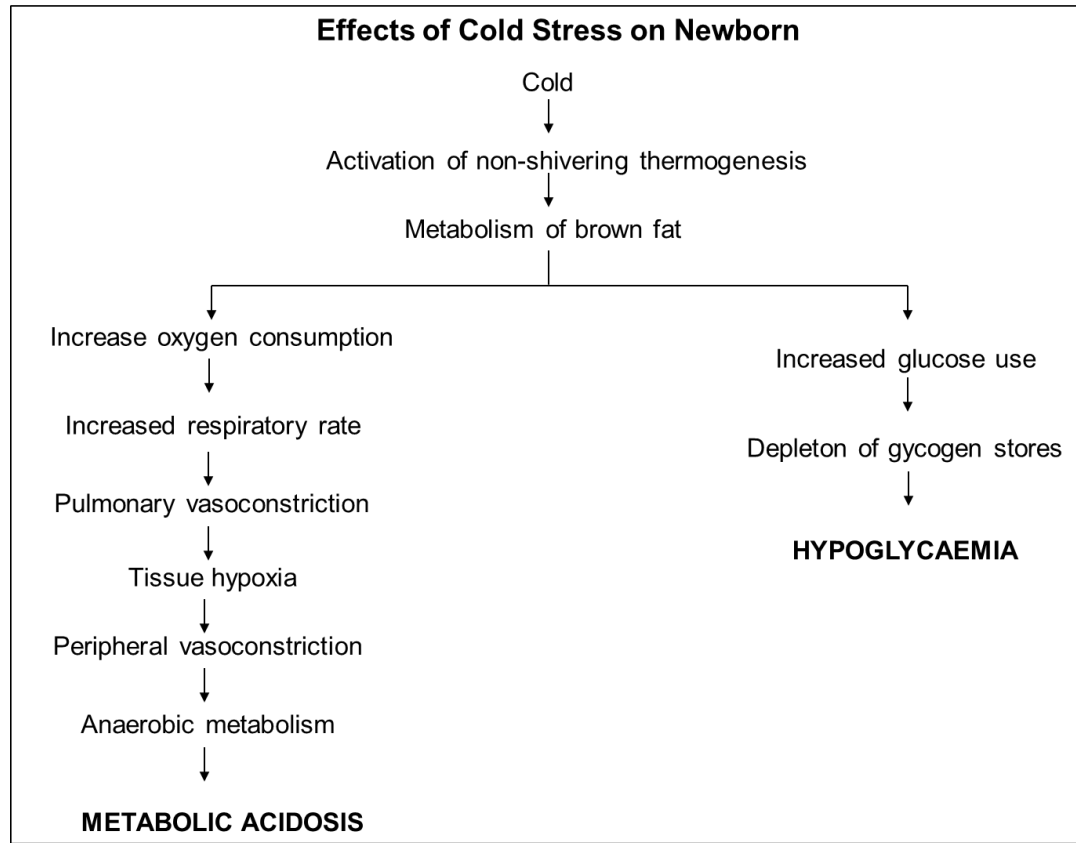
Figure 11.2 : Body Temperature in the newborn infant



Infants at risk for hypothermia

- Preterm, low birth weight babies
- Small-for-gestational age babies
- Infants who require prolonged resuscitation
- Infants who become acutely ill with infectious, cardiac, neurologic, and endocrine problems
- Infants with surgical problems – especially infants with open body wall defect (gastroschisis, omphalocele, exposed spinal defect), where heat loss is more rapid
- Infants who have decreased activity or are hypotonic from sedatives, analgesics, paralytics or anaesthetics

Figure 11.3 : Effects of Cold Stress on Newborn



Other harmful effects of hypothermia include impaired immune function, impaired coagulation and impaired surfactant production. If not treated, severe hypothermia can lead to cell damage and even death.

Signs of Hypothermia

- Pale, mottled skin that is cool to touch
- Acrocyanosis
- Respiratory distress
- Apnoea, bradycardia, central cyanosis
- Irritability then lethargy as hypothermia worsens
- Hypotonia
- Weak cry and suck
- Abdominal distension, vomiting, feeding intolerance
- Shivering in more mature babies

Treatment of Hypothermia

- Rewarm at a rate of 0.5 – 1°C/ hour.
- Rapid warming has been associated with apnoea, hypotension and rapid electrolyte (Ca⁺⁺, K⁺) shifts.
- During the rewarming process, monitor and record temperatures (skin, axillary and environmental) every 30 minutes. Heart rate, blood pressure and respiratory rate should also be monitored.

Additional Considerations

- Other methods to reduce heat loss include using double-walled incubators, humidifying the incubators, or using radiant warmers.
- Avoid hot water bottles, gloves filled with hot water or heat lamps because they may cause burns.
- Avoid warming linen in the microwave oven.

11.2.2 Hyperthermia

Hyperthermia is defined as body temperature more than 37.5°C

Causes of hyperthermia

- Environmental causes: excessive environmental temperature, overwrapping of the baby, loose skin temperature probe with an incubator or radiant warmer in servo-controlled mode, or a servo-controlled temperature set too high
- Infection
- Dehydration
- Maternal fever in labour
- Maternal epidural analgesia
- Drug withdrawal

Signs of Hyperthermia

- Reddened skin that is warm to touch
- Tachycardia
- Tachypnoea
- Irritability, lethargy, hypotonia, weak cry
- Poor feeding
- Apnoea
- Sweating in more mature babies
- Dehydration

Treatment of Hyperthermia

- Determine the cause of the hyperthermia. The most common cause in NICU is iatrogenic (remove external heat sources, check incubator and radiant warmers for appropriate functioning, assess temperature probe position for appropriate location on baby)
- Non-environmental causes of hypothermia (infection, dehydration or CNS disorder) should be considered and treated accordingly
- During the cooling process, monitor and record temperatures (skin, axillary and environmental) every 30 minutes

Reference

1. World Health Organization. (1997). Safe Motherhood: Thermal Protection of the Newborn: A Practical Guide.
2. Gomella TL, et al .Gomella's Neonatology : Management, Procedures, On-Call Problems, Diseases, and Drugs 8th ed. McGraw Hill: 2020 p.92-95
3. Gardner SL and Hernandez JA. Heat Balance : Merenstein & Gardner's Handbook of Neonatal Intensive Care 8th ed. Elsevier; 2016.p.105 -25
4. Karlsen K. The S.T.A.B.L.E programme 6th ed.

11.3 STABILISATION AND TRANSPORTATION OF THE NEWBORN

11.3.1 Cases Requiring Transport

- For neonatal or surgical care not available at the referring centre
- For transfer from labour room to NICU/SCN
- For transfer from one neonatal ward to another
- Health clinic to hospital

The principles of initial stabilisation of the neonate:

- **A**irway
- **B**reathing
- **C**irculation, **C**ommunication
- **D**rugs, **D**ocumentation
- **E**nvironment, **E**quipment
- **F**luids – electrolytes, glucose

11.3.2 Approach to Neonatal Transfer

COMMUNICATION

- Contact referral hospital and discuss with receiving staff about the case and request for advise
- Record time and details of discussion

STABILISATION

Airway

- Airway suctioning and maintenance of clear airway
 - Administer oxygen via cannula, headbox, bag and mask, or endotracheal tube
 - Monitor oxygen saturation with oximeter

Breathing

- Observe breathing effort and rate
- Support breathing by bagging
- Note effectiveness of manual respiratory support
- Monitor oxygen saturation with pulse oximeter
- Obtain a blood gas if available

Circulation

- Observe colour especially central area
- Suction airway and administer oxygen effectively

Drugs

- Administer drugs as required
- Correct hypoglycaemia after capillary blood sugar sampling
- Administer Vitamin K (if not given)

Environment

- Prewarm transport incubator setting at 35°C using mains power supply
- Place necessary articles inside
- Monitor temperature of baby closely
- Warm baby up to normal body temperature under the radiant warmer then place baby in the transport incubator

Fluids

- Set up intravenous infusion of dextrose 10%.
- Use a syringe pump if available and set the correct flow rate.

- If a syringe pump is unavailable, use paediatric chambers to deliver IV fluid at a correct drip rate. The maximum volume in a chamber is 40 mls only.
- Set up blood transfusion or plasma expander when required.

11.3.3 PREPARATION FOR NEONATAL TRANSFER

Transport team

- Inform team members: doctor, staff nurse, attendant, driver
- Inform team members of neonate's condition and stabilisation activities

Equipment

- The transport incubator temperature will be set at 35°C (or appropriate temperature according to the baby's gestation)
- When there is no incubator, wrap the neonate in warm towels/linen. However, it is impossible to observe abnormal respiratory functions without disturbing the neonate. This method prevents heat loss but will not warm up a low birth weight baby.
- Portable ventilator and adequate oxygen.
- Resuscitation equipment and drugs i.e. bag and mask, suctioning equipment, intubation equipment, drugs, syringes, and others (refer to "resuscitation equipment" and "medications").
- Intravenous fluids e.g. dextrose 10%.
- Monitoring equipment: appropriate equipments such as cardiorespiratory monitor, pulse oximeter, thermometer, and stethoscope.

Vehicle

- The ambulance should be in functioning order and have adequate equipments
- Secure the incubator and other equipments in place during the transfer

Parents

- Inform parents the need for transfer of the neonate
- Encourage one parent to accompany neonate
- Obtain written consent from parents for blood transfusion or emergency surgery
- Obtain mother's blood sample if she is not accompanying her child
- Allow neonate's mother to see and touch her baby
- Referral letter:
 - should include a complete and detailed history of maternal factors and neonatal problems

- treatment already carried out should be listed such as antibiotics, resuscitation given, immunisation, Vitamin K
- events that have occurred should be written in sequence
- date, time and name of the doctor should be written clearly

Records

- State the date and time of events in the progress notes
- Record a brief but concise account of the events before the transfer of the neonate.
- Record in admission book, census book and 24-hour report book

Checklist before departure

- The following should be ready to be sent with the baby:
 - cord blood specimen or baby's blood specimen
 - mother's blood specimen (10 ml clotted blood) labelled correctly with mother's full name and identity card number
 - referral letter
 - all X-rays and other investigation results
 - written consent of parent for the appropriate procedures
- Ensure the endotracheal tube, if required, is properly secured and at the appropriate level.
- The baby's condition should be reviewed just before transport and referral hospital should be reformed if the general condition has changed.
- Check that all equipment in the ambulance is functioning and has adequate power supply to last the journey.

During transport

- Connect ventilator or oxygen delivery system to ambulance supply, if available.
- Any electrical equipment should be plugged into the AC-DC converter in the ambulance.
- Monitor vital signs, IV fluids, and medication, and chart in the Neonatal Transport Chart.
- Where possible, observations should be done without disturbing the baby. Use monitoring equipment if available.
- Stop at the nearest health clinic or in a safe area if the condition of the neonate deteriorates or needs further resuscitation.
- Ensure airway is maintained by neutral neck position, suctioning as required, and correct position of the endotracheal tube.
- Check on adequacy of chest expansion, colour, and oxygen saturation, especially in the baby who is receiving assisted ventilation.

At the receiving hospital

- Hand over to the receiving nurse/medical officer the following information & records:
 1. Name and identity card number and full name of the mother
 2. Name of the baby, if available
 3. Sex of the baby
 4. Referral letter and other documents
 5. Records and observations during transport
 6. Medications and immunisation given
 7. Blood and other specimens
 8. Imagings e.g. x-ray films
 9. Introduce parents/relatives to the receiving staff
- Account for all equipment before leaving.

After returning from the referring hospital

- If parent(s) have not accompanied the baby, inform them about the condition of the child the ward/ hospital he or she is placed in, the contact number and the name of the doctor-in-charge.

Reference:

1. Protocol of Neonatal Nursing Procedure Pub. Ipoh Postgraduate Medical Society Ed. J.Ho, 1995
2. Paediatric Protocol for Malaysian Hospitals 4th edition 2018 (pp.74-81)

11.4 CRITERIA FOR VARIOUS LEVELS OF NEONATAL CARE**11.4.1 Levels of Neonatal Care**

LEVEL III – Intensive Care: For babies with problems requiring intensive care such as endotracheal intubation for assisted ventilation, intra- arterial blood pressure monitoring, continuous cardiorespiratory monitoring, parenteral nutrition, central venous catheterisation, transcutaneous blood gas and oxygen saturation monitoring and neonates requiring stabilisation following major surgery.

LEVEL IIb – Semi-intensive or High Dependency Unit: For babies with problems requiring close observation and intervention but not requiring intensive care. These babies may require peripheral intravenous therapy,

simplified cardiorespiratory monitoring, apnoea monitoring, oxygen not usually in excess of 40%, chronic oxygen dependency and surgical nursing.

LEVEL IIa – Special Care or Low Dependency Unit: For babies who could not be cared for at home. These babies include convalescent preterm neonates, or neonates requiring observation for transient problems, phototherapy, investigatory procedures and frequent feeding.

LEVEL I – Neonatal Care In Postnatal Wards: For uncomplicated maternal and neonatal cases where routine nursing care is provided and close observation is not required. Normally, these babies are placed together with their mothers in the postnatal wards and regarded as an inpatient.

Reference

Appendix II of Paediatric Services Operational Policy, Ministry of Health, 2012.

11.4.2 Indications for Admission to Various Levels of Care

LEVEL III - Intensive Care

1. Respiratory distress requiring ventilatory support or CPAP
2. Very low birth weight (VLBW) babies of birthweight (BW) < 1000 gm.
3. Moderate to severe neonatal encephalopathy
4. Severe birth trauma - intracranial haemorrhage
5. Duct dependent congenital heart disease which may be cyanotic or acyanotic; congestive heart failure; supraventricular tachycardia, arrhythmia.
6. Hypotension, shock
7. Need for resuscitation and inotropic support
8. Disseminated intravascular coagulation
9. Immediate post-op surgical patients
10. Necrotising Enterocolitis (> Grade 1)
11. Hydrops foetalis
12. Intractable hypoglycaemia
13. Persistent metabolic acidosis
14. Neonatal seizures
15. Any other baby whose clinical condition is considered to be unstable or require very close observation

LEVEL IIb - Semi-Intensive Care

1. Receiving NCPAP for some part of the day and ≥ 1000 gm in weight
2. Receiving parenteral nutrition and not fulfilling criteria for intensive care
3. Requiring monitoring for seizures
4. Requiring oxygen $> \text{FiO}_2$ 40%
5. Requiring continuous cardiorespiratory monitoring
6. Requiring frequent stimulation for apnoea
7. Requiring treatment for neonatal abstinence syndrome in acute period
8. Acute surgical cases

LEVEL IIa - Special Care

1. Babies $>1500\text{gm}$ to 2000gm birth weight
2. Babies < 35 weeks gestation
3. Large babies ie birth weight $> 4.2\text{kg}$
4. Large for gestational age (LGA) babies ie $\text{BW} > 90\text{th}$ centile
5. Small for gestational age (SGA) babies ie $\text{BW} < 10\text{th}$ centile
6. Babies with respiratory distress requiring $< 40\%$ oxygen
7. Babies with risk of meconium aspiration without respiratory distress on admission
8. Babies with Rhesus or ABO incompatibility
9. Babies with significant jaundice.
10. Babies with mild asphyxia or Apgar Score < 7 at 5 mins
11. Babies born to mothers with chorioamnionitis or pyrexia $> 38^\circ\text{C}$ or leaking liquor of more than 18 hours
12. Sepsis (fever, umbilical discharge, severe eye discharge) and congenital infection (e.g. maternal chickenpox)
13. Babies of diabetic mothers
14. Babies with more than one episode of hypoglycaemia (blood sugar $< 2.6\text{mmol/L}$)
15. Babies with birth trauma – mild subaponeurotic haemorrhage, Erb's palsy and fractures
16. Babies of drug addict mothers with no further withdrawal symptoms
17. Babies with multiple or serious congenital anomalies
18. Babies requiring IV drip
19. Babies requiring surgery and do not require intensive care
20. Unwell babies (e.g. poor feeding, lethargy, vomiting)
21. Babies born to HIV mothers and symptomatic babies of VDRL positive mothers
22. Stable babies with cardiac conditions

Note:

In conditions other than those listed above and if unsure, please consult Registrar or Specialist

LEVEL I - Neonatal Care in Postnatal Wards (these are babies that may be considered for nursing in the obstetric ward depending on local factors. The baby has to be monitored and transferred to appropriate level of care depending on the progress)

1. Borderline low birthweight (ie between 1.8 and 2.5 kg) babies who are otherwise well
2. Well babies of 35 to 37 weeks gestation who are 1.8 kg. and above
3. Large babies between 4 and 4.5 kg
4. LGA and SGA babies
5. Babies with meconium below cords during resuscitation with no respiratory distress or hyperinflation of the chest.
6. Babies with G6PD deficiency, Rhesus or ABO incompatibility and moderate jaundice (SB < 300 µmol/L except for babies with jaundice on day one of life) – phototherapy with monitoring
7. Asymptomatic babies with presumed sepsis needing antibiotic therapy
8. Asymptomatic babies born to VDRL positive mother
9. Babies with glucose 6-phosphate dehydrogenase (G6PD) deficiency
10. Babies of thyrotoxic mothers

11.5 Normal care

Normal care is the routine care of the healthy term baby who requires only the maintenance of body temperature, the establishment of feeding and hygiene care. This is usually provided in the obstetric ward or at home with the mother.

Infant's progress at every level should be monitored and transferred to the appropriate level as indicated by the patient's condition

Disclaimer : The adherence to the above guidelines shall depend on the local facilities and the availability of resources

CHAPTER 12

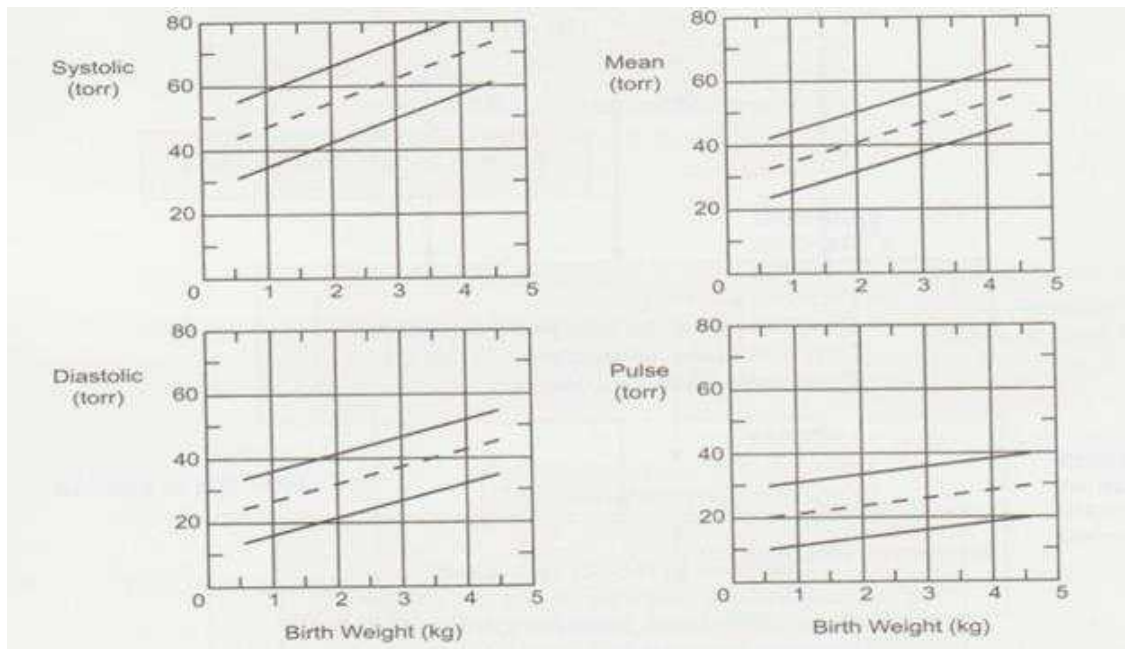
EARLY NEWBORN CARE

12.1 OVERVIEW OF NEWBORN EXAMINATION

- The aims of newborn examination should be fully explained and the results shared with the parents and recorded in the baby's home-based card.
- A complete examination of the baby should be done within 72 hours of birth.
- This examination should include a review of :
 - parental concerns
 - baby's medical history: family, maternal, antenatal and perinatal history
 - any previously plotted birth weight and head circumference
 - feeding adequacy
 - whether the baby has passed meconium and urine (and urine stream in a boy)
 - vitamin K and immunisation status
 - results of G6PD and cord TSH, if available
- A physical examination should also be carried out. This should include checking the baby's:
 - appearance including colour, breathing, behaviour, activity and posture
 - head (including fontanelles), face, nose, mouth including palate, ears, neck and general symmetry of head and facial features. Measure and plot head circumference
 - eyes; check opacities and red reflex
 - neck and clavicles, limbs, hands, feet and digits; assess proportions and symmetry
 - heart; check position, heart rate, rhythm and sounds, murmurs and femoral pulse volume, check oxygen saturation over lower limbs (if available)
 - lungs; listen for noisy breathing, check effort, rate and lung sounds
 - abdomen; check shape and palpate to identify any organomegaly; also check condition of umbilical cord
 - genitalia and anus; check for appearance of genitalia, undescended testes in males and anal patency
 - spine; inspect and palpate bony structures and check integrity of the skin
 - skin; note colour and texture as well as any birthmarks or rashes

- central nervous system; observe tone, behaviour, movements and posture. Elicit newborn reflexes ie Moro's reflex
 - hips; check symmetry of the limbs and skin folds (perform Barlow and Ortolani's manoeuvres)
 - cry; note sound
 - weight and length; measure and plot
 - gestational age; assess using Ballard Score (**Figure 12.1**) if the baby is of low birth weight or if gestation is uncertain
- Vital signs : normal range for newborn
 - Respiratory rate : 40- 60/min
 - Heart rate : 120 – 160 / min
 - Capillary refill time : < 3 seconds
 - Oxygen saturation : > 95% (for Term babies), 90-95% (for Preterm babies)
 - Blood pressure:

Figure 12.1 : Normal Range of Vital Signs for Newborn



Source: Blood pressure by birth weight (From Versmold HT, Kitterman JA, Phibbs RH et al:Paediatrics 67(5):607, 1981

Figure 12.2: Expanded New Ballard Score for determining gestational age by assessment of neuromuscular and physical maturity

Neuromuscular Maturity

Score	-1	0	1	2	3	4	5
Posture							
Square window (wrist)	>90°	90°	60°	45°	30°	0°	
Arm recoil		180°	140°–180°	110°–140°	90°–110°	<90°	
Popliteal angle	180°	160°	140°	120°	100°	90°	<90°
Scarf sign							
Heel to ear							

Physical Maturity

Skin	Sticky, friable, transparent	Gelatinous, red, translucent	Smooth, pink; visible veins	Superficial peeling and/or rash; few veins	Cracking, pale areas; rare veins	Parchment, deep cracking; no vessels	Leathery, cracked wrinkled
Lanugo	None	Sparse	Abundant	Thinning	Bald areas	Mostly bald	Maturity Rating
Plantar surface	Heel-toe 40-50 mm: -1 <40 mm: -2	>50 mm, no crease	Faint red marks	Anterior transverse crease only	Creases anterior 2/3	Creases over entire sole	
Breast	Imperceptible	Barely perceptible	Flat areola, no bud	Stippled areola, 1-2 mm bud	Raised areola, 3-4 mm bud	Full areola, 5-10 mm bud	Score
Eye/Ear	Lids fused loosely: -1 tightly: -2	Lids open; pinna flat; stays folded	Slightly curved pinna; soft; slow recoil	Well curved pinna; soft but ready recoil	Formed and firm, instant recoil	Thick cartilage, ear stiff	Weeks
Genitals (male)	Scrotum flat, smooth	Scrotum empty, faint rugae	Testes in upper canal, rare rugae	Testes descending, few rugae	Testes down, good rugae	Testes pendulous, deep rugae	-10 20
Genitals (female)	Clitoris prominent, labia flat	Clitoris prominent, small labia minora	Clitoris prominent, enlarging minora	Majora and minora equally prominent	Majora large, minora small	Majora cover clitoris and minora	-5 22
							0 24
							5 26
							10 28
							15 30
							20 32
							25 34
							30 36
							35 38
							40 40
							45 42
							50 44

12.2 COMMON FINDINGS IN NEWBORN EXAMINATION

Below are some of the conditions that may be found during routine newborn examination. The list, however, is far from complete. Please refer if in doubt.

12.2.1 Colour

- **Pallor** - associated with shock or low haemoglobin
- **Cyanosis** - It is important to distinguish between central cyanosis due to hypoxaemia (lips, buccal mucosa and peripheries are bluish) and peripheral cyanosis due to cold (only the feet and fingers are bluish). Circumoral cyanosis is common among newborn babies and has no diagnostic significance. Facial congestion may be due to a tight umbilical cord around the neck and parents should be reassured if the baby is pink centrally.
- **Mottling** – may be a response to hypothermia or associated with sepsis. If the baby is well, it may be a feature of cutis marmorata.
- **Plethora** - associated with polycythemia
- **Jaundice** - babies who are visibly jaundiced should have their bilirubin levels checked (TSB or TcB measurement) If the jaundice is detected within 24 hours of life, urgent referral is needed to rule out haemolysis or sepsis.

12.2.2 Skin

The following skin conditions are benign and self-limiting:

- **Erythema toxicum** - Most common newborn rash. Variable, irregular macular papular patches and sometimes vesicular lesions. Appears soon after birth and persists for a few days.
- **Milia** - pinpoint white papules of keratogenous material usually on nose, cheeks and forehead, last several weeks.
- **Miliaria** - obstructed eccrine sweat ducts. Pinpoint vesicles on forehead scalp and skinfolds. Clears within 1 week.
- **Transient neonatal pustular melanosis** - small vesiculopustules, generally present at birth, containing WBCs and no organisms. The intact vesicle ruptures to reveal a pigmented macule surrounded by a thin skin ring.
- **Mongolian blue spots** - Bluish discolouration commonly seen on lower back, buttocks or lower limbs. Can be mistaken for a bruise but unlike a bruise, it does not change in colour over a period of days. May persist for years.
- **Capillary naevi** - Pink macular discolouration over upper eyelids or forehead (salmon patch) or on back of neck (stork mark).

The following skin conditions may need referral for further assessments:

- **Septic spots** – these are yellow pus-filled spots with or without inflammation of the surrounding skin. May need antibiotic treatment.
- **Petechiae or Purpura** – this may indicate a bleeding disorder i.e. thrombocytopaenia or platelet dysfunction. Need urgent referral.
- **Cafe au lait spots** - suspect neurofibromatosis if there are many large spots.
- **Haemangioma (capillary, venous, arteriovenous)** - Large or multiple haemangiomas may cause circulatory problems or thrombocytopaenia and need urgent referral. Multiple haemangiomas may have visceral involvement. Haemangiomas on the face and neck, lumbosacral spine, axilla and perineum need referral for further assessment. Capillary haemangiomas usually increase in size in the first few months of life but will involute spontaneously.
- **Oedema** - generally unusual in the newborn. If generalised, may be associated with causes of hydrops fetalis. If oedema is localised to the neck, consider Turner syndrome or Down syndrome. Non-pitting oedema over the dorsum of both feet is suggestive of Turner or Noonan syndrome.

12.2.3

Head

- **Moulding**
- **Caput succedaneum** – should resolve in 2-3 days
- **Chignon** – commonly follows vacuum extraction
- **Cephalohaematoma**- a collection of blood between the periosteum and the bone and does not cross suture lines. Usually resolves in 2-3 months. Massaging should be avoided.
- **Subaponeurotic haemorrhage (subgaleal bleed)** -a boggy swelling of the head which may cross suture lines. The scalp appears diffusely swollen and the ears may be lifted forward. This requires urgent referral because the baby can bleed insidiously and rapidly go into hypovolaemic shock.
- Other swellings (i.e.; encephalocoele) may need referral for further assessment.

12.2.4

Eyes

- **Eye discharge** – common organisms causing eye discharge or conjunctivitis are staphylococci, chlamydia and gonococci. Bilateral copious purulent discharge with or without haemorrhage with accompanying oedema of the eyelids is typical of gonococcal conjunctivitis and should be urgently treated.

- **Cloudy cornea** - may indicate congenital glaucoma and needs urgent referral.
- **Subconjunctival haemorrhage** - commonly seen in normal babies.
- **Absent red reflex** - Hold the ophthalmoscope 6-8" from the eye. Use the +10 diopter lens. The normal newborn transmits a clear red colour back to the observer. Black dots may represent cataracts. The absence of a clear red reflex is indicative of cataract, glaucoma or retinoblastoma.

12.2.5

Ear and Nose

- **Pre-auricular skin tags** – benign, fairly common and usually inherited
- **Pre-auricular pit** – is a small opening near the front of the ear, which can be a blind-ended opening or have a sinus tract. If persistent or recurrent infection occurs, surgical excision may be indicated
- **Choanal atresia** may be manifested by respiratory distress or cyanosis (neonates are obligate nose breathers). To confirm patency, gently insert a soft NG tube through each nostril.

12.2.6

Mouth

- **Cleft lip and/or cleft palate**: may occur in isolation or in association with a syndrome
- **Epstein pearls** - small white papules found on the midline of the hard palate. These are benign and will eventually resolve.
- **Ranula** - small bluish white swelling on the floor of the mouth representing benign mucous gland retention cyst. This will resolve with time and is benign.
- **Tongue-tie** due to a short frenulum does not usually cause problems. Refer only if feeding difficulty or speech problems
- **Natal teeth** occur in 1/2,000 births and are mostly lower incisors. There is risk of aspiration and should be extracted if loosely attached.
- **Oral thrush** is a fungal infection of the mouth or throat caused by *Candida Albicans*. It is seen as white patches scattered over the tongue and the buccal mucosa, which cannot be easily wiped away. Treatment is with oral Nystatin.

12.2.7

Neck

- **Cystic hygroma** – a congenital malformation of the lymphatic system which may cause airway obstruction and need early surgical referral
- **Sternomastoid 'tumour'** - a firm fibrous mass in the sternomastoid muscle which may be associated with torticollis and can be treated with early physiotherapy.

12.2.8

Chest

- **Signs of Respiratory Distress** include:
 - Grunting – expiratory sound much like a short cry, occurs when glottis is closed during expiration
 - Nasal flaring – widening of the nostrils on inspiration
 - Retractions – muscles sucked in between the ribs to increase air flow, can be subcostal or intercostal
- **Stridor** - a high-pitched cry heard on inspiration. Laryngomalacia is the most common cause but may be due to other upper airway abnormalities
- **Breasts** in the newborn may be enlarged due to the effects of maternal oestrogen. A white discharge may be present and is normal

12.2.9

Heart

- **Murmur** – may be associated with congenital heart lesions, need referral for further assessment.
- **Cyanosis** – may be associated with congenital cyanotic heart disorders, pulmonary disorders or PPHN (persistent pulmonary hypertension of Newborn). Need to consult the Paediatric team urgently
- **Heart failure** - tachypnoeic, tachycardic, hepatomegaly, gallop rhythm, weak pulses – may be due to structural heart lesion, myocardial diseases, arrhythmia or extracardiac cause's i.e. severe anemia, neonatal thyrotoxicosis, fulminant sepsis. Need to consult the Paediatric team urgently

12.2.10

Abdominal Wall and Umbilicus

- **Omphalocele and Gastroschisis** - An omphalocele has a membrane covering (unless it has been ruptured during the delivery) whereas gastroschisis does not. Place cling wrap over exposed intestines and refer to a hospital with surgical facilities as soon as possible
- **Diastasis recti** – a gap between the two recti abdominis muscle and usually presents as a midline epigastric swelling which becomes more prominent during straining. In the newborn, the muscles are not fully developed and may not be sealed together at midline. This condition heals on its own and does not require treatment.
- **Single umbilical artery** – may be associated with other abnormalities
- **Omphalitis** – infection of the umbilicus and/ or surrounding tissues. It is characterised by tenderness, erythema, and induration of the umbilicus and surrounding tissues. If untreated, it may progress to systemic infection and death

- **Patent urachus** – is a communication between the bladder and the umbilicus, resulting in the urine coming from the umbilicus. Paediatric surgical referral is needed.
- **Umbilical hernia** – results from the weakness in the muscles of the abdominal wall or umbilical ring. Usually resolves in the first year of life, without treatment
- **Umbilical granuloma** - is composed of granulation tissue at the base of the umbilical cord, appearing after the umbilical cord has separated. It may be associated with serous or serosanguinous drainage and a tendency for easy bleeding with trauma. Treatment options include topical applications of silver nitrate, excision and cryosurgery.

12.2.11

Genitalia

Male genitalia

- **Micropenis** – penile length that is ≤ 2.5 cm in a term newborn (Term normal penis is 3.6 ± 0.7 cm stretched length).
- **Hypospadias** is abnormal location of the urethral meatus on the ventral surface of the penis, Epispadias is abnormal location of the urethral meatus on the ventral surface of the penis
- **Phimosis** - the inability to retract the foreskin or prepuce covering the glans of the penis. It can be normal in newborn and usually resolves around 5-7 years of age
- **Undescended testis** – can be unilateral or bilateral. Early referral to the Paediatric surgical team is recommended.
- **Hydrocoeles** are collections of fluids in the scrotum. They are common and usually disappear by 1 year old
- **Inguinal hernias** are more common in preterm babies. The bowel enters the scrotal sac through the patent processus vaginalis. Early referral to the Paediatric surgical team is recommended.

Female genitalia

- **Mucosal/vaginal tag** – commonly attached to the wall of the vagina and is of no clinical significance
- **Vaginal discharge** – whitish or blood tinged discharge is common due to maternal hormones and lasts for a few days, bloody discharge may be normal and secondary to maternal oestrogen withdrawal.

Indeterminate sex

- If the sex of the baby cannot be determined from physical examination, urgent referral is required to determine the sex early and to exclude medical emergencies such as congenital adrenal hyperplasia.

12.2.12 Spine

- **Skin stigmata** of neural tube defects : such as lipoma, haemangioma, pilonidal skin dimples with tufts of hair or no visible floor - should be referred for ultrasound.
- **Dermal sinuses** in the posterior midline area - usually communicates with the spinal canal and can lead to meningitis. Need urgent referral
- **Sacral dimples** are common and are of no consequence. Referral for ultrasound is required only for sacral dimple > 5 mm in diameter or > 2.5 cm from the anus.
- **Meningocele** – a neural tube defect in which there is incomplete closure of the posterior spine; the lumbar spine is the most common location. There is a risk of meningitis, especially if there is cerebrospinal fluid leakage.

12.2.13 Lower Limbs

- **Developmental Dysplasia of the Hips** – check for asymmetry of the thigh or gluteal creases and limb length discrepancy. Refer to the orthopaedic team if the Ortolani or Barlow test is positive (“clunk” sensation).
- **Congenital Talipes Equinovarus (CTEV)** - the foot is turned downward and inward, and the sole is directed medially. If this problem can be corrected with gentle force, it will resolve spontaneously. If not, orthopaedic treatment and follow up are necessary.

References

1. Gomella TL, et al .Gomella's Neonatology : Management, Procedures, On-Call Problems, Diseases, and Drugs 8th ed. McGraw Hill: 2020
2. Public Health England, Newborn and infant physical examination (NIPE) screening programme handbook <https://www.gov.uk/government/publications/newborn-and-infant-physical-examination-programme-handbook/newborn-and-infant-physical-examination-screening-programme-handbook>
3. Hj Muhammad Ismail et al., (2019) Paediatric Protocol for Malaysian Hospital 4th edition, Ministry of Health Malaysia

12.3

SCREENING FOR CONGENITAL HYPOTHYROIDISM

Congenital hypothyroidism is an uncommon but clearly identified and preventable cause of mental retardation. In Malaysia, the incidence is 1:2200 to 3000.

Infants born with congenital hypothyroidism usually lack clinical features in the first weeks or months of life and are usually discovered to have congenital hypothyroidism around the age of 2 – 6 months. Studies have shown that if detected and treated within the first week of life, it will result in average, normal or near-normal intellectual performance and growth.

Table 12.1: Causes and Birth Prevalence of Neonatal Thyroid Dysfunction

Disorder	Prevalence
Permanent disorder 1. Thyroid dysgenesis (agenesis, hypoplasia, ectopia) 2. Thyroid dyshormonogenesis 3. Hypothalamic-pituitary hypothyroidism 4. Generalised resistance to thyroid hormone	1: 4,500 1: 30,000 1: 100,000 Very rare
Transient disorder 1. Transient hypothyroxinemia (mainly premature infants) 2. Transient primary hypothyroidism (common in areas of iodine deficiency) 3. Transient hyperthyrotropinemia (predominantly seen in Japanese population)	1:2000 Variable Very rare

Often babies with congenital hypothyroidism appear normal at birth. However, the early features include :

- umbilical hernia,
- constipation,
- prolonged jaundice,
- poor feeding,
- inactivity and delayed bone age.

Late features of untreated congenital hypothyroidism

- macroglossia,
- coarse features,
- dry skin and hair,
- hoarse cry,
- delayed development,
- poor growth and mental retardation.

12.3.1 Newborn Screening for Congenital Hypothyroidism

Immediately after delivery, blood from the umbilical cord is collected and sent to the laboratory to screen for cord blood TSH (Refer to **Figure 12.3**). Babies with high TSH ($> 60\text{mIU/L}$) or borderline TSH ($20\text{-}60\text{mIU/L}$) with low FT4 ($< 15\text{pmol/l}$) will need to be retested.

Blood samples for confirmation (retesting) should be venous samples and should be taken from the baby after the third day of life. This is to avoid the TSH surge that occurs from half an hour after birth to 72 hours of age. Babies for retesting are those with high TSH ($> 60\text{mIU/L}$) or borderline TSH ($20\text{-}60\text{mIU/L}$) with low FT4 ($< 15\text{pmol/l}$).

12.3.2 Treatment

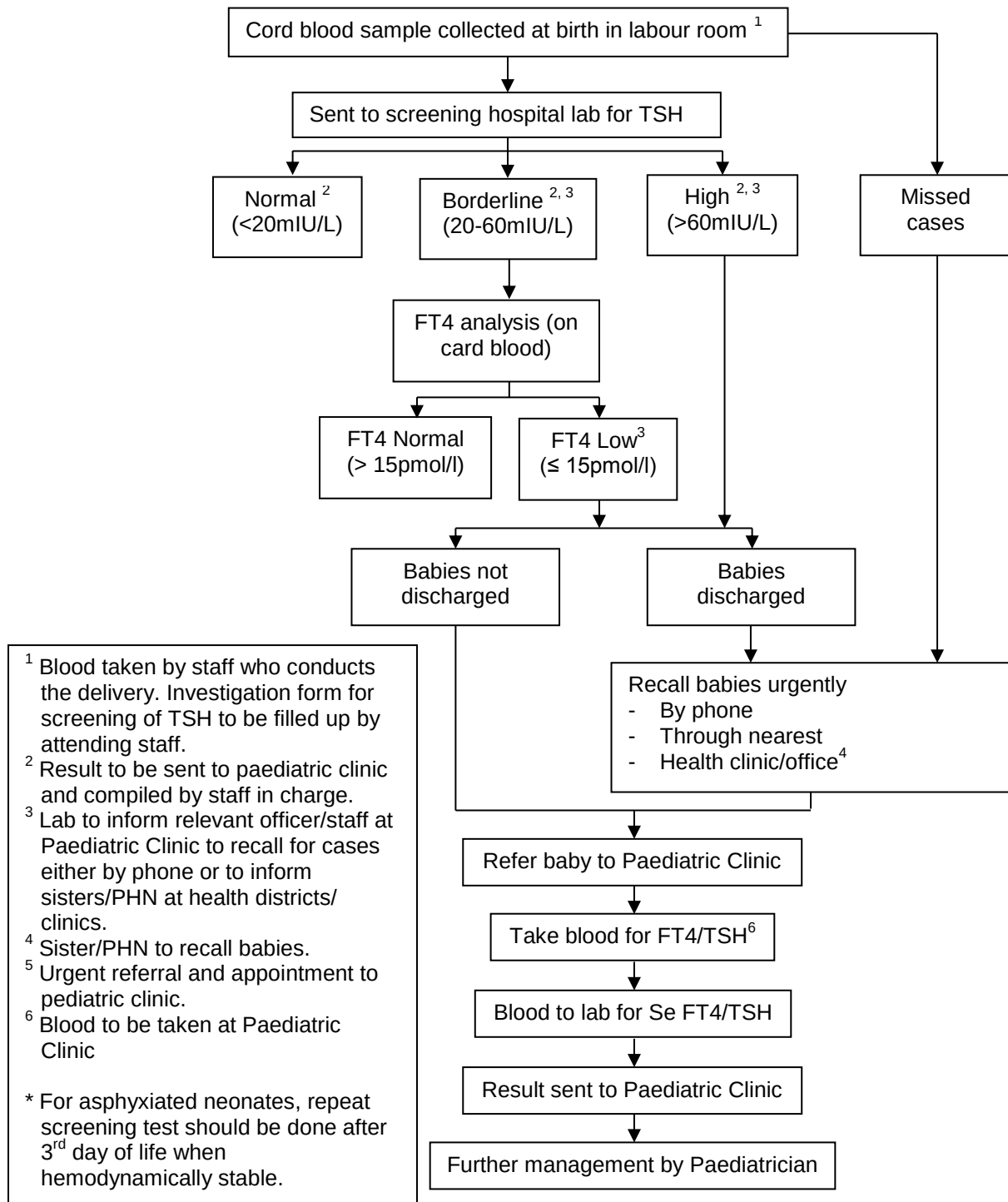
Treatment should begin immediately after diagnosis, and within 2 weeks of life. If features of hypothyroidism are present, treatment is to be started urgently. L-thyroxine is started at a dose of $10 - 15\text{ mcg/kg/dose}$ daily.

The goal of therapy is to restore euthyroid state by maintaining a serum FT4 level at the upper half of the normal age-related reference range. Ideally serum TSH levels should be between $0.5 - 2.0\text{ mIU/L}$ after the first month of life.

Table 12.2: Time interval for follow-up and thyroid function test

Age of patient	Intervals for Thyroid Function Test
After initiation of L-thyroxine	1 - 2 weeks (until normalization of results)
1 - 6 months	1 - 2 monthly
6 months – 3 years	3 – 4 monthly
>3 years until growth is complete	5 – 12 monthly
Note : More frequent intervals is needed if compliance is questionable or abnormal TFT values are obtained, and 4 – 6 weeks after any change in L-thyroxine dose/formulation	

Figure 12.3: Flowchart for Congenital Hypothyroidism Screening at Hospital with T4/TSH Screening Facilities



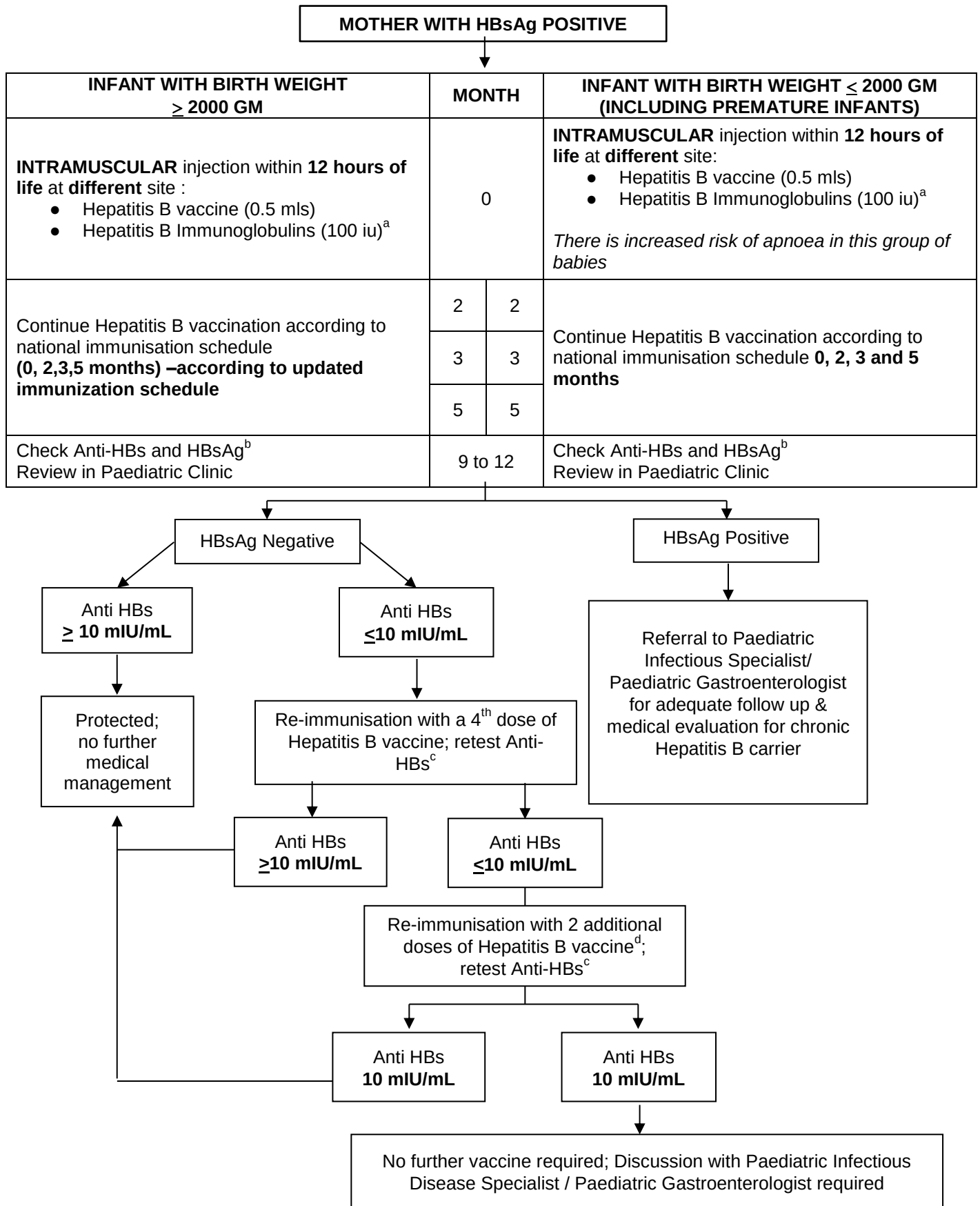
For further information, please refer to National Screening Programme for Congenital Hypothyroidism 2018, Family Health Development Division, MOH.

12.4 ADMINISTRATION OF HEPATITIS B PROPHYLAXIS, BCG VACCINATION AND VITAMIN K IN THE NEWBORN

12.4.1 Recommendations For Hepatitis B Vaccination And Hepatitis B Immunoglobulin In Newborns

- Hepatitis B vaccination is the mainstay effort in the prevention of HBV transmission.
- ALL infants weighing > 2 kg should be given 5mcg (0.5ml) (IM) Hep B vaccination at 0, 1, 6 months of life.
- Infants born to HBsAg-positive mother should receive Hepatitis B vaccine and 100iU (0.5ml) IM HepB immunoglobulin (HBIG) within 12 hours of birth, administered at different injection sites (e.g., anterolateral aspects of opposite thighs). Thereafter, to give Hepatitis B vaccination at 1, 6 months according to schedule. HBsAg and anti-HBs should be performed after completion of the vaccine series at age 9–12 months.
- For infants weighing <2kg, the birth dose should be given, but should not be counted as part of the vaccine series because of the potentially reduced immunogenicity of HepB vaccine in these infants; 3 additional doses of vaccine (for a total of 4 doses) should be administered beginning when the infant reaches age 1 month.
- Infants who are born to HBsAg-positive mothers and receive post-exposure prophylaxis may be breastfed beginning immediately after birth.

Figure 12.4 : Management of infant HBsAg positive mothers



^a Given at same limb as Vitamin K but at different area

^b Test at 9 to 12 months of age, generally at the next well child visit after completion of the primary series

^c Retest Anti-HBs 4 to 8 weeks after re-immunisation

^d Additional 2 doses given 1 to 2 months apart

Source: Redbook 2018 (Prevention of Hepatitis B Virus Infection in the United States: Recommendations of the Advisory Committee on Immunization Practices Recommendations and Reports / January 12, 2018 / 67(1);1–31)

Notes:

- All babies delivered at home should receive the above recommended regime from the nearest health facility.
- Presently it is not the MOH's policy to routinely screen all mothers for their Hepatitis B status.

12.4.2 Recommendations for BCG Vaccination in Newborns

- All newborns are to be given BCG soon after birth. This is usually carried out in well babies on the second day or just before discharge.
- For babies who are admitted directly to the neonatal ward after birth, BCG is often not given until the baby is due for discharge from the special care nursery. Being a live vaccine it is not recommended to be given within a neonatal intensive care unit where babies are ill or immature.
- There are no specific weight criteria for BCG vaccination. It has been shown that babies of 34-35 weeks post-conceptual age can be effectively vaccinated and comparable to vaccination at term.

12.4.3 Recommendations for Vitamin K Administration in The Newborn

- All newborns should receive a single intramuscular dose of Vitamin K within 1 hour of birth. Administer 0.5 mg if the baby weighs less than 1.5 kg or 1mg if the baby weighs more than 1.5 kg.
- If the injection has not been given in the labour room or other location of birth it must be given as soon as possible in the neonatal ward.
- For home deliveries, Vitamin K can also be given at the health clinics if the attending nurse has not given it immediately at birth.

Table 12.3 : National Immunisation Schedule

Age of Child	Vaccinations
At Birth	First BCG injection Birth dose: Hepatitis B
2 months	First dose: DTaP, IPV, Hib, Hepatitis B
3 months	Second dose: DTaP, IPV, Hib, Hepatitis B
4 months	First dose : Pneumococcal
5 months	Third dose : DTaP, IPV, Hib, Hepatitis B
6 months	Hepatitis B Measles (Sabah) Second dose: Pneumococcal
9 months	First dose MMR First dose: JE (Sarawak)
12 months	Second dose of MMR
15 months	Booster dose Pneumococcal
18 months	Booster dose: DTaP, IPV, Hib, Hepatitis B
21 months	Booster dose: JE (Sarawak)

In the cases of vaccine refusal, refer “*Garis panduan Imunisasi Kebangsaan Bayi dan Kanak-Kanak 2016*” by Ministry of Health.

12.5 COMMON NEONATAL ISSUES/CONDITIONS

12.5.1 Neonatal Jaundice

Excerpts from Malaysian CPG on Management of Neonatal Jaundice 2nd edition 2015 and the Integrated Plan for Management of Neonatal Jaundice 2nd Revision 2017.

Key messages:

- Neonatal jaundice (NNJ) is common in newborn babies. Severe NNJ can lead to acute & chronic bilirubin encephalopathy.
- NNJ within 24 hours of life is abnormal and needs urgent attention.
- Assess all babies for jaundice at every opportunity. Methods include visual assessment, transcutaneous bilirubinometer (TcB) or total serum bilirubin (TSB)
- The adequacy of breastfeeding, weight & hydration status of all babies should be assessed during the first week of life. Refer babies with weight loss $\geq 7\%$ of birth weight for further evaluation. See **Chapter 13** on weight monitoring
- Screen all babies for Glucose-6-phosphate dehydrogenase (G6PD) deficiency. Babies with G6PD deficiency should be admitted for the first five days of life.
- Start phototherapy when TSB reaches the phototherapy threshold. See **Table 12.4** below. The threshold is lower in preterm & low birth weight babies.
- Consider exchange transfusion (ET) when TSB reaches the ET threshold. This should follow a standardised protocol & be supervised by experienced personnel.
- Babies discharged <48 hours after birth should be seen by a health care provider in an ambulatory setting or at home within 24 hours of discharge.
- Continue breastfeeding in babies with jaundice. Provide adequate lactation support to all mothers, particularly those with preterm babies.
- Babies with acute bilirubin encephalopathy (ABE) should have long-term follow-up to monitor for neurodevelopmental sequelae. Auditory Brainstem Response testing should be done within the first three months of life.

Prevention of severe neonatal jaundice (antenatal and postnatal visits):

- Health education to parents on NNJ: prevention, detection and treatment
- Identification of risk factors for severe NNJ
- Early detection of NNJ

- Early referral for further assessment or treatment
- Breastfeeding should be encouraged. Supportive measures should be present to promote successful breastfeeding. Supplementation may be needed temporarily to ensure adequate hydration. Supplementing with water or dextrose water does not lower bilirubin level in healthy, jaundice and breastfeeding babies. See **Chapter 13** on weight monitoring
- All babies should be screened for G6PD deficiency. Ideally G6PD status should be known before discharge. If G6PD deficient/ intermediate, it is recommended that the babies are observed for at least 5 days. If they are discharged earlier, they should be followed-up closely.
- Babies of mothers with blood group “O” with a sibling who had severe neonatal jaundice should be observed for at least the first 24 hours of life.
- PredischARGE screening in hospitals using TcB or narrowing down to specific groups e.g. blood group “O” mother may be considered.

Risk factors for severe NNJ

Risk factors for severe NNJ should be identified during antenatal or postnatal as a guide to decide which group of babies to monitor closely, rather than to decide for treatment:

- Prematurity
- Sepsis
- Baby of diabetic mother
- Onset of jaundice before 24 hours of life
- A sibling with severe neonatal jaundice or exchange transfusion
- Inadequate breastfeeding/ dehydration (as shown by $\geq 10\%$ weight loss)
- Mothers with blood group O/ Rhesus negative
- G6PD deficiency
- Asphyxia
- Rapid rise of total serum bilirubin
- Cephalohaematoma or bruises

Indications for referral to hospital

1. Onset of jaundice within 24 hours of life
2. TSB reaches the phototherapy threshold according to the TSB level for phototherapy (**Table 12.4**).
3. Rapidly rising TSB:
 - a. More than $103\mu\text{mol/L/day}$ ($> 6\text{mg/dL/day}$) – indication for phototherapy

- b. More than 8.5µmol/L/hour (> 0.4mg/dL/hour) – indication for exchange transfusion
- 4. Clinical jaundice below umbilicus (if till the soles of the feet - urgent referral for possibility of ET)
- 5. G6PD deficiency (if not previously hospitalised)
- 6. Clinical symptoms/signs suggestive of sepsis

Effective phototherapy consists of:

- Phototherapy in the blue light range (wavelengths of about 400-500nm)
- A minimum irradiance of 15uW/cm²/nm for conventional phototherapy
- A minimum irradiance of 30uW/cm²/nm for intensive phototherapy
- A distance of light source not more than 30-50cm from top surface of the baby
- An adequately exposed baby.

Care of babies during phototherapy:

- Babies should be regularly monitored for vital signs including temperature, hydration status and urine output.
- Babies' eyes should be covered to prevent retinal damage
- Breastfeeding should be continued.

Discontinue phototherapy when:

- TSB is below the conventional phototherapy level. For healthy term babies who are 96 hours of life and older, discontinue phototherapy if only serum bilirubin is below 280µmol/L.

Follow-up

- All babies discharged less than 48 hours after birth should as far as possible be seen by a healthcare provider in an ambulatory setting or at home within 24 hours of discharge. On discharge, the local health worker should be informed and the parents advised accordingly.
- Babies with severe/ pathological jaundice who are discharged in the first 5 days of life, early follow-up is needed to detect rebound jaundice.
- Babies with acute bilirubin encephalopathy should have long-term follow-up to monitor for neurodevelopmental sequelae.
- Term and late preterm babies with TSB > 20mg/dL(342 µmol/L) and those who require exchange transfusion should have Auditory Brainstem Response (ABR) testing done, preferably before discharge or within the first month of life. They should continue the Audiology follow-up until 3 years old, to monitor for any development/ progression of hearing impairment.

- Healthy term and late preterm babies with non-haemolytic hyperbilirubinemia, normal hearing assessment, and TSB < 25mg/dL (428µmol/L) may be followed-up at the primary care level.

Postnatal Home visit

- Home visit for all newborns should be as schedule for postnatal care. Special attention for jaundice must be taken for the first five days of life.
- If the jaundice is detected but not indicated for admission or phototherapy, daily visits should be conducted to monitor the severity of jaundice, especially if a risk factor is present (see above).

Table 12.4: TSB levels for phototherapy & ET in babies' ≥35 weeks' gestation

Age	LOW RISK > 38 weeks & well		MEDIUM RISK >38 weeks with risk factors or 35-37 weeks + 6 days & well		HIGH RISK 35-37 weeks + 6 days with risk factors	
Hours of life	Conventional Phototherapy – TSB in mg/dL (µmol/L)	ET – TSB in mg/dL (µmol/L)	Conventional Phototherapy – TSB in mg/dL (µmol/L)	ET – TSB in mg/dL (µmol/L)	Conventional Phototherapy – TSB in mg/dL (µmol/L)	ET – TSB in mg/dL (µmol/L)
24	9 (154)	19 (325)	7 (120)	17 (291)	5 (86)	15 (257)
48	12 (205)	22 (376)	10 (171)	19 (325)	8 (137)	17 (291)
72	15 (257)	24 (410)	12 (205)	21 (359)	10 (171)	18.5 (316)
96	17 (291)	25 (428)	14 (239)	22.5 (385)	11 (188)	19 (325)
>96	18 (308)	25 (428)	15 (257)	22.5 (385)	12 (205)	19 (325)

- Start intensive phototherapy at a TSB level of 3 mg/dL (51 µmol/L) above the level for conventional phototherapy or when TSB increasing (increment rate) at >0.5 mg/dL/hour (8.5 µmol/L/hour).
- Risk factors are isoimmune haemolytic disease, G6PD deficiency, asphyxia & sepsis.

12.5.2 Prolonged Neonatal Jaundice

All babies with prolonged neonatal jaundice need to be clinically assessed and investigated. Infants with suspected biliary atresia have to be referred to the paediatric surgeon before 2 months of age for surgery. Below are the excerpts from the *Integrated Plan for Management of Neonatal Jaundice 3rd edition* and *Paediatric Protocols for Malaysian Hospital 4th edition*.

Definition

Visible jaundice (SB >85 µmol/L or 5 mg/dL) that persists beyond 14 days of life in a term baby (≥ 37 weeks) or 21 days in a preterm baby (≥35 weeks to < 37 weeks).

Causes of prolonged neonatal jaundice

- It may be unconjugated or conjugated hyperbilirubinaemia.
- Conjugated hyperbilirubinaemia is defined as the direct (conjugated) fraction of bilirubin more than 34 $\mu\text{mol/L}$ (2mg/dL), or more than 15% of the total bilirubin.

Table 12.5: Causes of Unconjugated and Conjugated Hyperbilirubinemia

Unconjugated Hyperbilirubinemia	Conjugated Hyperbilirubinemia*
Septicaemia/ Urinary Tract Infection Hypothyroidism Haemolysis Breast milk jaundice (most common, benign condition)	Biliary Atresia Neonatal hepatitis syndrome Choledochal Cyst Septicaemia/ Urinary Tract Infection Congenital Infection (TORCHES) Inborn Error of Metabolism TPN-induced cholestasis

**All babies with conjugated hyperbilirubinemia must be referred to a paediatric department urgently to exclude biliary atresia.*

Initial Approach and Management

- All babies MUST be assessed or screened for prolonged neonatal jaundice (PNNJ) at day 14 for term babies and day 21 for preterm babies.
- Even if the baby has resolved neonatal jaundice and reducing trend of serum bilirubin (SB), the baby must be reassessed for jaundice again at Day 14 of life.
- In the presence of prolonged neonatal jaundice (regardless of SB level), the baby must be referred to a medical officer (in any hospital or health clinic) the same day or the next working day for clinical assessment and further laboratory workup, using the PNNJ clerking sheet as a guide.
- Risk stratification into high, moderate or low risk groups are recommended (**Figure 12.5**). This is done by:
 - a. Clinical assessment
e.g. feeding method, weight, STOOL COLOUR and presence of hepatosplenomegaly **AND**
 - b. Important laboratory investigations
i.e. serum bilirubin with direct and indirect bilirubin.
- Subsequent management of these babies will depend on the risk groups.
- Refer to Paediatric Team if conjugated hyperbilirubinaemia, signs of obstructive jaundice (pale stool, dark urine and hepatosplenomegaly)

SB > 300 µmol/L, abnormal lab results, jaundice more than 2 months or any other features in the high or moderate risk category.

Figure 12.5: Flowchart for the Management of Prolonged Neonatal Jaundice in Babies ≥ 35 Weeks

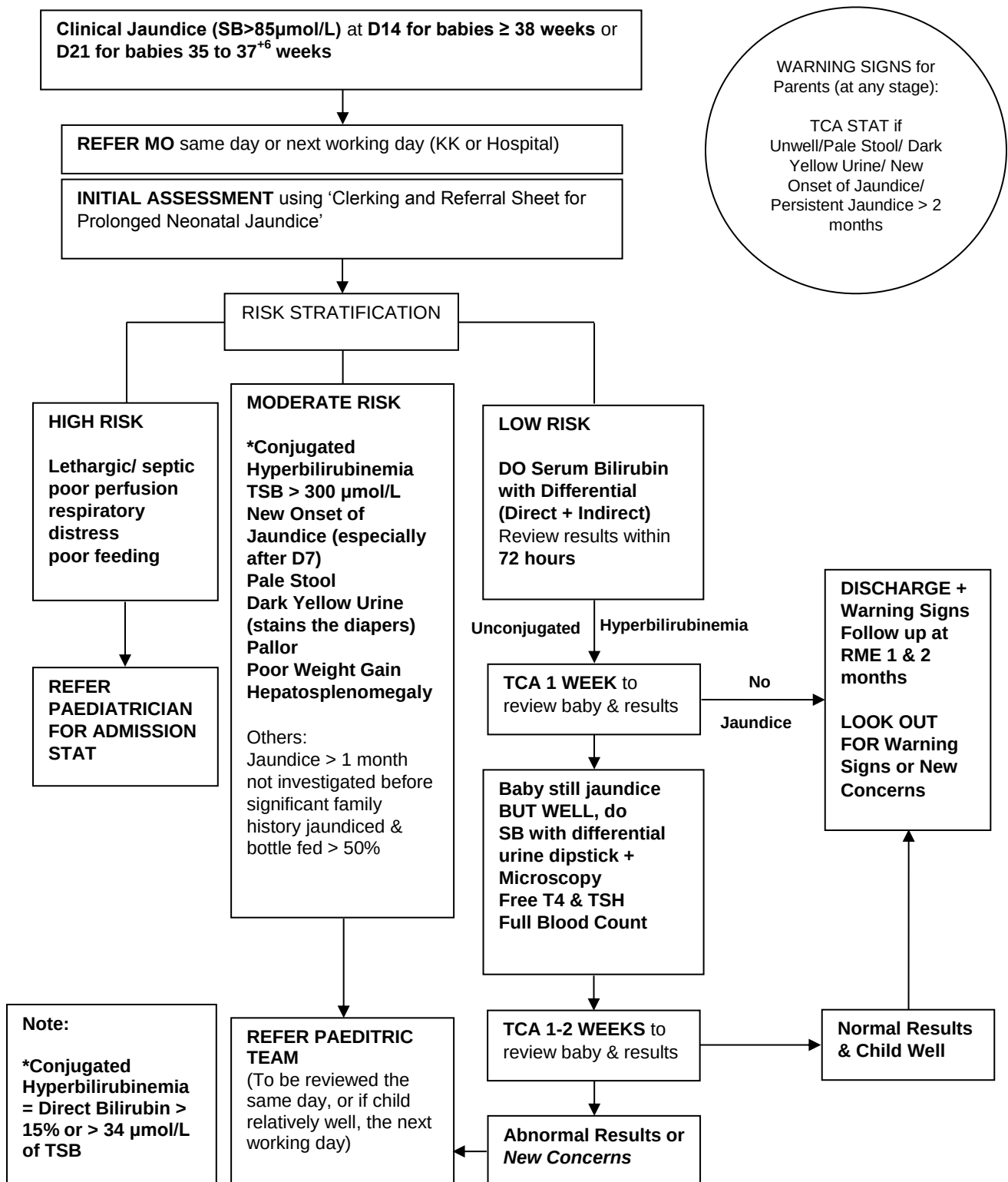


Table 12.6 : Management of Prolonged Neonatal Jaundice for Babies ≥ 35 weeks Gestation by Risk Groups at the point of diagnosis in any health facilities

HIGH RISK	MODERATE RISK	LOW RISK
Positive Clinical Features/ Lab Results • Ill/ Septic Looking • Respiratory distress • Poor feeding • Lethargy • Poor perfusion	Positive Clinical Features/ Lab Results • Conjugated Hyperbilirubinaemia • Severe Jaundice- TSB>300 µmol/L • New Onset Jaundice (esp after Day 7) • Pale Stools • Dark Yellow Urine (stains diapers) • Poor Weight Gain • Hepatosplenomegaly To also consider: • Bottle fed > 50% • Jaundice > 1 month not investigated before • Other suspected medical condition • Significant family history	Positive Clinical Features/ Lab Results • None • Well babies with good weight gain, exclusively breast fed (or > 50%), bright yellow stool with normal physical examination • Breast milk jaundice
		Management Can be managed and followed up at primary care level or hospitals without specialists. Term babies > 37 wks <u>Day 14:</u> • S. Bilirubin with Direct/ Indirect bilirubin, • Full Blood Count • UFEME + microscopy • Free T4, TSH
Management • Stabilise Airway, Breathing, Circulation • Refer to Paediatrician immediately	Management • Refer to Paediatric Team • Same day or next working day	Preterm babies ≥ 35 < 37 weeks <u>To work up 1 week later than term babies.</u>
Well, low risk babies DO NOT need heel prick capillary bilirubin till jaundice resolves. Warning signs* for parents and RME (routine medical examination) at 1 month and 2 months in health clinics, looking at the same clinical features will serve as a good safety net. Refer to Paediatric Team if conjugated hyperbilirubinaemia, warning signs*, SB> 300 µmol/L, abnormal lab results, jaundice more than 2 months or any features in the high or moderate risk category. *Unwell, pale stool, dark yellow urine, new onset of jaundice, persistent jaundice > 2 months		

12.5.3 Neonatal Hypoglycaemia

There is no single plasma glucose concentration or duration of hypoglycaemia that can predict permanent neurologic injury in high-risk infants. Clinical hypoglycaemia is defined as a plasma glucose concentration low enough to cause symptoms and/or signs of impaired brain function, **commonly accepted at <2.6 mmol/L in a term or preterm infant.**

- In term infants < 4 hours old, plasma glucose > 1.5 mmol/L is acceptable if the infant is well, asymptomatic and tolerating feeds and repeat glucose is > 2.6 mmol/L
- For infants > 48 hours old, it is recommended to keep plasma glucose level > 3.3 mmol/L to be above the threshold for neuroglycopenic symptoms.
- For infants with suspected congenital hypoglycaemia disorder or symptomatic infants, keep plasma glucose > 3.9mmol/L.

Infants who are at high risk of hypoglycaemia and require glucose screening:

- Symptoms of hypoglycaemia
- Large-for-gestational age (LGA) - even without maternal diabetes
- Intrauterine growth retardation/ small-for-gestational age (SGA)
- Perinatal stress: birth asphyxia, meconium aspiration syndrome, hypothermia, polycythaemia, sepsis
- Premature or postmature delivery
- Family history of a genetic form of hypoglycaemia
- Congenital syndromes (e.g.; Beckwith-Wiedemann), or abnormal physical features (e.g.; Midline facial malformations, microphallus).

Clinical signs of hypoglycaemia

- Jitteriness
- Cyanosis
- Seizures
- Apneic episodes
- Tachypnoea
- Weak or high-pitched cry
- Floppiness or lethargy
- Eye-rolling
- Poor feeding

Indications for Referral to Paediatric Team

- Babies of diabetic mothers on insulin treatment (or metformin)
- LGA (birth weight $\geq 4.2\text{kg}$) or SGA (birth weight $< 2\text{kg}$) babies
- Sick babies
- Symptomatic babies
- Blood glucose $< 1.5 \text{ mmol/L}$
- Blood glucose persistently low ($< 2.6 \text{ mmol/L}$)

Management of High Risk Infants

Blood glucose concentration should only be measured in symptomatic babies, or who are known to be at risk of hypoglycaemia.

1. Identify at risk babies
2. Well babies who are at risk
 - Immediate feeding within 1 hour of birth - first feed can be given in the labour room.
 - Supplement feeding until breastfeeding is established
 - Initial blood glucose should be done 30 minutes after the first feed.
3. Any sick babies:
 - Refer to Paediatric Team
 - Check blood glucose on admission
 - Set up IV dextrose 10% drip at the rate of 60ml/kg/day
4. Regular glucometer monitoring:
 - 3 - 6 hourly pre-feeding for 12 -24 hours

Management of Hypoglycaemia

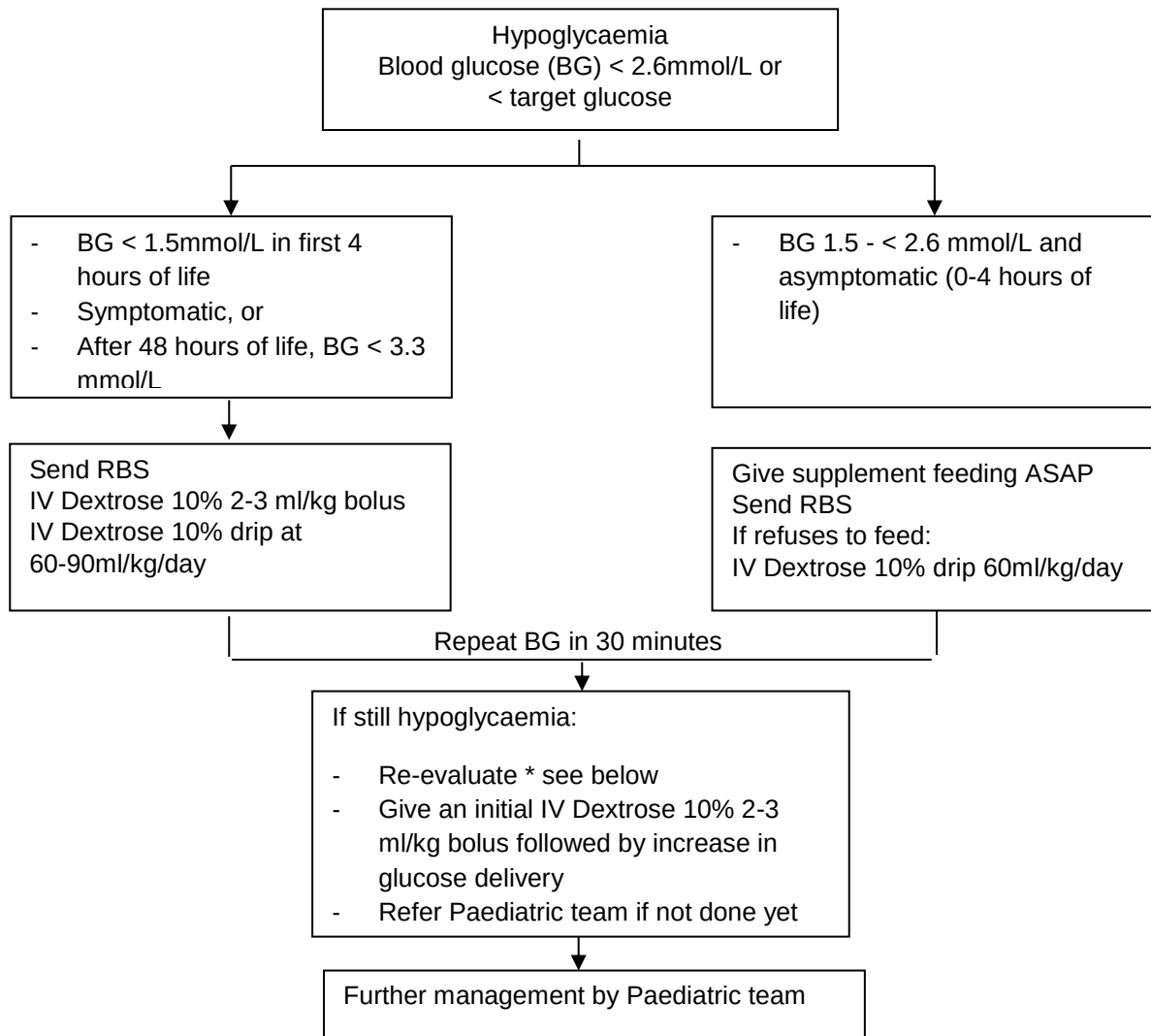
1. Repeat blood glucose (glucometer, dextrostix) and send for plasma glucose levels (RBS) stat.
2. Examine and document any symptoms.
3. Note when last feeding was given.
4. If on IV drip, check that IV infusion of glucose is adequate and running well.
5. If blood glucose is $< 1.5 \text{ mmol/L}$ in the first 4 hours of life or if the infant is symptomatic:
 - Inform Paediatric team stat
 - Quickly set up a peripheral intravenous line or umbilical venous line
 - Give IV Dextrose 10% at $2\text{-}3\text{ml/kg}$ bolus
 - Followed by IV Dextrose 10% drip at $60\text{-}90\text{ml/kg/day}$ (for day 1 of life).
 - If the infant is already on IV Dextrose 10% drip, consider increasing the rate or the glucose concentration (usually require $6\text{-}8\text{mg/kg/min}$ of glucose delivery load)

- Calculation of dextrose/ glucose load (or delivery rate):

$$\frac{\text{Rate (ml/hr)} \times \text{Dextrose concentration (\%)}}{\text{Weight} \times 6}$$

6. Within the first 4 hours of life, if blood glucose is 1.5 - 2.5mmol/L and asymptomatic:
 - Give supplementary feed [expressed breast milk (EBM) or formula] as soon as possible.
 - If blood glucose remains < 2.6 mmol/L and the infant refuses feeds, start IV dextrose 10% drip and refer to the paediatric team.
 - If the infant is already on IV Dextrose 10% drip, consider a stepwise increment of glucose infusion rate by 2 mg/kg/min until blood sugar is > 2.6 mmol/L.
 - If blood glucose is below target level, recheck blood glucose every 30 minutes.
 - Once blood glucose is above target level for 2 readings, monitor hourly x 2, then 2 hourly x 2, then to 3-6 hourly pre feeding, if blood glucose remains normal.
7. Start feeding when blood glucose remains stable and increase as tolerated. Reduce the IV Dextrose infusion rate 1 hour after the feeding increment.

Figure 12.7: Management of Persistent Hypoglycaemia



***NOTES**

- Once BG achieves >2.6mmol/L or target BG for 2 readings, monitor hourly twice, then 2 hourly twice, then 3-4 hourly
- If BG <2.6mmol/L or symptomatic, do critical sampling
- For those with risk of congenital hypoglycaemia disorders or symptomatic, aim for target BG > 3.9mmol/L

12.5.4

Common Skin Lesions

- a. Diaper rash (ammoniacal dermatitis)
- b. Candida diaper rash
- c. Seborrheic dermatitis

Diaper rash (ammoniacal dermatitis)

May be caused by prolonged contact with urine or faeces. It can also be an allergic reaction to the diaper material, creams, powders, wipes or detergents used in laundering cloth diapers.

The best treatment is prevention, by frequent diaper changes and by protection of the skin with a barrier product containing zinc oxide. The skin should be cleansed with warm water after voiding or passing motion. Avoid diaper wipes that contain alcohol. Cornstarch and baby powder should not be used as they provide a media for growth of bacteria and yeast. Don't use creams that contain steroids (cortisone or hydrocortisone).

Candida diaper rash

A fungal infection of skin in the diaper area may include buttocks, groins, thighs and abdomen. It is caused by the organism *Candida albicans*. It appears as a moist erythematous eruption often with white or yellow satellite pustules.

Treatment consists of antifungal cream or ointment such as nystatin applied to the rashes several times per day.

Seborrhoeic dermatitis

Seborrheic dermatitis affects the scalp, central face, and anterior chest. Seborrheic dermatitis also may cause mild to marked erythema of the nasolabial fold, often with scaling. The scales are greasy, not dry, as commonly thought. This rash has an erythematous background and a greasy yellow scale. It is common in hair-bearing areas of the body, especially the scalp and eyebrows. It is usually absent in the flexures. Scaling is prominent on the scalp producing the so-called 'cradle-cap'. It has a tendency to recur throughout infancy.

Treatment is with topical application of 1% sulphur + 1% salicylic acid in cream applied overnight and washed off the next day with a mild shampoo, used 3 times a week. Milder cases of cradle cap can be treated with topical olive oil. If severe, the whole body may be affected and should be referred to hospital.

12.5.5 Feeding Problems

Vomiting

Regurgitation or reflux

One of the most common symptoms in the neonatal period is regurgitation of milk during or shortly after feeding. It is usually seen soon after feeding. Unlike vomiting, the amounts are small and the baby does not seem to be in any distress, of normal hydration and thriving. Regurgitation begins in the first weeks of life and clears by one year of age. The cause is incomplete closure of the valve at the upper end of the stomach in the first few months of life. Causes include overfeeding, frequent change of formula, early use of supplementary food, improper feeding technique and posture.

The following tips can be given to the mother, if the regurgitation causes distress to the parents:

- Do not overfeed baby, especially if you are bottle-feeding
- Avoid pressure on the abdomen because it 'squeezes' the stomach. Check that the diapers are not tight when the baby is in a sitting position
- After feeding, try to hold or keep your baby in an upright position for 30-60 minutes
- If you think the regurgitation does not improve over time or the baby has other symptoms as given below, refer to hospital

Signs and symptoms of vomiting where baby must be referred to hospital:

- baby is lethargic, tachypnea
- not feeding well
- having a fever
- vomiting out greenish-coloured fluid (bile) or has a distended abdomen
- looks dehydrated or reduced urine output
- not putting on weight

Colic

Colic is a common problem, affecting 10-25% of all newborns. Colic describes episodes of crying that continuous for hours at times. It is quite common in babies below 3 months of age and the cause is not known. It may be associated with hunger, swallowing large amounts of air, overfeeding or inadequate carbohydrate intake. Colic usually begins at about 2 weeks of age and should resolve by 4 months of age. Breastfed babies are not as likely to get colic as babies who are formula-fed.

Parents are usually stressed and upset when the baby does not stop crying, no matter what they do. Provide reassurance after checking that the baby is healthy.

Some things that parents can do:

- Hold the baby in an upright position during feeding
- If being bottle fed, holding the bottle so that the milk covers the entire opening of the teat
- Burp the baby after feeding to reduce air in the stomach
- Try swaddling (wrapping) the baby in a blanket. Walking with the baby on the shoulder, or rocking the baby sometimes helps to quieten the baby
- A ride in the car, with the baby in a safety car seat, usually works well

IF YOU FEEL THE PARENTS ARE UNABLE TO COPE WITH THE STRESS OR IF THE BABY CANNOT BE COMFORTED IN A REASONABLE AMOUNT OF TIME, REFER TO HOSPITAL

12.5.6 Eye Problem

Nasolacrimal duct obstruction

Term and preterm newborn infants have the capacity to secrete tears (reflex tearing to irritants) but usually do not secrete emotional tears until 2-3 months of age.

Congenital obstruction is usually caused by an imperforate membrane at the distal end of the nasolacrimal duct.

Congenital nasolacrimal obstruction is the most common abnormality of the neonate's lacrimal apparatus. Incidence of this condition ranges between 2% and 6% of all newborn infants. The majority of nasolacrimal obstruction resolves spontaneously or with massage by 1 year of age.

Clinical presentation (usually with the first few weeks of life) :

- Persistent tearing. Need to rule out congenital glaucoma.
- Crusting or matting of the eyelashes (sticky eyes)
- Spilling of tears over the lower lid and cheek (a wet look in the involved eyes)
- Absence of conjunctival infection.
- Mucopurulent discharge

Complications

- Acute dacryocystitis
- Fistula formation
- Orbital or facial cellulitis

Management

- Conservative management – daily massage of nasolacrimal sac to rupture the membrane at the lower end of the duct. Technique – place index finger over common canaliculus and stroke downwards firmly.
- If mucopurulent discharge, antibiotic eye drops or ointment may be required.
- Eye should be cleaned with moist compresses.

Conjunctivitis

Inflammatory reactions resulting from infection of the conjunctiva by pathogenic organisms – e.g. *Neisseria gonorrhoea*, *Chlamydia trachomatis*, *Staphylococcus aureus*, enteric pathogen.

1. Neisseria Gonorrhoeal Conjunctivitis

Bilateral purulent conjunctival discharge within a few days of life. Onset of symptoms usually between the second and fifth day of life. Eye discharge on the first day of life is usually due to gonococcal infection.

Clinical presentation

Oedema of the eyelids, purulent discharge, redness of the conjunctiva.

Diagnostic findings

- Maternal history of sexually transmitted infections
- Physical examination – clinical signs of inflammation, purulent discharge.
- Laboratory – Gram negative diplococci on Gram stain of direct smear.
- Culture positive for *Neisseria gonorrhoea* from conjunctival surface or exudates.

Management

- Isolate baby
- Irrigate eyes with sterile normal saline solution hourly. Refer patient promptly to the hospital for further treatment
- Notify to health authorities concerned

2. Chlamydia Trachomatis Conjunctivitis

Unilateral or bilateral conjunctivitis onset between 5 and 14 days of age.

Clinical presentation - vary from mild conjunctivitis to intense oedema of the lids with purulent discharge.

Diagnostic findings - identification of Chlamydia antigen.

Management - Refer patients to the hospital for treatment.

12.5.7 Umbilical Cord Care

The cord stump remains the major means of entry of infections after birth. The umbilical cord stump usually drops off in 1-2 weeks. Until then, keep it clean and dry.

- The stump will dry and mummify if exposed to air without any dressing, binding or bandage
- Gently clean the area where the cord and the tummy meet at least once a day or when the area is damp.
- It will remain clean if it is protected with clean clothes and is kept from urine and soiling.
- If soiled, the cord can be washed with clean water and dried with clean cotton or gauze. If there is redness or the umbilical area is moist or smells, dip a cotton swab with the alcohol preparation provided by the staff at the time of discharge.
- There may be a spot of blood on the diaper when the stump falls off. If bleeding persists for more than a few days or is more than just spots, bring baby to see a doctor
- If you see pus, redness, or the baby cries when you touch the area, refer to the doctor.

WARNING SIGNS FOR BABIES

The following are warning signs of an ill baby who needs immediate attention by the doctor.

1. Central cyanosis**Table 12.8: Differentiating between central cyanosis from peripheral cyanosis**

	Central cyanosis	Peripheral cyanosis
Site	Tongue, lips, extremities	Extremities only. Tongue is pink.
Causes	Cardiac, pulmonary, profound sepsis	Mainly hypothermia causing peripheral vasoconstriction
Action required	Check SpO ₂ Immediate referral	Check Temperature Warm up the baby as necessary

2. Signs of respiratory distress

- Tachypnoea > 60/min.
- Grunting or stridor
- Intercostal and or subcostal recession
- Flaring of the nostrils
- Cyanosis
- Reduced air entry

3. Signs of sepsis**Table 12.9: Signs of sepsis**

Systems	Signs
Respiratory	Apnoea Tachypnoea Grunting, nasal flaring Recession
Cardiovascular	Tachycardia* Hypotension Bradycardia Poor perfusion
Central nervous	Temperature instability – hypothermia or fever*

Systems	Signs
	Lethargy* Hypotonia Irritability* Seizures
GIT	Feeding intolerance/ poor feeding* Abdominal distension Vomiting* Diarrhoea
Others	Jaundice Pallor Petechiae

*Early signs of sepsis



CHECKLIST APPROACH TO UNWELL CHILDREN UNDER FIVE YEARS THE SICK YOUNG INFANT AGE UP TO 2 MONTH			FH/ATUCUS- checklist 1/2020
Name: _____ Age: _____ Weight: _____ Temperature: _____ °C ASK: What are the child's problems? _____ Visit: 1st / 2nd / 3rd / 4th / 5th ASSESS (Circle all signs present)			
ASK	LOOK AND FEEL	REFER FOR ADMISSION IF PRESENT	
CHECK FOR VERY SEVERE DISEASE AND LOCAL BACTERIAL INFECTION			
- Is the infant not feeding well? - Does the infant vomit greenish vomitus? - Has the infant had convulsion?	- Count the breaths in one minute. _____ breath per minute. Repeat if 60 or more _____ Fast breathing? - Look for severe chest indrawing. - Look at umbilicus. Is it red or draining pus? - Look for skin pustule? - Look at young infant's movements Movement only when stimulated? No movement at all? - Check temperature: * Fever ($> 37.5^{\circ}\text{C}$) * Low body temperature (below 35.5°C)	- Not able to feed - Feeding poorly - Greenish vomitus - Convulsions - Fast breathing - severe chest indrawing - Fever ($> 37.5^{\circ}\text{C}$) or low body temperature (35.5°C) - Movement only when stimulated or no movement at all	
CHECK FOR JAUNDICE (Jaundice present / No Jaundice)			
- When did the jaundice first appear? Before 24 hours of life / after 24 hours of life - Is the infant more than 14 days old?	- Look for level of jaundice * Jaundice palms and sole * Jaundice below umbilicus * Jaundice above umbilicus - Look at stool colour - Check TSB if jaundice present	- Jaundice appearing before 24hrs of age - Jaundice palm and sole at any age - TSB above photolevel - Pale stool	
DOES THE CHILD HAVE DIARRHOEA? (YES / NO)			
- For how long? _____ Days - Is there blood in the stools?	- Look at the young infant's general condition. Move only when stimulated or not move at all? Restless or irritable? - Look for sunken eyes. - Pinch the skin of the abdomen. Does it go back: Slowly OR Immediately	2 or more signs * movement only when stimulated or no movement * restless or irritable * sunken eyes * skin pinch goes back slowly	
CHECK THE YOUNG INFANT'S IMMUNIZATION STATUS (IMMUNIZATION SCHEDULE)			
AGE Birth BCG 1 month	VACCINE Hep B 1 Hep B 2	**Vit K	
<i>If the infant has no indications to refer urgently to hospital</i> CHECK FOR FEEDING PROBLEM OR LOW WEIGHT			
- Is the infant breastfed? Yes _____ No _____ - If Yes, how many times in 24 hours? _____ times - Does the infant usually receive any other foods or drinks? If Yes, please specify _____ If Yes, how often? _____ - Check feeding hygiene and preparation			
ASSESS BREASTFEEDING			
- Has the infant breastfed in the previous hour? - If infant has not fed in the previous hour, ask the mother to put her infant to the breast. Observe the breastfeed for 4 minutes. - Is the infant able to attach? To check attachment, look for : - More areola seen above infant's top lip than below bottom lip Yes _____ No _____ - Mouth open wide open Yes _____ No _____ - Lower lip turned outwards Yes _____ No _____ - Chin touching breast Yes _____ No _____ not well attached good attachment - Is the infant suckling effectively (that is, slow deep sucks, sometimes pausing) not suckling effectively suckling effectively - Clear a blocked nose if it interferes with breastfeeding.			
Assess other problem			

Reference: Training Manual on Approach to Unwell Children under 5 years

CHAPTER 13

BREASTFEEDING AND WEIGHT MONITORING

13.1 BREASTFEEDING**13.1.1 Promoting Breastfeeding**

WHO/UNICEF Global Strategy for Infants and Young Child Feeding (2003) recommended that infants should be exclusively breastfed for six months to achieve optimal growth development and health. This recommendation is in line with the Ministry of Health Malaysia Breastfeeding Policy.

All hospitals under the Ministry of Health have achieved the Baby Friendly Hospital Initiative (BFHI) Status. It is recognised that there is a need to ensure continuity of practises as listed in the 10 steps plus 3 items to successful Breastfeeding. To ensure the sustainability of this programme, all hospitals will be assessed every 2 to 3 years by the accredited BFHI assessors.

13.1.2 Early Initiation of Breastfeeding

Immediate and uninterrupted skin-to-skin contact and initiation of breastfeeding within the first hour after birth are important for the establishment of breastfeeding. Early suckling is important for stimulating milk production and establishing the maternal milk supply. All mothers should be supported to initiate breastfeeding as early as possible after birth. Early initiation of breastfeeding will ensure the success of breastfeeding for 6 months and also will ensure the baby gets enough colostrum.

Initiation of breastfeeding at birth checklist for normal and full term delivery

- Put to breast as soon as possible after birth for skin-to-skin contact
- Initiate breastfeeding in the first hour of life
- Check baby's positioning and attachment
- Keep baby warm

13.1.3 Support with Breastfeeding

Mothers should receive practical support which includes providing emotional and motivational support, imparting information and teaching concrete skills to enable them to initiate and maintain breastfeeding and manage common breastfeeding difficulties.

13.1.4 Breastfeeding Problems and Management

Problems	Remedial Measures
Sore nipple	• Help mother to improve attachment and positioning • Show the mother how to feed in different feeding position • Treat skin condition or remove source of irritation • Treat candida both on mother's nipple and baby's mouth
Breast engorgement	• Warm compress or warm shower • Massage neck and back • Light massage of breast • Help mother to relax • Provide supportive atmosphere • Cold compress on the breast if oedema present • If baby is able to suckle : - Feed frequently, help with positioning • If baby is not able to suckle : - Express the breast
Block duct and Mastitis	• Advise mother : - Breastfeeding frequently - Proper attachment - Gentle massage the blocked and tender area down toward the nipple before and during the feeds - Apply a moist, warm cloth to the area before a breastfeed - Check that her clothing especially her bra, does not have a tight fit • Analgesia to reduce pain • Antibiotic therapy if indicated
Breast abscess	• Needs surgical incision and antibiotic • Continue breastfeeding if: - Incision far enough from areola and does not interfere - Mother tolerate pain - Otherwise express milk from affected side - Continue breastfeeding from unaffected breast
Insufficient milk	• Start early feeding • Frequent demand feeding • Good breastfeeding technique – good attachment • Express breast milk every two to three hourly if baby is not able to suck well • Not to use pacifier or bottle feeding

13.1.5 Checklist for Initiation of Breastfeeding at Birth (Normal, Full Term Delivery)

- Put to breast as soon as possible after birth for skin to skin contact
- Initiate latching to the breast within half to 1 hour
- Check baby's attachment and suckling
- Keep baby warm

13.1.6 Checklist for Assessing Progress of Breastfeeding (Prior to Discharge)

- Correct body position
- Correct attachment and sucking
- Feeding on demand
 - No limits on frequency or duration
 - Night feeding
- Emotional bonding
 - Secure, confident hold
 - Face to face attention from mother
 - Close contact with mother
- Breastfeeding problems are attended to:
 - E.g. inverted nipples; breast engorgement

13.1.7 Checklist for Indicators of Sufficient Breast Milk Intake (At Second Week of Life)

- Weight gain :
 - Good weight gain according to growth chart, or 25 gram/day
 - In the first 7 - 10 days of life, babies lose weight. Babies regain their birth weight by the second week, double this by 5 months of age, and triple the birth weight by 1 year of age
 - At day 5-6, if weight loss is more than 10% - refer to hospital for possible hypernatraemic dehydration
 - if weight loss is between 5 – 10 % - refer baby to medical officer at nearest health clinic for review. Baby to be monitored until weight gain is obtained
- Stools: 5 to 10 yellow milk stools per day
- Urine: 6 to 8 wet diapers per day, urine looks clear, not dark or concentrated
- Baby's behaviour: Baby acts hungry at times and appears satisfied after feeding, and generally calm and relaxed during feeding.

- Baby's general condition: Baby has normal skin colour, mucous membrane wet with good skin turgor, alert, and good tone.
- Number of feeds: At least 8-12 feeds per day.
- Length of feeds: Feeding for 5-40 minutes at most feeds
- End of feeds: Baby lets go spontaneously, or does so when breast is gently lifted.
- Sucking pattern: Initial rapid sucks changing to slower sucks with pauses and soft swallowing.

13.1.8 Care for Mothers at Discharge

Mothers need sustained support to continue breastfeeding. Coordinate discharge so that parents and their babies have timely access to ongoing support and care

- Reevaluate mothers's knowledge and skills on breastfeeding
 - Correct body position
 - Correct attachment and sucking
 - Breastfeeding on cues or baby-led feeding
- Understand the importance of breastfeeding exclusively for six months
- Breastfeeding problems addressed to, e.g. inverted nipples; breast engorgement
- Inform mother where to get help if she has breastfeeding problem – Healthcare Facilities, NGO & Breastfeeding Support Group
- Notification to health staff of all hospital discharges especially high-risk cases

13.2 ACCEPTABLE MEDICAL REASONS FOR SUPPLEMENTATION TO BABIES BELOW SIX MONTHS OF AGE

13.2.1 Infant's conditions

- Infants who should not receive breast milk or any other milk except specialised formula. This applies to babies with inborn errors of metabolism eg:
 - Classic galactosemia : a special galactose-free formula is needed
 - Maple syrup urine disease : a special formula free of leucine, isoleucine and valine is needed
 - Phenylketonuria : a special phenylalanine-free formula is needed (some breastfeeding is possible, under careful monitoring)

- ii. Infants for whom breast milk remains the breastfeeding options but who may need other food in addition to breast milk for a limited period
 - Very low birth weight infants (those born weighing less than 1500g)
 - Very preterm infants, i.e. those born less than 32 week gestational age
 - Newborn infants who are at risk of hypoglycaemia by virtue of impaired metabolic adaptation or increased glucose demand (such as those who are preterm, small for gestational age or who have experienced significant intrapartum hypoxic/ischaemic stress, those who are ill and those whose mothers are diabetic) if their blood sugar fails to respond to optimal breastfeeding or breast milk feeding;
 - Infants younger than 6 months who, in spite of frequent and effective sucking and in the absence of illness, show persistent growth faltering (as demonstrated by a flat or downward growth curve).

The mothers of the above babies must be encouraged to establish early bonding with their babies and given more support with breastfeeding.

13.2.2

Maternal Conditions

The mother who is affected by any of the conditions mentioned below should receive treatment according to standard guidelines.

- i. **Mother who should avoid breastfeeding PERMANENTLY**
 - HIV infection
 - HTLV-I (Human T-cell Leukaemia virus)
- ii. **Mother who should avoid breastfeeding TEMPORARILY**
 - Severe illness that prevents a mother from caring for her infant, for example septicaemia
 - Herpes simplex virus type 1 (HSV-1): direct contact between lesions on the mother's breasts and the infant's mouth should be avoided until all active lesions have resolved
- iii. **Mother who CAN CONTINUE BREASTFEEDING, although the health problems may be of concern**
 - Breast abscess: breastfeeding should continue on the unaffected breast; feeding from the affected breast can resume once the abscess has been drained and antibiotic treatment has started

- Mother with Hepatitis B : infants should be given hepatitis B vaccine, within the first 48 hours or as soon as possible thereafter
- Hepatitis C infection in the mother is not a contraindication to breastfeeding. There is no current data to suggest that HCV is transmitted by human breast milk.
- Mastitis : If breastfeeding is very painful, milk must be removed by expression to prevent progression of the condition;
- Tuberculosis : If active pulmonary disease or miliary TB has been recently diagnosed, it has been recommended that mother and baby be separated until mother is sputum negative or mother has been given anti TB therapy (usually at least two completed weeks) then mother can breastfeed. The newborn should receive isoniazid prophylaxis for 6 months followed by BCG

When breastfeeding has to be temporarily delayed or interrupted, mothers should be helped to establish or maintain lactation e.g through hand expression of milk, in preparation for breastfeeding to be resumed.

13.2.3 Medications and other substances that can adversely affect the breastfed infant

Maternal medication

Risks are greater during the first 2 months on high dosages of medications (as therapy or with abuse). Monitor infants for adverse effects. The use of low doses usually requires no special precautions in older infants.

- Sedating psychotherapeutic drugs, anti-epileptic drugs and opioids and the combinations may cause side effects such as drowsiness and respiratory depression in neonates; use less sedating alternative and low dosages whenever possible;
- Sulphonamides, chloramphenicol, tetracyclines – small risk of side effects : use alternative drugs if possible
- Oestrogen (including oestrogen containing contraceptives), thiazide diuretics, - may reduce milk supply, use alternative drugs.

- Mothers on antithyroid drugs e.g. propylthiouracil can be allowed to breastfeed but monitor babies closely with thyroid function tests.
- Radioactive iodine (RAI) in therapeutic doses should be avoided given that safer alternatives are available; a mother may resume breastfeeding about two months after receiving this substance with measured low milk radioactivity;
- Excessive use of topical iodine or iodophors (e.g., povidone-iodine), especially on open wounds or mucous membranes, can result in thyroid suppression or electrolyte abnormalities in the breastfed newborn and should be avoided;
- Cytotoxic chemotherapy requires that a mother stop breastfeeding during therapy.

Substance use

Mother should be advised not to use these substances and given opportunities and support to abstain. Mothers who choose not to cease their use of these substances or who are unable to do so, should seek individual advice on the risks and benefits of breastfeeding, depending on their individual circumstances. For the mother who uses these substances in short episodes, consideration may be given to avoiding breastfeeding temporarily during this time.

- Maternal use of nicotine often decreases the duration of breastfeeding, and can adversely affect the infant, but breastfeeding is preferable to formula feeding in mothers who smoke. Infants should not be exposed to tobacco smoke.
- Alcohol taken before breastfeeding can cause infant sedation and reduced milk intake.
- Abuse of amphetamines, cocaine and related stimulants may produce harmful effects on babies who are breastfed especially if the infant is additionally exposed to inhalation of smoked drugs.

The following are usually safe in usual dosage:

- Analgesics – short courses of paracetamol, acetylsalicylic acid, ibuprofen, occasional doses of morphine and pethidine
- Antibiotics – penicillin, cloxacillin and related drugs, erythromycin, metronidazole
- Antihistamines, antacids, digoxin, insulin, bronchodilators, corticosteroids, antihelmintics, chloroquine, antituberculous drugs.
- Nutritional supplements e.g. iron, iodine and vitamins.

Adapted from UNICEF/ WHO (January 2008) Section 4: Hospital Self-Appraisal and Monitoring, Baby Friendly-Friendly Hospital Initiative. Revised, updated and expanded for integrated care.

References:

1. Implementation Guidance. Protecting, Promoting And Supporting Breastfeeding In Facilities Providing Maternity And Newborn Services: The Revised Baby-Friendly Hospital Initiative. WHO/UNICEF 2018
2. Acceptable medical reasons for use of breast-milk substitutes, WHO/UNICEF 2009
3. BABY-FRIENDLY HOSPITAL INITIATIVE Revised, Updated and Expanded for Integrated Care. Section 3. Breastfeeding Promotion and Support in a Baby Friendly Hospital. A 20-Hour Course for Maternity Staff. WHO/UNICEF 2009

13.3 WEIGHT MONITORING IN NEWBORN TERM INFANTS

1. A term baby loses weight in the first 7-10 days of life.
2. After day 7-10, the expected rate of weight gain is 15-25g/day.
3. The baby should regain birth weight by Day 14 of life.
4. A baby who loses more than 10% of their birth weight must be referred to a medical officer immediately.
5. Although it is the current practice to weigh the baby on day 5 and day 14-16, it is recommended that a baby is also weighed at any opportunity and especially when there is a concern with weight or feeding. This is because weight is the only objective measure for adequacy of feeding and we also could detect earlier if a baby is not doing well or not feeding well. Studies have also shown that a baby that lost more than 7% of birth weight in the first few days of life has a higher risk of severe neonatal jaundice.
6. Breastfeeding education and assessment on the adequacy of feeding should be carried out at every encounter.

Table 13.1: Management Plan related to Weight Monitoring of Newborn Babies

Group of babies	Management Plan
All babies	To weigh at 1 st visit, Day 5, Day 15, Day 20 and when there is a concern with weight or feeding.
Babies with weight loss > 7%	To reinforce on breastfeeding education To assess breastfeeding technique, supplement if needed. If the baby is generally well, to review the baby again in 1-2 days. To refer to a medical officer if there any concerns.
Babies with weight loss > 10% of birth weight	To refer to a medical officer on the same day

Table 13.2: Assessment of Adequacy of Feeding in Newborn Babies

Parameter	Normal
Urine output	At least 5-6 heavy wet nappies in 24 hours
Appearance and frequency of stools	At least 2 in 24 hours; normal appearance
Baby's colour, alertness and tone	Normal skin colour, alert, good tone
Weight	Weight loss not more than 10% of birth weight
Number of feeds in the last 24 hours	At least 8-12 feeds
Baby's behaviour during feeds	Generally calm and relaxed
Sucking pattern during feeds	Initial rapid sucks changing to slower suck with pauses and soft swallowing
Length of feed	Feeding for 5-40 minutes at most feeds
End of the feed	Baby lets go spontaneously, or does so when breast is gently lifted
Baby's behaviour after feeds	Content after most feeds

Source: Breastfeeding assessment form, UNICEF.

CHAPTER 14

SPECIFIC PERINATAL CONDITIONS RELATED TO MATERNAL COMORBIDITIES

CHAPTER 14: SPECIFIC PERINATAL CONDITIONS RELATED TO MATERNAL COMORBIDITIES

14.1 INFANT OF MOTHER WITH DIABETES

- Infant of mother with DM/GDM may have many of the morbidities or complications as below:

System	Conditions
Prenatal risks	▪ Sudden foetal death in third trimester ▪ Premature birth with or without caesarean delivery ▪ Intrapartum asphyxia ▪ Birth trauma secondary to macrosomia
Growth	▪ Macrosomia ▪ Intrauterine growth retardation
Cardiac	▪ Restrictive cardiomyopathy ▪ Congenital heart disease e.g.; VSD
Central nervous system	▪ Anencephaly, ▪ Spina bifida ▪ Microcephaly ▪ Sacral agenesis and/ or caudal regression syndrome
Gastrointestinal	▪ Atresias ▪ Small left colon syndrome ▪ Hirschsprung disease
Urinary tract	▪ Urinary tract abnormalities
Hematologic	▪ Hyperviscosity or polycythaemia
Metabolic	▪ Hypoglycaemia ▪ Hypocalcemia ▪ Hypomagnesemia ▪ Hyperbilirubinemia
Respiratory	▪ Respiratory Distress Syndrome (RDS)

Source: PediatricEducation.org

- Refer to paediatric team at birth.
- At birth, assess for congenital anomalies, birth injuries and respiratory distress.
- Initiate early feeding and monitor for hypoglycaemia and observe vital signs (Refer **Chapter 12** - Management of Neonatal Hypoglycaemia).

14.2 INFANT OF MOTHER WITH THALASSAEMIA

- If both parents are thalassaemia carriers, to refer to Paediatric team postnatally for further management and follow-up plans
- If only one parent is thalassaemia carrier, for Paediatric clinic review at 6-12 months of age.

14.3 INFANT OF MOTHER WITH AUTOIMMUNE DISEASE (eg: SLE)

- To refer to paediatric team at birth.
- Infants of mother with autoimmune disease, especially if mother has anti-SSA/Ro or anti-SSB/La antibodies, may present with congenital complete heart block or other conduction problem, and neonatal lupus erythematosus.
- At birth, assess heart rate, ECG for conduction problems, and take FBC, KIV FBP.

14.4 INFANT OF MOTHER WITH OTHER BLOOD DISORDER (eg: ITP, THROMBOCYTOPENIA etc)

- To refer to the paediatric team at birth. Assess the baby, send FBC and KIV FBP.

14.5 INFANT OF MOTHER WITH THYROID DISEASE

- To refer to the paediatric team at birth.
- Several conditions could arise in the newborn, depending on the type of maternal thyroid disease: neonatal thyrotoxicosis (tachycardia and goitre), congenital hypothyroidism, transient neonatal hypothyroidism or hyperthyroidism.
- Trace cord TSH before discharge and repeat T4TSH after 72 hours of life, and Day 10-14 of life if needed.
- Refer to Paediatric Protocols for Malaysian Hospitals 4th Edition (2019) for further information and the references as below.

14.6

INFANT OF MOTHER WITH SYPHILIS

- To refer to the paediatric team at birth.
- Take baby's blood for VDRL or RPR at birth.
- Assess for clinical features of congenital syphilis at birth: non-immune hydrops, IUGR, jaundice, hepatosplenomegaly, rhinitis, skin rash, pseudoparalysis of extremity.
- Management of the baby will depend on the mother's VDRL or RPR status (to be labelled as having recent infection, it must be reactive), timing of maternal treatment and stage of maternal infection, and the clinical condition of the baby.
- Refer to Paediatric Protocols for Malaysian Hospitals 4th edition (2019) for further information.

14.7

INFANT OF MOTHER WITH HIV

- To refer to the paediatric team before birth, to assess the mother's condition and to arrange for the antiretroviral (ARV) prophylaxis and care of the newborn.
- After assessing at birth for other evidence of congenital infections and symptoms of drug withdrawal (if maternal history suggestive), babies must receive ARV prophylaxis as soon as possible after birth, and also to undergo blood tests and further counselling on feeding and follow-up.
- Standard precautions must be observed in the care of the baby.
- All routine vaccinations can be given according to schedule.
- Several interventions have proven effective in reducing vertical transmission: a) total substitution of breastfeeding with infant formula, b) elective caesarean section, c) ARV prophylaxis.
- To follow BFHI procedures (AFASS) if infant formula is opted for baby.
- Refer to Paediatric Protocols for Malaysian Hospitals 4th edition (2019)

14.8

INFANT OF MOTHER WITH ACTIVE TUBERCULOSIS

- To refer the paediatric team at birth and to isolate and admit the baby to special care nursery.
- Defer BCG immunisation.
- The infant should be evaluated for congenital TB. Prophylactic TB treatment should be given to babies born to mothers with active pulmonary TB (except those diagnosed more than two months before delivery who have documented smear negative before delivery) once congenital TB is ruled out.

- BCG is only given months later once prophylaxis has been completed and the tuberculin test is negative.
- Congenital TB is rare but should be suspected if the infant born to a tuberculous mother fails to thrive or is symptomatic.
- Baby can receive expressed breast milk from mother even on medication as the amount of drug ingested by the baby is minimal.
- If the mother is already on treatment (preferably after 2 weeks) and is non-infective, the baby can be breastfed.
- Women who are receiving isoniazid and are breastfeeding should receive pyridoxine.
- Symptomatic mothers can hold and be with their baby with double masks.
- To refer to CPG Management of Tuberculosis 2012 3rd edition (**Table 14.1** below) and Paediatric Protocols for Malaysian Hospitals 4th edition for further information.

Table 14.1: Prophylaxis for infants with maternal TB

Active PTB diagnosed before delivery			Active PTB diagnosed after delivery	
> 2 months before		< 2 months before	<2 months after	> 2 months after
Smear negative just before delivery	Smear positive just before delivery	-	-	-
No prophylaxis for infant	Give prophylaxis : Isoniazid for 6 months OR isoniazid for 3 months followed by TST		Give prophylaxis : Isoniazid for 6 months OR isoniazid + rifampicin for 3 months	
BCG at birth	Defer BCG at birth, give after stopping isoniazid		Reimmunised with BCG after stopping isoniazid	If BCG given at birth, no need to immunise

Recommendations:

- BCG should not be given to babies on prophylactic tuberculosis (TB) treatment.
- Prophylactic TB treatment should be given to babies born to mothers with active pulmonary TB except those diagnosed more than two months before delivery who have documented smear negative before delivery.

14.9 INFANT OF MOTHER WITH MENTAL DISORDERS

- Assess the stability of the mother's psychiatric condition and her social or family support. Refer to the social worker or child protector if the welfare of the baby is of concern.
- If possible, proper rooming in with mother for a few days before discharge.
- Another family member needs to be briefed on the condition and management or follow-up of the newborn before discharged.

14.10 INFANT OF TEENAGE OR SINGLE MOTHER

- Assess the social or family support of the mother.
- Refer to a social worker if not done yet.
- Settle placement of baby before discharge.
- Another family member needs to be briefed on the condition and management or follow-up of the newborn before discharged.

14.11 INFANT OF MOTHER WITH SUBSTANCE ABUSE

- To refer to Paediatric team at birth for risk of neonatal withdrawal/abstinence syndrome
- Assess the social or family support of the mother.
- Refer to a social worker if not done yet.
- Babies may be born to opiate-dependent mothers OR mothers on drugs associated with withdrawal symptoms (generally milder).
- Further information by Neonatal Guidelines 2017-2019, NHS:
 - a) Timescale of withdrawal symptoms in babies:
 - i. Signs of withdrawal from opiates (e.g.; heroine) can occur < 24 hours after birth
 - ii. Signs of withdrawal from opioids (prescribed drugs eg methadone) can occur 3-4 days to 2 weeks after birth.
 - iii. Multiple drug use can delay, confuse and intensify withdrawal signs.
 - b) Minor signs: tremors when disturbed, tachypnoea, pyrexia, sweating, yawning, sneezing, nasal stuffiness, poor feeding, regurgitation, loose stools, sleeping < 3 hrs after feed.
 - c) Major signs: convulsions, profuse vomiting or diarrhoea, inability to coordinate sucking, baby inconsolable after 2 feeds
- Usual management:
 - Do NOT give naloxone to babies
 - Care of baby as any other baby

- Assess for signs of withdrawal (e.g. with Modified Finnegan Neonatal Abstinence Score), other risk of infection, child protection issues, social history etc.
- Babies with signs of withdrawal will need close observation and may need treatment e.g. morphine/ phenobarbital (mother on other drugs as well)/ chlorpromazine (mother on benzodiazepines) etc.

14.12 INFANT OF MOTHER WITH GROUP B STREPTOCOCCAL (GBS) INFECTION OR RISK OF GBS ASCENDING INFECTION

- To refer to the paediatric team at birth.
- GBS systemic infection in the newborn carries high mortality. Early onset GBS (EOGBS) infection is defined as isolation of GBS from blood, cerebrospinal fluid (CSF), or another normally sterile site from birth through 6 days of age. EOGBS infections are acquired vertically through exposure to GBS from the maternal gastro-intestinal and genito-urinary tract.
- The use of intravenous intrapartum antibiotic prophylaxis to prevent early-onset GBS disease in infants has been proven through clinical trials and well-designed observational studies.
- Late-onset infections occur in infants aged > 1 week, with most infections evident during the first 3 months of life (Prevention of Perinatal Group B Streptococcal Disease Revised Guidelines from CDC, 2010)
- Infected infants become symptomatic shortly after birth, majority by 12 to 24 hours of age. Signs and symptoms of neonatal infection can be non-specific and include: temperature instability, lethargy, irritability, poor peripheral perfusion, mottling, pallor, jaundice, sclerema, tachypnoea, respiratory distress, apnoea, tachycardia, hypotension, hypoglycaemia, hyperglycaemia, metabolic acidosis
- Refer section 5.7.6 for indication for intrapartum antibiotic prophylaxis (IAP) and its regimen
- Neonatal management of baby with risk of GBS infection:
 - If baby is unwell at birth, shows any signs of sepsis or maternal chorioamnionitis, for septic workup (FBC and blood culture. CXR and LP if necessary) and start IV antibiotics (IV C-Penicillin and Gentamicin) immediately. This will fall under Paediatric care.
 - If baby is born term and stable without maternal chorioamnionitis (regardless of whether had completed at

least 2 doses of antibiotics) to observe for 24-48 hours and to give warning signs especially on sepsis before discharge. Baby could be monitored either in SCN or postnatal ward with paediatric review and blood tests or empirical antibiotics may be needed.

- iii. If baby is born preterm (<37 weeks), to refer to the paediatric team, may need full septic workup and IV antibiotics.

14.13

INFANT OF MOTHER WITH RECENT VARICELLA ZOSTER INFECTION

- Infants born to mothers who develop varicella zoster infection between **7 days before delivery or 7 days after delivery** should receive prophylaxis.
- Infants in this group need to be referred to the paediatric team immediately to receive Varicella Zoster immunoglobulin (VZIG) as soon as possible, after delivery or within 96 hours of initial exposure. Attenuation of disease might still be achieved with administration of VZIG up to 10 days after exposure.
- This is because infants who develop varicella within this time frame have mortality as high as 20% as they have not acquired maternal protecting antibodies.
- For infants born to mothers who develop varicella between 5 days before and 2 days post-delivery, add IV acyclovir 15 mg/kg/dose over 1 hour every 8hrly (total 45 mg/kg/day) for 5 days.
- On sending home, warn parents to look out for new vesicles or babies being unwell, for 28 days after exposure. If so, parents to bring the infant to the nearest hospital as soon as possible (62% of healthy such neonates given VZIG after birth)
- If vesicles develop, give Acyclovir 15 mg/kg/dose over 1 hour every 8hrly (total 30-45 mg /kg/day) for 7-10 days.
- Women with varicella at time of delivery should be isolated from their newborns, breastfeeding is contraindicated. The newborn baby can receive expressed breast milk in the meantime and breastfeeding commences when all the mother's lesions have crusted.
- Neonates with varicella lesions should be isolated from other infants but not from their mothers.
- Refer to Paediatric Protocols for Malaysian Hospitals 4th edition (2019) for further information.

14.14 INFANT OF FAMILY HISTORY OF NEONATAL DEATH

To refer to the paediatric team at birth for further assessment, especially if there is a history of inborn error of metabolism, primary immunodeficiencies disorders or other serious hereditary disease.

Reference:

1. Paediatric Protocol for Malaysian Hospitals 4th edition (2019)
2. Neonatal Guidelines 2017-2019, by the Bedside Clinical Guidelines Partnership, Staffordshire, Shropshire and Black Country Neonatal Operational Delivery Network and Southern West Midlands Neonatal Operational Delivery Network, NHS.
3. Health Technology Assessment, Childhood Immunization 2001
4. Centers for Disease Control and Prevention (CDC): Prevention of Hepatitis B infection 2018
5. Prevention of Perinatal Group B Streptococcal Disease. Revised Guidelines from CDC, 2010
6. Malaysian CPG on Management of Tuberculosis 3rd Edition, 2012

SECTION C: PERINATAL NUTRITION CARE

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CHAPTER 15

PERINATAL NUTRITION CARE

15.1 INTRODUCTION TO PERINATAL NUTRITION CARE (PNC)

The Perinatal Nutrition Care (PNC) section outlines nutrition advice as a guide for health professionals to educate mothers starting from pre pregnancy up to postpartum period. Management of PNC are accustomed towards a holistic approach in which nutrition advice for mothers should consist of comprehensive information to support optimal nutrient intake according to mother health conditions.

These include basic advice on pre pregnancy nutrition as it is vital to have an optimal nutrition status before getting pregnant as it will give impact to the health of the mother and fetus in various stages of pregnancy. Any women with chronic diseases must have an optimal control of the diseases and in optimal nutritional status to undergo pregnancy. It must be a well-planned pregnancy. This may require a referral to a dietitian for medical nutrition therapy (MNT) intervention for optimal nutritional status and disease control.

To undergo pregnancy, any woman with chronic conditions must have excellent disease control and an optimal nutritional status. It has to be a meticulously planned pregnancy. For optimal nutritional status and illness control, a referral to a dietitian for medical nutrition treatment (MNT) intervention is necessary. A dietitian will need to perform a detailed nutrition assessment including past and current medical history; social history; anthropometric measurement; biochemical and clinical evaluation; dietary intake in order to prescribe proper MNT. It will be dealt with on a one-to-one basis.

Dietetic referral may be necessary at any stages of perinatal care for any chronic or acute disease in order to achieve or maintain optimal nutrition and health.

Pregnant women have an increased nutrient requirement throughout the pregnancy to achieve optimal maternal and foetal health. There is an increased need for energy, protein and micronutrients such as iron, iodine, folate, and vitamins such as A and C intake. The increased nutrient intake extends to the postpartum mothers.

PNC also includes the management of Gestational Weight Gain (GWG) which uses the pre pregnancy BMI categories to determine the GWG range according to the Institute of Medicine, IOM 2009 guidelines. Certain pregnancy conditions such as anaemia, diabetes and hypertension that need specific nutrient recommendation have been added in this topic.

Special consideration must be made for women with chronic or acute illness, any kind of malnutrition (excess or deficient macro or micronutrients), any electrolyte imbalance, adolescent pregnancy should be given special attention because they are at risk for poor maternal and foetal outcomes.

Existing Ministry of Health document e.g. Recommended Nutrition Intake (2017), Dietary Guidelines for Pregnant and Lactating Mothers (2019), CPG On Management of Diabetes in Pregnancy, Medical Nutrition Therapy Guideline for Type 2 Diabetes Mellitus 2005, Medical Nutrition Therapy Guideline for Type 2 Diabetes Mellitus Second Edition and other related international guidelines and policy on nutrition such as WHO Nutritional Anaemias; Tools For Effective Prevention And Control, Guidelines on Weight Gain And Pregnancy, Institute of Medicine (IOM) 2009 and WHO Recommendations on Antenatal Care for a Positive Pregnancy Experience 2016 are used as references.

15.2 PROMOTING HEALTHY EATING FOR PREGNANCY

This section outlines the key aspect of maternal nutrition for all women. All women deserve to be healthy throughout their lifetime. Menstruation, childbirth, entering menopause and aging pose challenges to women's nutrition and health.

Therefore, women need to be encouraged to prepare themselves first specifically in the nutrition aspect and to have an optimum health status before pregnancy. It is important for women to enter pregnancy with healthy body weight (normal BMI range) and in good nutritional status.

15.2.1 The importance of pre pregnancy nutrition

Women who are well-nourished and healthy will have a higher chance of having healthier offspring. Therefore, nutrition information should be disseminated to women in reproductive age between 15-49 years old on various platforms long before pregnancy takes place. Pre pregnancy nutrition care should take into consideration their health condition,

socioeconomic status, lifestyle habits, history of previous pregnancy, and other history that might affect their pregnancy.

Table 15.1: Improving maternal nutrition with special nutrition care focus on three critical stages of pregnancy

SUPPORTIVE ENVIRONMENTAL CONDITIONS	PRECONCEPTION	PREGNANCY	POSTPARTUM
▪ Appropriate provision of healthcare services for optimum nutrition for maternal and infant health ▪ Appropriate education of health professionals on diet-related issues ▪ Specific guidance on clinical care pathways to manage and prevent underweight, overweight and obesity ▪ Incorporating of e-health with shared access among the different groups of health care professionals	▪ Dietary advise ▪ Physical activity advise ▪ Weight optimisation for women with a BMI ≤ 18.5 or 25.0 ▪ Assessment of micronutrient status	▪ Dietary advise ▪ Physical activity advise ▪ Assessment of micronutrient status ▪ Appropriate GWG ▪ Appropriate education of future parents (antenatal classes)	▪ Dietary advise ▪ Physical activity advise ▪ Weight optimisation for women with a BMI ≤ 18.5 or 25.0 ▪ Support for exclusive breastfeeding for the first 6 months ▪ Assessment of micronutrient status

Source: WHO Good Maternal Nutrition, The Best Start In Life, 2016

Women need to start eating healthy and balanced meals as soon as they plan to conceive or get married. Poor eating patterns and low intake of protein, vegetables, fruit, legumes and milk or milk products could increase the risk of having micronutrient deficiencies that might result in poor pregnancy conditions and outcomes. Women should be advised on healthy eating in pre pregnancy stage with other healthy lifestyle advice as below:

- Know your health status once planning to get pregnant or married.
- Aim to have a normal BMI before getting pregnant.
- Start eating a balanced diet consists of a variety of food using Malaysian Healthy Plate Concept #sukusukuseparuh for every meal.
- Limit intake of high-sugar food, high-fat food, processed food, and sugary drinks and beverages.
- Limit intake of caffeinated drinks such as tea, coffee, carbonated drinks, and cocoa drinks.
- Start taking a folic acid pill three months before getting pregnant.
- Take the initiative to check your haemoglobin status (Hb) and consult a doctor if Hb level is below 12g/dL.
- Incorporate a healthy lifestyle including staying active, stopping smoking, and managing your stress.

15.2.2 Nutritional requirements during pregnancy

The ability of pregnant women to provide nutrients and oxygen for her baby is a critical factor for foetal health and its survival. Failure in supplying adequate amounts of nutrients to meet foetal demand can lead to foetal malnutrition. The fetus responds and adapts to undernutrition but by doing so it permanently alters the structure and function of the body. Maternal overnutrition also has long-lasting and detrimental effects on the health of the offspring. There is growing evidence that maternal nutrition can induce epigenetic modifications of the foetal genome (Martin-Gronert & Ozanne 2006).

Therefore, it is important to ensure pregnant women get all the nutrient requirements through everyday meals. Recommendations for healthy eating during pregnancy are based on the Dietary Guidelines for Pregnant and Lactating Mothers (2021), Malaysia Dietary Guidelines (2020) and Recommended Nutrition Intake (2017).

a. Pregnancy requirement on selected nutrients:

Below is the requirement of each selected nutrients:

Table 15.2: Pregnancy requirements on selected nutrients

	NUTRIENTS	REQUIREMENT
1	Total Energy Intake (TEI)	• Meet energy requirement range for women from 1600kcal/day for sedentary lifestyle women or can up to 1800kcal/day for moderate-active women. • Obtain additional energy requirement during pregnancy: ○ First trimester: +80kcal/ day ○ Second trimester: +280kcal/ day ○ Third trimester: +470kcal/ day
2	Macronutrients: Protein	• Recommended additional protein intake: ○ First trimester: +0.5g/day ○ Second trimester: + 8g/day ○ Third trimester: + 25 g/day • Eat more protein from animal source such as fish, poultry, egg and meat. • Include legumes in daily meals or snacking.
3	Macronutrients: Fat	• Eat 25-30% of total energy intake (TEI) based on dietary fat during pregnancy. • Eat according to fat distribution: ○ Saturated fatty acids : < 10% TEI

	NUTRIENTS	REQUIREMENT
		○ Monosaturated fatty acids : 12% to 15% TEI ○ n-6 polyunsaturated fatty acids: 5% to 7% TEI ○ n-3 polyunsaturated fatty acids: 0.3% to 1.2% TEI
4	Macronutrients: Carbohydrate	● Eat 50-65% of total energy intake (TEI) based on carbohydrate source during pregnancy ● Limit sugars intake to <10% from TEI (equal to 10 teaspoon based on 2,000 kcal) ● Eat 20-30g dietary fibre per day ● Choose food low Glycemic Index (GI)
5	Micronutrient: Iron	● Diets alone could not provide enough iron as iron requirement increased tremendously in pregnancy for the formation of red blood cells. ● Take daily oral iron supplementation with 30mg to 60mg of elemental iron to prevent maternal anaemia. ● Eat a variety of protein from animal sources that are rich in heme-iron to support optimum iron needs by the body (refer Nutritional Anaemia Section for more info).
6	Micronutrient: Folate	● Folic acid requirements for pregnancy are 600 µg/day. ● WHO suggested daily oral supplement of 400µg (0.4mg) folic acid for at least four weeks before, and 12 weeks after conception to reduce the risk of NTDs. ● Pregnant women who have high risk of NTDs are recommended to take a 5000 µg (5 mg) folic acid supplement for the same period of time. ● Precautions should be taken when giving high doses of folic acid supplementation for healthy women as general as there were growing evidences on excessive intake of folic acid supplement during pregnancy and the effects on offspring's NTDs, allergy/respiratory problems, cancer and behaviour problems. Eat variety of food sources that are rich in folate such as legumes, green leafy vegetables and fortified food product daily.
7	Micronutrient: Iodine	● Iodine is an essential trace element required for the synthesis of the thyroid hormones. Iodine requirement during pregnancy is increased to provide the needs of the foetus and to compensate for the increased loss of iodine in the urine. ● Increasing iodine intake up to 200µg/ day during pregnancy reduces risk of pregnancy loss, infant mortality, and irreversible cognitive impairment in children.

	NUTRIENTS	REQUIREMENT
		• Eat variety of food source that rich in iodine such as seaweeds, seafood, seawater fish, egg, milk, milk products and food enriched with iodine such as cereal and iodized salt • Store iodised salt in air tight containers and avoid exposure to moisture, heat and sunlight. • Put a small amount of iodised salt at the end of cooking or after extinguishing the fire and cover the food to avoid iodine from evaporating. • Iodized salt needs to be used sparingly (less than 1 teaspoon flat /day).Increased intake of salt could lead to other health problems.
8	Micronutrient: Calcium	• Women who chronically consume low amounts of calcium (<500 mg/day) may be at risk for increased bone turnover during pregnancy. If a diet does not provide enough calcium, then the body steals it from the bones. Dietary calcium intake has a negative correlation with bone resorption markers. High calcium intake is associated with improved calcium balance, perhaps providing a protective effect against bone loss during pregnancy. • The requirement for calcium is 1000mg/day for pregnant and breastfeeding women aged 20 years and above.

Note:

This is only a selection of nutrients that has been highlighted for pregnant women. For further readings on other nutrients that are equally important, such as Vitamin C, Vitamin B1, B2, B6, B12, Vitamin A and etc, do check Recommended Nutrient Intakes for Malaysia (2017) to get more info.

b. How to achieve a sufficient nutrient intakes in pregnancy

- Eat moderate, variety and balanced meals everyday.
- Include all five (5) food groups in meals and use Malaysian Food Pyramid as a guide on serving size which are:
 - Group 1: Vegetables
 - Group 2: Fruits
 - Group 3: Cereal
 - Group 4 : Fish, poultry, egg, meat and legume
 - Group 5: Milk and milk product

- Use Healthy Malaysia Plate Concept #sukusukuseparuh as a guide to plan each meal for a day (Breakfast, Lunch & Dinner).
- Add nutrient-dense food as a snacking in between meals such as baked bean, sandwiches, chicken salad etc.
- Take 2 servings of milk or milk products such as yoghurt, cheese and other food high in calcium (green vegetables, cereal, sardines, anchovies, beans and beans products including yellow dhal, tofu, tempe and soybean milk every day).
- Limit intake of sugar, salt and fat
- Limit intake of high-sugar foods and beverages.
- Limit caffeinated drinks such as tea, coffee or cocoa.
- It is best not to drink alcohol during pregnancy.
- Use healthier cooking methods such as steaming, roasting, grilling, boiling, and braising to reduce the fat intake.
- Use iodised salt less than 5g (1 teaspoon flat) per day.
- Limit intake of food high in salt such as sauces and processed food.
- Drink plenty water (6 to 8 glasses daily)

c. Consume clean and safe foods and beverages

- Choose and purchase fresh foods
- Prepare foods using clean appliances
- Cook foods with the right temperature
- Store foods using appropriate, clean and safe containers to ensure food safety
- Ensure pregnant women and lactating women are safe from environmental pollution such as lead, mercury and cadmium
- Select restaurants/ out-of-home premises with a clean environment within and outside the premise, those with clean food handlers, as well as with sufficient facilities such as sinks and toilets

d. WHO recommendation on micronutrient supplement for pregnancy

These recommendations are applied to pregnant women and adolescent girls within the context of routine ANC.

Table 15.3 : WHO recommendations on micronutrient supplement for pregnancy

A. Nutritional Interventions		
Recommendation		Type of Recommendation
Dietary interventions	A.1.1: Counselling about healthy eating and keeping physically active during pregnancy is recommended for pregnant women to stay healthy and to prevent excessive weight gain during pregnancy	Recommended
	A.1.2: In undernourished populations, nutrition education on increasing daily energy and protein intake is recommended for pregnant women to reduce the risk of low-birth-weight neonates.	Context-specific recommendation
	A.1.3: In undernourished populations, balanced energy and protein dietary supplementation is recommended for pregnant women to reduce the risk of stillbirths and small-for-gestational-age neonates.	Context-specific recommendation
	A.1.4: In undernourished populations, high-protein supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Iron and folic acid supplements	A.2.1: Daily oral iron and folic acid supplementation with 30 mg to 60 mg of elemental iron and 400 mcg (0.4 mg) of folic acid is recommended for pregnant women to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth.	Recommended
	A.2.2: Intermittent oral iron and folic acid supplementation with 120 mg of elemental iron and 2800 mcg (2.8 mg) of folic acid once weekly is recommended for pregnant women to improve maternal and neonatal outcomes if daily iron is not acceptable due to side-effects, and in populations with an anaemia prevalence among pregnant women of less than 20%.	Context-specific recommendation
Calcium supplements	A.3: In populations with low dietary calcium intake, daily calcium supplementation (1.5–2.0 g oral elemental calcium) is recommended for	Context-specific recommendation

A. Nutritional Interventions		
Recommendation		Type of Recommendation
	pregnant women to reduce the risk of pre-eclampsia.	
Vitamin A supplements	A.4: Vitamin A supplementation is only recommended for pregnant women in areas where vitamin A deficiency is a severe public health problem to prevent night blindness.	Context-specific recommendation
Zinc supplements	A.5: Zinc supplementation for pregnant women is only recommended in the context of rigorous research.	Context-specific recommendation (research)
*Multiple micronutrient supplements	A.6: Antenatal multiple micronutrient supplements that include iron and folic acid are recommended in the context of rigorous research. <i>*This recommendation updates in 2020 and supersedes the WHO recommendation found in the WHO ANC guideline issued in 2016.</i>	Context-specific recommendation – research
Vitamin B6 (pyridoxine) supplements	A.7: Vitamin B6 (pyridoxine) supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Vitamin E and C supplements	A.8: Vitamin E and C supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.	Not recommended
Vitamin D supplements	A.9: Oral vitamin D supplementation is not recommended for all pregnant women to improve maternal and perinatal outcomes. <i>*This recommendation updates in 2020 and supersedes the WHO recommendation found in the WHO ANC guideline issued in 2016.</i>	Not recommended
Restricting caffeine intake	A.10: For pregnant women with high daily caffeine intake (more than 300 mg per day), lowering daily caffeine intake during pregnancy is recommended to reduce the risk of pregnancy loss and low-birth-weight neonate.	Context-specific recommendation

Source: WHO Recommendations on Antenatal Care for A Positive Pregnancy Experience 2016 (update 2020)

15.2.3 Postpartum nutrition for recovery and lactation

Giving birth may affect mothers' health in every aspect. Therefore, optimum intake of nutrients is critical for recovery of childbirth and also as preparation for lactation. Both of actions require a good and various balance meal to provide all the important nutrients to the body.

Table 15.4: Nutrition requirements for postpartum recovery and lactation

	NUTRIENTS	REQUIREMENT
1	Total Energy Intake	• Meet energy requirements range for women from 1600 kcal/day for sedentary lifestyle women or can up to 1800 kcal/day for moderate-active women. • Obtain additional energy needs of 500 kcal/day for the first 6 month if exclusively breastfeeding.
2	Macronutrients: Protein	• Add 19 g/day protein during the first 6 month and lower it to 13 g/day the subsequent 6 months respectively of breastfeeding. • Eat more lean meat such as fish and chicken. • Include legumes, milk and milk product ini daily meal or snacking.
3	Macronutrients: Fat	• Eat 25-30% of total energy intake (TEI) based on dietary fat during pregnancy. • Eat according to fat distribution: ○ Saturated fatty acids : < 10% TEI ○ Monosaturated fatty acids : 12% to 15% TEI ○ n-6 polyunsaturated fatty acids: 5% to 7% TEI ○ n-3 polyunsaturated fatty acids: 0.3% to 1.2% TEI
4	Micronutrient: Iron	• Iron is a critical micronutrient that needs to be restored in postpartum stages due to the blood loss and is important for lactation. • Iron requirement for the first 12 months of postpartum period should follow non-pregnant women needs as below: ○ 29mg/day with bioavailability of 10% ○ 20mg/day if the bioavailability is 15% • If the mother is lactating, an additional 1.1mg iron requirement is needed everyday (WHO/FAO, 2004). • Eat a variety of protein food sources that are rich in iron to support optimum iron needs by the body.
5	Micronutrient: Iodine	• Iodine requirement in postpartum is 200 mg/day and essential for proper function of mother thyroid hormones. • At the same time, infants depend on breast milk as the

	NUTRIENTS	REQUIREMENT
		main source of iodine to continue the development of their brains and nervous system. • Therefore, it is critical to have an optimum iodine intake for the postpartum period to support iodine demands in breastmilk in which the iodine level respectively also highly depends on maternal iodine storage. • Eat a variety of food sources that are rich in iodine such as seaweeds, seafood, seawater fish, egg, milk, milk products and food enriched with iodine such as cereal and iodised salt. • Use iodised salt properly (storage and cooking) in order to get the benefits of iodine. Iodised salt needs to be used sparingly (less than 1 teaspoon flat /day). Increased intake of salt could lead to other health problems.
6	Micronutrient: Calcium	• Postpartum mothers who are lactating need to ensure adequate calcium intake for their health and also lactation. • Eat calcium-rich food to achieve 1000mg/ day requirement for women aged 20 years and above.

Notes:

- To achieve optimal intake of nutrients from foods, refer to pregnancy advice on 'How to achieve sufficient nutrient intakes in pregnancy'.
- This is only a selection of nutrients that has been highlighted for postpartum stages. For further readings on other nutrients that are equally important, refer Recommended Nutrient Intakes for Malaysia (2017).
- Refer to Dietary Guidelines for Pregnant and Lactating Mothers (2019) for details.

15.3 MANAGING GESTATIONAL WEIGHT GAIN

15.3.1 Maternal obesity and underweight

Women of childbearing age living in Malaysia nowadays are significantly heavier than in the past. The National Health and Morbidity Survey (NHMS) 2019 showed that 54.7% women aged 18 and above were overweight or obese compared to 48.9% in 2015 and 46.0% in 2006.

This is alarming because overweight and obese women are more likely to have health issue during pregnancy and may reflect of poor pregnancy outcome. These women will have a higher risk of premature birth, hypertension, diabetes during pregnancy and increased risk of developing large-for-gestational age infants.

Underweight women remain a concern as they also have a risk of having micronutrient deficiency for certain nutrients such as iron, folate, calcium or iodine that can affect their health and development of fetus. They also have higher chances to have a preterm baby in which later on have a higher risk of malnutrition episodes if their child's nutrient intake is not being taken care off.

15.3.2 Excessive or inadequate gestational weight gain (eGWG/iGWG)

All women tend to have the same risk to gain weight whether excessive or inadequate than the recommended range during pregnancy regardless of their pre pregnancy BMI status. This GWG rate is a guide to determine the risk for health complications among pregnant mothers.

Range of gestational weight gain varies according to pre pregnancy BMI status. Poor gestational weight gains, whether excessive (eGWG) or inadequate (iGWG) than the recommended range can risk the mother to develop complications throughout the pregnancy which can affect baby's health.

Physiologically, women will gain weight during pregnancy to cater energy and nutrient requirements for the development of fetus and as well for breastfeeding preparation. Distribution of the weight consists of placenta, fetus and amniotic fluid account for approximately 35% of the total pregnancy weight gain. The rest comes from increment of blood and fluids, tissues of the breast and uterus and fat stores. Throughout pregnancy, maternal body weight is used as a general indicator of the health of the mother and development of a healthy fetus. Weight gain in pregnancy normally occurs in the second and third trimesters with minimal weight gain in the first trimester.

a. Risk of excessive gestational weight gain (eGWG):

- Mother: Increase the risk of gestational diabetes, pre-eclampsia and postpartum weight retention (PPWR) which further increases the risk of obesity.
- Infant: Increase risks of neonatal hypoglycaemia, low birth weight or macrosomia in which increases the risk of other infant and birth complications.

b. Risk of inadequate gestational weight gain (iGWG):

- Mother: Less likely to initiate breastfeeding.
- Infant: Increase risk of preterm birth, low birth weight and foetal distress.

Therefore, women who wish to get pregnant in any stages of pregnancy should be educated about weight management as below:

- Pre pregnancy stage:
 - Try to achieve optimum weight before getting pregnant.
- During pregnancy stage:
 - Monitor weight gain by using the range that is specifically according to pre pregnancy BMI status of the mother according to the IOM 2009 indicator.
 - Women should not avoid gaining weight or try to lose weight.
 - Eat a balanced diet with nutrient-dense meals everyday.
- Postpartum stage:
 - Try to reduce postpartum weight retention (PPWR) by proper weight and diet management.
- In any stage, physical activity should be routinely done accordingly.

15.3.3 Gestational Weight Gain Range

Malaysia is using the standard range from Weight Gain during Pregnancy Reexamining Guidelines, Institute of Medicine (IOM), United States, 2009. There is no data available to establish a recommended range for the local population. These ranges vary according to the pre pregnancy BMI status and types of pregnancy.

a. Women with Single Pregnancy

Table 15.5: Gestational Weight Gain Range for Single Pregnancy

Pre pregnancy BMI (kg/m ²)	Total Weight Gain in (kg) in first trimester	Mean and range of weight gain in second and third trimester	Total Weight Gain (kg)
Underweight < 18.5	0.5 – 2.0 kg	0.5kg (0.44-0.58kg) per week	12.5 – 18.0
Normal 18.5 – 24.9		0.4kg (0.35-0.50kg) per week	11.5 – 16.0
Overweight 25.0 – 29.9		0.3kg (0.23-0.33kg) per week	7.0 – 11.5
Obese ≥ 30.0		0.2kg (0.17-0.27kg) per week	5.0 – 9.0

b. Women with Twin Pregnancy

Women carrying twins or multiple fetuses have more maternal tissue and higher foetal weight, and should therefore gain more weight. There is a little information on weight gain for women carrying 3 or more babies. Further assessment should be done accordingly.

Table 15.6: Gestational Weight Gain Range for Twin Pregnancy

Pre pregnancy BMI (kg/m ²)	Total Weight Gain (kg)
Underweight < 18.5	
Normal 18.5 – 24.9	17.0 – 25.0
Overweight 25.0 – 29.9	14.0 – 23.0
Obese ≥ 30.0	11.0 – 19.0

Notes:

- There is insufficient information to develop guidelines for underweight women carrying twins (IOM, 2009).
- Refer to Weight Gain During Pregnancy Reexamining the Guidelines, Institute of Medicine (IOM), 2009 for more information (p: 76)

15.3.4 Monitoring of Gestational Weight Gain

a. At booking visit:

- Detailed prepregnancy weight, height and BMI status.
- Weigh the mother and record it in Pregnancy Weight Monitoring Form (KIK (a) & (b)) (**Appendix 15-3**)
 - If pre pregnancy weight could not be obtain, use booking BMI status (≤12 weeks) as a proxy to select appropriate GWG range.
 - If booking visit > 12 weeks, ask the mother about their pre pregnancy BMI or weight. If mother could not recall, the nurse should estimate their BMI category. Estimation BMI category purpose is to monitor the rate of weight gain throughout the pregnancy is in the appropriate range given. TheGWG graph is not suitable to be used.
- For maternal underweight (BMI <18.5 kg/m²) and obese (BMI > 30.0 kg/m²) mother, further assessment should routinely be

done on history or family background with chronic diseases, cancer due to obesity or thyroid problem. Clinical investigation should consist of comorbid conditions due to obesity, sleep apnea and the risk of VTE. Investigation such as OGTT, ECG, renal profile and others should be done if needed.

- Referred to nutritionist for nutrition management for maternal underweight ($<18.5 \text{ kg/m}^2$) and maternal obesity ($>30.0 \text{ kg/m}^2$) without underlying causes.
- Referred to dietitian for counselling if poor GWG is due to gestational diabetes and any other related diseases.

b. At subsequent visits:

- Weigh the mother and record it in Pregnancy Weight Monitoring Form (KIK (a) & (b)).
- Calculate rate of GWG per week by looking at the differences between current weight and weight during previous visit.
- Calculate total gestational weight gain by the differences of current weight compared to pre pregnancy weight / booking weight (POG < 12 weeks).
- Explain to mothers about the trend of their weight gain and monitor their weight by eating a balanced diet and keep physically active with appropriate antenatal exercise.
- Refer or discuss with a medical officer if there is excessive GWG (eGWG) or inadequate GWG (iGWG) for further action.
- Nutrition management of GWG should be a holistic approach and take into account all the underlying health problems such as GDM, PIH etc and not to be managed separately. Therefore refer to a dietitian if mother weight gain is due to this factor.
- Refer to nutritionist if inappropriate GWG is solely due to anemia or dietary pattern problem. (Use Borang Rujukan Runding Cara Pemakanan di Klinik Kesihatan Pindaan 2019).

15.3.5 Postpartum Weight Management

Usually women will gain weight during the first four days after delivery, but by the fifth postpartum day, most will have begun to lose weight. An above-average rate of weight loss immediately will occur among women who experienced hypertension or preeclampsia during pregnancy—probably because of postpartum diuresis.

Women normally lose weight through the first 4 to 6 months; however, some women gain weight, even if they are breastfeeding. After a period of rapid

weight loss in the first few weeks, the average rate of weight loss by lactating women is 0.5 to 1.0 kg per month through the first sixth month.

If weight loss is an appropriate goal, the woman should be encouraged to set a reasonable body weight goal, and she should be assisted in developing a healthy dietary plan for achieving that goal. For overweight breastfeeding women, the maximum suggested rate of weight loss after the first month postpartum is about 2 kg/month.

The total energy intake by these women should not be less than 1,600 kcal/day, to allow adequate intake of protein, vitamins and minerals. Although this level of energy intake may seem high to some women who are familiar with reducing diets, it takes into account the energy required for breastfeeding. Liquid diets and weight loss medications are not recommended (Institute of Medicine US Committee on Nutritional Status during Pregnancy and Lactation, 1992)

15.4 NUTRITION CARE FOR SPECIFIC CONDITIONS IN PREGNANCY

An existing medical condition, or one that is acquired during pregnancy, is a risk factor generally for both mother and infant. Medical nutrition therapy (MNT) is often required for conditions that are associated with pregnancy.

Therefore, this section will highlight several conditions during pregnancy such as nutritional anaemia, gestational diabetes mellitus and hypertension as a basic guide for health professionals.

Usually, collaboration within health care providers (multidiscipline) is required, and referral to other providers needed in some instances.

Nutrition management of pregnant mothers should be a holistic approach and take into account all the underlying health problems. Dietary intervention should be as one comprehensive nutrition message that caters to all related health problems with a practical diet plan to follow.

15.4.1 Nutritional anaemia

- Anaemia is a condition in which the haemoglobin (Hb) concentration in the blood is lower than normal. WHO defines anaemia in pregnancy as a haemoglobin concentration of less than 11 g/dL at any stage of pregnancy. Anaemia in pregnancy has adverse impact on both maternal and foetal outcomes such as increased risk of maternal / prenatal, low birth weight and preterm delivery. It also affects everyday working productivity due to reduced oxygen

transport in the body, which then translates to conditions such as difficulties to concentrate and fatigue.

- Anaemia develops through three main mechanism which are:
 - Ineffective erythropoiesis (when the blood makes too few red blood cells)
 - Haemolysis (when red blood cells are destroyed)
 - Blood loss
- Root cause of anaemia:
 - Nutrient insufficient – Iron, Vitamin A, B2, B6, Folate etc.
 - Diseases – Infectious or chronic disease such as HIV, malaria, TB & parasite infection.
 - Genetic haemoglobin disorders (Sickle cell disorder/ thalassemia etc).

- **Iron Deficiency Anaemia**

Iron deficiency anaemia is a type of anaemia in which the supply of iron to bone marrow is inadequate and total body iron is deficient to support optimal erythropoiesis for the development of red cell mass. It occurs following prolonged negative iron balance and is characterised by a small pale erythrocyte.

During pregnancy, increased maternal iron is needed to provide the demands of the growing foetus and placenta, increased erythrocyte mass and expanded maternal blood volume especially in the third trimester.

Iron deficiency anemia in pregnancy can be caused by many factors including an iron-deficient diet, gastrointestinal issues affecting absorption, or a short interpregnancy interval. Other causes of anaemia include parasitic diseases, micronutrient deficiencies, and genetically inherited hemoglobinopathies.

Figure 15.1 : Indices for Assessing Iron Status at Various Stages of Pregnancy

	IRON STATUS			
	NORMAL	STAGE I	STAGE II	STAGE III
		Storage depletion (low Fe storage)	Mild functional Fe deficiency (Fe-deficient)	Functional Fe deficiency (Fe-deficient anaemia)
STORAGE IRON				
TRANSPORT IRON				
FUNCTIONAL IRON				
HAEMATOLOGICAL CHANGES	Normal	↓ Ferritin	↓ Ferritin; ↓ SI; ↓ Tsat; ↓ Hep; ↑ TIBC; ↑ sTfR; ↑ rbZP	↓ Ferritin; ↓ SI; ↓ Tsat; ↓ Hep; ↓ Hb; ↓ MCV; ↓ MCH; ↑ sTfR; ↑ rbZP

REPLETE
DEPLETED

Fe: iron; Hb: haemoglobin; Hep: hepcidin; MCH: mean corpuscular hemoglobin;
MCV: mean corpuscular volume; rbZP: red blood cell zinc protoporphyrin; SI: serum iron;
sTfR: serum transferrin receptor; TIBC: total iron-binding capacity; Tsat: transferrin saturation.

Source: *Nutritional anaemias: tools for effective prevention and control*. Geneva: World Health Organization; 2017

- Iron Availability in Food**

Several dietary factors have been identified which positively or negatively influenced the absorption of the dietary iron. The absorption of heme iron from animal sources such as meat, chicken, egg, fish and other seafood are very effective compared to non-heme iron from vegetable sources such as peria, kangkung, spinach and legum. The absorption of non-haem iron from a meal depends upon the net effect of factors enhancing iron absorption (ascorbic acid and organic acids; meat, chicken, fish and other seafood; fermented vegetables, fermented soy sauces). There are several foods containing inhibiting factors for iron absorption such as phytates and inositol phosphates; iron-binding polyphenols; calcium; soy proteins and vegetable proteins.

Table 15.7: Factors influencing dietary iron absorption:

CATEGORY	FACTORS
Haem iron absorption	a) Amount of dietary haem iron, especially from meat. b) Content of calcium in meals (e.g. from milk, cheese). c) Food preparation (i.e. time, temperature)
Non-haem iron absorption	Amount of potentially available non-haem iron (includes adjustment for fortification iron and contamination iron)
Balance between the following enhancing and inhibiting factors:	Enhancing factors: a) Ascorbic acid (e.g. certain fruit juices, fruits, potatoes, and certain vegetables) b) Meat, fish and other seafood c) Fermented vegetables (e.g. sauerkraut), fermented soy sauces, etc.
	Inhibiting factors: a) Phytate and other lower inositol phosphates (e.g. bran products, bread made from high-extraction flour, breakfast cereals, oats, rice — especially unpolished rice, pasta products, cocoa, nuts, soya beans, and peas) b) Iron-binding phenolic compounds (e.g. tea, coffee, cocoa, certain spices, certain vegetables, and most red wines) c) Calcium (e.g. from milk, cheese) d) Soya

Source: WHO/FAO (2004)

15.4.2 Diabetes in pregnancy

- The recommendations for diabetes in pregnancy are based on CPG Management of Diabetes in Pregnancy (2017) and Medical Nutrition Therapy Guidelines for Type 2 Diabetes Mellitus 2nd Edition (2014).
- Pregnant women with the following conditions should be referred to a dietitian for early initiation of Medical Nutrition Therapy (MNT):
 - at risk for GDM
 - pre-existing diabetes
 - at diagnosis of GDM
 - at initiation of insulin therapy
 - postpartum care
- Medical nutrition therapy (MNT) consists of a nutritional diagnosis and therapy, which includes dietary intervention and counselling.

- Energy prescription should be individualised based on pre pregnancy BMI, weight gain, foetal growth pattern, physical activity, food and blood glucose records.
 - For women with normal pre pregnancy BMI, energy prescriptions (approximately ranges from 30-35 kcal/kg body weight depending on activity level) should be given as per normal pregnancy based on the Recommended Nutrient Intakes for Malaysia.
 - For obese women, a 30-33% of calorie restriction of their estimated energy needs (approximately 25 kcal/kg body weight) can be prescribed to reduce the rate of GWG without inducing maternal ketosis and compromising foetal growth or birth weight. The GWG must be closely monitored, and the additional energy needs during pregnancy should be modified accordingly.
- Main focus of MNT in diabetes during pregnancy is on carbohydrate (CHO)-controlled meal plan. The aim is to achieve and maintain optimum glycaemic levels and appropriate GWG, while meeting essential nutrients to promote positive maternal and foetal outcomes.
- A minimum of 175 g CHO daily has been recommended.
- A general recommendation of macronutrients distribution to promote a balanced diet and influence glycemic control would be:
 - Carbohydrate : 45-60% of total energy requirement
 - Protein : 15-20% of total energy requirement
 - Fat : 25-35% of total energy requirement

The steps and sequence of carbohydrate prescription and distribution are shown in **Figure 15.2**.
- Both the amount and type of CHO influence glycaemic control. The type of CHO is best described using the glycaemic index (GI) concept. A low-GI diet was more effective in reducing insulin requirement compared with high-GI diet. The components of CHO-meal plan include the following:
 - Monitoring of total CHO intake using grams, exchange list, household or hand measures as long as it is practical for women to comprehend and follow. An example of carbohydrate exchanges in food as illustrated in **Figure 15.3** and **Figure 15.4**.
 - Distributing total CHO exchanges according to SMBG, lifestyle and medications.
 - Choosing the appropriate type of CHO which is lower in GI.

- Sucrose (e.g. sugars) intake must be counted as part of the total CHO intake; excess sucrose intake contributes to calories and may cause excessive GWG.
- Non-nutritive sweeteners do not impact glycaemic level. However, it should not exceed the acceptable recommended daily intake.

Figure 15.2 : Carbohydrate prescription and distribution steps and sequence

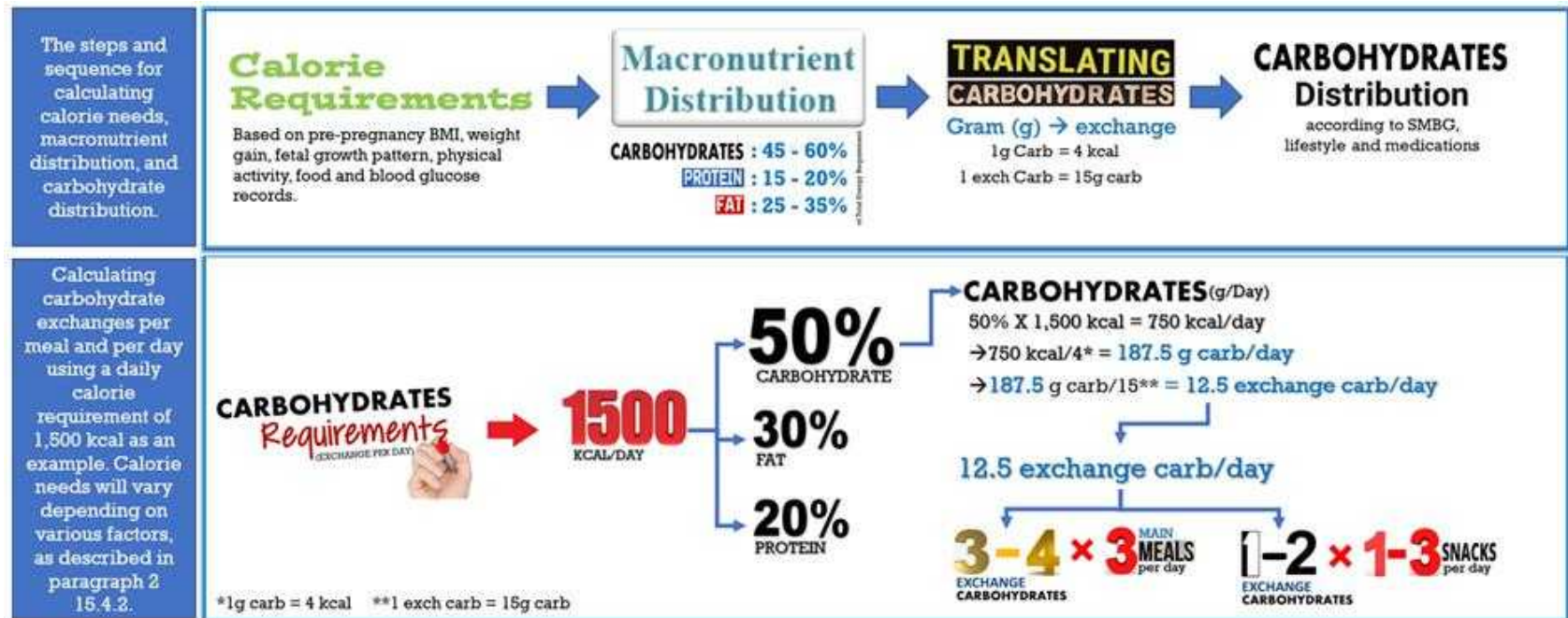
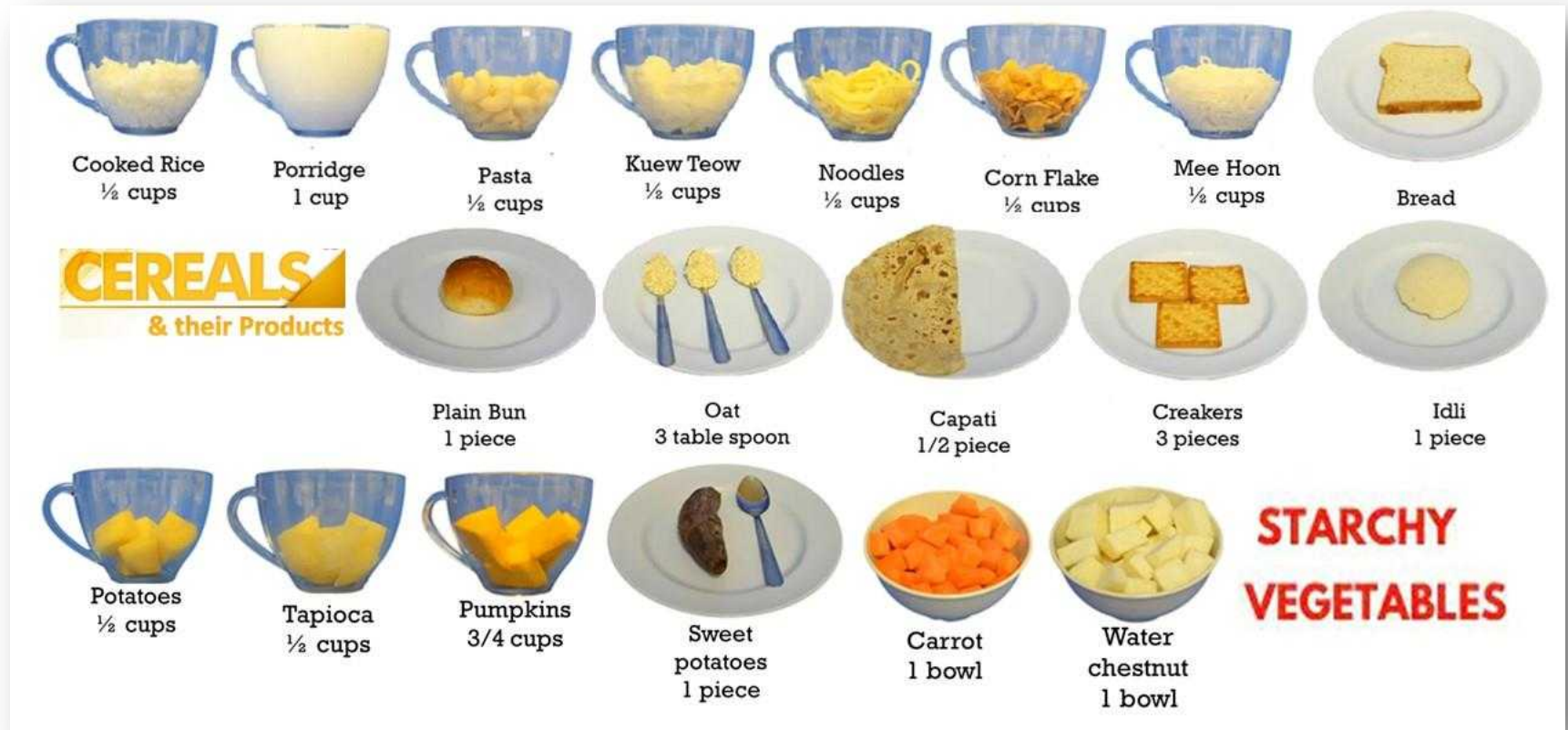


Figure 15.3: Carbohydrate exchange for cereals, cereal products and starchy vegetable



Notes: Each photo represents 1 exchange carbohydrate equivalent to 15g carbohydrate.

Figure 15.4: Carbohydrate exchange for fruits and milk



Notes: Each photo represent 1 exchange carbohydrate equivalent to 15g carbohydrate.

a. Medical Nutrition Therapy and Nutrition-related Implications: Assessment

Nutrition assessment consist of anthropometric, biochemical, clinical and dietary. The nutrition goals for preeclampsia are the same as those for a typical pregnancy. There are a few additional factors to consider, however.

- Anthropometric: Rapid weight gain (e.g., >1kg/week or 3kg/month) may suggest preeclampsia
- Individual and diagnosis-specific biochemical and clinical evaluation.
- Dietary:
 - Fibre, vitamins & minerals (vegetable and fruits)
 - Calcium (dairy, non-dairy, and supplements)
 - Sodium

b. Medical Nutrition Therapy and Nutrition-related Implications: Intervention

- Maternal weight gain is one of the important indicators for monitoring. Rapid weight gain >2kg/week may suggest / increase risk of preeclampsia.
- In general, nutrition interventions may be able to help prevent preeclampsia (or its progression), and the dietitian can assist in dietary interventions through the early detection of symptoms. Once the condition has progressed, however, medical therapies that are beyond the scope of nutrition practice are often needed.
- Patients with hypertension are encouraged to eat more fruits, vegetables, whole grain and low-fat dairy foods. DASH (Dietary Approaches to Stop Hypertension) recommends eating 4 to 5 servings of fruits, 4 to 5 servings of vegetables and 1 serving of whole grain per day.
- Fat consumption should be limited to 30–35 percent of total energy consumption. Limit consumptions of food high in saturated fat, cholesterol, and trans-fats. Choose healthy source of protein, fish, poultry, and nuts.
- The sodium needed is the same as for a normal person, which is 2000 mg per day. Restriction of sodium intake is sometimes recommended to reduce risk of preeclampsia. However, a

Cochrane Database Syst Rev. found no evidence to support this recommendation (Arvizu M et.al, 2020).

- Low dietary calcium may increase risk of hypertension during pregnancy. Pregnant mothers are encouraged to take adequate dietary calcium intake 1000-1500mg per day. Dairy products are rich sources of protein and calcium. Choose fat-free (skim) or low fat (1%) liquid or powdered milk, fat free or low fat yoghurt and low fat cheese. Condensed milk, evaporated milk and non-dairy creamer are not recommended. Alternatively, for those individuals with lactose intolerance, calcium enriched soy milk and other calcium rich foods such as green leafy vegetables can be taken as an alternative.
- In populations with low dietary calcium intake, WHO recommends daily calcium supplementation (1.5 g–2.0 g oral elemental calcium) is recommended for pregnant women to reduce the risk of pre-eclampsia. Calcium intakes greater than 4000 mg per day can cause calcium toxicity.
- Supplementation with vitamins C and E has also been suggested to reduce the risk of pre-eclampsia but a recent study showed no benefit. Another study showed a link between a food pattern high in fibre, potassium, magnesium, and calcium and lower rates of preeclampsia.

15.4.4 Adolescent pregnancy

- The growing adolescent has distinct nutritional needs. While no longer experiencing the rapid changes associated with early childhood, adolescents undergo a period of rapid growth and maturation as the ability to reproduce is gained. Therefore adolescents who become pregnant need to be given extra attention on their nutrient intake.

Table 15.8: Nutrient requirement for adolescent pregnancy

	NUTRIENTS	REQUIREMENT
1	Total Energy Intake (TEI)	• Meet energy requirement range for sedentary to moderate active lifestyle adolescent: ○ Age 13-15 years old: 1580 – 1810 kcal/day ○ Age 16-<18 years old:1660-1890 kcal/day • Obtain additional energy requirement during pregnancy: ○ First trimester: +80kcal/ day ○ Second trimester: +280kcal/ day ○ Third trimester: +470kcal/ day
2	Micronutrient: Calcium	Adolescents pregnant and lactating age 13 to 19 years old have to achieve calcium requirement 1300mg/ day.
3	Micronutrient: Iodine	Adolescents pregnant and lactating age 13 to 19 years old have an increased requirement of iodine up to 200mg everyday. This is essential for proper function of mother thyroid hormones and at the same time, infants depend on breast milk as the main source of iodine to continue the development of their brains and nervous system.

a. Gestational Weight Gain for Adolescent

Monitor adolescent GWG using the same standard range from Weight Gain during Pregnancy Reexamining Guidelines, Institute of Medicine (IOM), United States, 2009. The Pre pregnancy BMI for adolescents should be determined by the usage of adult BMI. This will result in a wider range of GWG for adolescents in pregnancy to fulfil the requirement of energy and micronutrient intake.

b. Dietary advise for pregnant adolescent

General pregnancy dietary advice applies for adolescent pregnancy with emphasis on calcium intake. Health practitioners need to consider the developmental characteristics of adolescents in order to ensure the adolescent could understand nutrition education that is being given. Things to consider is as such:

- Simple use of word / message.
- Use visual / pictographic material.
- Focus on types of food to increase the intake of certain micronutrients.

- Individualise messages and work within the pregnant adolescent's current eating habits.
- Consider to include family members in the discussion
- Highlight the negative impact on certain food to the pregnancy and foetus such as fast food, high sugar food and beverages and low nutrient snack food.
- Focus on weight gain patterns for optimal birth outcomes.

15.5 BREASTFEEDING

15.5.1 The importance of breastfeeding

- Breastfeeding is important to children, mothers and families. Breastfeeding protects the infant's health. Children who are not breastfed are more likely to be:
 - Ill or to die from infection such as diarrhoea and gastrointestinal infections, and chest infection
 - Underweight and not grow well, if they live in poor circumstances
 - Overweight and to have later heart problems, if they live in rich circumstances
- Breastfeeding is important to mother, women who do not breastfeed are more likely:
 - To develop anaemia and to retain fat deposited during pregnancy, which may result in later obesity
 - To become pregnant soon after the baby's birth
 - To develop breast cancer
 - To have hip fractures in older age
- In addition:
 - Breastmilk is readily available and free. It needs no preparation or storage
 - Breastfeeding is simple, with no equipment or preparation needed
 - If a baby is not breastfed, the family will need to buy replacement milk for the baby and find time to prepare feeds and keep feeding equipment clean
 - If a baby is not breastfed, there may be loss of income through a parent's absence from work to care for an ill child
- Mother's milk is all a baby needs:
 - Exclusive breastfeeding is strongly recommended for the first six months. The baby does not need water, other fluids, or foods during this time

- Breastfeeding continues to be important after the first 6 months even when other foods are given to the baby
- A mother's milk is especially suited for her own baby and changes from day to day, month to month, and feed to feed to meet the baby's needs. The baby learns the tastes of the family foods through the flavours of breastmilk
- Mother's milk is unique (special). Human milk is a living fluid that actively protects against infection. Artificial formula provides no protection from infections

15.5.2 Practices that can help breastfeeding to go well

- Hospital practices can help breastfeeding to go well. These practices include to:
 - Have a companion with you during labour, which can help you to be more comfortable and in control
 - Avoid labour and birth interventions such as sedating pain relief and caesarean sections unless they are medically necessary
 - Have skin-to-skin contact immediately after birth, which keeps the baby warm and gives an early start to breastfeeding
 - Keep the baby beside you (rooming-in or bedding-in), so that your baby is easy to feed as well as safe
 - Learn feeding signs in your baby so that feeding is baby-led rather than to a schedule
 - Feeding frequently helps to develop good milk supply
 - Breastfeeding exclusively with no supplements, bottles or artificial teats
 - It is important to learn how to position and attach the baby for feeding. The hospital staff will teach the mother how to breastfeed. Most women can breastfeed and help is available if needed

15.5.3 Information for HIV positive mothers on breastfeeding

- All pregnant women are offered voluntary and confidential HIV counselling and testing. If a woman is HIV-infected there is a risk of transmission to the baby during the pregnancy and birth, as well as during breastfeeding. If the pregnant woman knows that she is HIV-positive then she can make informed decisions.
- About 5-15% of babies (one in 20 to one in seven) born to women who are HIV infected will become HIV-positive through breastfeeding.

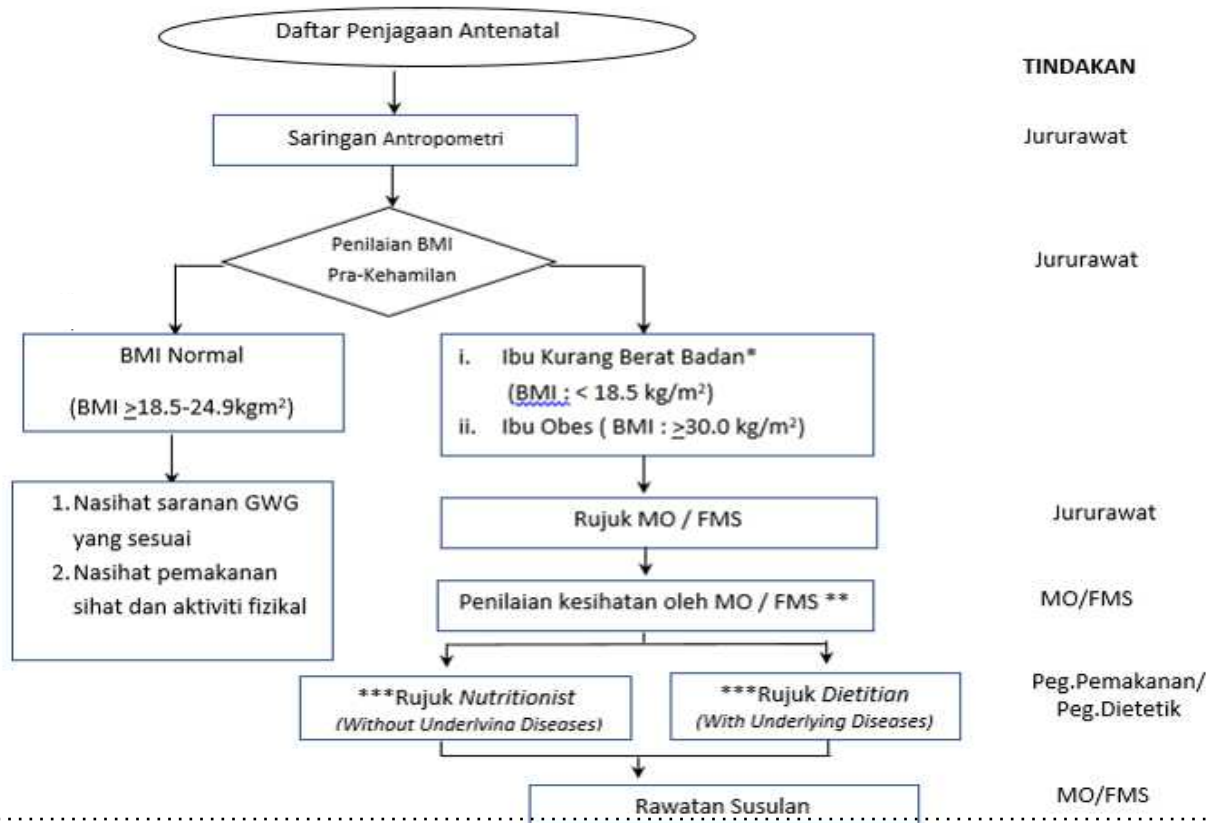
This means most infants born to women who are HIV-positive will be infected through breastfeeding.

- In some settings, the risk to the child of illness and death from not exclusively breastfeeding is higher than the risk of HIV transmission from breastfeeding. One of the reasons that individual counselling is so important is that it gives the mother the information they need to make the informed choices about how to feed their babies in their own situations.
- The majority of women are not infected with HIV. Breastfeeding is recommended for:
 - Women who do not know their status, and
 - Women who are HIV-negative

Reference:

1. Implementation Guidance – Protecting, promoting and supporting Breastfeeding in facilities providing maternity and newborn services : the revised Baby-Friendly Hospital Initiative – UNICEF/WHO2018

Carta Alir Pemantauan Berat Badan Wanita Hamil Semasa Pemeriksaan Antenatal Pertama



NOTA:

* Sekiranya ibu berada dalam kategori BMI kurang berat badan, penilaian perlu dibuat untuk mengenalpasti punca / penyakit berkaitan (contoh: TB, hipertiroidism, kanser dll) dan rawatan/ intervensi berdasarkan diagnosa semasa.

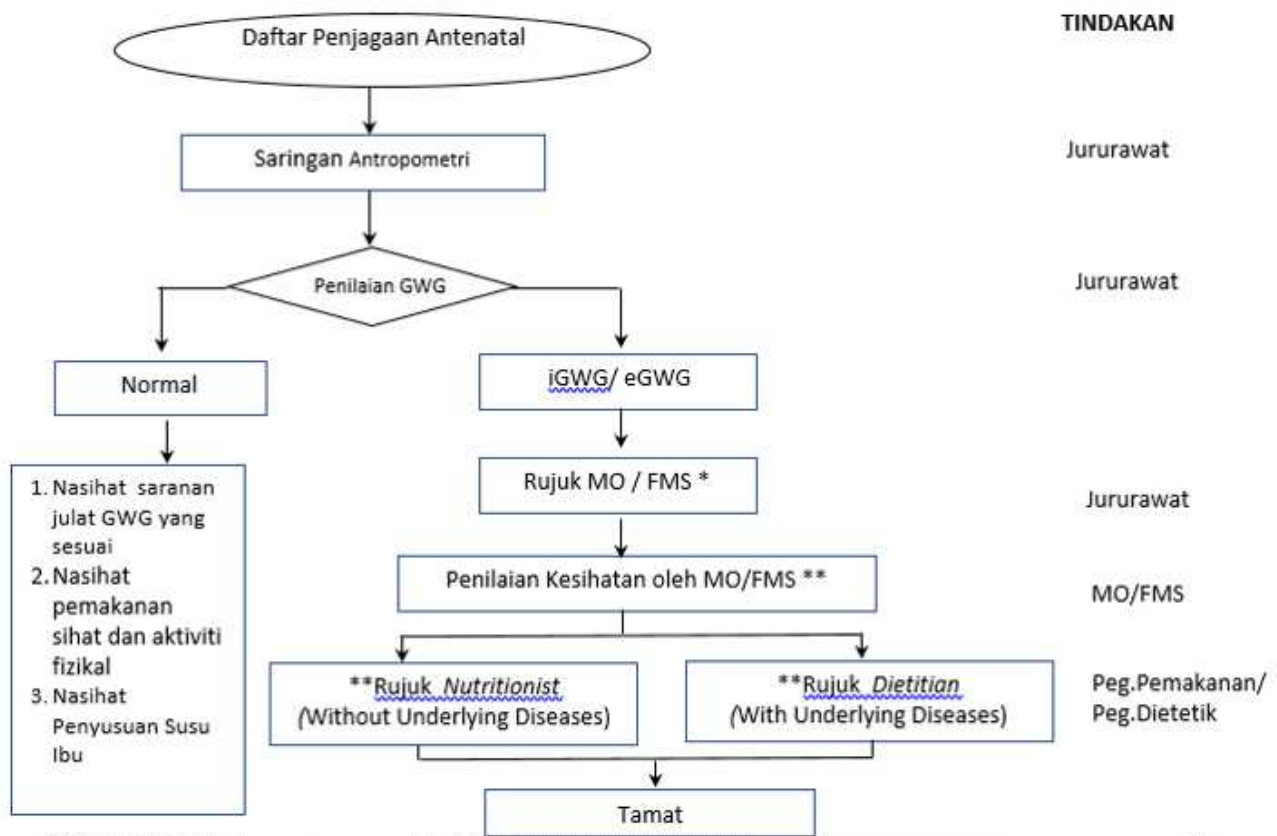
*Bagi kes ibu obes, peningkatan berat badan semasa hamil secara berlebihan perlu diberi perhatian. Penilaian teliti perlu dilakukan bagi menentukan punca penyebab/ penyakit berkaitan seperti risiko pre-eklampsia, GDM dan makrosomia/ polyhydramnios

** Penilaian oleh MO/FMS:

- 1 Sejarah
 - tabiat pemakanan dan aktiviti fizikal
 - perihai keluarga: obesiti, DM, HPT, dislipidaemia, penyakit kardiovaskular, kanser berkaitan obesiti, tiroid
2. Pemeriksaan klinikal
 - komorbid berkaitan obesiti, sleep apnea
 - penilaian risiko VTE
3. Siasatan:
 - OGTT
 - ECG
 - Lain-lain

*** Boleh melibatkan juga penglibatan oleh Pegawai Fisioterapi dll yang bersesuaian.

Carta Alir Pemantauan Berat Badan Wanita Hamil Semasa Pemeriksaan Antenatal Ulangan



NOTA

* Penilaian oleh MO/FMS:

1. Sejarah
 - tabiat pemakanan dan aktiviti fizikal.
 - perihal keluarga: obesiti, DM, HPT, dislipidaemia, penyakit kardiovaskular, kanser berkaitan obesiti, tiroid.
2. Pemeriksaan klinikal
 - komorbid berkaitan obesiti, sleep apnea.
 - penilaian risiko VTE.
3. Siasatan:
 - OGTT, ECG atau lain-lain ujian jika perlu (RP, LFT, TFT, Echo, dll).
4. Sekiranya berlaku eGWG, penilaian teliti perlu dilakukan bagi menentukan punca penyebab dan risiko penyakit berkaitan seperti:
 - Risiko pre-eklampsia.
 - Kemungkinan gestational diabetes.
 - Makrosomia/ polyhydramnios.
5. Sekiranya ibu mengalami iGWG penilaian perlu dilakukan untuk mengenalpasti punca penyebab dan risiko penyakit berkaitan (ie TB, hipertiroidism, kanser dll) dan rawatan/ intervensi berdasarkan diagnosa semasa.

**

1. Boleh melibatkan Pegawai Fisioterapi sekiranya bersesuaian.
2. Ibu mengalami iGWG boleh dirujuk kepada PSP untuk didaftarkan dalam program Makanan Tambahan Susu Tepung Penuh Krim (STPK) sekiranya memenuhi kriteria kelayakan bantuan.

**PEMANTAUAN PENINGKATAN BERAT BADAN IBU ANTENATAL
(GESTATIONAL WEIGHT GAIN - GWG)**

Tinggi : . meter
 Berat ibu sebelum hamil: . kg

*Nota: Jika tiada maklumat berat ibu sebelum hamil, berat ibu semasa booking POG ≤12/52 boleh digunakan.
 Jika POG > 12/52, kategori BMI boleh didapati secara anggaran berdasarkan maklumat ibu.*

Nilai BMI sebelum hamil . kg/m²

Status BMI sebelum hamil : Kurang Berat Badan / Normal / Lebih Berat Badan / Obes (Bulatkan yang berkenaan)

Jadual Berat Badan Sepanjang Kehamilan

Bil (a)	Tarikh Lawatan (b)	Minggu Gestasi (c)	Berat Badan (kg) (d)	Peningkatan Berat Badan (kg) (e)	Kadar Peningkatan Berat Badan kg/minggu (f)
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
Jumlah Peningkatan Berat Badan (kg) Sepanjang Kehamilan					



**PANDUAN PEMAKANAN BAGI IBU HAMIL YANG MENGALAMI ANEMIA
KEKURANGAN ZAT BESI (IRON DEFICIENCY ANAEMIA-IDA)**

PERTANYAAN	PENERANGAN
Apakah anemia dan bagaimana ia boleh terjadi?	Anemia merupakan keadaan di mana kepekatan hemoglobin (Hb) dalam sel darah merah adalah rendah. Terdapat pelbagai faktor yang boleh menyebabkan kepekatan Hb ini menjadi rendah iaitu: • Kekurangan mikronutrien terutama zat besi (IDA), folat atau vitamin B6. • Penyakit berjangkit atau kronik seperti HIV, malaria atau jangkitan parasit. • Gangguan genetik hemoglobin seperti talasemia atau <i>sickle cell disorder</i>. Anemia Kekurangan Zat Besi (IDA) berlaku akibat bekalan zat besi yang kurang dalam tubuh badan untuk membentuk hemoglobin dalam sel darah merah secukupnya. Anemia akan dikesan melalui paras hemoglobin (Hb) dalam darah yang kurang daripada 11 gm% semasa hamil. Pemeriksaan profil darah selanjutnya perlu dijalankan bagi mengesahkan anemia yang berlaku berpunca daripada kekurangan zat besi.
Apakah kesan anemia?	Sebelum hamil: • Mudah penat, tidak bersemangat, cepat mengantuk, kurang tumpuan dan fokus, dan mudah tercungap (<i>breathless</i>). • Risiko untuk mendapat anemia semasa hamil lebih tinggi dan boleh menjejaskan kesihatan bayi yang akan dikandung kelak. Semasa hamil: • Meningkatkan risiko kesihatan kepada ibu dan bayi yang dikandung seperti kelahiran pra-matang dan bayi lahir kurang berat. • Meningkatkan risiko kepada kematian ibu anak akibat pelbagai komplikasi. Semasa berpantang dan menyusui: • Menyebabkan ibu menjadi terlalu penat, lemah, hilang fokus dan tidak bertenaga untuk menguruskan diri dan anak. • Ibu lebih cenderung mengalami tekanan perasaan.

PERTANYAAN	PENERANGAN
Bagaimana untuk mencegah dan merawat?	Untuk meningkatkan paras Hb dan seterusnya mengatasi masalah IDA ini, ibu hamil boleh meningkatkan pengambilan zat besi melalui: 1. Pengambilan makanan kaya dengan sumber zat besi. 2. Pengambilan pil zat besi dan vitamin C. <i>*Di Klinik Kesihatan, sekiranya kaedah di atas tidak dapat membantu meningkatkan paras Hb ibu, pegawai perubatan mungkin akan menyarankan rawatan lain contohnya seperti parenteral (IV/IM).</i>
Apakah jenis makanan yang perlu ibu ambil untuk bantu atasi masalah ini?	1. Tingkatkan kekerapan pengambilan makanan kaya zat besi seperti: Zat besi heme dari sumber haiwan seperti: Kerang, ikan bilis, hati ayam/lembu, ayam, daging, telur, ikan seperti cencaru, kembung, tenggiri, tongkol. Zat besi non-heme dari sumber tumbuhan seperti: kacang kuda, kacang hijau, pucuk mamin, peria, pegaga gajah, ulam raja, bayam, kangkung, pucuk paku, taueh kasar, celery dan lain-lain 2. Tingkatkan kekerapan pengambilan makanan mengandungi Vitamin C untuk bantu tingkatkan penyerapan zat besi seperti: Buah - Jambu, limau, betik, limau kasturi, mangga, nanas, cempedak dan lain-lain buah-buahan. Sayur – Bunga kobis, brokoli, tomato, bayam dan lain-lain sayuran hijau. 3. Elakkan pengambilan makanan dan minuman yang tinggi kandungan kafein dan tanin untuk mengelakkan gangguan penyerapan zat besi kesan interaksi antara makanan/suplemen zat besi seperti berikut: Kopi, teh, koko, wain merah. 4. Jarakkan pengambilan makanan dan minuman di bawah dengan pengambilan pil atau makanan kaya zat besi bagi mengelakkan gangguan penyerapan zat besi: Susu, soya, keju, atau lain-lain produk makanan tinggi kalsium.

PERTANYAAN	PENERANGAN
Bagaimana cara pengambilan pil zat besi?	- Suplemen yang biasanya dibekalkan oleh Klinik Kesihatan kepada semua ibu hamil anemia bagi membantu mengatasi masalah anemia zat besi merangkumi empat jenis pil iaitu: - Zat besi (<i>ferrous fumarate</i>), asid folik, Vitamin C, dan Vitamin B Kompleks. - Makan pil zat besi sejam sebelum makan atau dua jam selepas makan bagi penyerapan zat besi yang optima. Dapatkan nasihat lebih lanjut daripada anggota kesihatan sekiranya mempunyai alahan atau masalah dalam pengambilan pil zat besi. - Makan pil /suplemen tersebut bersama air kosong atau bersama jus buah yang mengandungi Vit C membantu penyerapan zart besi. - Sekiranya pihak klinik kesihatan membekalkan suplemen zat besi seperti Iberet, Zincofer, Obimin atau lain-lain, sila rujuk kepada arahan di risalah produk atau dapatkan nasihat farmasi bagi pengambilan yang betul.

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